

Program overview

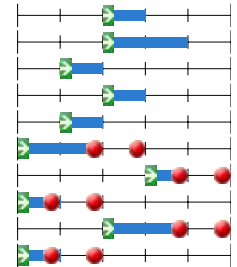
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Year 2022/2023
Organization Aerospace Engineering
Education Master Aerospace Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master LR 2022							
Master Aerospace Engineering 2022							
Track 1: Aerodynamics and Wind Energy							
Profile I: Aerodynamics							
AE4115	Experimental Simulations	3					
AE4120	Viscous Flows	3					
AE4130-22	Aircraft Aerodynamics	4					
AE4135	Rotor / wake Aerodynamics	4					
AE4136-22	CFD 2: Discretization techniques	3					
AE4140	Gas Dynamics	3					
AE4180	Flow Measurement Techniques	3					
AE4202	CFD for Aerospace Engineers	3					
AE5051	Internship	15					
AE5122	Thesis Aerodynamics & Wind Energy	45					
WI3150TU	Partial Differential Equations A	3					
WM0324LR	Ethics and Engineering for Aerospace Engineering	3					
Preferred Electives Profile Aerodynamics							
AE4010	Research Methodologies	2					
AE4117	Fluid-Structure Interaction	4					
AE4138-18	CFD 4: Uncertainty Quantification	3					
AE4139	CFD 3: Large Eddy Simulation	3					
AE4143	Hypersonic Aerodynamics	3					
AE4185	Fluid Flow Data Processing and Visualization	3					
AE4240	Advanced Aircraft Design I	4					
AE4260A	Fundamental of Aeroacoustics	2					
AE4260B	Experimental Applications of Aeroacoustics	2					
AE4262	Combustion for propulsion and power technologies	4					
AE4431	Aircraft noise and emissions	3					
AE4463P	Advanced Aircraft Noise Modeling and Measurement	3					
AE4ASM506	Fundamentals of Aeroelasticity	3					
AE4ASM514TU	Continuum Mechanics	4					
AE4T40	Airborne Wind Energy	3					
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W21-14	Wind Turbine Aeroelasticity	2					
AE4W30	Wind Resource and Wind Farm Yield	4					
AE4W31	Floating Offshore Wind Energy	3					
ME45001	Advanced Heat Transfer	4					
ME45030	Turbulence	5					
ME46060	Engineering Optimization: Concepts and Applications	3					
WI3151TU	Partial Differential Equations B	3					
WI4014TU	Numerical Analysis	6					
WI4201	Scientific Computing	6					
WI4212	Advanced Numerical Methods	6					
Profile II: Wind Energy							
AE4120	Viscous Flows	3					
AE4135	Rotor / wake Aerodynamics	4					
AE4202	CFD for Aerospace Engineers	3					
AE4ASM506	Fundamentals of Aeroelasticity	3					
AE4T40	Airborne Wind Energy	3					
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W09	Wind Turbine Design	5					
AE4W13	Site Conditions for Wind Turbine Design	3					
AE4W21-14	Wind Turbine Aeroelasticity	2					
AE5051	Internship	15					
AE5122	Thesis Aerodynamics & Wind Energy	45					
WM0324LR	Ethics and Engineering for Aerospace Engineering	3					
Preferred Electives Profile Wind Energy							
AE4010	Research Methodologies	2					
AE4115	Experimental Simulations	3					
AE4117	Fluid-Structure Interaction	4					
AE4130-22	Aircraft Aerodynamics	4					
AE4136-22	CFD 2: Discretization techniques	3					
AE4138-18	CFD 4: Uncertainty Quantification	3					

AE4139	CFD 3: Large Eddy Simulation
AE4180	Flow Measurement Techniques
AE4260A	Fundamental of Aeroacoustics
AE4260B	Experimental Applications of Aeroacoustics
AE4463P	Advanced Aircraft Noise Modeling and Measurement
AE4W30	Wind Resource and Wind Farm Yield
AE4W31	Floating Offshore Wind Energy
ME45001	Advanced Heat Transfer
ME45030	Turbulence
WI3150TU	Partial Differential Equations A

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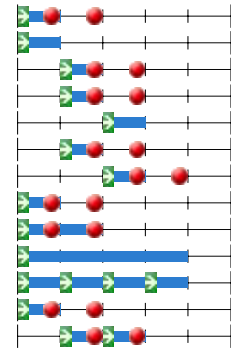


Track 2: Control and Operations

Profile I: Control and Simulation

AE4301	Automatic Flight Control System Design
AE4301P	Exercise Automatic Flight Control System Design
AE4302	Avionics and Operations
AE4304	Stochastic Aerospace Systems
AE4304P	Stochastic Aerospace Systems Practical
AE4316	Aerospace Human-Machine Systems
AE4322	Piloted Flight Simulation
AE4431	Aircraft noise and emissions
AE4441-16	Operations optimisation
AE5051	Internship
AE5322	Thesis Control and Operations
WI2056LR	Systems Theory
WM0324LR	Ethics and Engineering for Aerospace Engineering

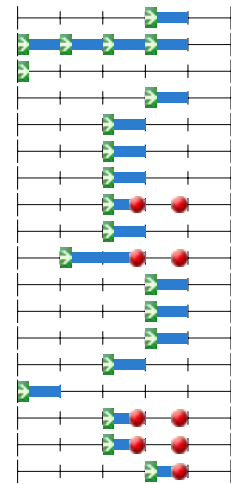
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Preferred Electives Profile Control & Simulation

AE4006	Sustainability in Engineering
AE4010	Research Methodologies
AE4270	Control and Operations Project
AE4311	Nonlinear and Adaptive Flight Control
AE4313-20	Spacecraft Attitude Dynamics and Control
AE4314-21	Helicopter Performance, Stability and Control
AE4316P	Advanced Topics in Aerospace Human-Machine Systems
AE4317	Autonomous Flight of Micro Air Vehicles
AE4320	System Identification of Aerospace Vehicles
AE4321-15	Air Traffic Management
AE4323	Real-time Distributed Flight and Space Simulation
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications
AE4351	Robust Flight Control
AE4352	Mathematical and human-inspired decision making
AE4422-20	Agent-based Modelling and Simulation in Air Transport
CS4240	Deep Learning
RO47019	Intelligent Control Systems
SC42035	Integration Project Systems and Control

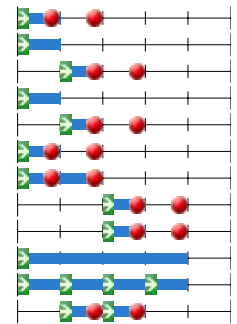
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Profile II: Sustainable Air Transport

AE4301	Automatic Flight Control System Design
AE4301P	Exercise Automatic Flight Control System Design
AE4302	Avionics and Operations
AE4422-20	Agent-based Modelling and Simulation in Air Transport
AE4423-20	Airline Planning and Optimisation
AE4431	Aircraft noise and emissions
AE4441-16	Operations optimisation
AE4462-17	Aircraft Emissions and Climate Effects
AE4465	Maintenance Modeling & Analysis
AE5051	Internship
AE5322	Thesis Control and Operations
WM0324LR	Ethics and Engineering for Aerospace Engineering

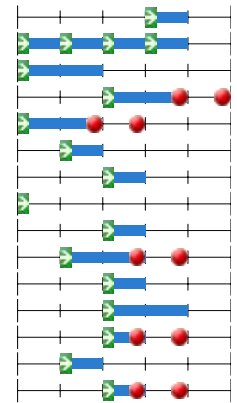
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Preferred Electives Profile Sustainable Air Transport

AE4006	Sustainability in Engineering
AE4010	Research Methodologies
AE4130-22	Aircraft Aerodynamics
AE4135	Rotor / wake Aerodynamics
AE4238	Aero Engine Technology
AE4260A	Fundamental of Aeroacoustics
AE4260B	Experimental Applications of Aeroacoustics
AE4270	Control and Operations Project
AE4320	System Identification of Aerospace Vehicles
AE4321-15	Air Traffic Management
AE4352	Mathematical and human-inspired decision making
AE4439	Air Transport Operations Modelling Project
AE4446	Airport and Cargo Operations
AE4463P	Advanced Aircraft Noise Modeling and Measurement
WI4062TU	Transport, Routing and Scheduling

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Track 3: Spaceflight

Profile I: Space Engineering

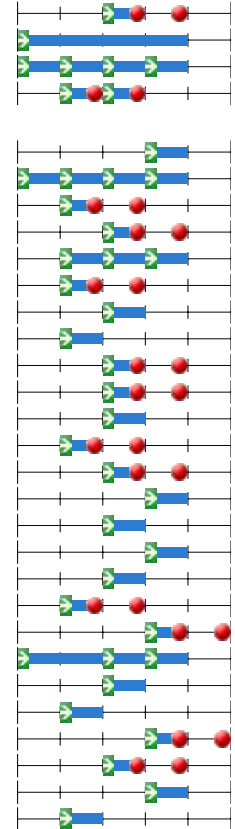
AE4874 I	Fundamentals of Astrodynamics	4	
AE4880	Optical Space Sensors	4	
AE4890-11	Planetary Sciences I	4	
AE4S01	Thermal Rocket Propulsion	4	
AE4S01P	Exercise Thermal Rocket Propulsion	2	
AE4S07	Micropropulsion	4	
AE4S10	Microsat Engineering	4	
AE4S12	Space Systems Engineering	3	
AE4S15	Space Embedded Systems	3	
AE4S52-21	Collaborative Space System Design Project	5	
AE5051	Internship	15	
AE5822	Thesis Space	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Space Engineering			
AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4499	Space Project	0	
AE4ASM526	Spacecraft Thermal Design	3	
ET3604LR	Electronic Circuits	3	
ME46060	Engineering Optimization: Concepts and Applications	3	
Profile II: Spaceflight Dynamics			
AE4866-1	Propagation and Optimisation in Astrodynamics	5	
AE4868-1	Numerical Astrodynamics	4	
AE4870A	Rocket Motion	3	
AE4870B	Re-Entry Systems	3	
AE4872	Satellite Orbit Determination	6	
AE4874 I	Fundamentals of Astrodynamics	4	
AE4876-11	Planetary Sciences II	4	
AE4890-11	Planetary Sciences I	4	
AE4894	Practical Astrodynamics	3	
AE4S12	Space Systems Engineering	3	
AE5051	Internship	15	
AE5822	Thesis Space	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Spaceflight Dynamics			
AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4143	Hypersonic Aerodynamics	3	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4499	Space Project	0	
AE4889	Special Topics in Astrodynamics	2	
AE4895	Measurement Strategies for Planetary Science Missions	3	
WI4771TU	Object Oriented Scientific Programming with C++	3	
Profile III: Planetary Exploration			
AE4866-1	Propagation and Optimisation in Astrodynamics	5	
AE4868-1	Numerical Astrodynamics	4	
AE4870B	Re-Entry Systems	3	
AE4872	Satellite Orbit Determination	6	
AE4874 I	Fundamentals of Astrodynamics	4	
AE4876-11	Planetary Sciences II	4	
AE4880	Optical Space Sensors	4	
AE4890-11	Planetary Sciences I	4	
AE4S12	Space Systems Engineering	3	
AE5051	Internship	15	
AE5822	Thesis Space	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Planetary Exploration			
AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4499	Space Project	0	
AE4893	Physics of Planetary Interiors	4	
AE4895	Measurement Strategies for Planetary Science Missions	3	
AE4S52-21	Collaborative Space System Design Project	5	
ME46060	Engineering Optimization: Concepts and Applications	3	
Track 4: Aerospace Structures and Materials			
Profile I: Materials and Manufacturing			
AE4ASM001	Design of lightweight structures I: Composites & Metals	3	
AE4ASM002	Designing Materials with Aerospace Specific Properties	3	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3	
AE4ASM005	Fatigue of Structures & Materials	3	
AE4ASM101TU-22	Polymer Science	4	
AE4ASM110-22	Polymer Composite Manufacturing	4	
AE4ASM515-22	Characterization of materials and components	4	

AE4ASM521	Additive Manufacturing	3	
AE5051	Internship	15	
AE5722	Thesis Aerospace Structures & Materials	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Materials and Manufacturing			
AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4ASM103	Functional Coatings	3	
AE4ASM105	Trinity Exercise	4	
AE4ASM106	Stability & Analysis of Structures I	3	
AE4ASM108	Experimental Techniques & NDT	3	
AE4ASM109	Design & Analysis of Composite Structures I	5	
AE4ASM503	Sheet Metal Forming	3	
AE4ASM504	Structural Integrity and Maintenance	3	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4ASM508	Design of Self-healing materials	3	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE4ASM513	Forensic Engineering	3	
AE4ASM514TU	Continuum Mechanics	4	
AE4ASM516	Material Selection in Mechanical Design	3	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
AE4ASM520	Industrial Composite Manufacturing	3	
AE4ASM521	Additive Manufacturing	3	
AE4ASM522	Applied Aircraft Aeroelasticity	3	
AE4ASM523	Design of Spacecraft and Launcher Structures	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
Profile II: Design and Safety of Structures			
AE4ASM001	Design of lightweight structures I: Composites & Metals	3	
AE4ASM002	Designing Materials with Aerospace Specific Properties	3	
AE4ASM003	Linear Modeling (incl. F.E.M.)	3	
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3	
AE4ASM005	Fatigue of Structures & Materials	3	
AE4ASM106	Stability & Analysis of Structures I	3	
AE4ASM108	Experimental Techniques & NDT	3	
AE4ASM109	Design & Analysis of Composite Structures I	5	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE5051	Internship	15	
AE5722	Thesis Aerospace Structures & Materials	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Design and Safety of Structures			
AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4ASM101TU-22	Polymer Science	4	
AE4ASM103	Functional Coatings	3	
AE4ASM105	Trinity Exercise	4	
AE4ASM110-22	Polymer Composite Manufacturing	4	
AE4ASM503	Sheet Metal Forming	3	
AE4ASM504	Structural Integrity and Maintenance	3	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM508	Design of Self-healing materials	3	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE4ASM513	Forensic Engineering	3	
AE4ASM514TU	Continuum Mechanics	4	
AE4ASM515-22	Characterization of materials and components	4	
AE4ASM516	Material Selection in Mechanical Design	3	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
AE4ASM520	Industrial Composite Manufacturing	3	
AE4ASM521	Additive Manufacturing	3	
AE4ASM522	Applied Aircraft Aeroelasticity	3	
AE4ASM523	Design of Spacecraft and Launcher Structures	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
Profile III: Structures for Space			
AE4870A	Rocket Motion	3	
AE4870B	Re-Entry Systems	3	
AE4ASM001	Design of lightweight structures I: Composites & Metals	3	
AE4ASM002	Designing Materials with Aerospace Specific Properties	3	
AE4ASM003	Linear Modeling (incl. F.E.M.)	3	
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3	
AE4ASM005	Fatigue of Structures & Materials	3	
AE4ASM106	Stability & Analysis of Structures I	3	
AE4ASM523	Design of Spacecraft and Launcher Structures	3	

AE4ASM524	Spacecraft Structures Development	3
AE5051	Internship	15
AE5722	Thesis Aerospace Structures & Materials	45
WM0324LR	Ethics and Engineering for Aerospace Engineering	3

Preferred Electives Profile Structures for Space

AE4006	Sustainability in Engineering	4
AE4010	Research Methodologies	2
AE4ASM101TU-22	Polymer Science	4
AE4ASM103	Functional Coatings	3
AE4ASM105	Trinity Exercise	4
AE4ASM108	Experimental Techniques & NDT	3
AE4ASM109	Design & Analysis of Composite Structures I	5
AE4ASM110-22	Polymer Composite Manufacturing	4
AE4ASM503	Sheet Metal Forming	3
AE4ASM504	Structural Integrity and Maintenance	3
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
AE4ASM506	Fundamentals of Aeroelasticity	3
AE4ASM508	Design of Self-healing materials	3
AE4ASM510	Design & Analysis of Composite Structures II	3
AE4ASM511	Stability & Analysis of Structures II	3
AE4ASM513	Forensic Engineering	3
AE4ASM514TU	Continuum Mechanics	4
AE4ASM515-22	Characterization of materials and components	4
AE4ASM516	Material Selection in Mechanical Design	3
AE4ASM517	Aircraft Manufacturing Laboratory	6
AE4ASM520	Industrial Composite Manufacturing	3
AE4ASM521	Additive Manufacturing	3
AE4ASM522	Applied Aircraft Aeroelasticity	3
AE4ASM525	Materials for Space	3
AE4ASM526	Spacecraft Thermal Design	3
AE4S10	Microsat Engineering	4



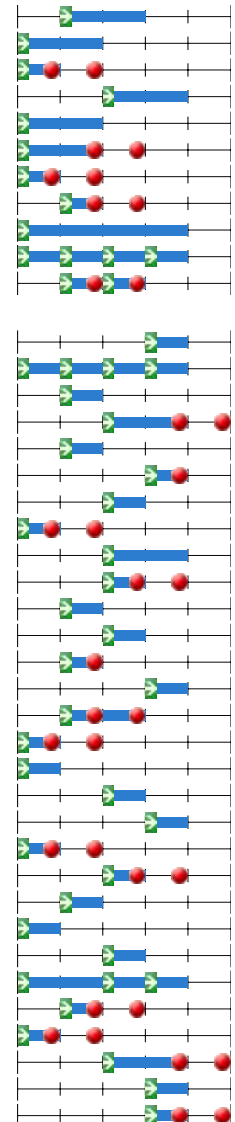
Track 5: Flight Performance and Propulsion

Profile I: Flight Performance

AE4115	Experimental Simulations	3
AE4130-22	Aircraft Aerodynamics	4
AE4202	CFD for Aerospace Engineers	3
AE4204	Knowledge Based Engineering	4
AE4205	MDO for Aerospace Applications	4
AE4238	Aero Engine Technology	4
AE4240	Advanced Aircraft Design I	4
AE4ASM506	Fundamentals of Aeroelasticity	3
AE5051	Internship	15
AE5222	Thesis Flight Performance & Propulsion	45
WM0324LR	Ethics and Engineering for Aerospace Engineering	3

Preferred Electives Profile Flight Performance

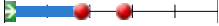
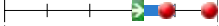
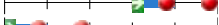
AE4006	Sustainability in Engineering	4
AE4010	Research Methodologies	2
AE4120	Viscous Flows	3
AE4135	Rotor / wake Aerodynamics	4
AE4136-22	CFD 2: Discretization techniques	3
AE4138-18	CFD 4: Uncertainty Quantification	3
AE4139	CFD 3: Large Eddy Simulation	3
AE4140	Gas Dynamics	3
AE4180	Flow Measurement Techniques	3
AE4206	Turbomachinery	3
AE4260A	Fundamental of Aeroacoustics	2
AE4260B	Experimental Applications of Aeroacoustics	2
AE4261	Internal Flows	3
AE4262	Combustion for propulsion and power technologies	4
AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
AE4301	Automatic Flight Control System Design	3
AE4301P	Exercise Automatic Flight Control System Design	1
AE4314-21	Helicopter Performance, Stability and Control	4
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
AE4431	Aircraft noise and emissions	3
AE4462-17	Aircraft Emissions and Climate Effects	4
AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
AE4ASM003	Linear Modeling (incl. F.E.M)	3
AE4ASM109	Design & Analysis of Composite Structures I	5
AE4ASM517	Aircraft Manufacturing Laboratory	6
ET4117	Electrical Machines and Drives	4
ME45001	Advanced Heat Transfer	4
ME45025	Introduction to Multiphase Flow	5
ME46060	Engineering Optimization: Concepts and Applications	3
SET3085	Hydrogen Technology	4



Profile II: Propulsion and Power

AE4130-22	Aircraft Aerodynamics	4	
AE4202	CFD for Aerospace Engineers	3	
AE4205	MDO for Aerospace Applications	4	
AE4206	Turbomachinery	3	
AE4238	Aero Engine Technology	4	
AE4261	Internal Flows	3	
AE4262	Combustion for propulsion and power technologies	4	
AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5	
AE5051	Internship	15	
AE5222	Thesis Flight Performance & Propulsion	45	
ME45001	Advanced Heat Transfer	4	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Propulsion and Power			
AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4115	Experimental Simulations	3	
AE4120	Viscous Flows	3	
AE4136-22	CFD 2: Discretization techniques	3	
AE4138-18	CFD 4: Uncertainty Quantification	3	
AE4139	CFD 3: Large Eddy Simulation	3	
AE4140	Gas Dynamics	3	
AE4180	Flow Measurement Techniques	3	
AE4204	Knowledge Based Engineering	4	
AE4240	Advanced Aircraft Design I	4	
AE4260A	Fundamental of Aeroacoustics	2	
AE4260B	Experimental Applications of Aeroacoustics	2	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4431	Aircraft noise and emissions	3	
AE4462-17	Aircraft Emissions and Climate Effects	4	
AE4463P	Advanced Aircraft Noise Modeling and Measurement	3	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM109	Design & Analysis of Composite Structures I	5	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
AE4S01	Thermal Rocket Propulsion	4	
ET4117	Electrical Machines and Drives	4	
ME45025	Introduction to Multiphase Flow	5	
ME46060	Engineering Optimization: Concepts and Applications	3	
MS43310	Materials at High Temperature	4	
SET3085	Hydrogen Technology	4	
Track 6: European Wind Energy Master Track Rotor Design			
Profile I: Aerodynamics			
AE4135	Rotor / wake Aerodynamics	4	
AE4180	Flow Measurement Techniques	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4W09	Wind Turbine Design	5	
AE4W13	Site Conditions for Wind Turbine Design	3	
AE4W21-14	Wind Turbine Aeroelasticity	2	
AE5912	Thesis Wind Energy Rotor Design	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Courses at DTU			
Preferred Electives Profile Aerodynamics			
AE4010	Research Methodologies	2	
AE4117	Fluid-Structure Interaction	4	
AE4W31	Floating Offshore Wind Energy	3	
AE5055	Professional Training	6	
WM1115TU	Dutch Elementary 1	3	
Profile II: Structures and Composites			
AE4ASM105	Trinity Exercise	4	
AE4ASM109	Design & Analysis of Composite Structures I	5	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4W09	Wind Turbine Design	5	
AE4W21-14	Wind Turbine Aeroelasticity	2	
AE5912	Thesis Wind Energy Rotor Design	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Courses at DTU			
Preferred Electives Profile Structures and Composites			
AE4010	Research Methodologies	2	
AE4117	Fluid-Structure Interaction	4	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE5055	Professional Training	6	
WM1115TU	Dutch Elementary 1	3	
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AE4006	Sustainability in Engineering	4	
AE4010	Research Methodologies	2	
AE4115	Experimental Simulations	3	
AE4117	Fluid-Structure Interaction	4	
AE4120	Viscous Flows	3	
AE4130-22	Aircraft Aerodynamics	4	
AE4135	Rotor / wake Aerodynamics	4	
AE4136-22	CFD 2: Discretization techniques	3	
AE4138-18	CFD 4: Uncertainty Quantification	3	
AE4139	CFD 3: Large Eddy Simulation	3	
AE4140	Gas Dynamics	3	
AE4143	Hypersonic Aerodynamics	3	
AE4180	Flow Measurement Techniques	3	
AE4185	Fluid Flow Data Processing and Visualization	3	
AE4204	Knowledge Based Engineering	4	
AE4206	Turbomachinery	3	
AE4238	Aero Engine Technology	4	
AE4240	Advanced Aircraft Design I	4	
AE4260A	Fundamental of Aeroacoustics	2	
AE4260B	Experimental Applications of Aeroacoustics	2	
AE4261	Internal Flows	3	
AE4262	Combustion for propulsion and power technologies	4	
AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5	
AE4270	Control and Operations Project	4	
AE4301	Automatic Flight Control System Design	3	
AE4301P	Exercise Automatic Flight Control System Design	1	
AE4311	Nonlinear and Adaptive Flight Control	4	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4314-21	Helicopter Performance, Stability and Control	4	
AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3	
AE4317	Autonomous Flight of Micro Air Vehicles	4	
AE4320	System Identification of Aerospace Vehicles	4	
AE4321-15	Air Traffic Management	4	
AE4323	Real-time Distributed Flight and Space Simulation	3	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4351	Robust Flight Control	4	
AE4352	Mathematical and human-inspired decision making	3	
AE4422-20	Agent-based Modelling and Simulation in Air Transport	4	
AE4431	Aircraft noise and emissions	3	
AE4439	Air Transport Operations Modelling Project	2	
AE4446	Airport and Cargo Operations	4	
AE4462-17	Aircraft Emissions and Climate Effects	4	
AE4463P	Advanced Aircraft Noise Modeling and Measurement	3	
AE4465	Maintenance Modeling & Analysis	4	
AE4889	Special Topics in Astrodynamics	2	
AE4893	Physics of Planetary Interiors	4	
AE4895	Measurement Strategies for Planetary Science Missions	3	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM101TU-22	Polymer Science	4	
AE4ASM103	Functional Coatings	3	
AE4ASM105	Trinity Exercise	4	
AE4ASM106	Stability & Analysis of Structures I	3	
AE4ASM108	Experimental Techniques & NDT	3	
AE4ASM109	Design & Analysis of Composite Structures I	5	
AE4ASM110-22	Polymer Composite Manufacturing	4	
AE4ASM503	Sheet Metal Forming	3	
AE4ASM504	Structural Integrity and Maintenance	3	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4ASM508	Design of Self-healing materials	3	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE4ASM513	Forensic Engineering	3	
AE4ASM514TU	Continuum Mechanics	4	
AE4ASM515-22	Characterization of materials and components	4	
AE4ASM516	Material Selection in Mechanical Design	3	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
AE4ASM520	Industrial Composite Manufacturing	3	
AE4ASM521	Additive Manufacturing	3	
AE4ASM522	Applied Aircraft Aeroelasticity	3	
AE4ASM523	Design of Spacecraft and Launcher Structures	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
AE4ASM526	Spacecraft Thermal Design	3	
AE4S01	Thermal Rocket Propulsion	4	
AE4S10	Microsat Engineering	4	
AE4S52-21	Collaborative Space System Design Project	5	

AE4T40	Airborne Wind Energy	3	
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4	
AE4W21-14	Wind Turbine Aeroelasticity	2	
AE4W30	Wind Resource and Wind Farm Yield	4	
AE4W31	Floating Offshore Wind Energy	3	
CS4240	Deep Learning	5	
ET3604LR	Electronic Circuits	3	
ET4117	Electrical Machines and Drives	4	
ME45001	Advanced Heat Transfer	4	
ME45025	Introduction to Multiphase Flow	5	
ME45030	Turbulence	5	
ME46060	Engineering Optimization: Concepts and Applications	3	
MS43310	Materials at High Temperature	4	
RO47019	Intelligent Control Systems	4	
SC42035	Integration Project Systems and Control	5	
SET3085	Hydrogen Technology	4	
WI3150TU	Partial Differential Equations A	3	
WI3151TU	Partial Differential Equations B	3	
WI4014TU	Numerical Analysis	6	
WI4062TU	Transport, Routing and Scheduling	3	
WI4201	Scientific Computing	6	
WI4212	Advanced Numerical Methods	6	
WI4771TU	Object Oriented Scientific Programming with C++	3	
WM1115TU	Dutch Elementary 1	3	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Master LR 2022

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 1: Aerodynamics and Wind Energy	
MSc Coordinator	Dr.ir. A.H. van Zuijlen
Contact for students	Programme coordinator Aerodynamics: A.H. van Zuijlen Programme coordinator for Wind Energy: F. Avallone
Introduction 1	The master Track Aerodynamics and Wind Energy combines fundamental and applied research disciplines of aerospace and wind-power systems focusing on development and optimization. The track is of interest to students who want to acquire expertise with advanced aerodynamic tools (experimental and numerical modeling techniques) or desire experience with the applications of aerodynamics in design and sustainable energy conversion. The Track offers two Profiles: Aerodynamics and Wind Energy. The first is supported by the Aerodynamics research group, while the second is supported by the Wind Energy research groups.
Program Goals	Learning objectives of the MSc Track: The objective of this track is to provide students with the opportunity to become a specialist with specific knowledge in analysis of aerodynamic systems, and the methods used for their application in design. The students will obtain a thorough fundamental basis in aerodynamics as well as in modern techniques to investigate such systems.
Program Structure 1	In the first year of the MSc, the student follows core courses from the Aerodynamics and Wind Energy program, which is designed to provide a fundamental background in fluid mechanics and its applications. Aerodynamics analysis by means of experimental, theoretical and computational methods is also taught in the first year. Concurrently, the student follows three or four elective courses associated with a selected profile. After completing the courses and an internship, the student joins one of the research groups for a literature study and final thesis. The literature study and final thesis are performed under the supervision of a scientific staff member, working on a project of current relevance to the field. In some cases, the thesis work may be carried out in conjunction with a company or a research institute. The supervisor and project should be chosen early in the MSc to aid the selection of pertinent courses, nevertheless they can be defined at latest during the second part of the course period.

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Aerodynamics

Program Coordinator	Dr.ir. A.H. van Zuijlen
Introduction 1	The Aerodynamics Profile is concerned with the understanding and control of flows associated with aerospace vehicles. On the fundamental side, it considers the design of measurement and computational methods for the provision of detailed flow descriptions. On the applied side, it considers the design of flow control devices and techniques to maximise vehicle performance. The course program of the profile focuses on the knowledge which must be acquired to understand modern flow analysis. This includes courses in gasdynamics and viscous flows, computational fluid dynamics, flow measurement techniques and experimental simulations. It also includes practical courses which establish the context of this knowledge, including aircraft and rotor aerodynamics. The final thesis, which combines the literature study and thesis project, can be performed within the Aerodynamics research group, our partner research centres, or industry. The topics cover a large area, from the development of new computational methods for difficult flow problems, to the design of high-speed flow measurement systems, to the application of such methods and systems in the design of aerospace vehicles. A key advantage of the profile is its focus on fundamental knowledge. As flows are ubiquitous in engineering systems, much of the experience gained in the profile is easily transferable to other fields. Graduates of the Aerodynamics profile can therefore expect a wide range of employment opportunities.

AE4115	Experimental Simulations	3
Responsible Instructor	T. Sinnige	
Instructor	Prof.dr.ing. G. Eitelberg	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling. Fractional analysis, symmetry and similarity, application to specific problems from aerodynamics, BL-approximations, S/BL interactions; 3 Dimensional analysis: Buckingham's Pi-theorem, rigorous methods of obtaining dimensionless parameters, simplifications of Navier-Stokes equations; 4 Subsonic and supersonic wind tunnels: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 5 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 6 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 7 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 8 Design of experiments: factorial designs, response-surface models; 9 Reynolds-number effects and simulation; 10 Pre-test support session; 11 Engine integration: engine simulators, performance of those. Thrust reverser testing for re-ingestion and stability and control; 12 Acoustic testing. Noise generation mechanisms for submerged bodies, edges and jets. Appropriate scaling. Definition of sound pressure levels. Measurement requirements and devices. Far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session. (Excursion) Ground test facilities of DNW (organized by AWEP society, limited places available).	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports and oral exam.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter. The lectures are supported by an excursion to the testing facilities of DNW, organized by the AWEP society.	

AE4120	Viscous Flows	3
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Responsible Instructor	M. Kotsonis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and (to limited extend) viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Typically Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during at the beginning of Q1).	
Assessment	The assessment is done through 5 personal assignments.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE3-W02 (4P720 TUe) plus basic aerodynamics.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade element/momentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsen's Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4136-22	CFD 2: Discretization techniques	3
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results.	
Study Goals continuation	After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Special Information	M.I.Gerritma@tudelft.nl	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	Prof.dr. F. Scarano	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written (proctored online) exam and report on final assignment.	
Permitted Materials during Tests	none	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Officially registered and enrolled as a full time Master student Aerospace Engineering.

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

Enrolment / Application

Special Information

Set-up

AE5122	Thesis Aerodynamics & Wind Energy	45
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The thesis project engages the student in innovative research on a topic related to wind energy, or experimental, numerical or applied aerodynamics. The student addresses a specific research question, and is responsible for the planning and execution of the project. The project may be carried out in industry, but most are carried out within the Wind Energy or Aerodynamics research groups, where the student is normally engaged as part of a research team working within a particular area.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student receives regular guidance from one or more members of the scientific staff.	
Assessment	Assessment is performed by an expert committee, based on a final report, public presentation, and private defence session.	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (WI3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (WI3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and Maple practical work (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam	
Special Information	The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	The contents of WI3150TU + WI3151TU is the same as TW2070.	
Co-Instructor	Dr.ir. W.T. van Horssen	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Aerodynamics

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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6	Variational inverses	[Inversion]	Finalize []	--																																						
7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Dr. S.J. Hulshoff	
Responsible Instructor	Prof.dr.ir. S. Hickel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4185	Fluid Flow Data Processing and Visualization	3
Responsible Instructor	W.J. Baars	
Responsible Instructor	Dr. D. Modesti	
Responsible Instructor	Dr. N.A.K. Doan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is a general introduction to data processing techniques used in fluid mechanics to analyse unsteady flows. Specifically, techniques related to the flow visualization and the postprocessing of numerical data will be presented. The overall content of the course is: 1. Review of fundamentals of turbulent flows (energy cascade, Reynolds-averaged equations, Reynolds stress, turbulent kinetic energy, vorticity equation, turbulence spectrum, energy budget, ...) 2. Characteristics of flow datasets (Nyquist criterion, sampling and resolution, first principle of visualization, vortex/Q criteria, ...) 3. Statistical analysis in flows - Deterministic/random data - Averaging (ensemble, condition, phase) - Probability: PDFs, statistical moments, conditional PDFs - Correlation function - Ergodic hypothesis - Statistical convergence and uncertainty 4. Temporal and spectral data processing - Fourier transform - Physical/spectral space and multi-dimensional FFTs - Auto/cross spectra - Integral length/time scale - Convolution (filtering) and transfer function - RANS/LES averaging/filtering 5. Modal analysis and coherent structure education - Base of Proper Orthogonal Decomposition (and some extensions)	
Study Goals	At the end of this course, the student should be able to: - Visualize fluid mechanics datasets using relevant quantities - Analyse fluid mechanics datasets using statistical methods - Apply temporal and spectral analysis methods to analyse flow datasets - Postprocess fluid mechanics datasets using modal-based methods	
Education Method	The course will be a combination of lectures, scheduled tutorials and homework assignments.	
Assessment	The course will be assessed with a final assignment (80%) and an oral exam (20%).	

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015 Optional textbooks: - Airplane Design, Prt I-VII, Roskam, J, DARCorp, 1990-2003 - Synthesis of Subsonic Airplane Design, Torenbeek, E, Kluwer Academic Publishers, 1982 - Fluid Dynamic Drag, Hoerner, S. F., Hoerner Fluid Dynamics, 1965 - Fluid Dynamic Lift, Hoerner, S. F. Hoerner Fluid Dynamics, 1965	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Course Contents	Dr. D. Ragni Prof.dr.ir. M. Snellen 0/0/2/0 3 3 Exam by appointment English	
Study Goals	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources. The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resourcen/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	A. Gangoli Rao	
Responsible Instructor	Dr. I. Langella	
Instructor	Prof.dr. D.J.E.M. Roekaerts	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Radiation and transport processes 4. Premixed and diffusion flames 5. Turbulent reactive flows 6. Combustion experiments and diagnostics 7. Laboratory Tutorial & Exercise 8. Practical combustion systems for aerospace applications	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation- and adiabatic flame temperature. Apply principles of thermodynamics to analyze reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames. Derive equations related to turbulent reacting flows and chemistry interaction. Explain laser diagnostics and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions. In addition, the student is required to complete project assignments and perform a laboratory exercise. The project assignments and associated numerical problems are relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with laminar flame speed measurements of the simplest hydrocarbon fuel, methane. The student will gain hands-on experience in assembling the experimental setup, acquiring data, performing data analysis and writing a technical report.	
Course Relations	Aero Engine Technology	
Literature and Study Materials	Slides and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION);	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the student must pass the oral examination, the evaluation of the laboratory exercise, and successfully complete the assignments given during the course. The work on the laboratory exercise and assignments should be summarized in reports, to be written by a single student or in a group of two. The grade is dictated by the result of the oral examination (50%) and the grades obtained in the assignments and lab exercise (50%).	

AE4431	Aircraft noise and emissions	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Responsible Instructor	Prof.dr. D.G. Simons	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The aircraft noise part of this course is required for AE4463P 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics - Dutch aircraft noise model NRM - acoustic imaging Education method: The practicing is in the form of concrete assignments, which are individually graded.	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and terts band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. (Acoustic imaging: Principle of beamforming. Imaging aircraft noise data.) Aircraft noise sources: Propeller noise (mechanism and directional properties) and blade passage frequency. Turbo engine noise and directionality of the fan and exhaust jet noise source. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise (and the modelling of it). (Aero-acoustics (Lighthill theory).) The topics in parentheses (acoustic imaging, airframe noise modelling and aero-acoustics) are part of the reader, but do not belong to this course and will be treated in the follow-up course AE4463P. Aircraft emissions and climate effects Aircraft emissions: Gas emissions from aircraft engines. Complete and actual combustion. Combustion products from burning kerosene. Emission Index. Carbon dioxide emissions of transport (aviation and, for comparison, road traffic). Atmospheric radiation physics: Concept of radiation flux. Solar spectrum and solar constant. Radiative properties of natural surfaces. Thermal emission. Black body radiation and the laws of Planck, Wien, Stefan-Boltzmann and Kirchoff. Atmospheric transmission and emission. Effects of ozone. Climate effects: A climate model for the Earth. Greenhouse effect. Radiative forcing. Example calculations of global warming due to various carbon emission scenarios. Contribution of aviation. Feedback in the climate system.	
Study Goals	Understand the relationship between aviation and the environment.	
Education Method	Lectures and exercises.	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions Readers:	

	Course AE4431 Aircraft Noise and Emissions Part A: Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022 Course AE4431 Aircraft Noise and Emissions Part B: Introduction to aircraft emissions and climate effects Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022
Assessment	Written exam (students may bring their own formula sheet to the exam)
Set-up	Lectures and exercises

AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week	0/2/0/0	
x/x/x/x	This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	AE4260 B (Experimental Applications of Aeroacoustics)	
Expected prior knowledge	AE4431 (Aircraft noise and emissions)	
Course Contents	AE4463P aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) will be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B and F are required. A choice can be made between C, D and E.	
Study Goals	The course aims at providing the students with an thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4ASM506		Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker		
Contact Hours / Week x/x/x/x	0/4/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.		
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.		
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.		
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.		
Assessment	1. Written exam. 2. Take home assignment.		

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: describe the components and configurations of wind turbines describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. identify the drivers of dynamic behaviour and calculate their typical values describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Special Information	Assessment may differ due to COVID-19 measures. Changes to the assessment method will be communicated via Brightspace.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	W. Yu	
Gast Instructor	Dr.ir. A.C. Viré	
Gast Instructor	Prof. R. Beuker	
Gast Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MANDATORY: AE4ASM506 (Aeroelasticity). The student should have completed or is attending this course. An equivalent course can be accepted upon discussion with the instructor.	
Course Contents	The course will cover main aerodynamic and structural phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. Possible resonances on wind turbines. What are the excitation frequencies, what are the natural frequencies? 2. What are wind turbine aeroelasticity tools based on? Linearised tools in frequency domain and nonlinear time domain simulation tools. How accurate are these tools and where are inaccuracies coming from? 3. Wind turbine mode shapes, whirling modes. 4. Wind turbine instabilities. 5. Dynamic inflow and dynamic stall.	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand which aeroelastic principles are important for wind turbines. 2. explain the contribution of the different wind turbine components on the resulting time responses. 3. apply the knowledge to program their own models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments, computer demonstrations as well as instruction. In 2021, all lectures will be online. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction/office hours, during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. Weekly instruction is likely to be on campus.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	3mE Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	At the end of this course you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	Brightspace/MATLAB exercises and tests (optional) during education period. The course is concluded through an optimization project and report. Deadlines for handing in the final report will be announced at the start of the course.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and Maple practical work (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and practical work.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Practical work and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Remarks	The contents of WI3150TU + WI3151TU is the same as TW2070.	
Judgement	The practical work is mandatory, and as long as the practical work is not completed no final grading will be given.	
Co-Instructor	Dr.ir. W.T. van Horssen	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports.	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The assessment consists of three parts: homework exercises deadline start of Q2 leads to grade G1, take home exam deadline half of January grade G2 and a written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High-resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	In the first period a set of exercises is given. These should be worked out, strict deadline 26-3-2019 (Grade: G₁). The exercise should be handed in individually. In the middle of April a take home exam including practical exercises is given. This exam can be done by groups of two students. The report of this exam should be returned to us before 11-6-2019 (Grade: G₂). Thereafter an appointment can be made for an oral examination over Chapter 18 (including exercise 18.1, 18.2, and 18.3), 19, and 20 (Grade: G₃). Note that material discussed in Chapter 1 to 9, which is relevant for the material in Chapter 18 to 20 can also be part of the oral examination. The oral exam must be passed with a 6 or higher. Then, provided that G₃ is larger than or equal to 6, the final grade is computed by the formula: $(G_1 + G_2 + 2 G_3)/4$. disclaimer: information may change depending on the developments around the coronavirus.	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Wind Energy	
Program Coordinator	Dr.ir. D.A.M. De Tavernier
Introduction 1	Profile II Wind Energy The Wind Energy profile focuses on methods and systems of energy extraction from wind. Both wind turbine and kite power systems are addressed. The profile offers courses on atmospheric wind conditions, rotor aerodynamics, wind turbine design, design of rotor blades and kite systems and of offshore wind farms. Graduation projects range from numerical wind field modeling and wind turbine aerodynamics to design aspects of large offshore wind farms, kites and other non conventional wind energy systems. Projects related to energy extraction from other moving media such as water currents and waves are also possible. After successful completion of the MSc programme you: - Will have in-depth knowledge of low speed aerodynamics and its application to energy extracting devices such as wind turbines and kites; - Will have thorough system knowledge regarding these energy extraction devices; - Will be able to apply advanced analysis and design techniques in at least one of the following areas: wind turbines, wind farms or kite power systems; - Will have insight in related disciplines such as boundary layer meteorology, structural design and dynamics of structures, control and operations. - Will have knowledge of the noise generation mechanisms in wind-turbines, of sound propagation in realistic terrain and atmospheric conditions, and the implications of noise reduction on wind-farm economy.

AE4120	Viscous Flows	3
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Responsible Instructor	M. Kotsonis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE3-W02 (4P720 TUE) plus basic aerodynamics.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade element/momentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsen's Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written (proctored online) exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: describe the components and configurations of wind turbines describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. identify the drivers of dynamic behaviour and calculate their typical values describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Special Information	Assessment may differ due to COVID-19 measures. Changes to the assessment method will be communicated via Brightspace.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control Blade materials and assessment of fatigue damage Wind data and description The use of standards for load calculations	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): AE4W13 Site conditions for wind turbine design AE4135 Rotor / wake aerodynamics AE4W21-14 Wind turbine aeroelasticity ET4117 Electrical machines and drives OE44135 Offshore wind support structures	

AE4W13	Site Conditions for Wind Turbine Design	3
Responsible Instructor	Dr.ir. W.A.A.M. Bierbooms	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Course Contents	Introduction Atmospheric Boundary Layer Meteorology Fluid Dynamics Turbulence Wind power meteorology Measurement instruments Statistics Methods in turbulence Offshore wind climate External conditions in standards Wave and current climate and loading Example analysis site data Turbulence spectra Satellite remote sensing techniques Simulation of turbulent wind fields and gusts	
Study Goals	After successfully completing the course, students will be able to: Discuss the relevance of particular site data for wind turbine design, operation and siting. Analyse simple problems related to wind power meteorology Interpret scientific dissemination by experts on main points Apply analysis methods to convert basic site data to wind turbine design parameters	
Education Method	Lecture + assignments + compulsory practical	
Course Relations	Follow up courses AE4-W09, AE4135	
Literature and Study Materials	www.eos.ubc.ca/books/Practical_Meteorology Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier. Lecture notes and reader	
Prerequisites	AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Assessment	The written exam consists out of two parts, one for each period (scheduled in the exam period). The final grade is a weighted average of these 2 exams and the grade for one of the assignments (which one will be specified during the course). In order to take part in Exam2 one should join the compulsory practical The resit is an oral exam (to be scheduled with the teacher) on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).	
Set-up	The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	W. Yu	
Gast Instructor	Dr.ir. A.C. Viré	
Gast Instructor	Prof. R. Beuker	
Gast Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MANDATORY: AE4ASM506 (Aeroelasticity). The student should have completed or is attending this course. An equivalent course can be accepted upon discussion with the instructor.	
Course Contents	The course will cover main aerodynamic and structural phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. Possible resonances on wind turbines. What are the excitation frequencies, what are the natural frequencies? 2. What are wind turbine aeroelasticity tools based on? Linearised tools in frequency domain and nonlinear time domain simulation tools. How accurate are these tools and where are inaccuracies coming from? 3. Wind turbine mode shapes, whirling modes. 4. Wind turbine instabilities. 5. Dynamic inflow and dynamic stall.	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand which aeroelastic principles are important for wind turbines. 2. explain the contribution of the different wind turbine components on the resulting time responses. 3. apply the knowledge to program their own models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Officially registered and enrolled as a full time Master student Aerospace Engineering.

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

Enrolment / Application

Special Information

Set-up

AE5122	Thesis Aerodynamics & Wind Energy	45
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The thesis project engages the student in innovative research on a topic related to wind energy, or experimental, numerical or applied aerodynamics. The student addresses a specific research question, and is responsible for the planning and execution of the project. The project may be carried out in industry, but most are carried out within the Wind Energy or Aerodynamics research groups, where the student is normally engaged as part of a research team working within a particular area.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student receives regular guidance from one or more members of the scientific staff.	
Assessment	Assessment is performed by an expert committee, based on a final report, public presentation, and private defence session.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Wind Energy

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4115	Experimental Simulations	3
Responsible Instructor	T. Sinnige	
Instructor	Prof.dr.ing. G. Eitelberg	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling. Fractional analysis, symmetry and similarity, application to specific problems from aerodynamics, BL-approximations, S/BL interactions; 3 Dimensional analysis: Buckingham's Pi-theorem, rigorous methods of obtaining dimensionless parameters, simplifications of Navier-Stokes equations; 4 Subsonic and supersonic wind tunnels: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 5 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 6 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 7 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 8 Design of experiments: factorial designs, response-surface models; 9 Reynolds-number effects and simulation; 10 Pre-test support session; 11 Engine integration: engine simulators, performance of those. Thrust reverser testing for re-ingestion and stability and control; 12 Acoustic testing. Noise generation mechanisms for submerged bodies, edges and jets. Appropriate scaling. Definition of sound pressure levels. Measurement requirements and devices. Far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session. (Excursion) Ground test facilities of DNW (organized by AWEP society, limited places available).	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports and oral exam.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter. The lectures are supported by an excursion to the testing facilities of DNW, organized by the AWEP society.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and (to limited extend) viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Typically Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during at the beginning of Q1).	
Assessment	The assessment is done through 5 personal assignments.	

AE4136-22	CFD 2: Discretization techniques	3
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results.	
Study Goals continuation	After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Special Information	M.I.Gerritma@tudelft.nl	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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6	Variational inverses	[Inversion]	Finalize []	--																																						
7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Dr. S.J. Hulshoff	
Responsible Instructor	Prof.dr.ir. S. Hickel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	Prof.dr. F. Scarano	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Course Contents	Dr. D. Ragni Prof.dr.ir. M. Snellen 0/0/2/0 3 3 Exam by appointment English Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resourcen/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	AE4260 B (Experimental Applications of Aeroacoustics)	
Expected prior knowledge	AE4431 (Aircraft noise and emissions)	
Course Contents	AE4463P aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) will be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B and F are required. A choice can be made between C, D and E.	
Study Goals	The course aims at providing the students with an thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments, computer demonstrations as well as instruction. In 2021, all lectures will be online. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction/office hours, during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. Weekly instruction is likely to be on campus.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	3mE Department Process & Energy	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (WI3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (WI3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and Maple practical work (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Remarks	The contents of WI3150TU + WI3151TU is the same as TW2070.	
Co-Instructor	Dr.ir. W.T. van Horssen	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 2: Control and Operations

MSc Coordinator	Dr.ir. C.C. de Visser
Introduction 1	Control and Operations is focused on the through-life operations associated with aerospace industry. It studies the avionics systems of individual aircraft, flight control and flight deck design, airline operations and support (maintenance), air traffic control and air traffic management, airports, as well as the operations associated with air transport as a whole. By selecting one of the three C&O profiles, the student chooses an educational programme that contains a particular balance of these elements. Theory and practice are combined in exciting and challenging fields of study for talented and ambitious students. Our graduates easily find their way in finding a good job in engineering, research, consulting, and management. The C&O profiles incorporate a research project that builds on the learning gained from the taught modules. C&O research projects focus on academic quality and the contribution to the body of knowledge in the field. There is also a strong recognition of the importance of understanding problems coming from industry, through industrial collaboration, and experimental validation of the novel operational concepts in real-life scenarios. The Track offers two Profiles: I. Control and Simulation II. Sustainable Air Transport
Program Goals	Learning objectives The focus in C&O lies on first establishing a sound theoretical and framework and then to sharing our department's expertise in applying this knowledge in practical problems.
Program Structure 1	The Track consists of three Profiles that share a common core of 16 ECTS. Each profile programme consists of a 'Profile core' of mandatory courses, ranging between 16 and 17 ECTS, supplemented by Electives (13 to 20 ECTS) that can be adjusted to either the subject of the final thesis work, or to the student's own taste. The MSc thesis project is an individual research assignment, conducted 'in house' or at an appropriate external organization partner.

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Control and Simulation

Program Coordinator Dr.ir. J. Ellerbroek

Introduction 1

The C&S profile focuses on design, analysis and testing of automatic and manual guidance, navigation, and control (GNC) systems. Our aim is to enhance safety, sustainability and performance of aerospace vehicles by creating better interfaces, procedures and automation. The profile builds on systems and control theory, stochastic dynamic systems and signals, flight simulation, and modelling human cognition and manual control. For their electives and thesis students can focus on autonomous guidance and control, micro aerial vehicles, the air traffic management system, or human-in-the-loop control systems. Students can work with state-of-the-art facilities to close the loop of theory and application, using the laboratory aircraft, the SIMONA flight simulator, the ATC lab, or the CyberZoo indoor flight-test area.

For autonomous systems, students can focus on the design of highly-automated, intelligent and (semi-)autonomous aerial vehicles. They can learn advanced control theory, global optimisation approaches, state and parameter estimation techniques, and modern dynamic modelling techniques. Aeronautical research examples are fault tolerant and re-configurable flight control, flight envelope clearance and protection, vehicle dynamics model identification, and advanced flight test instrumentation systems. Space examples are rendezvous/docking vehicle control, GNC for formation flying spacecraft, modelling of spacecraft with flexible structures and liquid sloshing, and optimal terminal area energy management of re-entry vehicles.

Collision avoidance and inherent safety form a red line through the Micro Aerial Vehicle (MAV) research and education, with applications such as collaborative, vision-based, and acoustic avoidance. The low Reynolds conditions of MAVs, which cause unsteady aerodynamic effects, are investigated to better understand and predict how new MAV designs behave. Here we focus on special MAV designs, such as flapping or morphing wing MAVs. For sense and avoid, we study an MAV system as a whole: not only the MAV, but also the human operator(s) and the context (including surrounding buildings and other air traffic). An example is single-operator control of a swarm of MAVs.

With Air Traffic Management (ATM), students will learn about both the conventional air traffic management system, as well as challenges and approaches for new airspace users such as drones and electric vehicles (Urban Traffic Management and mixed operations). Research examples are design and evaluation of new (emission-reducing) procedures for conventional aviation, development of conflict detection and resolution algorithms (both AI and analytical), airspace design, complexity analysis, data science in ATM, and analysis of communication, navigation, and surveillance applications.

For human-in-the-loop systems, students will learn to understand and model human perception and skill-based control behaviour, using modern identification techniques and classical control. Research examples are the design of haptic force-feedback systems for aerospace (and other) vehicles, modelling human visual-vestibular perception and control, neuromuscular system dynamics and biodynamic coupling effects, the optimal tuning of flight simulator visual and motion cues, aircraft handling qualities and advanced fly-by-wire systems, 3D flight guidance displays, and the tele-operation of unmanned aerial vehicles.

Learning objectives:

1. Knowledge on modern methods for analysis of control problems and synthesis of control systems
2. Knowledge of modelling human perception and control behaviour, including modern methods for identification
3. Skills in the use of modern control and simulation programs
4. Skills in the programming, setup and execution of experiments in flight simulators or in flight tests
5. Insight in related disciplines such as artificial intelligence, statistics, sensor fusion, distributed computing, fast-time and real-time simulation

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory a. Laplace transformation b. Elementary closed loop systems c. Transfer functions 2. Poles and zeros a. First order systems b. Second order systems c. Pole placement for simple systems 3. Root locus method a. Characteristic equation b. Angle and magnitude conditions 4. State space formulation a. Controllability, observability b. Ackerman's formula c. LQR 5. Basic controllers: P,PI,PD,PID 6. Frequency response (Bode plots) 7. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 8. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 9. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 10. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 11. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 12. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 13. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance - substantiate the function of a Flight Control System(FCS) in civil aviation.	

	- apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 AE3212-I AE2235-I	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace	
Assessment	Exercise	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4317 AE4318 AE4319 AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. - Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 4. Navigation equations. 5. Flight deck instruments and integrated systems. 6. The Flight Management System (FMS). 7. Inertial Navigation Systems. 8. Radio navigation systems (ADF, VOR, DME). 9. Landing guidance systems (ILS, GPS). 10. Communication, Navigation, Surveillance (CNS). 11. Satellite navigation systems (GPS). 12. Air Traffic Management (ATM). 13. The Future Air Navigation System (FANS). (see week arrangement)	
Course Contents	This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	(Digital) written exam (closed-book)	

AE4304	Stochastic Aerospace Systems	3
Responsible Instructor	Prof.dr.ir. M. Mulder	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4306 AE4307	
Expected prior knowledge	AE2235-I Aerospace System and Control Theory AE2235-II Instrumentation and Signals AE4301 Automatic Flight Control Systems Design	
Parts	Week arrangement See Detailed Description of Course 1. Introduction 2. Chapter 2 of the lecture notes. 3. Chapter 3. 4. Chapter 3. 5. Chapter 4. 6. Chapter 4. 7. Chapter 5.	
Course Contents	The course AE4304 covers ONLY the first five chapters of the lecture notes, the practical assignment AE4304P covers chapters six to eight. Chapter 9 (Etkin's 4 point model) serves as background reading. So, the lecture AE4304 (and its exam) covers: 1. Introduction (aircraft do respond to atmospheric turbulence, effects on flight control system design). 2. Scalar stochastic processes (probability theory, joint probability density functions, covariance and correlation functions, stochastic processes, ergodic processes). 3. Spectral analysis of stochastic processes in continuous time (Fourier analysis, power spectral densities, analysis of dynamic linear system responses in frequency domain). 4. Spectral analysis of stochastic processes in discrete time (discrete time Fourier transform, Fast Fourier Transform, spectral estimates-smoothing). 5. Multivariable stochastic processes (covariance function matrix and spectral density matrix, multi-variable system responses in the frequency and in the time domain). The practical assignment AE4304P (Matlab or Python) covers: 6. Description of atmospheric turbulence (physical mechanisms, stochastic models of atmospheric turbulence, the two fundamental correlation functions, von Karman en Dryden spectra, models in the time domain). 7. Symmetric aircraft response to atmospheric turbulence (symmetrical aerodynamic forces and moments due to turbulence, gust derivatives, equations of motion of aircraft flying in symmetrical atmospheric turbulence). 8. Asymmetric aircraft response to atmospheric turbulence (elementary two-dimensional fields of turbulence, asymmetrical aerodynamic forces and moments, asymmetrical gust derivatives, equations of motion).	
Study Goals	Introduction to stochastic processes, spectral analysis, understanding the physics of aircraft responses to atmospheric turbulence, derivation of equations of motion of symmetrical and asymmetrical responses to atmospheric turbulence.	
Education Method	Lecture	
Literature and Study Materials	J.A. Mulder, J.C. van der Vaart, W.H.J.J. van Staveren, M. Mulder: Aircraft responses to atmospheric turbulence, Lecture notes AE4304 , November 2019. All lecture slides will be made available electronically via the Brightspace.	
Assessment	Written (with formula page)	
Remarks	The written examination (on probability and stochastic processes, chapters 1-5 of the lecture notes) is followed by the assignment ae4304P (related to the theories of chapters 6-8 of the lecture notes).	
Set-up	Lectures, MATLAB/PYTHON demonstrations.	

AE4304P	Stochastic Aerospace Systems Practical	1
Responsible Instructor	D.M. Pool	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE3212-I Flight Dynamics AE2235-I Systems and Control Theory AE2235-II Signal Analysis and Telecommunication Principles AE4301	
Course Contents	Application of MATLAB or Python software to aircraft specific turbulence responses: 1. Calculation of aircraft time-histories due to both symmetrical and asymmetrical, longitudinal, lateral and vertical turbulence components. 2. Calculation of analytical transfer functions, frequency response functions, and auto- and cross Power Spectral Density (PSD) functions of state- and output variables (e.g. acceleration levels). 3. Numerical calculation of frequency response functions, and auto- and cross Power Spectral Density (PSD) functions of state- and output-variables. 4. Calculation of (co)variance- and correlation-functions of aircraft state-and output-variables. 5. The effects of Automatic Flight Control Systems on the aircrafts responses on atmospheric turbulence.	
Study Goals	Introduction to both time- and frequency-domain identification and simulation techniques using MATLAB or Python. The techniques are applied to example aircraft (amongst others Cessna Citation 500).	
Education Method	General Practice Exercise	
Literature and Study Materials	Lecture Notes ae4304, by J.A. Mulder, J.C. van der Vaart, M. Mulder: Aircraft Responses to Atmospheric Turbulence, November 2016. Recommended literature: D.E Newland, Random Vibrations and Spectral Analysis, Longman, New York, 1984 3rd. ed ISBN 0582215846.	
Assessment	Take-home assignment.	
Set-up	Take-home assignment. Deadline April 1. At the end of the lecture period, the assignment will be available.	

AE4316	Aerospace Human-Machine Systems	4
Responsible Instructor	Dr.ir. M.M. van Paassen	
Instructor	Dr.ir. C. Borst	
Instructor	Prof.dr.ir. M. Mulder	
Instructor	D.M. Pool	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Follow-up course (elective) AE4316P. That course focuses on advanced topics in aerospace human-machine systems by having small groups of students participate in an experiment (in either a manual or supervisory control setting), collect data and analyze their own generated data. In other words, experience the theory behind human-machine systems in practice!	
Expected prior knowledge	Control theory (stability and Bode plots); signals AE2235 Aerospace Signals, Systems & Control, AE4301 Automatic Flight Control System Design Stochastic signals; AE4304 Stochastic Aerospace Systems Flight instruments and procedures; AE4302 Avionics and Operations	
Course Contents	This course focuses on the various aspects of actual and future flight deck and air traffic control human-machine interfaces. It provides an extensive theoretical as well as practical knowledge on the specific characteristics of human behavior, such as human control behavior (i.e., cybernetics), human perception (visual & haptic), human mental processing, cognitive factors, and human-automation interaction in manual and supervisory control tasks.	
Study Goals	Overall, the student will have a working knowledge of human operator (pilot) characteristics that are relevant for the design and evaluation of human-machine systems. Specific study goals: The student: * Is able to classify different types of human behaviour according to Rasmussen's rule-skill and knowledge taxonomy * Is able to predict performance and human behaviour in manual control tasks, using McRuer's cross-over theory, and is able to reason on the effect of task variables (motion, haptics, etc.) * Knows the physiology and characteristics of human sensory systems and actuation processes (visual, vestibular, proprioceptive senses and neuromuscular system) as relevant for human behaviour, and is able to predict the implications of these properties for human perception and behaviour, and relate this to design choices for displays and manipulators. * can describe the pitfalls of automation (i.e., ironies of automation) and is familiar with various taxonomies on levels & stages of automation. * can analyse accident and incident reports, find latent and active errors and classify these with Rasmussen's SRK taxonomy and identify Reason's error shaping factor. The student understands the wider context of human error (Dekker's "new view"). * is familiar with workload and situation awareness, and knows which methods are used to measure these properties * understands the nature of human cognition, can distinguish between different views and models of cognition and knows when these are applicable * can describe the differences and similarities between interface design approaches that aim to support both system and human (control & cognitive) performances * is familiar with workload theory, knows different metrics for workload, and when these are applicable	
Education Method	Lectures with activating elements. Week arrangement: 1. Introduction + SRK 2. Crossover model I 3. Crossover model II 4. Crossover Model II 5. Visual systems (the eye and visual motion perception) 6. Vestibular system 7. Neuromuscular system 8. Human-Automation Interaction 9. Human Error 10. Cognition 11. Ecological Interface Design 12. Display Design Principles 13. Applications of HMS theory 14. Workload (self study module, no lecture) The course also includes a demonstration session in the human machine laboratory, in which students will encounter experimentation settings and experience manual control tasks.	
Assessment	Written exam. The exam typically consists of 6 questions, covering a selection of the course's topics.	

AE4322	Piloted Flight Simulation	4
Responsible Instructor	Ir. O. Stroosma	
Guest Instructor	Dr.ir. M.M. van Paassen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	AE4323 (Real-time Distributed Flight and Space Simulation)	
Expected prior knowledge	AE2235 (Aerospace Signals, Systems & Control, or equivalent) AE3212 (Aerospace Flight Dynamics, or equivalent) AE4302 (Avionics and Operations, or equivalent) AE4316 (Aerospace Human-Machine Systems, or equivalent)	
Course Contents	Lecture topics, not necessarily in chronological order 1) Introduction to piloted flight simulation, including fidelity considerations 2) Use of piloted flight simulation, including training and qualification 3) Simulator sub-systems 4) Modelling of vehicle dynamics 5) Real-time software engineering 6) Distributed simulation 7) Motion bases, including cueing 8) Visual systems, including image generation 9) Control loading	
Study Goals	1) You can explain the use cases of piloted flight simulators and the resulting (sub-)system requirements. 2) You can explain the working principles of piloted flight simulator subsystems (vehicle model, real-time and distributed software, motion system, visual system, and control loading). 3) You can identify the design options for a flight simulators functions, distinguish strong and weak points, as well as relate these to the simulators intended use. 4) You can identify, analyze, and evaluate piloted flight simulator systems in an operational context, e.g. in a flight training centre or a simulator manufacturing plant. 5) You can identify and explain the different ways in which the fidelity of a simulator (system) can be evaluated (physical, perceptual, behavioural). 6) You can use common methodologies used for assessing simulator fidelity in industry (QTG, OMCT, numerical stability).	
Education Method	Lectures, simulator demonstrations, and excursions to industry.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace. Recommended but not required literature: D. Allerton, Principles of Flight Simulation, Wiley, 2009, ISBN 978-0-470-75436-8	
Assessment	Written examination and excursion report.	

AE4431	Aircraft noise and emissions	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Responsible Instructor	Prof.dr. D.G. Simons	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The aircraft noise part of this course is required for AE4463P 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics - Dutch aircraft noise model NRM - acoustic imaging Education method: The practicing is in the form of concrete assignments, which are individually graded.	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and terts band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. (Acoustic imaging: Principle of beamforming. Imaging aircraft noise data.) Aircraft noise sources: Propeller noise (mechanism and directional properties) and blade passage frequency. Turbo engine noise and directionality of the fan and exhaust jet noise source. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise (and the modelling of it). (Aero-acoustics (Lighthill theory).) The topics in parentheses (acoustic imaging, airframe noise modelling and aero-acoustics) are part of the reader, but do not belong to this course and will be treated in the follow-up course AE4463P. Aircraft emissions and climate effects Aircraft emissions: Gas emissions from aircraft engines. Complete and actual combustion. Combustion products from burning kerosene. Emission Index. Carbon dioxide emissions of transport (aviation and, for comparison, road traffic). Atmospheric radiation physics: Concept of radiation flux. Solar spectrum and solar constant. Radiative properties of natural surfaces. Thermal emission. Black body radiation and the laws of Planck, Wien, Stefan-Boltzmann and Kirchoff. Atmospheric transmission and emission. Effects of ozone. Climate effects: A climate model for the Earth. Greenhouse effect. Radiative forcing. Example calculations of global warming due to various carbon emission scenarios. Contribution of aviation. Feedback in the climate system.	
Study Goals	Understand the relationship between aviation and the environment.	
Education Method	Lectures and exercises.	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions Readers:	

	Course AE4431 Aircraft Noise and Emissions Part A: Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022 Course AE4431 Aircraft Noise and Emissions Part B: Introduction to aircraft emissions and climate effects Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022
Assessment	Written exam (students may bring their own formula sheet to the exam)
Set-up	Lectures and exercises

AE4441-16	Operations optimisation	4
Responsible Instructor	Ir. P.C. Roling	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course aims at providing the students with knowledge and experience to set-up and analyze linear and nonlinear optimization problems. The course covers the following topics: 1. Introduction to Operations Research, examples of OR models for air transport. 2. Linear programming (LP) models and the simplex Method. 3. Sensitivity analysis and Duality. 4. Transportation and assignment problems. 5. Network optimization. 6. Mixed Integer Linear programming. 7. Nonlinear programming.	
Study Goals	At the end of the course, the students will be able to: 1. Understand the theory behind basic linear and non-linear optimization problems. 2. Model a problem as a linear program, a (mixed) integer program or a network optimization problem; 3. Verify the model using self created test data set. 4. Apply a sensitivity analysis.	
Education Method	The course consists of lectures (2hrs every week) and studio classroom sessions (2hrs every week), where the theory is practiced. Students need to read up on the subject before the studio classroom sessions.	
Books	Introduction to Operations Research Frederick S. Hillier and Gerald L. Lieberman, Tenth Edition Mc Graw-Hill International Edition	
Assessment	The final grade consists of two parts: 50% individual exam (computer-based open book); 50% an assignment (groups of 2-3 students). Both partial grades must be 6.0 or higher.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5322	Thesis Control and Operations	45
Responsible Instructor	Prof.dr.ir. M. Mulder	
Responsible Instructor	Prof.dr. D.G. Simons	
Instructor	Dr.ir. C.C. de Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The final graduation project.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Frequent meetings with the graduation supervisors, where feedback is given on the students progress and performance, and also suggestions for improvement.	
Assessment	At the end of the graduation project, students are required to deliver two things. First of all, they hand in a report that summarizes the work being done in the project. Second, the graduate presents the results of her/his project for an academic audience. Then, the student will be invited to be part of the one-hour 'ir-examen', where she or he is questioned by her/his supervisors and most of the times also external experts on the project as a whole. The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum.	

WI2056LR	Systems Theory	4
Responsible Instructor	Dr.ir. J.G. Maks	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	You are required to be familiar with the following concepts of linear algebra: elementary row operations, vector and matrix algebra, subspace, rank, characteristic polynomial, eigenvalue, algebraic and geometric multiplicity of eigenvalue, eigenvector, basis transformation, matrix exponential. You are also required to be familiar with the theory of systems of linear first order differential equations and with the theory of the Laplace transform.	
Course Contents	During this course the following topics will be covered: State-space representation of input-output system (both for continuous-time and discrete-time case). Linearization of a system. Solution of a linear system (both for continuous-time and discrete-time case). Impulse response and step response of a linear system (both for continuous-time and discrete-time case). Asymptotic stability, BIBO stability (both for continuous-time and discrete-time case). Controllability and observability (both for continuous-time and discrete-time case). Kalman decomposition. State feedback (both for continuous-time and discrete-time case). State reconstruction by observer (both for continuous-time and discrete-time case). System description in frequency domain. Composition of systems in frequency domain. Realization of transfer function.	
Study Goals	After a successful completion of the course you will be able to model an input-output system by a state space model (both for continuous-time and discrete-time case). linearize a system around a given solution. determine whether an equilibrium point of a linear system is asymptotically stable, weakly stable or unstable (both for continuous-time and discrete-time case). compute the solution of a linear time-invariant system (both for continuous-time and discrete-time case). compute the impulse response and the step response of a linear time-invariant system (both for continuous-time and discrete-time case). determine whether or not a linear system is controllable (both for continuous-time and discrete-time case). determine whether or not a linear system is observable (both for continuous-time and discrete-time case). construct a Kalman decomposition of a linear system. design a feedback control (if it exists) which makes an unstable system stable or one which reduces the effect of disturbing signals (both for continuous-time and discrete-time case). design an observer (if it exists) which produces an approximation of the state of the system such that the error converges to zero (both for continuous-time and discrete-time case). represent a linear system in the frequency domain. construct various realizations of a given transfer function.	
Education Method	Lectures	
Literature and Study Materials	Mathematical Systems Theory, 4th edition, VSSD, Delft, 2011, by G.J. Olsder, J.W. van der Woude, J.G. Maks and D. Jeltsema.	
Assessment	Written exam of 3 hours.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Control & Simulation

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-construct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism. There are five deadlines for handing in each year.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4270	Control and Operations Project	4
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Dr. O.A. Sharpans'kykh	
Contact Hours / Week x/x/x/x	///	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course code is a placeholder for miscellaneous student assignments within the Control and Operations track. For example, a lecturer would like to have some data processed or have a nice simulation model that can be used for demonstration purposes in the lectures. Through this course code, we now have a way to compensate students for their efforts by considering it as an elective of 4 ECTS. If there are such assignments available, they will be announced.	
Study Goals	n/a, depends on the nature of the assignment	
Education Method	n/a	
Assessment	A (short) report needs to be written about the assignment. The content needs to be discussed with the supervisor of the assignment.	
Elective	Yes	

AE4311	Nonlinear and Adaptive Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I AE4301 AE4301P AE4312/AE4320	
Course Contents	Safety and survivability of modern civil and military aircraft when airframe and structural failures have occurred in flight have become a key issue in the aerospace industry nowadays. This course introduces nonlinear and adaptive control techniques to enhance reconfigurabilities of aircraft flight control systems. Advanced nonlinear and adaptive control techniques such as Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion (INDI), Adaptive Backstepping (ABS) and Incremental Backstepping are described together with applications to fault tolerant and reconfigurable aircraft flight control system designs. Unconventional yet available control effectors are used through control allocation algorithms to achieve required performance.	
Study Goals	Students will be able to acquire the skills in designing fault tolerant and reconfigurable flight control systems using Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion, Adaptive Backstepping (ABS) and Incremental Backstepping control techniques.	
Education Method	Lectures altered with assisted computer exercises	
Literature and Study Materials	Handouts of lecture slides (available on Brightspace)	
	Recommended literature: J.-J.E. Slotine & Weiping Li, Applied non-linear control, Englewood Cliffs : Prentice-Hall, 1991, ISBN 0-13-040890-5 M. Krstic, et al, Nonlinear and adaptive control design, John Wiley & Sons, Inc., ISBN 0-471-12732-9	
Assessment	Individual Take-home assignment	
Remarks	Hand-in exercise and a report of the design of an advanced controller. The report must be handed in at the request date given by the course instructor.	
Set-up	Course contents Introduction to advanced flight control Nonlinear aircraft flight control based on Nonlinear Dynamic Inversion Adaptive Nonlinear Dynamic Inversion Incremental Nonlinear Dynamic Inversion Aircraft fault tolerant flight control based on Adaptive and Incremental Nonlinear Dynamic Inversion Nonlinear control based on backstepping Nonlinear aircraft flight control based on backstepping Adaptive Backstepping Incremental Backstepping Aircraft fault tolerant flight control based on Adaptive and Incremental Backstepping Control allocation	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hover performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aerospace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course. One final take-home assignment at the end of the course.	
Special Information	A helicopter flight in SIMONA simulator will be given to the students following this course. Note: This may differ due to COVID-19 measures. Changes to this will be communicated via Brightspace.	
Set-up	Online education and Classical lecturing	

AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3
Responsible Instructor	D.M. Pool	
Instructor	Dr.ir. C. Borst	
Instructor	Dr.ir. M.M. van Paassen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is an elective extension to the AE4316 (core) course for the C&S master profile. Prior knowledge from AE4316 is expected and mandatory.	
Course Contents	This course further explores the various aspects of supervisory control and manual control (cybernetics) and their applications in aerospace as introduced in the AE4316 lecture series. Through a number of lectures in the first weeks, this course provides detailed practical information on the current advanced state-of-the art methods used for measuring, analyzing, and modeling human performance and behavior in supervisory and manual control tasks and how such methods are applied in applications relevant to aerospace. Furthermore, through extensive practical work performed in small groups, students will gain practical experience with the advanced analysis of human-machine system performance through an experiment they need to plan and perform themselves. In this experiment, which will be performed in one of the simulator setups available at Control & Simulation (i.e., the Air Traffic Management Laboratory, the Human-Machine Laboratory or the SIMONA Research Simulator), students will collect data for an experiment of their choice, analyze the collected data, and reflect on the conclusions drawn from their experimental measurements.	
Study Goals	1. Formulate a good research question and hypotheses for a supervisory or manual control experiment. 2. Demonstrate understanding of the direct link between an experiments independent, dependent and control variables and the data analysis that can be performed on its data. 3. Analyze data from a supervisory or manual control experiment using state-of-the-art data analysis techniques and statistical tests. 4. Compare experimental findings with theoretical constructs and elementary models of human performance or control behavior. 5. Reflect on the scientific conclusions drawn from a set of human subject experiment data and their reliability. 6. Reflect on the consequences of experiment design choices on the scope and validity of human subject experiment results.	
Education Method	Classroom lectures, twice a week in the first three lecture weeks (3.1-3.3). After that, students will work in groups on the design and execution of an experiment they will perform in one of the simulation facilities at Control & Simulation.	
Literature and Study Materials	The presentation slides of all lectures will be made available through Brightspace Recommended reading for this course: 1. Jagacinski, Richard J., and John Flach. Control theory for humans: Quantitative approaches to modeling performance. Taylor & Francis, 2011. 2. Wickens, Christopher D., Lee, John D., Liu, Yili, and Gordon Becker, Sallie E. An Introduction to Human Factors Engineering, Second Edition, Pearson Education Inc., 2004. 3. Field, Andy. Discovering statistics using SPSS. Sage publications, 2009.	
Assessment	No written exam. Instead, grading for AE4316P will be based on the quality of two group deliverables: an Experiment plan (deadline end week 3.4, 20% of final grade) and a final Scientific Article (deadline end of Q3, 80% of final grade).	
Elective	Yes	
Set-up	Classroom lectures will be provided in the first three weeks. The following topics will be addressed in the lectures: 1. Introduction and experiment descriptions. 2. Human subject experiment design. 3. Experimental data analysis and statistics. 4. Cognitive Work Analysis. 5. Multi-channel manual control behavior identification. In addition to the lectures, students work in small groups on a practical assignment. This effort includes: developing an experimental plan, collecting and analyzing data, and writing a scientific article.	

AE4317	Autonomous Flight of Micro Air Vehicles	4
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. C. de Wagter	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Programming experience (any language) If the student has no previous background in C / C++, we expect the student to do an online tutorial before / at the beginning of the course. For example: Desired prior knowledge: Programming experience with C / C++ AE4393 (Avionics and Operations)	
Course Contents	This course covers the challenges and existing state-of-the-art methods for enabling autonomous flight of Micro Air Vehicles (MAVs), ranging from 20-gram flapping wings to 1 kg quad rotors. The emphasis is on computationally efficient, bio-inspired approaches to MAV autonomous flight.	
Study Goals	The theoretical knowledge will be applied by the students in the practical assignment, in which student groups program quad rotors in order to avoid obstacles in TU Delft's Cyberzoo. (1) Students can explain the differences between MAVs and larger aircraft in terms of control, design, aerodynamics, and electronics. (2) Students can explain how the above-mentioned differences affect the task of achieving autonomous flight with MAVs. (3) Students can list and explain Artificial Intelligence (AI) and control techniques useful for autonomous flight of Micro Air Vehicles (MAVs). a) Bio-inspired robotics b) Vision-based navigation (4) Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task. (5) Students can reproduce the current state of regulations for MAV operations. In addition, they can assess and discuss on the safety and ethical aspects of MAV operations.	
Education Method	The course will consist of theory (7 lectures) and a practical assignment (6 practical sessions and a competition session). Brief description practical assignment: Student teams will develop an efficient, purely vision-based approach to autonomously navigate through an obstacle field. The algorithm will have to run onboard the Parrot AR drone 2, a commercially available quad rotor equipped with a range of sensors. Importantly for the project, it has a single frontal camera. Hence, the developed algorithm concerns a monocular vision solution to obstacle avoidance.	
Assessment	The assessment of the course will consist for 50% of the theoretical exam, and 50% of the practical assignment. Assessment of the practical assignment is based on the performance of the drone on the task, the followed approach as explained in a presentation, and the code produced by the team.	

AE4320	System Identification of Aerospace Vehicles	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	D.M. Pool	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301.	
Parts	Week arrangement 1. Introduction. A global overview of the complete system identification cycle applied to aerospace vehicles, from sensor data acquisition and state estimation to aircraft aerodynamic model identification and validation. Introduction to sensor data acquisition and management, state and parameter estimation, and the two-step method for aerodynamic model identification. Finally an overview is give of advanced modelling techniques. 2. State estimation. Introduction to the essential role of the Kalman Filter (KF) in aircraft state estimation. The extension of the standard KF to nonlinear systems will be introduced in the form of the Extended Kalman Filter/Smoothen (EKF/EKS), and iterated extended Kalman filter (IEKF). Applications of the Kalman filters in aerospace navigation, attitude determination, on-board sensor calibration, sensor integration, and data fusion will be discussed. 3. Parameter estimation. Introduction to the process of parameter estimation and its role in creating accurate aerodynamic models of aircraft. Introduction to four of the most widely used parameter estimation techniques; Least squares (LS), Weighted Least squares (WLS), Total Least Squares (TLS), and Maximum Likelihood (ML) parameter estimation. Additionally, recursive parameter estimation methods like Recursive Least Squares (RLS) will be introduced. 4. Advanced optimization techniques. Some problems cannot be solved adequately with standard linear optimization techniques. In that case, a nonlinear optimization method must be used. In this part of the course, a powerful global nonlinear optimization technique based on interval analysis will be introduced. 5. Advanced modelling methods: Neural Networks. Neural Networks are a class of black-box function approximators that can be used to model nonlinear systems like Micro Air Vehicles and high performance aircraft. The application of neural networks to aircraft system identification is discussed, and it will be shown how neural networks can be used inside a framework for parameter estimation. 6. Advanced modelling methods: Multivariate Splines. Multivariate Simplex B-Splines are a powerful type of white-box function approximators that can be used to create accurate models of aircraft with highly nonlinear aerodynamics such as MAVs and high performance aircraft. A number of unique methods for assessing the quality of the created models are discussed, and pointers towards future research are given. 7. Demonstration: Aircraft system identification with multivariate splines. An in-depth demonstration will be given of the process of Aircraft system identification with multivariate splines using real flight test data from the Cessna Citation II laboratory aircraft. The goal is not only to create an accurate flight dynamics model of the Citation II, but also to apply various model validation techniques to assess the quality of the created model.	
Course Contents	Accurate aerodynamic models play a crucial role in the design and operation of flight simulators and flight control systems. The creation of accurate aerodynamic models from CFD, wind tunnel, and flight test data has historically been a highly challenging task. This is a direct result of the nonlinear nature of aircraft (aero)dynamics and the fact that not all aircraft states can be measured directly. As a consequence, the aerospace vehicle parameter identification problem constitutes a joint parameter-state estimation problem. It is the aim of this course to provide the student with a complete overview of the system identification cycle as it is currently applied to aerospace systems, and to introduce the student to the current state of the art in the field of aerodynamic model identification. The course consists of 7 parts covering the entire system identification cycle. In the first part of the course, the process of data acquisition using on-board sensors including accelerometers, gyroscopes, GPS and various air data sensors is discussed. It will be demonstrated that measurements made by real-world sensors are contaminated with noise, and are sometimes biased. Additionally, some aircraft states, like the true angle of attack, cannot be measured directly and must be reconstructed by combining sensor measurements. The second part of the course introduces the concept of state estimation in which prior physical knowledge of the system is used in a Kalman filter (KF) to estimate the true aircraft states from the measured states. Next to the ordinary Kalman filter, the Extended Kalman filter (EKF) and the Iterated extended Kalman filter (IEKF) will be introduced. In the third part of the course various methods for the estimation of model parameters will be discussed. It will be shown how the results from the state estimation are used in combination with a parameter estimator. This part not only focuses on offline batch parameter estimation methods like ordinary least squares (OLS), weighted least squares (WLS), total least squares (TLS), and maximum likelihood (ML) estimators, but also on recursive parameter estimation methods like recursive least squares (RLS) that can be used online. Special attention will be paid on the process of choosing a parameter estimator that is right for the job. The fourth part of the course introduces an advanced global nonlinear optimization method based on interval analysis. In aerospace system identification, so-called non-convex and nonlinear optimization problems are often encountered. Such problems can be solved with global nonlinear optimization methods like interval analysis. The fifth and sixth part of the course focuses on two advanced model structures that can be used in combination with the earlier introduced parameter estimation methods. In the fifth part of the course the neural network black-box function approximator is introduced. It is shown how neural networks are used to approximate scattered multidimensional data. In the sixth part of the course a new method for aerodynamic model identification based on multivariate simplex B-splines is introduced. This method was recently developed at the Faculty of Aerospace Engineering of the TU-Delft, and has a number of advantages over existing methods. For example, the simplex B-splines have a transparent model structure, are general in any number of dimensions, and can be computed efficiently in real-time.	

Study Goals	In the final part of the course, all theory introduced in the first six parts is used in a demonstration that uses real flight data from a real-life system like the Cessna Citation II laboratory aircraft, or a Micro Air Vehicle to identify a multivariate spline based aerodynamic model. The created model is validated using various techniques. Finally, pointers are given towards further research in the field of aerodynamic model validation. After completing the course, students will have a thorough understanding of the theoretical elements of aircraft system identification. Students will be able to apply these elements to go from real CFD, wind tunnel, and flight test data to an accurate aerodynamic model that can be used in simulator or flight control system applications.
Education Method	Lectures.
Literature and Study Materials	Q.P. Chu, Modern flight test technologies and system identification, lecture notes, Faculty of Aerospace Engineering, Delft University of technology. C.C. de Visser, Global Nonlinear Model Identification with Multivariate Splines, PhD thesis, Faculty of Aerospace Engineering, Delft University of technology. Recommended literature V. Klein and E. A. Morelli, Aircraft System Identification, AIAA, 2006. J.A. Mulder, J.K.Sridhar, J.Breeman, Nonlinear Analysis and Manoeuvre Design, Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 2, , May 1994 R.E. Maine, K.W.Illiff, AGARD Flight Test Techniques Series on Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 1, , December 1986 . Eric Walter and Luc Pronzato, Identification of parametric models, Springer, ISBN 3-540-76119-5.
Assessment	Individual Take-home assignments

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written (or on-line remote exam, depending on COVID-19 situation) (50%) and assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4323	Real-time Distributed Flight and Space Simulation	3
Responsible Instructor	Ir. O. Stroosma	
Instructor	Ing. E.H.H. Thung	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE1205 (Programming & Scientific Computing in Python, or equivalent introduction to programming) AE4322 (Piloted Flight Simulation)	
Course Contents	Lecture topics, not necessarily in chronological order 1) C++ introduction 2) Real-time software engineering 3) Delft University Environment for Communication and Activation (DUECA) 4) Modelling tools 5) Verification and Validation 6) Software project management Programming assignments, individually and in a team. The course is completed with a practical project in which students program, demonstrate and document a distributed simulation with an aerospace theme. Attendance in the project meetings is mandatory.	
Study Goals	1) The student can identify the software issues related to real-time, distributed simulation. 2) The student can actively apply the theory of rigid body dynamics in real-time simulation. 3) The student can combine the modelling of dynamical systems with numerical integration routines. 4) The student can use quaternions to represent rigid body attitudes. 5) The student can implement simulation models using different software languages and tools (C++, Simulink). 6) The student can create 3D models for computer visualisation and optimise these for real-time simulation. 7) The student can (in a team) implement a real-time simulation on a distributed platform, using advanced software tools.	
Education Method	Lectures, programming instruction, and programming assignments.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace.	
Assessment	Individual assignment during 2nd/3rd week and team project report. Attendance at the team project meetings in week 3 and later is mandatory.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Self-supervised learning - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4351	Robust Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	S. Theodoulis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I WI2056LR AE4301 AE4301P	
Course Contents	The vast majority of aerospace vehicles, spanning from UAVs and rotorcrafts to military or commercial aircraft and all the way up to space launchers, need to fly in a secure, predictable and repeatable way. To achieve this, the on-board flight control system needs to be designed to withstand (i.e. to be robust against) a whole array of uncertainties, disturbances, noise, rapidly-varying flight conditions and imperfect components. Furthermore, the designer needs to provide with solid guarantees and verifiable certificates that the flight control system must meet. This course is aimed at offering to the students hands-on theoretical and practical knowledge on the modelling, design and analysis of robust flight control laws, paying special attention to the industrial applicability of the tools albeit with a research twist.	
Study Goals	This course builds upon traditional control design tools already acquired by the students (such as pole placement, root locus or Bode diagram techniques, etc.), extending them to modern multivariable optimal and robust control techniques extensively used in industry and research. The path towards a robust flight controller goes through three steps: modelling, design and finally analysis. The modelling part of the course deals with uncertainty modelling that may stem from various sources such as poorly known aerodynamics, varying flight conditions or imperfect actuators and sensors. The linear fractional representation (LFR) concept for linear systems is introduced eventually leading to a detailed description of what we know about what we don't know. The design part of the course gradually moves from optimal control methods (such as LQR, LQG, LQG/LTR, H2) towards robust control methods (such as Hinf, Hinf by loop shaping) in a state or output feedback configuration. Controllers of predefined structure (such as PID, lead/lag, etc.) but designed using robustness considerations are also presented. The final analysis part of the course tests the controllers designed for the uncertain model of the system and evaluates in a systematic way their robust stability and performance. This is done via traditional tools such as multivariable stability margins but most importantly using the structural singular value (SSV) and the underlying mu analysis theory and tools. To illustrate the above concepts the longitudinal and lateral dynamics of a highly maneuverable fighter aircraft are considered and various control and stability augmentation systems are designed. The results and various simulations are done in MATLAB/Simulink.	
Education Method	Lectures together with assisted computer exercises in MATLAB/Simulink	
Books	Some books that may be used from the students for reference are: - Multivariable Feedback Control: Analysis and Design, Ian Postlethwaite and Sigurd Skogestad - Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems, Brian L. Stevens, Eric N. Johnson, and Frank L. Lewis - Robust Multivariable Control of Aerospace Systems, Declan Bates and Ian Postlethwaite	
Assessment	Take-home assignment	

AE4352	Mathematical and human-inspired decision making	3
Responsible Instructor	Dr. A. Jamshidnejad	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course you will learn about model predictive control (MPC) as well as human-inspired decision making (particularly, how fuzzy logic is used for decision making)	
Study Goals	After finishing this course successfully, the student should be able to: - Formulate model predictive control (MPC) problem for mathematical decision making - Develop fuzzy logic controllers for human-inspired decision making - Implement MPC controllers and fuzzy logic controllers for case study problems	
Education Method	Lectures	
Assessment	Group project (report + presentation of results)	

AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
Responsible Instructor	Dr. O.A. Sharfans'kykh	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Introduction. Agents and Multiagent systems. Agent-based modelling architectures. Examples from air transportation. Emergence in Multiagent systems. Agent-based simulation. Agent-based modeling and simulation tools. Agent-based coordination, planning, and scheduling in air transportation. Nature-inspired approaches to solve optimization problems. Swarm intelligence. Adaptive behavior and learning in agent-based systems. Collaborative decision making in air transportation. Negotiation, auctions, game-theoretic approaches. Agent-based model analysis: sensitivity, uncertainty, robustness. Validation of agent-based models.	
Study Goals	The student has to be able: - to formulate a practical air transportation problem as an agent or a multiagent system model; - to identify appropriate agent-based methods and to apply them; - to implement agent-based models; - to perform agent-based simulation, interpret and analyze simulation results; - to be able to apply agent-based optimization techniques.	
Education Method	Lectures and practical sessions	
Assessment	70% of the final grade for the course is the grade for a group assignment; 30% is the grade for the individual defense.	
Tags	Adventurous Algorithmics Analysis Artificial intelligence Group work Intensive Modelling Programming Small groups Transport & Logistics Transport phenomena	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy).	
Course Contents	Note: Without some affinity/experience with python (and numpy) programming it might still be possible to pass the course, but will become quite difficult. In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics, following the book; building the backbone. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ Research papers will be made available through Brightspace.	
Assessment	Assignments are based on PyTorch: https://pytorch.org/ 1. 0.1 weight: Presentation: during the seminar a small group of students presents a paper. You will have to present once. 2. 0.9 weight: Exam about the papers, assignments, and the theory. 2. Pass/Fail: Project: in a small group of students you work on a deep learning paper reproducibility project. Final grade: (pass project) * (0.1 * Presentation + 0.9 * Exam) *** Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) ***	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of control theory	
Course Contents	Theory and application of intelligent control system i.e. the combination of machine learning and control theory. Topics: * Introduction to nonlinear control * Introduction to intelligent control: model learning, control learning * Fuzzy sets and systems * Artificial neural networks, deep learning (Reservoir Computing, CNN, Geometric Deep Learning) * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of real-world applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely fuzzy logic, iterative control, Gaussian processes, artificial neural networks, etc. LO2 Implement 'intelligent control' techniques in MatLab and Simulink or Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Group work Lineair Algebra Mathematics Matlab Modelling	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42035	Integration Project Systems and Control	5
Responsible Instructor	D. Boskos	
Instructor	S.C. Bregman	
Instructor	Dr. L. Laurenti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	SC42015 Control Theory SC42155 Modeling of Dynamical Systems SC42095 Control Engineering SC42025 Filtering and Identification	
Course Contents	The course is based on practical laboratory sessions, in which students will gain hands-on experience with the application of control theory to real-world systems. Matlab and Simulink are used as the primary software environment for the design, analysis and real-time implementation of the algorithms. Students work in groups of two in the lab, with a setup of their choice. The setups include for instance the inverted pendulum, helicopter model, inverted wedge, rotational pendulum, flexible link, magnetic levitation.	
Study Goals	The goal of this course is to integrate and apply the theoretical knowledge gained in the compulsory courses of the Systems and Control curriculum. These course cover the subjects of Control Theory, Physical Modeling, Filtering and Identification. The concepts and tools to be used include mechanistic modeling (based on principles like mass balances, Lagrange equations, etc.), filtering and estimation (e.g., Kalman filtering), linear control design and performance analysis, system identification in open and closed loop. It is assumed that students already know these concepts or are able to look them up in the literature. No theoretical lectures are given in this course.	
Education Method	Project, demonstration	
Assessment	There is no written exam. The final grade is determined on the basis of a written report (50% of the grade), the final discussion of the results with the lecturers (50%) and the performance during the entire course period (rounding off the grade).	
Remarks	Old course code: SC52035	
Department	3mE Department Delft Center for Systems and Control	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Sustainable Air Transport

Program Coordinator	Dr. I.C. Dedoussi
Program Coordinator	Dr. O.A. Sharpans'kykh
Introduction 1	In operating airlines and airports, a large number of (competing) requirements are encountered. These comprise minimizing environmental effects (noise, air quality and climate effects) and costs, and maximizing resilience and, of course, safety. To deal with all requirements two elements are important. First, you need to be able to quantify all these so-called Key Performance Indicators (value, environmental impact, cost, capacity, efficiency and safety). You will learn to deal with questions like How can we define how efficient operations are? or What is the contribution of aviation to greenhouse gasses in the atmosphere?. Simulations, using appropriate modelling techniques, are an essential element in this. Three SAT profile core courses aim to provide you with the relevant theories. - The SAT profile course Agent-based modelling and simulation in Air Transport introduces you to the important and widely used concept of agent-based modelling of complex distributed dynamic systems. - Through the course Aircraft Emissions and Climate Effects you will be able to quantify the effects of aviation on the atmosphere, both near-the airport and globally (climate effects). - Finally, the Stochastic Processes and Simulation course brings in the concept of randomness, accounting for the unpredictability of daily life. The second element draws on this ability to quantify the Performance Indicators and aims to actually optimize airline and airport operations given the requirements agreed upon by the relevant stakeholders. The SAT profile course Airline Planning & Optimization expands upon the track course on operations optimization, now putting a special focus on airlines. Transfer of the developed tools and insights to daily life should be realized, for example through the use of decision support systems or monitoring strategies. The increasing availability of (big) data, ranging from airline network data and aircraft maintenance monitoring to noise and pollution measurements near airports, together with innovative AI and machine learning methods provide new insights and opportunities in the field of air transportation. Both modelling and such data-driven approaches are covered throughout the courses in this profile. Although all courses apply theories to the case of aircraft, airports and airlines, the theories are general and transferable to other fields.
Program Goals	The SAT profile learning objectives are: 1. Understanding of the key operational challenges faced now and in the future by airlines and airports. 2. Understanding the theories needed to quantify the performance indicators related to the operation of airports and airlines (environmental impact, cost, capacity, efficiency, and safety). 3. Apply these theories for modelling, measurements, and analysis in applications related to the operation of existing and future air transport systems. 4. Developing the ability to transfer the methods to society both through technological advancements (decision support systems) and communication.

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory a. Laplace transformation b. Elementary closed loop systems c. Transfer functions 2. Poles and zeros a. First order systems b. Second order systems c. Pole placement for simple systems 3. Root locus method a. Characteristic equation b. Angle and magnitude conditions 4. State space formulation a. Controllability, observability b. Ackerman's formula c. LQR 5. Basic controllers: P,PI,PD,PID 6. Frequency response (Bode plots) 7. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 8. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 9. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 10. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 11. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 12. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 13. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance - substantiate the function of a Flight Control System(FCS) in civil aviation.	

	- apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 AE3212-I AE2235-I	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace	
Assessment	Exercise	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4317 AE4318 AE4319 AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. - Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 4. Navigation equations. 5. Flight deck instruments and integrated systems. 6. The Flight Management System (FMS). 7. Inertial Navigation Systems. 8. Radio navigation systems (ADF, VOR, DME). 9. Landing guidance systems (ILS, GPS). 10. Communication, Navigation, Surveillance (CNS). 11. Satellite navigation systems (GPS). 12. Air Traffic Management (ATM). 13. The Future Air Navigation System (FANS). (see week arrangement)	
Course Contents	This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	(Digital) written exam (closed-book)	

AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
Responsible Instructor	Dr. O.A. Sharfans'kykh	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Introduction. Agents and Multiagent systems. Agent-based modelling architectures. Examples from air transportation. Emergence in Multiagent systems. Agent-based simulation. Agent-based modeling and simulation tools. Agent-based coordination, planning, and scheduling in air transportation. Nature-inspired approaches to solve optimization problems. Swarm intelligence. Adaptive behavior and learning in agent-based systems. Collaborative decision making in air transportation. Negotiation, auctions, game-theoretic approaches. Agent-based model analysis: sensitivity, uncertainty, robustness. Validation of agent-based models.	
Study Goals	The student has to be able: - to formulate a practical air transportation problem as an agent or a multiagent system model; - to identify appropriate agent-based methods and to apply them; - to implement agent-based models; - to perform agent-based simulation, interpret and analyze simulation results; - to be able to apply agent-based optimization techniques.	
Education Method	Lectures and practical sessions	
Assessment	70% of the final grade for the course is the grade for a group assignment; 30% is the grade for the individual defense.	
Tags	Adventurous Algorithmics Analysis Artificial intelligence Group work Intensive Modelling Programming Small groups Transport & Logistics Transport phenomena	

AE4423-20	Airline Planning and Optimisation	4
Responsible Instructor	Dr.ir. B.F. Lopes dos Santos	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	It is assumed that students are familiar with modelling and optimization techniques. It is assumed that students have experience in code development in Python or MatLab (or any other programming language).	
Course Contents	This course provides students with knowledge to analyse planning problems related to airline operations and to develop modeling approaches to solve these problems. The focus is on the relationship between planning models and their operational implications. It starts with a general overview of the airline demand analysis, followed by the study of the planning framework which airlines operate in. This planning framework includes strategic decisions, namely fleet planning and network development, tactical decisions, such as scheduling and maintenance planning.	
Study Goals	At the end of the course, the students should be able to: Obj1: identify the main strategic, tactical, and operational problems of an airline; Obj2: get familiarized with the development and implementation of some modeling techniques: (a) mix-integer linear programming, (b) multi-commodity flow networks, (c) time-space networks, (d) Markov decision programming. Obj3: get familiarized with the development and implementation of some solution techniques: (a) branch-and-bound, (b) column generation, (c) dynamic programming. Obj4: identify, formulate and solve airline strategic and tactical planning problems: (a) airline networks development, (b) fleet planning, (c) frequency planning, (d) aircraft assignment and routing planning, (e) crew scheduling, and (f) maintenance planning. Obj5: identify an airline operations problem, analyse and solve it; Obj6: explain the implications of airline planning decisions and report them in an academic manner.	
Education Method	The course will work according to a blended learning approach. Online lectures will be prepared for 9 content sessions. Discussion and Q&A sessions of 1 hour will be planned per week and there will be tutorial lectures to discuss the assignment(s). The students are assessed according to the grades of a group assignment (60%) and an individual online exam (40%).	
Literature and Study Materials	References literature: - Belobaba, P., Odoni, A., Barnhart, C. (2009), "The Global Airline Industry", John Wiley & Sons Ltd, West Sussex, UK. [ISBN: 978-0-470-74077-4] - Lectures notes and additional material posted on Brightspace during the course. Additional Literature: - Doganis, R. (2010), "Flying Off Course: Airline economics and Marketing", 4th Edition, Routhledge: London, UK. [ISBN: 0-415-44737-2] - Bazargan, M. (2010), "Airline Operations and Scheduling", 2nd Edition, Ashgate Publishing Company, Surrey, UK. [ISBN: 978-0-7546-7900-4] - Clark, P. (2007), "Buying the Big Jets: Fleet Planning for Airlines", Ashgate Publishing Company, Hampshire, UK. [ISBN: 978-0-7546-7090-2]	
Prerequisites	The students should have completed the AE4441 course, or the ME44206 course, or an equivalent operations research course. The students should also be familiar with programming. For python courses, students can follow TIL6010, AE1205, or TI3105TU.	
Assessment	Group assignments - divided into two parts (60%) Final exam (40%) The students need to have a grade equal to or higher than 5.5 in both examination elements. The group assignments are done by groups of (maximum) 3 students.	
Permitted Materials during Tests	None	

AE4431	Aircraft noise and emissions	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Responsible Instructor	Prof.dr. D.G. Simons	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The aircraft noise part of this course is required for AE4463P 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics - Dutch aircraft noise model NRM - acoustic imaging Education method: The practicing is in the form of concrete assignments, which are individually graded.	
Expected prior knowledge		Basic knowledge on calculus and first year physics.
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and tert's band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phono and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. (Acoustic imaging: Principle of beamforming. Imaging aircraft noise data.) Aircraft noise sources: Propeller noise (mechanism and directional properties) and blade passage frequency. Turbo engine noise and directionality of the fan and exhaust jet noise source. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise (and the modelling of it). (Aero-acoustics (Lighthill theory).) The topics in parentheses (acoustic imaging, airframe noise modelling and aero-acoustics) are part of the reader, but do not belong to this course and will be treated in the follow-up course AE4463P. Aircraft emissions and climate effects Aircraft emissions: Gas emissions from aircraft engines. Complete and actual combustion. Combustion products from burning kerosene. Emission Index. Carbon dioxide emissions of transport (aviation and, for comparison, road traffic). Atmospheric radiation physics: Concept of radiation flux. Solar spectrum and solar constant. Radiative properties of natural surfaces. Thermal emission. Black body radiation and the laws of Planck, Wien, Stefan-Boltzmann and Kirchoff. Atmospheric transmission and emission. Effects of ozone. Climate effects: A climate model for the Earth. Greenhouse effect. Radiative forcing. Example calculations of global warming due to various carbon emission scenarios. Contribution of aviation. Feedback in the climate system.	
Study Goals		Understand the relationship between aviation and the environment.
Education Method		Lectures and exercises.
Literature and Study Materials		Lecture slides, exercises with worked-out solutions Readers:

	Course AE4431 Aircraft Noise and Emissions Part A: Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022 Course AE4431 Aircraft Noise and Emissions Part B: Introduction to aircraft emissions and climate effects Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022
Assessment	Written exam (students may bring their own formula sheet to the exam)
Set-up	Lectures and exercises

AE4441-16	Operations optimisation	4
Responsible Instructor	Ir. P.C. Roling	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course aims at providing the students with knowledge and experience to set-up and analyze linear and nonlinear optimization problems. The course covers the following topics: 1. Introduction to Operations Research, examples of OR models for air transport. 2. Linear programming (LP) models and the simplex Method. 3. Sensitivity analysis and Duality. 4. Transportation and assignment problems. 5. Network optimization. 6. Mixed Integer Linear programming. 7. Nonlinear programming.	
Study Goals	At the end of the course, the students will be able to: 1. Understand the theory behind basic linear and non-linear optimization problems. 2. Model a problem as a linear program, a (mixed) integer program or a network optimization problem; 3. Verify the model using self created test data set. 4. Apply a sensitivity analysis.	
Education Method	The course consists of lectures (2hrs every week) and studio classroom sessions (2hrs every week), where the theory is practiced. Students need to read up on the subject before the studio classroom sessions.	
Books	Introduction to Operations Research Frederick S. Hillier and Gerald L. Lieberman, Tenth Edition Mc Graw-Hill International Edition	
Assessment	The final grade consists of two parts: 50% individual exam (computer-based open book); 50% an assignment (groups of 2-3 students). Both partial grades must be 6.0 or higher.	

AE4462-17		Aircraft Emissions and Climate Effects	4
Responsible Instructor	Dr. V. Grewe		
Instructor	Dr. I.C. Dedoussi		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics. Aircraft noise and emissions (AE4431) highly recommended.		
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories		
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.		
Education Method	Lectures and exercises		
Literature and Study Materials	Lecture slides and hand-outs		
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.		
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.		

AE4465	Maintenance Modeling & Analysis	4
Responsible Instructor	Dr.ir. B.F. Lopes dos Santos	
Responsible Instructor	Dr. M. Lourenço Baptista	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Aircraft maintenance is one of the most important operational activities of the aviation industry. It covers all the planning and execution steps that ensure that the airline fleet is safe and in proper working conditions to fly, being a key part to achieving certification and compliance of air travel. This course aims to help students develop skills necessary to: 1 - Acquire the theoretical concepts necessary to model the maintenance needs of present and future aircraft 2 - Analyze data with both classical reliability methods and advanced predictive techniques in order to predict and forecast the present and future maintenance needs of aeronautical equipment On the successful completion of the course, the student should be able to: I Preventive maintenance aka Reliability engineering (classical reliability methods): 1. List, relate, and structure concepts from preventive maintenance 2. Apply life data analysis tools (e.g. Weibull distribution) to determine aircraft system and component reliability characteristics. 3. Apply stochastic demand arrival (e.g. point processes) models to determine and predict aircraft system and component reliability characteristics. II Predictive maintenance: 1. List, relate, and structure concepts from predictive maintenance 2. Select and apply forecasting techniques to model the reliability of complex systems 3. Select and apply predictive models (e.g. physics-of-failure, data-driven) models to address aeronautical problems that involve the estimation of the remaining useful life Apply simple models from physics to predict failure behavior Apply data-driven techniques from artificial intelligence such as neural networks (machine learning) to produce predictive results 4. Apply appropriate methodologies to evaluate a predictive maintenance solution (measures such as mean absolute error, root mean square error, scoring functions).	
Study Goals	This course has two main learning objectives, described below. 1. Apply concepts, models and strategies from preventive maintenance, including lifetime distribution models to model the reliability of non-repairable systems stochastic point processes to model the reliability of repairable systems 2. Understand and apply concepts, models and strategies from predictive maintenance, including forecasting techniques to model the reliability of complex systems machine learning models to model the reliability of complex systems evaluation techniques to assess the reliability of complex systems.	
Education Method	The content of the course will be introduced via on-campus lectures. The student will also have access to exercises and quizzes on Brightspace. There will be assignments to be solved in group or individually. The assignments may involve small coding problems, open questions, or mathematical exercises. The main bibliography for this course will be: The presentation slides of all lectures will be made available through Brightspace. Reference literature (covering the third part of the course): Goebel, K., M. Daigle, A. Saxena, S. Sankararaman, I. Roychoudhury, and J. R. Celaya. "Prognostics: the science of predictions." ISBN-13 (2017): 978-1539074830.	
Assessment	The final grade of the course will be composed of the individual exam grade (60%) and individual or group assignments (40% - 4 to 6 assignments). The exercises and quizzes on Brightspace are formative only. It is required that students need to have a grade equal to or higher than 5.5 in the individual exam and assignment grade to pass the course. Transition AE4426-19 -> AE4465 1. The course Stochastic Processes and Simulation, profile course for SAT, is cancelled. 2. For the profile courses, the replacement is Maintenance Modelling & Analysis (AE4465, 4 EC). 3. For students from the 2021 2022 cohort and earlier who have not yet completed AE4426-19, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4426-19. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4426-19 is not completed in 2022 2023, students will need to complete AE4465.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Officially registered and enrolled as a full time Master student Aerospace Engineering.

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

Enrolment / Application

Special Information

Set-up

AE5322	Thesis Control and Operations	45
Responsible Instructor	Prof.dr.ir. M. Mulder	
Responsible Instructor	Prof.dr. D.G. Simons	
Instructor	Dr.ir. C.C. de Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The final graduation project.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Frequent meetings with the graduation supervisors, where feedback is given on the students progress and performance, and also suggestions for improvement.	
Assessment	At the end of the graduation project, students are required to deliver two things. First of all, they hand in a report that summarizes the work being done in the project. Second, the graduate presents the results of her/his project for an academic audience. Then, the student will be invited to be part of the one-hour 'ir-examen', where she or he is questioned by her/his supervisors and most of the times also external experts on the project as a whole. The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Sustainable Air Transport
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AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-construct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and (to limited extend) viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Typically Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during at the beginning of Q1).	
Assessment	The assessment is done through 5 personal assignments.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE3-W02 (4P720 TUe) plus basic aerodynamics.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade element/momentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsen's Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4238	Aero Engine Technology	4
Responsible Instructor	A. Gangoli Rao	
Instructor	Ir. M. Pini	
Instructor	F. Yin	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery"; "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Turbomachinery - o New Engine /propulsion Concepts - o Turbine Cooling - o Inlets and Exhaust - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an hybrid course in which the lectures are pre-recorded and some for tutorials will be solved in the classroom. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Analyse the cooling requirements and compute the effect of cooling on engine performance 14. Evaluate the calorific values for various fuels 15. Understand the basics of combustion and emission formation in Aero engines 16. Understand the working of gas turbine combustor 17. Understand the basic principles of gas turbine controls and evaluate different strategies for control 18. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a hybrid teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen
	2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Reader	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on several homework assignments (5-6) given during the course and a final exam. All assignments together will form 50% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resourcen/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4270	Control and Operations Project	4
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Dr. O.A. Sharpans'kykh	
Contact Hours / Week x/x/x/x	///	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course code is a placeholder for miscellaneous student assignments within the Control and Operations track. For example, a lecturer would like to have some data processed or have a nice simulation model that can be used for demonstration purposes in the lectures. Through this course code, we now have a way to compensate students for their efforts by considering it as an elective of 4 ECTS. If there are such assignments available, they will be announced.	
Study Goals	n/a, depends on the nature of the assignment	
Education Method	n/a	
Assessment	A (short) report needs to be written about the assignment. The content needs to be discussed with the supervisor of the assignment.	
Elective	Yes	

AE4320	System Identification of Aerospace Vehicles	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	D.M. Pool	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301.	
Parts	Week arrangement 1. Introduction. A global overview of the complete system identification cycle applied to aerospace vehicles, from sensor data acquisition and state estimation to aircraft aerodynamic model identification and validation. Introduction to sensor data acquisition and management, state and parameter estimation, and the two-step method for aerodynamic model identification. Finally an overview is give of advanced modelling techniques. 2. State estimation. Introduction to the essential role of the Kalman Filter (KF) in aircraft state estimation. The extension of the standard KF to nonlinear systems will be introduced in the form of the Extended Kalman Filter/Smotherer (EKF/EKS), and iterated extended Kalman filter (IEKF). Applications of the Kalman filters in aerospace navigation, attitude determination, on-board sensor calibration, sensor integration, and data fusion will be discussed. 3. Parameter estimation. Introduction to the process of parameter estimation and its role in creating accurate aerodynamic models of aircraft. Introduction to four of the most widely used parameter estimation techniques; Least squares (LS), Weighted Least squares (WLS), Total Least Squares (TLS), and Maximum Likelihood (ML) parameter estimation. Additionally, recursive parameter estimation methods like Recursive Least Squares (RLS) will be introduced. 4. Advanced optimization techniques. Some problems cannot be solved adequately with standard linear optimization techniques. In that case, a nonlinear optimization method must be used. In this part of the course, a powerful global nonlinear optimization technique based on interval analysis will be introduced. 5. Advanced modelling methods: Neural Networks. Neural Networks are a class of black-box function approximators that can be used to model nonlinear systems like Micro Air Vehicles and high performance aircraft. The application of neural networks to aircraft system identification is discussed, and it will be shown how neural networks can be used inside a framework for parameter estimation. 6. Advanced modelling methods: Multivariate Splines. Multivariate Simplex B-Splines are a powerful type of white-box function approximators that can be used to create accurate models of aircraft with highly nonlinear aerodynamics such as MAVs and high performance aircraft. A number of unique methods for assessing the quality of the created models are discussed, and pointers towards future research are given. 7. Demonstration: Aircraft system identification with multivariate splines. An in-depth demonstration will be given of the process of Aircraft system identification with multivariate splines using real flight test data from the Cessna Citation II laboratory aircraft. The goal is not only to create an accurate flight dynamics model of the Citation II, but also to apply various model validation techniques to assess the quality of the created model.	
Course Contents	Accurate aerodynamic models play a crucial role in the design and operation of flight simulators and flight control systems. The creation of accurate aerodynamic models from CFD, wind tunnel, and flight test data has historically been a highly challenging task. This is a direct result of the nonlinear nature of aircraft (aero)dynamics and the fact that not all aircraft states can be measured directly. As a consequence, the aerospace vehicle parameter identification problem constitutes a joint parameter-state estimation problem. It is the aim of this course to provide the student with a complete overview of the system identification cycle as it is currently applied to aerospace systems, and to introduce the student to the current state of the art in the field of aerodynamic model identification. The course consists of 7 parts covering the entire system identification cycle. In the first part of the course, the process of data acquisition using on-board sensors including accelerometers, gyroscopes, GPS and various air data sensors is discussed. It will be demonstrated that measurements made by real-world sensors are contaminated with noise, and are sometimes biased. Additionally, some aircraft states, like the true angle of attack, cannot be measured directly and must be reconstructed by combining sensor measurements. The second part of the course introduces the concept of state estimation in which prior physical knowledge of the system is used in a Kalman filter (KF) to estimate the true aircraft states from the measured states. Next to the ordinary Kalman filter, the Extended Kalman filter (EKF) and the Iterated extended Kalman filter (IEKF) will be introduced. In the third part of the course various methods for the estimation of model parameters will be discussed. It will be shown how the results from the state estimation are used in combination with a parameter estimator. This part not only focuses on offline batch parameter estimation methods like ordinary least squares (OLS), weighted least squares (WLS), total least squares (TLS), and maximum likelihood (ML) estimators, but also on recursive parameter estimation methods like recursive least squares (RLS) that can be used online. Special attention will be paid on the process of choosing a parameter estimator that is right for the job. The fourth part of the course introduces an advanced global nonlinear optimization method based on interval analysis. In aerospace system identification, so-called non-convex and nonlinear optimization problems are often encountered. Such problems can be solved with global nonlinear optimization methods like interval analysis. The fifth and sixth part of the course focuses on two advanced model structures that can be used in combination with the earlier introduced parameter estimation methods. In the fifth part of the course the neural network black-box function approximator is introduced. It is shown how neural networks are used to approximate scattered multidimensional data. In the sixth part of the course a new method for aerodynamic model identification based on multivariate simplex B-splines is introduced. This method was recently developed at the Faculty of Aerospace Engineering of the TU-Delft, and has a number of advantages over existing methods. For example, the simplex B-splines have a transparent model structure, are general in any number of dimensions, and can be computed efficiently in real-time.	

Study Goals	In the final part of the course, all theory introduced in the first six parts is used in a demonstration that uses real flight data from a real-life system like the Cessna Citation II laboratory aircraft, or a Micro Air Vehicle to identify a multivariate spline based aerodynamic model. The created model is validated using various techniques. Finally, pointers are given towards further research in the field of aerodynamic model validation. After completing the course, students will have a thorough understanding of the theoretical elements of aircraft system identification. Students will be able to apply these elements to go from real CFD, wind tunnel, and flight test data to an accurate aerodynamic model that can be used in simulator or flight control system applications.
Education Method	Lectures.
Literature and Study Materials	Q.P. Chu, Modern flight test technologies and system identification, lecture notes, Faculty of Aerospace Engineering, Delft University of technology. C.C. de Visser, Global Nonlinear Model Identification with Multivariate Splines, PhD thesis, Faculty of Aerospace Engineering, Delft University of technology. Recommended literature V. Klein and E. A. Morelli, Aircraft System Identification, AIAA, 2006. J.A. Mulder, J.K.Sridhar, J.Breeman, Nonlinear Analysis and Manoeuvre Design, Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 2, , May 1994 R.E. Maine, K.W.Illiff, AGARD Flight Test Techniques Series on Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 1, , December 1986 . Eric Walter and Luc Pronzato, Identification of parametric models, Springer, ISBN 3-540-76119-5.
Assessment	Individual Take-home assignments

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written (or on-line remote exam, depending on COVID-19 situation) (50%) and assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4352	Mathematical and human-inspired decision making	3
Responsible Instructor	Dr. A. Jamshidnejad	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course you will learn about model predictive control (MPC) as well as human-inspired decision making (particularly, how fuzzy logic is used for decision making)	
Study Goals	After finishing this course successfully, the student should be able to: - Formulate model predictive control (MPC) problem for mathematical decision making - Develop fuzzy logic controllers for human-inspired decision making - Implement MPC controllers and fuzzy logic controllers for case study problems	
Education Method	Lectures	
Assessment	Group project (report + presentation of results)	

AE4439	Air Transport Operations Modelling Project	2
Responsible Instructor	Dr. O.A. Sharpans'kykh	
Instructor	Dr.ir. B.F. Lopes dos Santos	
Instructor	A. Bombelli	
Instructor	Dr. M.A. Mitici	
Instructor	Dr. M. Lourenço Baptista	
Instructor	Dr. L. Li	
Instructor	Ir. P.C. Roling	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	All mandatory ATO courses.	
Course Contents	The course has a project format. It is meant to deepen knowledge and skills in mathematical and computational modelling in application to specific air transport problems. The course can be done under supervision of any ATO lecturer, who is willing to supervise such a project. Specific project themes can be very diverse, but should be in line with the research areas of ATO. A specific theme can be defined in consultation with a supervisor. It is advisable to work on a project in Q3 or Q4, when the basic modelling techniques have been learned in Q1 and Q2.	
Study Goals	n/a, depend on the project	
Education Method	The course has a project format.	
Assessment	A report needs to be written about the project. The content needs to be discussed with the supervisor.	

AE4446	Airport and Cargo Operations	4
Responsible Instructor	A. Bombelli	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course consists of the following lectures: Lecture 1 Introduction Lecture 2 System planning Lecture 3 Airside capacity and runway design Lecture 4 Landside operations Lecture 5 Gate planning Lecture 6 Air cargo supply chain Lecture 7 Bin packing problems in air cargo operations Lecture 8 Cross-docking and ULD build-up scheduling Lecture 9-14 Guest lectures	
Study Goals	A short quiz is given after lectures 1 till 8. A bonus point for the test can be earned by taking part in these quizzes. At the end of the course, students will be able to: - Understand and critically assess key airport characteristics. - Analyze and describe an airport in terms of operations. - Understand the air cargo supply chain in terms of stakeholders and operations - Understand and reproduce mathematical models describing different operations occurring inside or around airports	
Education Method	The course consist of about 8 base lectures and 6 guest lectures (based on availability of guests). Students work in a group of 4 students to assess and reproduce airport/cargo related research and write a report. They also have to review about 3-4 other groups work. A closed book computer test is used to test and grade the individual knowledge of the students.	
Literature and Study Materials	Slides and other information and supplementary material are posted on Brightspace	
Books	Airport Systems: Planning, Design, and Management, Richard de Neufville and Amadeo Odoni, McGraw-Hill	
Assessment	40 % Individual computer test (must be 5.5 or higher) 40 % Group report (must be 5.5 or higher) 10 % Group review of other groups reports 10 % Homework quizzes	
Permitted Materials during Tests	Non graphical calculator	

AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	AE4260 B (Experimental Applications of Aeroacoustics)	
Expected prior knowledge	AE4431 (Aircraft noise and emissions)	
Course Contents	AE4463P aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) will be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B and F are required. A choice can be made between C, D and E.	
Study Goals	The course aims at providing the students with an thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	- The student is able to recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - The student is able to solve the shortest path problem and the transportation problem as well as some small flow shop problems. - The student is able to solve the travelling salesman problem by the Branch and Bound algorithm. - The student knows several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - The student knows some basic theorems concerning the mentioned problems and is able to prove some of these theorems. - The student has knowledge about methods to solve large scale problems, especially shortest path and vehicle routing problems. - The student has basic knowledge about scheduling problems and complexity theory - The student is able to implement several of the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam.	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr. K.S. Postek	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 3: Spaceflight

MSc Coordinator	Ir. R. Noomen
Introduction 1	The Space MSc track (old name: Spaceflight MSc track) is offered to graduates that are looking for an academic programme that prepares them well for an exciting career in the international space sector. The master curriculum covers the full life cycle of space missions, from conceptual design to implementation and utilisation. This track focuses on space engineering, spaceflight dynamics and planetary exploration. Astronautics is quite different from aeronautics because each space mission is uniquely designed to perform a specific task related to its operational or scientific objectives. This requires an end-to-end approach where the objectives drive the design of the mission and data processing is an integral part of the mission. Within this track, you are offered opportunities to participate in ongoing engineering and scientific projects. You will develop skills to carry out an engineering or research project independently and individually. This will be achieved by a highly focused MSc thesis project under supervision of a staff member. In addition, you will acquire broad knowledge in the field of space and its applications. These objectives will make you a well-rounded aerospace engineer with generally applicable engineering and research skills and with a clear focus area. The track consists of three profiles: Space Engineering, Spaceflight Dynamics and Planetary Exploration. Each profile has a different focus, but they are also interrelated. On the one hand you will become an all-round space professional, but on the other hand you will acquire generic skills that enable you to pursue a career in a broad spectrum of industrial and research environments.
Program Goals	Learning objectives of the MSc Track You will develop skills to carry out an engineering or research project independently and individually. This will be achieved by a highly focused MSc thesis project under strict supervision. In addition, you will acquire broad knowledge in the field of space and its applications. These objectives will make you a broad aerospace engineer with generally applicable engineering and research skills and with a clear focus area. Experience has shown that all students who selected this profile have made a successful career in industry, knowledge institutes, agencies and academia both within as well as outside the aerospace world.
Program Structure 1	The track consists of three profiles that share a common core of 14 ECTS. The track core contains three space courses covering satellite orbits, planetary sciences, and space systems engineering, plus a compulsory courses on ethics. These courses provide a strong basis for any space engineer. The profile core courses are intended to broaden this basis depending on the student interest in the engineering, dynamics or exploration aspects of space missions. The courses of Profile I (Space Engineering, a minimum of 17 ECTS out of a total of 26 ECTS to be selected) go deeper into the propulsion, instrumentation, thermal characteristics and design of spacecraft and rockets. Profile II (Spaceflight Dynamics) also requires a minimum of 17 ECTS, now out of a total of 28 ECTS; it highlights the flight mechanics, simulation, optimization, data processing and applications content. Profile III (Planetary Exploration) also demands a minimum of 17 ECTS out of a total of 26 ECTS; very similar to profile II it addresses instrumentation, flight mechanics, simulation, optimization, data processing and applications, but now with the focus on objects in the Solar System and beyond. In addition, students will also select elective courses, part of which will be spent on courses on Personal and Professional Development (PPD). The program is completed with a mandatory internship (or a Joint Interdisciplinary Project, JIP) of 15 ECTS and the thesis project (45 ECTS). The choice of the elective courses is done in consultation with the potential supervisor of the thesis project theme, the profile coordinator of the profile of interest, and/or the track coordinator; preferably they are tailored for the area of interest of the student, with an outlook already for a thesis theme (or even specific project). The participating chairs in this MSc Spacetrack provide a wide choice of recommended electives, but it is also possible, in consultation with the potential thesis supervisor and approved by the profile/track coordinator, to choose elective courses from other chairs within and outside the faculty or even Delft university.

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Space Engineering	
Program Coordinator	Ir. B.T.C. Zandbergen
Introduction 1	Profile I "Space Engineering" Short description: this profile combines multidisciplinary engineering fields to realise high-performance space systems and system components. Areas of interest include the engineering of space missions, space vehicles and instruments, sensors, actuators, mechanisms, propulsive means, vehicle control, distributed space systems, and Systems Engineering. In Space Engineering, applying state-of-the-art technologies is daily business. The Space Engineering profile aims to educate future academic engineers in end-to-end space engineering. You will learn how to engineer complex space vehicles: launch vehicles, satellites and deep space probes, as well as their subsystems, instruments and components using a systematic approach. You will gain expertise on every phase of a spacecraft system development including design, development, integration, verification/testing and operation. You will also learn about management aspects of engineering including scheduling and planning, and about working in multi-disciplinary design teams to formulate spacecraft system solutions. This will enable you to apply your gained knowledge also to other complex technical systems outside of aerospace engineering.

AE4874 I	Fundamentals of Astrodynamics	4
Responsible Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ae4866 ae4868 ae4870 ae4872 ae4876 ae4889 ae4894 ae4895	
Summary	Thorough treatment of the fundamentals of astrodynamics; topics are treated in a mathematically consistent way. You will acquire fundamental insights in the theory and methods of astrodynamics as well as working knowledge to compute and analyze spacecraft trajectories.	
Course Contents	Introduction to astrodynamics; two-body and many-body problems; relative motion; coordinates, reference frames, orbital elements, and time; geocentric, cislunar, and interplanetary flight; introduction to orbital perturbations.	
Study Goals	You will be able to: 1. Provide full and accurate descriptions of the fundamental concepts and phenomena of astrodynamics; 2. Explain the relationships between astrodynamics concepts & phenomena and the mathematics & physics theory that describes them; 3. Apply the relevant relationships, models, and methods for astrodynamics analysis; 4. Derive and manipulate the equations of motion for a spacecraft subject to gravitational and other forces; 5. Apply the equations of motion, their solutions, and related equations to analyze the absolute and relative positions and velocities of spacecraft as a function of time. Additional detail will be available in the course syllabus.	
Education Method	Lectures, self study	
Literature and Study Materials	Fundamentals of Astrodynamics, Karel Wakker, 2015. ISBN: 978-946-186-4192 available for download from the TU Delft Library / Institutional Repository also available to order as university reader in hardcopy Astrodynamics course website Orbital Mechanics for Engineering Students, Howard Curtis, 2013, 3rd edition. ISBN: 978-008-097-7478 (print) 978-008-097-7485 (ebook) available for download from the TU Delft Library also available to order from booksellers as hardcopy	
Assessment	Written exam	

AE4880	Optical Space Sensors	4
Responsible Instructor	Ir. J.M. Kuiper	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course is organised from two perspectives: one is the space instrumentation engineering: Sensor physics, anatomy and system design requirements Classes of instruments Geometrical optics Optical design and performance Optical components Opto-mechanical design Detection subsystems Radiometric performance analysis Spectral performance analysis On-ground characterization and calibration Manufacturing, assembly, alignment, test Instrument sizing The other line is the design and performance of different classes of instruments. Spectrometers Optical imagers Radiometers (Space) Interferometers Laser Satellite Communication Deployable optics Miniaturized instruments	
Course Contents	Space instrumentation is a discipline aimed at creating state-of-the-art sensors capable of measuring the environments of the Earth, the planets and their satellites, the Sun, and interplanetary space. This course surveys the physical principles and engineering of instrumentation used throughout the many related fields of space science. Upon completion of the course the students will have a firm grasp of the principles and techniques used to sense and measure photons. The course focuses on the design and engineering process of electro-optical space instruments, starting from the design requirement up till the on-ground functional and performance testing and calibration.	
Study Goals	After the course the student should be able: - To list the candidate measurement techniques for given observation requirements - To list and describe for each type of instrument: - o The specific applications and their importance - o The various elements that make up an instrument and their main function(s) - o The different options that exist to perform this function - o The important performance parameters - To explain how the performance requirements translate into design parameters and operational scenarios - To perform instrument engineering to a level of detail, sufficient to accomplish a feasible concept for the instrument (i.e. conceptual definition and justification of its optical-thermal-mechanical-electrical and detector design including budgeting, sizing and sampling) - To indicate and justify the design drivers - To give the rationale for the critical engineering aspects (e.g. alignment accuracy, straylight, stability, sensitivity to contamination or radiation, testability, manufacturability) - To set up an elementary instrument simulator to predict instrument performance and sensitivities to variations and error sources - To draft an on-ground test flow - To draft a concept for in-orbit calibration - To determine interface requirements on the hosting platform, the orbit and its operation - To give an outlook for future developments and applications - Be able to make a breakdown of a SatCom (laser sat communication) chain and its link budget - Understand the basics of a DST (deployable space telescope) system design and its thermo-mechanical boundary conditions - Understand the basics of a Space Interferometer for exo-planetary research	
Education Method	Lectures are partly given by professionals from space industry. To bridge the knowledge gap in optics from high school level to MSc level (in the absence of relevant preparatory BSc lectures), a series of in-class optical exercises will be given. To support this part the lectures will be partly given on basis of "flipping the classroom". This means that some basic optical lectures on BSc level will be provided with in advance as slides. These will be released on Brightspace at the end of January and need to be studied by the students themselves before the lectures start. In the follow-up classroom exercise sessions questions can be raised on these parts. Elements are: Group assignments of in-class optical exercises. Both the assignments and related homework are part of the formal close-out of the course, in combination with a written exam.	
Literature and Study Materials	Mandatory study material: - BSc level lecture slides to be released in January 2020 as an introduction to the course. - Lecture slides and exercises during the course. - Laser SatCom capita selecta are lectured with slide material only and relevant examples (worked out problems) like link budgets. - The same holds for the topic Deployable Optics - The same holds for the topic Space Interferometers - Reader AE4-880 Space Instrumentation Engineering. Capita selecta of the reader serve as supporting material. Only limited parts of the reader apply to the actual content of the course. These parts will be addressed during the lectures and given in a study reference matrix: to be made actual and released on Brightspace during the course.	
Practical Guide	The Brightspace site AE4880 contains an abundance of information about the course, the content, as well as enrichment data, with links to supporting websites	
Assessment	2 Elements: A) Written exam + B) Group assignments: optical exercises + presentation. If the mark for the exam is 5.0 or higher, the final grade is the weighted average of 80% * mark A + 20 % mark B. These numbers are valid in normal conditions which excludes COVID-19 or similar conditions.	

Permitted Materials during Tests	The exam is an open-book exam. The student is allowed to bring his printed Lecture Slides and Reader to the exam. including the personal notes he made at the lectures (on the slide sheets only). He is NOT allowed to use a laptop, to bring in the exercises, assignments or homework problems, nor their answers, nor previous exams.
Enrolment / Application	Students have to enroll in the first course week or via the Brightspace site of the course.
Remarks	Self activation by the in-class and homework assignments is the key to a successful pass of the exam.
Set-up	The course is organised along two learning ways: one is the space instrumentation engineering that is lectured by the lecturer or guest lecturers. The other line is the design of different classes of instruments. This part is provided by the students in the form of assignments and practicing homework problems.

AE4890-11	Planetary Sciences I	4
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr. D.M. Stam	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Instructor	Dr.ir. W. van der Wal	
Responsible for assignments	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4876 Planetary Sciences II	
Parts	Planetary Sciences: - Origin and Evolution of our Solar System - Planetary Surfaces and Interiors - Planetary Atmospheres - Planetary Magnetism - Planetary Ring Systems - Asteroids and Comets - Extrasolar Planets - Recent Topics in Planetary Science Current topics in planetary science with a focus on a specific topic each time the course is offered, such as Mars, Venus, Galilean satellites, comets, recent discoveries, spacecraft observations, or other developments.	
Course Contents	Planetary science is a major interdisciplinary field, combining aspects of astrophysics with geology, geophysics, meteorology, atmospheric and space science. This close relationship to geophysics, atmospheric and space sciences implies that the study of the planetary bodies in our Solar System and beyond offers the unique opportunity for comparison available to Earth scientists. This course teaches the concepts in the planetary physical sciences and solar system properties. The learning process is greatly enhanced by involving students in solving related problems.	
Study Goals	After the course the student should be able to: 1. Describe the primary physical processes in solar and planetary systems and the properties involved 2. Apply modelling techniques that are commonly used to describe the primary physical processes in solar and planetary systems 3. Demonstrate the use of physical principles to derive knowledge on the state and evolution of (extra-)solar system bodies from observations 4. Assess the quality of evidence of current knowledge of (extra-)solar system bodies 5. Assess the habitability of a (extra-)solar system body based on inferences of its properties and orbit	
Education Method	Classical oral lectures and homework assignments.	
Literature and Study Materials	Jack J. Lissauer and Imke de Pater, Fundamental Planetary Science, Cambridge University Press, ISBN: 9780521618557, 2013. PowerPoint/PDF files of the lectures and lecture notes.	
Books	Jack J. Lissauer & Imke de Pater, Fundamental Planetary Science, Cambridge Univ. Press, 2013.	
Assessment	The final grade for the planetary sciences course is determined from: - mark of a written exam - mark for mandatory homework assignments The mark for the written exam needs to be 5.0 or higher in all cases, no exceptions. If the mark for the exam is 5.0 or higher, grading for this course follows from averaging two parts with equal (so 0.5) weight: the mark for the classroom/take home exercises and the mark for the exam. No opportunity will be given to redo the classroom/take home exercises within the same academic year. The grade for the homework assignments will remain valid for the next consecutive academic year, and so will annul the year thereafter	
Exam Hours	3 hours written exam	
Permitted Materials during Tests	Closed-Book exam, Non-programmable calculator is allowed.	
Set-up	Oral lectures and classroom/take home assignments.	

AE4S01	Thermal Rocket Propulsion	4
Responsible Instructor	Ir. B.T.C. Zandbergen	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Propulsion material contained in the BSc courses AE1222-II and AE2230-I. Course material will be made available via Brightspace.	
Course Contents	The course focuses on thermo-(chemical) rocket propulsion system analysis and design. Topics dealt with include: 1. Fundamentals of (thermo-chemical) rocket propulsion; 2. Ideal rocket motor/nozzle: Ideal performances, optimum thrust, characteristic velocity and thrust coefficient, and quality factors; 3. Nozzles: Types of nozzles (conical, bell, etc.), nozzle dimensions, flow divergence, boundary layers, under- and over-expansion, and Summerfield criterion; 4. Chemical propellants: Molar mass, specific heat ratio and adiabatic flame temperature calculation for gas mixtures (based on known reaction equation), mass density, dynamic viscosity, thermal conductivity; 5. Chemical equilibrium calculations, hemical equilibrium flow, frozen flow and chemical kinetics; Introduction to program for calculation of chemical equilibrium gas composition and gas properties (in tutorial); 6. Heat transfer & cooling: Convection, radiation and conduction; 7. Cooling: Thermal insulation, ablation, radiation, film, dump and regenerative cooling; 8. Liquid rocket engine combustor design: Steady state internal ballistics, liquid injection, operating pressure, chamber pressure drop, and characteristic length; 9. Solid rocket motor combustor design*: solid regression, grain shape and internal ballistics including operating pressure, necessary condition(s) for stable operation, pressure sensitivity for initial temperature and change in Klemmung, local conditions (flow velocity, pressure, etc.)), and two phase flow; 10. Hybrid rocket motor combustor design*: Solid regression, grain shape, (quasi-)steady operation (operating pressure, and local conditions (flow velocity, pressure, etc.)); 11-13 Liquid propellant storage and feed systems including gas-pressure and pump fed systems, motor cycles and propellant distribution; 14. Capita Selecta * Only one of the two will be dealt with. This varies from year to year.	
Study Goals	At the end of this course, the student shall be able to perform important steps in the analysis and design of thermo-(chemical) rocket propulsion systems using basic methods that allow for taking into account fluid flow, heat addition, propellant thermo-chemistry, heat transfer and cooling, liquid, solid or hybrid ballistics, (liquid) propellant feeding, and propellant storage.	
Education Method	Lecture + exercises (for feedback); During the lectures extensive use is made of existing data to allow for verification of the methods presented.	
Literature and Study Materials	Lecture notes "Thermal Rocket Propulsion" + hand-outs. Supporting information and tools are available through Brightspace. Recommended literature: G.P. Sutton, Rocket propulsion elements, Wiley Inter Science, 6th edition (or higher).	
Assessment	Assessment is through a number of small homework assignments (20% of final grade) and an hybrid (written and online) exam (80% of final grade). In case of a fail grade there is one resit exam opportunity that makes up 100% of the final grade (no homework assignments). Resit exam, like the normal exam, is a hybrid exam and will be scheduled in Q3 (exam period).	
Remarks	Enrolment for the course is via Brightspace. The course will be made available on Brightspace from early September.	
Set-up	Lectures (14 x 2 hours) possibly with 1 tutorial on the use of RPA-Lite and/or NASA CEA for rocket motor performance estimation.	

AE4S01P	Exercise Thermal Rocket Propulsion	2
Responsible Instructor	Dr. B.V.S. Jyoti	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This practical builds on the course material taught in the course AE4S01. It is important that the student has a basic experience in predicting rocket motor performance using the theory taught in AE4S01.	
Course Contents	Thermal rocket propulsion analysis and design practical. The work centres about preparing, executing, analysing and reporting a rocket motor functional test. The aim of this test is to determine how well the motor performs and determine motor quality as well as the quality of its main components (chamber and nozzle). For a given motor* students are asked to: 1. predict motor performances from theoretical analysis/simulations performed; 2. devise a test plan including at least a description of the item to be tested, test objective(s), test approach with short description of planned test(s), test equipment and instrumentation, required calibrations, pass/fail criteria, test risks/issues, test deliverables); 3. prepare the test (assemble motor and install it on the thrust bench) and perform test readiness review; 4. test the motor; 5. discuss the test data obtained and draw inferences; 6. compare test data with simulation results and draw inferences; 7. report work done. * Last year we conducted a test on either a small solid rocket motor or a water rocket.	
Study Goals	To improve the students skills in designing and analyzing thermal rocket propulsion systems. To provide students the basic skills (and associated knowledge) of functionally testing a rocket motor/thruster. Following this practical, the student is able to: - explain the main elements of a test plan, - devise a test plan, - predict motor performance (needed for the generation of the test plan), - conduct a functional test of a rocket motor, - analyze rocket motor test data to determine the quality of the motor and its main components (chamber and nozzle), - draw inferences on the quality of the motor and its main constituents.	
Education Method	Practical. Lap-top in the lectures (teaching with data)	
Literature and Study Materials	Literature and study materials are published via Brightspace.	
Assessment	Based on written material as well as participation in lab practical. Grade is pass/fail.	
Set-up	Practical is scheduled partly in Q2 and partly in Q3 (about 50-50 division).	

AE4S07	Micropropulsion	4
Responsible Instructor	A. Cervone	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	2 3 4	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No mandatory requirements on previously taken courses. Advised: Propulsion and Power (BSc) or Spacecraft Technology (BSc, minor), Thermal Rocket Propulsion (MSc).	
Parts	The first part of the course (1 ECTS) will consist of online study material (available on Brightspace) covering the basics of micro-propulsion for small satellites. At the end of the first part, the student will be asked to take an individual test (1 hour, multiple-choice questions) to check their understanding of the material studied in Brightspace. Students who pass the test with a sufficient grade will be admitted to an individual project (3 ECTS), chosen by the student from a list of options regularly updated by the responsible instructor and aligned with the current micro-propulsion research at the SSE section.	
Course Contents	(1) Fundamental theory and state-of-art of micro-propulsion systems for small satellites; (2) Down-scaling and manufacturing of miniaturized propulsion systems and components; (3) Individual project (variable topic and goals, see below).	
Course Contents Continuation		
Study Goals	Understand the basics of micro-propulsion systems for small satellites and their fundamental differences to larger scale space propulsion. Apply the basic theory to identify the most important requirements for a micro-propulsion system, starting from the relevant top-level mission and satellite requirements. Understand and apply down-scaling rules for the miniaturization of propulsion and fluidic systems. Know, compare and apply (if required by the specific individual project taken by the student) the main available manufacturing techniques for micro-propulsion systems and components, their range of applicability and their advantages/drawbacks. Critically analyze the available state-of-art micro-propulsion options, identify their peculiar characteristics and ranges of applicability, discuss and justify the most suitable one(s) based on given mission and satellite requirements. Actively contribute to solve the challenges and achieve the goals of the current micro-propulsion research in the SSE section, by performing the tasks of a specific project, chosen among the ones proposed by the responsible instructor. Acquire hands-on skills on one or more of the following (depending on the specific individual project selected by the student): micro-propulsion laboratory and testing activities; design of micro-propulsion systems and components; micro-propulsion verification/optimization by means of analytical or numerical tools; or combinations of the above.	
Education Method	Self study (using the study material made available by the responsible instructor on Brightspace). Individual projects (supervised by the responsible instructor or other members of the micro-propulsion research team).	
Literature and Study Materials	All study material (preparation material for the exam, project descriptions, lists of reference documents, etc.) will be made available on Brightspace.	
Assessment	1) Interim test (after taking the online part of the course): 1 hour, multiple choice questions, taken individually in front of the responsible instructor (in a date/time agreed with him). With a grade of 6.0 or higher, the student will be admitted to the individual project. With a grade lower than 6.0, the student will have to wait no less than 4 weeks before re-taking the interim test. With a grade higher than 6.0, the student is allowed to ask for a re-sit to improve the grade: this re-sit shall however be scheduled no less than 4 weeks after the first test. In line with the current general rules of the faculty, only two test opportunities will be allowed per student per Academic Year (first exam + one re-sit). 2) Individual project: chosen from a list prepared and regularly updated by the responsible instructor. For all projects, a description will be provided to the students including: general background, tasks, research question(s) to be answered, expected deliverables and grading criteria. The project is designed for a total individual effort of 3 ECTS (= 84 hours) and will be considered successful when graded 6.0 or higher. With a grade lower than 6.0, the student will be given an opportunity (with an exact deadline) to do additional work in order to improve the deliverables and bring them to a sufficient level. Kick-Off date for the project will be considered as the date when the student is provided by the responsible instructor with the required input material and general instructions on the selected project, as indicated in the project description. After this Kick-Off date, it is NOT allowed to the student, under any circumstance, to change the project choice. The only constraint on the project duration is that the final deliverables have to be provided to the responsible instructor not later than 3 calendar months after the Kick-Off date. A student who has not provided yet the deliverables will receive a reminder about it approximately two calendar weeks before the deadline. A student who doesn't meet the 3-months deadline will be given an additional extension of one calendar month to complete the project, but in this case the final course grade of that student will be reduced by 0.5 points. If a student doesn't provide the deliverables within this additional one month extension, or if a student has a project grade lower than 6.0 after the extra opportunity given for improving the deliverables and bring them to a sufficient level, the project will be considered a fail and the following will apply: -) The student will have to start a completely new project, not earlier than 3 calendar months after the previous one has been failed; -) The final course grade of that student will be reduced by 1.0 points. No exceptions to the above rules for schedule and deadlines will be granted, independently on the reason for the exception request. Final course grade: 25% interim test grade, 75% project grade.	
Permitted Materials during Tests	The only allowed materials during the interim test are a calculator and (when applicable) a formula sheet provided by the responsible instructor. No other materials are allowed, including lecture notes or internet-connected devices.	
Enrolment / Application	Students have to enroll via the Brightspace page of the course, study the online material by following the instructions given in the same page, and inform the responsible instructor as soon as they are ready to take the interim test, deciding with him the date and time of the test. The test date shall be within 15 working days from the time when the responsible instructor is contacted by the student, except specific cases when this is not possible (e.g. holidays or business trips of the responsible instructor during the 15 working days period).	
Set-up	The course will continuously be available to students, who can decide when to start it based on their own study and schedule needs. The list of individual projects will regularly be updated in order to be always aligned with the ongoing micro-propulsion research activities in the SSE section. Various types of projects will be offered to best meet the preferences and characteristics of different students, ranging from lab/testing activities to design tasks, analytical/numerical work, or combinations of them. No specific limitations will be given on the duration of the individual projects, provided that the expected 84 hours effort is put by the student in it and that all project tasks are completed within a maximum of 3 months after the project Kick-Off date. After the student provides to the responsible instructor the deliverable items at the end of the project, the responsible instructor will check these deliverables and send his comments and remarks to the student. The student will then be allowed to revise the deliverables based on the comments received, and this revised version will be graded. No further iterations on the deliverables	

after this one will be allowed. The end date of the project will be considered the day when the student has sent the first draft of the required deliverables, provided that these deliverables are integral, logically formulated and complete.

AE4S10	Microsat Engineering	4
Responsible Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Lectures from experts on various aspects of small satellites and group/individual projects for practicing.	
Course Contents	The course covers the engineering of small satellites. The concept and definition of micro-, nano-, pico-, and femto-satellites is introduced along with the past, present, and future of small satellites in general. The course comprises seven to eight lectures, partly from professionals in the space industry. The lectures are intended to familiarise the students with current industry trends and state-of the art research in small satellite engineering. The lectures vary each year and potential lecture topics include communications, AOCS, onboard computer, distributed space systems, miniature payloads, small satellite missions, etc.	
Study Goals	The course will enable students to design and engineer small satellite missions. The high-level learning objectives of the course are: The students shall be able to explain key aspects of small satellite missions explain and describe elements in the architecture of a spacecraft mission discuss in detail at least one element in the architecture of a spacecraft mission present and defend their design of an element in the architecture of a spacecraft mission show familiarity with current trends in space industry and small satellite research	
Education Method	Lectures (Provided partly by professionals from the space industry). Group work (Students work on assignment in groups; Coaching will be provided from experts to guide the students). Individual work (Students work on small individual assignments).	
Literature and Study Materials	Optional: Wertz J.R., Larson W.J. (eds.); Space Mission Analysis and Design; 3rd edition Kluwer Academic Publishers (2003). Mandatory: Lecture Notes AE4-S10 MicroSatellite Engineering.	
Prerequisites	General mathematical, physical and engineering knowledge from completed BSc study. In particular, AE1101 + AE1102 + AE3201, or similar	
Assessment	There will be two assessment components A Group Assignment and an individual assignment. Group assignment The final demonstration or report will be graded on the following aspects: Planning Presentation, argumentation and defence of the design Functionality and Usability of the deliverable Context of Small satellite (How well has the student team incorporated current small satellite research and industry trends/needs in the design) Independence Creativity Individual assignment This assignment will be graded on the following aspects: Understanding and critical review of articles in a specified technical field of small satellites	
Permitted Materials during Tests	N/A	
Enrolment / Application	Students have to enroll via the BrightSpace site of the course. Professionals are welcome to the course. Conditions for participation and registration are available with the secretary of the Chair of Space Systems Engineering (www.sse.lr.tudelft.nl).	
Set-up	The course comprises lectures, guest lectures, team sessions, group work and individual work.	

AE4S12	Space Systems Engineering	3
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. Specifically, AE1110 + AE1111 + AE2111 + AE3211, or similar	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 2 a. Identifying stakeholder needs 2 b. Generating, evaluating and selecting concepts 2 c. Tutorial 2 d. From stakeholder expectations to technical system requirements 2 e. Logical decomposition and design solution 2 f. Estimating lifecycle costs 2 g. Managing technical risk 3. Integrating Systems 3 a. Integrating the system 3 b. Verifying and validating 3 c. System roll-out & lessons learnt 4. Systems Engineering Approach and Process 4a. Model-Based Systems Engineering 4b. Managing the Systems Engineering Process 4c. Managing the technical effort 4d. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers advanced Space System Engineering demonstrated through many examples from past, current and future space programs. It introduces a process which allows the creation of successful products, such as space applications, space missions and systems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems. The interrelationships between systems engineering and project management, programmatics, cost and risk in the development of space projects are demonstrated at various occasions during the course. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply methods and tools which help to create successful products.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are four high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to use Systems Engineering methods and tools to solve practical real-world problems when engineering a space system. 3. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 4. Participants shall be able to analyze Systems Engineering topics.	
Education Method	Lectures* Tutorial* Individual Homeworks to practise material Group Assignment, to be done by students in small groups The Group Assignment together with graded Individual Homeworks comprise the formal course close-out.	
Literature and Study Materials	Optional: Larson W., Kirkpatrick D., Sellers J.J., Thomas L.D., Verma D. (eds.); Applied Space Systems Engineering, Space Technology Series, McGrawHill (2009) Required: Lecture Slides AE4-S12 Space Systems Engineering.	
Assessment	6 graded Individual Homeworks + 1 Group Assignment. Acceptance of 8 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the secretary of the Chair of Space Systems Engineering (www.tudelft.nl/ae/sse).	
Set-up	The course comprises lectures, tutorials, group work and individual homework. Elements indicated by * are related to the potential impact of the pandemic on the course in 2022/2023 and may be subject to change in the course of the academic year.	

AE4S15	Space Embedded Systems	3
Responsible Instructor	Dr. A. Menicucci	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basics of programming and electronics	
Course Contents	The first part of the course will consist of lectures covering the following topics: 1. Introduction to embedded systems 2. Hardware and software elements 3. Space environment and effects on embedded systems 4. Fault tolerant techniques in space application 5. Radiation testing The second part of the course will be dedicated to a Group Assignment: students, in groups of 2-4, will build and program an embedded system for a possible space application which will feature fault tolerant techniques	
Study Goals	The students will learn to: Explain the basic elements of embedded systems and their characteristics. Explain the different radiation environments encountered in space and their effects on electronics components. Design and implement (hardware and software) an embedded system based on microcontrollers. Apply fault tolerance techniques both on software and hardware parts. Define the system requirements and select the components Evaluate the performances of the selected components to ensure they satisfy the system requirements.	
Education Method	Lectures, partly given by professionals from space area Group assignment, allowing actual hands-on application of the concepts learned. The group assignment will be carried out in period 4 but mostly off-campus. The assignment hand-in deadline will be agreed with the students based on their internships constraints.	
Assessment	Written exam + 1 group assignment	
Permitted Materials during Tests	Lectures notes and books. Wireless communication with electronic devices is NOT allowed.	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Responsible Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a multidisciplinary Master course at TU Delft/Faculty of Aerospace Engineering (AE). In CSDP students of primarily the Spaceflight track work in groups (6-12 students) on a space mission/spacecraft design identified by interested parties (this may be a space company, space institute, or other) for a total of 112 working hours (4 EC) over a period extending 2-3 months under supervision of faculty staff. Additionally, students spend maximum 28 hr (1 EC) on workshops that provide information and training on the use of a typical process management tool like SCRUM (and/or Kanban) and a Concurrent Design Platform / tool (like e.g Excel and/or CDP-4). The course aims to develop student skills in multidisciplinary team projects from a challenge-driven perspective. For the students, CSDP brings forward unique characteristics such as: a starting point in a value-driven approach to problem formulation and decision-making, concurrent and collaborative systems engineering to solve engineering problems thereby taking into account realistic constraints and allowing for training of interpersonal skills and/or workplace competencies (teamwork, planning and organizing, creative thinking, etc.). Topics vary year by year and depend on interest of engineers & researchers to define great engineering challenges for a motivated team of students.	
Study Goals	- Students get advanced experience in the use of Concurrent Engineering (CE), as opposed to the more classical sequential engineering (SE), in a design project. Students will learn not only about how to apply CE, but also about the benefits of CE (as opposed to SE, i.e. the waterfall method) and its limitations; - Students will advance their ability in systems engineering/collaborative while also improving their ability in cooperation, collaboration and communication.	
Education Method	Students will perform a design project as part of a multidisciplinary team. Each group is guided by a mentor. Parallel to the design project, students are instructed on the use of CDP4 and SCRUM (and/or Kanban). For each team regular design sessions will be held in a collaborative learning environment. In these design sessions, teams are given the opportunity for brainstorming (requirements generation, creating first solutions), demonstrating progress, setting tasks, and reviewing the design accomplishments. It is the purpose that each team organizes themselves a team of experts (from industry, academia or other) that are willing to advise the team in their project. At various instants, the work accomplished will also be externally reviewed by other CSDP teams as well as staff from the faculty and external experts with as aim to identify strong/weak points in the design.	
Assessment	The design projects are graded based on a design report, a design log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic (e.g. communications, modelling/experimenting and critical and analytical thinking) and personal effectiveness competencies.	
Enrolment / Application	There is no free enrollment for this course. In Q1, an information session will be held informing students of Space Engineering on the projects available and on how they can apply for a project. Students can then apply and only if they are accepted, they can enroll.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Officially registered and enrolled as a full time Master student Aerospace Engineering.

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

Enrolment / Application

Special Information

Set-up

AE5822	Thesis Space	45
Responsible Instructor	Ir. R. Noomen	
Instructor	Dr.ir. P.N.A.M. Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	Every MSc program in aerospace engineering is completed with an MSc thesis which comprises 45 ECTS. It is intended to provide candidates with an opportunity to carry out an in-depth investigation in a particular area of interest and to make a substantive contribution to knowledge in the area. It is expected that the thesis will include a comprehensive and critical synthesis of the relevant literature and will also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution.	
Study Goals	Learning objectives MSc Thesis During the MSc Thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. Major requirements are duration (nominally 7 months) and innovation. The thesis project can be done at the Faculty of Aerospace Engineering, but also at external organizations (e.g. companies, other universities). In any case, the topic has to be approved by the responsible supervisor before the student is allowed to start with his/her project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Thesis supervisor(s) will provide assistance and direction to the student until the thesis has been defended and accepted. Regular progress/review meetings/consultations with the thesis supervisors are required.	
Assessment	At the end of the MSc thesis project, students are required to defend their thesis before an examination committee. To this end, the student is required to present the thesis in the form of a written report and an oral presentation. The written report should describe the work performed in the thesis project. The oral presentation is to be held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion, and is followed by a private discussion with the examination committee. For assessment of the thesis work, the following aspects are considered: content and scientific & engineering approach, report, presentation & defence, creativity & initiative, and project management. The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty will introduce an updated schematic using rubrics for this, more information will be provided at a later moment.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Space Engineering
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AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments	

AE4499	Space Project	0
Responsible Instructor	Ir. R. Noomen	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Responsible Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	(New) MSc students with a deficit in knowledge on space topics, or who want to do a project with limited duration on a space-related topic.	
Expected prior knowledge	None specific.	
Course Contents	This project covers miscellaneous student assignments within the Space track. Assignments can be a specific research or design assignment or may relate to building a simulation model that can be used for demonstration purposes in lectures. Depending on the size of the exercise, it can be honored with up to 4 ECTS.	
Study Goals	Depends on the specific project.	
Education Method	Homework assignment.	
Books	Depending on the assignment, it could be a general textbook on spaceflight. Examples: SMAD (Wertz et al), The New SMAD (Wertz et al), Spacecraft Systems Engineering (Fortescue et al) and/or the ADSEE course readers and slides. Other input material is also possible, and not necessarily restricted to literature only.	
Reader	Lecture material of space courses BSc AE (depending on the actual assignment)	
Prerequisites	Completed BSc from a degree in engineering.	
Assessment	Grading of deliverables as agreed upon in project statement.	
Enrolment / Application	Email to Profile Coordinator of relevant profile: Barry Zandbergen for Space Engineering, Ron Noomen for Spaceflight Dynamics, or Sebastiaan de Vet for Planetary Exploration.	

AE4ASM526		Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin		
Contact Hours / Week x/x/x/x	0/0/0/4		
Education Period	4		
Start Education	4		
Exam Period	none		
Course Language	English		
Course Contents	Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Both passive as well as active thermal control systems are discussed. Passive devices include: radiators, baffles, heat pipes, multi layer insulation, coatings and finishes. Active devices include: louvers, controlled heaters, cryogenic coolers, Peltier elements and fluid loops.		
Study Goals	The student is able to generate a thermal design of a satellite thermal control system using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - distinguish design hardware options - generate a (concept of a) thermal design - generate analysis models for hand calculation methods and/or computer tools - analyze a thermal design using hand calculation methods and/or computer tools - distinguish thermal environmental test methods		
Education Method	Lectures + homework Lectures are given by an experienced thermal design engineer from Airbus DS NL. Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Asnwers to the questions may be given by fellow students and/or the lecturer(s).		
Literature and Study Materials	Lecture notes available on Brightspace.		
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%)		

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam will be multiple choice. The logistics will depend on the Corona regulations.	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	At the end of this course you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	Brightspace/MATLAB exercises and tests (optional) during education period. The course is concluded through an optimization project and report. Deadlines for handing in the final report will be announced at the start of the course.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Spaceflight Dynamics

Program Coordinator Ir. R. Noomen

Introduction 1 Profile II "Spaceflight Dynamics"

This profile addresses many aspects of satellite missions: from concept to application, and from launch to end-of-life. It aims at delivering skilled space mission engineers and scientists. You will learn about techniques and methods to track, predict, optimize and very precisely compute satellite trajectories, both for spacecraft around and beyond Earth and for launch and (re-)entry vehicles. Space debris is another main research element. You will also learn about methods and tools to exploit satellites for a wide range of scientific and societally relevant issues and applications, including monitoring all kinds of aspects that are related to Earth's climate such as sea level and the ice caps. This profile is generally theoretical in nature.

AE4866-1	Propagation and Optimisation in Astrodynamics	5
Responsible Instructor	D. Dirksen	
Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4868-1 (Numerical Astrodynamics) is required; AE4870B (Re-Entry Systems) is recommended; knowledge of basics of atmospheric flight mechanics is expected.	
Course Contents	1. Introduction. Explanation of course assignment. Introduction to software. 2. Numerical propagation. Formulations of equations of motion. Selection of propagation method. 3. Overview of formulation of acceleration models. Relation between acceleration models and environment models. Influence of acceleration model selection on simulation fidelity/runtime. 4. Introduction to local optimization methods. Constraints in optimization problems. Robustness of optimal solutions. Optimal control. Examples of local optimization in astrodynamics. 5. Global optimization methods (genetic algorithms, particle swarm, etc.). Multi-objective optimization. Examples of global optimization in astrodynamics 6. Pre/post-processing of results. Analysis of numerical propagation/optimization results. Use of numerical tools (e.g. interpolation).	
Study Goals	1. Select and define the settings of a numerical astrodynamics simulation problem, based on top-level description and requirements. 2. Describe the relation between models of the physical environment and the propagated dynamical behaviour. 3. Define and set up local and global optimization problems for typical astrodynamics problems 4. Assess the quality of numerical orbit propagation for typical astrodynamics problems 5. Assess the quality of optimization results for typical astrodynamics problems	
Education Method	Lectures (partly pre-recorded), workshops and open-office hours	
Computer Use	Laptop computer with minimum 4 GB RAM	
Course Relations	AE4870A (Rocket Motion); AE4870B (Re-Entry Systems); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I)	
Literature and Study Materials	Selected articles; lecture notes; lecture slides (details to be announced and distributed during the course).	
Assessment	The assessment is a combination of two individual and one group take-home assignments. The individual assignments are thematically related to the group assignments, and are used as a starting point for the group work . For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 50% Project (individual grade): 25% Project (group grade): 25% Each of these three grades must be ≥ 5.0 to pass the course. In the assignments, students will apply the material covered in the course to the analysis of a real astrodynamics problem. Weekly practical sessions will be held to discuss questions and issues with homework, the practical assignment and the numerical implementation. The resit of the course will be done during the summer period. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	NOTE: To take part in the course, students must submit their preferences for an assignment topic. These must be submitted through Brightspace on the first Wednesday of the course's first lecture week, at the latest. Only enrolling for the course on Brightspace is NOT sufficient to indicate you wish to take part in the course. Instructions for submitting a topic preference will be posted on Brightspace one week before the start of the course, at the latest. Not submitting your topic preference in a timely manner means that you will NOT be able to take part in the course. It is required that students have followed the course AE4868 (Numerical Astrodynamics), and have submitted all assignments, and can still reasonably pass that course, before taking part in this course.	

AE4868-1	Numerical Astrodynamics	4
Responsible Instructor	D. Dirks	
Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4870B (Re-Entry Systems) is recommended; knowledge of basics of atmospheric flight mechanics is expected.	
Course Contents	Basic orbit propagation/integration methods and options; simulation options in Tudat. Defining input for orbit propagation Architectural design of Tudat software. Creation of user-defined modules for e.g. guidance * Linking analytical and numerical design methods in astrodynamics Details of numerical integration methods: multi-stage, multi-step and extrapolation methods (fixed and variable step size) Selection of numerical integration methods; quantification of integration errors	
Study Goals	Set up and run a numerical orbit propagation using the TUDAT software suite from a given set of simulation settings Critically evaluate and analyze numerical orbit propagation results Interpret numerical propagation results, and relate numerical results to theoretical predictions Select numerical integration scheme for a given astrodynamics problem	
Education Method	Lectures on Tudat, Dynamical Modelling, Analytical vs. Numerical Mission Design and Numerical integration Practical sessions/open-office hours,	
Computer Use	Laptop computer with at least 4 GB of RAM	
Course Relations	AE4866 (Propagation and Optimization in Astrodynamics); AE4870A (Rocket Motion); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4874 II (Astrodynamics II)	
Assessment	Individual take-home assignments. Assignments will consist of setting up, running, analyzing, and documenting several astrodynamics simulations in Tudat. The resit of the course will be done during the summer period. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	

AE4870A	Rocket Motion	3
Responsible Instructor	Ir. M.C. Naeije	
Instructor	Ir. W.J.F. Simons	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	Course Overview	
Study Goals	1. Introduction and recap previous courses on orbital mechanics 2. Fundamentals of rocket motion 3. Launch trajectories in a homogeneous gravity field, in vacuum and in an atmosphere (2D); vertical flight, constant pitch angle, gravity, and sounding rockets. 4. Theory of the multi-stage rocket; optimal mass distribution. 5. Ballistic flight over the Earth; 2D, and 3D over spherical (non-rotating and rotating Earth). 6. Launch systems design, unconventional launchers like air launch 7. Summary and question hour (workout exam question example) This course introduces the fundamentals of rocket motion by studying the motion of rockets under different circumstances. Emphasising on analytical solutions of the equations of motion, it will provide insight into the qualitative and quantitative aspects of launch trajectories of rockets in a homogeneous gravitational field, the performance of single and multi-stage rockets, and the exoatmospheric ballistic flight over the Earth. It discusses launcher design considerations and unconventional launch systems like air launch. At the end of the course you will be able to assess the performance of existing space launchers, and set up the (preliminary) design of such systems for specific applications. You also should be able to design a rocket motion simulator (e.g. in Matlab).	
Education Method	Lectures	
Literature and Study Materials	To be studied: Lecture slides AE4870A Rocket Motion Lecture notes AE4-870A Rocket Motion (selected chapters) Further reading: Turner, Martin J.L., Rocket and Spacecraft propulsion: principles, practice and new developments. 3rd Edition. ISBN 978-3-540-69202-7 (selected parts)	
Assessment	Closed-book written exam	

AE4870B	Re-Entry Systems	3
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	IMPORTANT NOTICE: Please read the restrictions under "Assessment" +++++ The course aims at giving a complete overview of re-entry systems, mainly from a mission point of view, but also from the system point of view. To begin with the mission, the fundamental principles of mechanics (Newton's Laws) are applied to derive the equations of motion for translational and rotational motion. Subsequently, simplifications are applied to give a description of planar ballistic, gliding and skipping flight, where the focus will be on analytical solutions for quantifying maximum mechanical and thermal loads. Next, typical mission applications, such as planetary entry and descent with parachutes and/or propulsion systems are discussed, as well as terminal-area flight of winged entry vehicles. To extend the mission applications, attention is also paid to the atmospheric flight aspects during an aero-gravity assist, aerocapture and aerobraking. As many re-entry missions rely on the application of guidance, navigation and control systems, the fundamentals of each of these three sub-systems will be discussed and applied in simple examples, such that after the course the student has a starting point for further study. The relation between vehicle configurations (capsule, winged, lifting body, etc.) and mission aspects, such as flight time and (maximum) system loads is a typical system aspect that will pass review. Finally, all presented theory will be combined in a so-called development plan for building a simulator that can be used for performance analysis of both controlled and uncontrolled re-entry vehicles. How to use such a simulator efficiently is depending on what kind of simulation and analysis technique will be used. Therefore, the course is concluded with an overview of such techniques like Monte-Carlo Analysis, and how to extract the required information from the simulated data.	
Course Contents Continuation	Lecture Topics (the actual topics covered may vary per year): #01 Introduction #02 Entry Environment and Aeroheating #03 Fundamentals of Motion #04 Ballistic entry #05 Gliding Entry #06 Skip Entry #07 Guidance, Navigation and Control #08 Planetary Entry and Descent #09 Advanced Descent and Landing Systems #10 Terminal Area Energy Management	
Study Goals	At the end of this course, the student will be able to (depending on topics covered): 1. Identify the influence of the planetary environment on the motion of and loads on an entry vehicle 2. Derive the equations of planar motion 3. Identify the three unpowered entry mechanisms (ballistic, gliding and skip entry), and derive the corresponding equations of motion 4. Apply a sizing methodology to design an entry descent system (parachute and/or propulsion) 5. Understand the fundamental functions of a Guidance, Navigation and Control System and design a simple system 6. Name the characteristic phases of the Terminal Area flight, and derive the equations for maximum range and maximum dive (steady-state approximation)	
Education Method	Classical lectures (2 x 2 hrs per week). The course is structured in a number of blocks, which each deal with an aspect of re-entry missions and systems. Many real-life examples will be used to highlight certain theoretical aspects, and to show the applicability of the presented theory. Typical exam questions will be used to prepare the student for the exam by testing his/her insight in the presented material. There may be homework assignments to deepen understanding of the material. This will be announced at the beginning of the course. See further details under assessment.	
Course Relations	AE3212-I - Aerospace Flight Dynamics and Simulation AE3212-II - Simulation, Verification and Simulation AE4870A - Rocket Motion	
Literature and Study Materials	1. Lecture Notes and Lecture Slides (Mandatory) Selected references for additional study: 1. Anderson, J. (1989). Hypersonic and High Temperature Gas Dynamics. McGraw-Hill. 2. Bertin, J. (1994). Hypersonic Aerothermodynamics. AIAA. 3. Cornelisse, J., Schoeyer, H., and Wakker, K.F. (1979). Rocket propulsion and spaceflight dynamics. Pitman, London. 4. Hankey, W. (1988). Re-Entry Aerodynamics. AIAA. 5. Mulder, J., van Staveren, W., van der Vaart, J., de Weerd, E., In 't Veld, A. and Mooij, E. (2013). Aerospace Flight Dynamics and Simulation. Delft University of Technology, Lecture Notes AE3-212-I. 6. Regan, F. and Anandakrishnan, S. (1993). Dynamics of Atmospheric Re-Entry. AIAA. 7. Vinh, N. (1981). Optimal trajectories in atmospheric flight. Elsevier. 8. Vinh, N., Busemann, A., and Culp, R. (1980). Hypersonic and planetary entry flight mechanics. The University of Michigan.	
Assessment	Closed-book, written examination. It will be decided at the beginning of the course whether homework assignments are part of the assessment. If that is the case, homework will count for 40% of the grade and the exam for 60%. Both partial grades have to be 6 or higher to pass the course. IMPORTANT NOTICE: The homework and/or exam is only open to students of the following profiles: 1) Spaceflight Dynamics 2) Planetary Exploration 3) Structures in Space Capacity permitting, students from other profiles can send a personal participation request to the course manager	

AE4872	Satellite Orbit Determination	6
Responsible Instructor	Dr.ir. E.J.O. Schrama	
Instructor	Dr.ir. P.N.A.M. Visser	
Instructor	Ir. J.A.A. van den IJssel	
Instructor	C. Siemes	
Responsible for assignments	C. Siemes	
Contact Hours / Week x/x/x/x	0/4/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Course contents 1 Dynamics Introduction to dynamics - o Planetary Gravity field - o Tides and the three-body problem - o Hill radius and Roche limit - o Relation to planetary sciences and astrodynamics Solving Equations of motion - o reformulate orbit problem as a system of ordinary differential equations - o efficiency and accuracy of numerical integration methods - o implementation of numerical integration methods 2 Observations techniques and reference systems Observation techniques - o Laser, Doppler and Camera observations - o Refraction, Electromagnetism, radio- and optical technology - o Tropospheric and ionospheric refraction - o Relativity and the definition of time, - o Classification of time systems (UTC, TAI, etc) - o Light-time effect - o Quality of clocks (Allan Variance behaviour of clocks) Reference systems - o Local and global coordinate systems - o Definition of geoid and reference ellipsoid, height systems - o Precession and nutation, polar motion, polar wander. - o Newton or Einstein, consequences for reference systems 3 Statistics Random variables, probability density functions, moments, hypothesis testing Least squares minimisation - o unconstrained linear parameter estimation, - o data weighting - o nonlinear parameter estimation. Rank deficient equation systems - o compatibility conditions - o general and homogeneous solutions - o constrained linear parameter estimation Mechanisation of parameter estimation algorithms - o Choice of algorithms 4 Orbit determination Perturbation analysis and variational problems - o state transition matrix for initial state vector problems - o partial derivatives for dynamical parameters Parameter estimation - o Identification of parameters - o batch least squares - o Kalman filter, theory and implementation 5 Applications Global Navigation Satellite Systems: - o Technology and terminology, - o various data processing strategies and available software - o Modelling deformation of the solid Earth, - o the Earths gravity field and thermospheric density Satellite laser ranging and Doppler tracking via DORIS: - o technology and terminology, results and applications Observing changes in the cryosphere with satellites Hydrology and Oceanography observed with satellites 6 Homework assignments An exercise related to dynamics, observation systems or reference systems An exercise related to GNSS applied to orbit determination An exercise related to Kalman filtering 7 Numerical Practical Exercises with the Ghost software, typical examples are to solve: - o initial value problems (state vector estimation) - o problems with parameters in a dynamic model (drag parameter estimation) - o problems with time bias parameters	
Study Goals	The candidate should be able 1) To explain the physical and mathematical context of orbit determination (OD) 2) To construct transformations between various coordinate and time systems that play a role in OD	

	3) To examine error sources in satellite tracking data and implement error mitigation strategies 4) To make use of parameter estimation methods in the context of tracking data for OD 5) To apply relevant statistical techniques within the framework of OD 6) To discuss scientific applications of satellite missions that depend on precise OD 6.1) GNSS techniques to model the deformation of the Earth 6.2) Satellite gravimetry to model the gravity field of the Earth 6.3) Satellite altimetry to model the ocean topography 6.4) Satellite Altimeter missions to measure variations of land and sea ice 7) To compute and evaluate precise satellite orbits and dynamic model parameters with state-of-the-art software 8) To work with Linux commands such as ftp, to setup and adapt Python or MATLAB scripts
Education Method	We provide lectures and classroom instructions during quarter 2. In quarter 3 there is a numerical practical to introduce the topics mentioned under the course contents on an MSc level for aerospace engineering students. In quarter 2 all lecture material will be made available, instruction hours will be organised to complement the material presented during the lectures, also, the instruction hours will be used to introduce homework assignments which are part of this course. In quarter 3 we run a numerical practical. During this practical software tools will be used to compute precise satellite orbits and which comes with the estimation of parameters with a state-of-the-art software package (GHOST). In addition a quality assessment of the results will be part of the practical.
Computer Use	Use will be made of Python or MATLAB (preferred) during the assignments, for the numerical practical use will be made of Linux and Fortran.
Literature and Study Materials	Lecture notes will be provided on BrightSpace and they containing most of the material used in this course. Optional reference material: Statistical Orbit Determination, Schutz, Tapley and Born, Academic Press, ISBN: 9780126836301 Montenbruck and Gill, Satellite Orbits, Springer Verlag, doi: doi: 10.1007/978-3-642-58351-3
Assessment	Additional information could be provided via BrightSpace The final grade will come together from: 1) Written examination results in a grade 2) Several individual assignments result in a weighted average grade 3) Numerical practical results in "pass" or "fail". The rules are as follows: 1) The numerical practical result should be: pass 2) The written examination result should be at least a 5,0 3) In order to participate in the numerical practical you should have passed the homework assignments with at least a 5,0 4) No claims can be made with regard to homework assignment results or numerical practical results obtained before the previous course year. For the course in 2022-2023 it means that the results from 2021-2022 are still accepted, however, but we do not accept results from 2020-2021 or earlier years. Note 1: Due to the covid19 pandemic the examination method could be online via remote proctoring in such a way that you obtain the equivalent form of a three hour individual written exam. Note 2: The numerical practical and the assignments were provided in an hybrid format in the academic year 2021-2022, it depends on the university policy and the situation of the country during the pandemic whether we need to stay online in 2022-2023.
Exam Hours	Conform the regulations there are two written exam opportunities within an academic year. These are three hour written exams at the end of quarter 2 and 3.
Permitted Materials during Tests	We provide a closed book examination, required information for the exam (like physical constants or a selected number of equations) will be made available on the examination form.
Tags	Analysis Elektricity & Magnetism Lineair Algebra Matlab Numeric Methods Physics Signals and Systems Technology

AE4874 I	Fundamentals of Astrodynamics	4
Responsible Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ae4866 ae4868 ae4870 ae4872 ae4876 ae4889 ae4894 ae4895	
Summary	Thorough treatment of the fundamentals of astrodynamics; topics are treated in a mathematically consistent way. You will acquire fundamental insights in the theory and methods of astrodynamics as well as working knowledge to compute and analyze spacecraft trajectories.	
Course Contents	Introduction to astrodynamics; two-body and many-body problems; relative motion; coordinates, reference frames, orbital elements, and time; geocentric, cislunar, and interplanetary flight; introduction to orbital perturbations.	
Study Goals	You will be able to: 1. Provide full and accurate descriptions of the fundamental concepts and phenomena of astrodynamics; 2. Explain the relationships between astrodynamics concepts & phenomena and the mathematics & physics theory that describes them; 3. Apply the relevant relationships, models, and methods for astrodynamics analysis; 4. Derive and manipulate the equations of motion for a spacecraft subject to gravitational and other forces; 5. Apply the equations of motion, their solutions, and related equations to analyze the absolute and relative positions and velocities of spacecraft as a function of time. Additional detail will be available in the course syllabus.	
Education Method	Lectures, self study	
Literature and Study Materials	Fundamentals of Astrodynamics, Karel Wakker, 2015. ISBN: 978-946-186-4192 available for download from the TU Delft Library / Institutional Repository also available to order as university reader in hardcopy Astrodynamics course website Orbital Mechanics for Engineering Students, Howard Curtis, 2013, 3rd edition. ISBN: 978-008-097-7478 (print) 978-008-097-7485 (ebook) available for download from the TU Delft Library also available to order from booksellers as hardcopy	
Assessment	Written exam	

AE4876-11	Planetary Sciences II	4
Responsible Instructor	Dr. D.M. Stam	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. A. Menicucci	
Instructor	Ir. M.C. Naeije	
Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr.ir. W. van der Wal	
Instructor	Dr. B. Brandl	
Instructor	C. Siemes	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Core course Planetary Sciences I forms a proper base for the Planetary Sciences II course.	
Parts	Planetary Sciences II is a follow-up, but elective course for core course Planetary Sciences I.	
Course Contents	Intertwined lectures by topical experts give deeper insight and understanding into planetary sciences topics that were only touched upon in Planetary Sciences I, with an emphasis on applied technologies, previous, current and (near) future space missions, and data analysis. Lecture topics focus on: Planetary missions, atmospheric dynamics, magnetic fields, ocean dynamics & oceanography, comets, radiation & space weather, climates & tides, gravitational fields, astronomical missions, and astrobiology. In small groups (4-5 students), students work on the design of a planetary space mission, with a science goal of their choice, supervised by a mentor. Under normal circumstances, the course includes a day at ESTEC, the European Space Agencies facilities at Noordwijk, with a tour of the premises, presentations by ESA mission scientists, and presentations of the planetary space missions designed by the students.	
Study Goals	After completion, the student should be able to reproduce the theoretical backgrounds of the topics taught, and to put them to practice in real life planetary observation problems. The student should also be able to explain the road map from measured quantities to physical quantities, to think about new developments and to come up with new solutions, i.e., understand and being able to apply the course material.	
Education Method	Oral lectures (if necessary, online), homework exercises, planetary mission project, and tutorials.	
Literature and Study Materials	Recommended literature: Fundamental Planetary Science by J. J. Lissauer & I. de Pater, Cambridge University Press, ISBN: 978-0521-61855-7 (paperback) This book is the same book as the one that is used in Planetary Sciences 1. Other books with background information (not required for taking Planetary Sciences 2): C.M.R. Fowler, The solid earth: introduction to Global Geophysics, 2nd ed., Cambridge University Press, 2005, ISBN 0 521 89307 0 W.M. Kaula, Theory of satellite geodesy, Blaisdell, 1966, Open University, Ocean Circulation & Waves, Tides and Shallow-Water Processes editions, Butterworth-Heinemann, 1999 A.E. Gill, Atmosphere-Ocean Dynamics, Int'l geophysics series, vol. 30, Academic Press, INC. (LONDON) LTD., ISBN 0 12 283522 0	
Assessment	Weekly, individual, homework exercises are graded and count towards the final grade. The homework grade must be higher than 5.5. The remainder of the grade is formed by the Planetary Mission Project: In groups of 4-5 students, a mission is designed to investigate a planetary science topic chosen by the students themselves. Each group is guided by a mentor. The Planetary Mission Project is graded through a poster to be presented by each group at the ESTEC day (or, if the ESTEC facilities are closed to outside visitors, though an oral presentation at an online meeting), and a report written by the group. The final grade is computed as follows: 30% homework exercises + 70% planetary mission project	
Remarks	There is no written exam for AE4876-II! Under normal circumstances, towards the end of the course, there will be a visit of a day to ESTEC, the facilities of ESA in Noordwijk. There, the students will attend presentations by ESA personnel about a number of missions. The students will present and defend their poster on the Planetary Mission project, and they get excursions within ESTEC. Attendance of this day is mandatory. The date will be chosen such that there will be minimal to no interference with other lectures.	

AE4890-11	Planetary Sciences I	4
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr. D.M. Stam	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Instructor	Dr.ir. W. van der Wal	
Responsible for assignments	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4876 Planetary Sciences II	
Parts	Planetary Sciences: - Origin and Evolution of our Solar System - Planetary Surfaces and Interiors - Planetary Atmospheres - Planetary Magnetism - Planetary Ring Systems - Asteroids and Comets - Extrasolar Planets - Recent Topics in Planetary Science Current topics in planetary science with a focus on a specific topic each time the course is offered, such as Mars, Venus, Galilean satellites, comets, recent discoveries, spacecraft observations, or other developments.	
Course Contents	Planetary science is a major interdisciplinary field, combining aspects of astrophysics with geology, geophysics, meteorology, atmospheric and space science. This close relationship to geophysics, atmospheric and space sciences implies that the study of the planetary bodies in our Solar System and beyond offers the unique opportunity for comparison available to Earth scientists. This course teaches the concepts in the planetary physical sciences and solar system properties. The learning process is greatly enhanced by involving students in solving related problems.	
Study Goals	After the course the student should be able to: 1. Describe the primary physical processes in solar and planetary systems and the properties involved 2. Apply modelling techniques that are commonly used to describe the primary physical processes in solar and planetary systems 3. Demonstrate the use of physical principles to derive knowledge on the state and evolution of (extra-)solar system bodies from observations 4. Assess the quality of evidence of current knowledge of (extra-)solar system bodies 5. Assess the habitability of a (extra-)solar system body based on inferences of its properties and orbit	
Education Method	Classical oral lectures and homework assignments.	
Literature and Study Materials	Jack J. Lissauer and Imke de Pater, Fundamental Planetary Science, Cambridge University Press, ISBN: 9780521618557, 2013. PowerPoint/PDF files of the lectures and lecture notes.	
Books	Jack J. Lissauer & Imke de Pater, Fundamental Planetary Science, Cambridge Univ. Press, 2013.	
Assessment	The final grade for the planetary sciences course is determined from: - mark of a written exam - mark for mandatory homework assignments The mark for the written exam needs to be 5.0 or higher in all cases, no exceptions. If the mark for the exam is 5.0 or higher, grading for this course follows from averaging two parts with equal (so 0.5) weight: the mark for the classroom/take home exercises and the mark for the exam. No opportunity will be given to redo the classroom/take home exercises within the same academic year. The grade for the homework assignments will remain valid for the next consecutive academic year, and so will annul the year thereafter	
Exam Hours	3 hours written exam	
Permitted Materials during Tests	Closed-Book exam, Non-programmable calculator is allowed.	
Set-up	Oral lectures and classroom/take home assignments.	

AE4894	Practical Astrodynamics	3
Responsible Instructor	J.G. De Teixeira da Encarnacao	
Instructor	D. Dirkx	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The topics covered in this course are chosen to expose you to the practical side of Astrodynamics. These topics are: based on elementary algorithms, straightforward to implement numerically, related practical assignments are to be numerically implemented from-scratch, i.e., without resorting to existing, black-box, or proprietary libraries or toolboxes, and make use of data that are publicly available. Each academic year (refer to Brightspace) a selection of topics will be made from: 1. Open satellite orbit data: Two-Line Elements (TLEs), JPL Ephemerides, space-track.org 2. Applications and transformation of references frames and element types 3. Numerical integration of trajectories 4. Orbit perturbations and incorporation into numerical orbit integration 5. Basics of numerical optimisation 6. Arc-based methods 7. Shape-based low-thrust methods 8. Mission planning: repeat orbits, coverage, and footprints	
Study Goals	By the end of this course, you will be able to: 1. Explain different algorithms that model astrodynamics. 2. Develop numerical algorithms that solve the practical assignments from scratch by applying the acquired knowledge, without resorting to existing libraries or toolboxes. 3. Analyse the results obtained by the computer code you implemented. 4. Validate your results against alternative approaches. 5. Evaluate the strengths and weaknesses of the algorithms used.	
Education Method	The learning material will be distributed through Brightspace. During class, we will largely focus on solving example problems to prepare and guide you to handle the practical exercises effectively. During this course, you will: a. recall knowledge and theory gained in this and other Astrodynamics-related courses, b. extend their understanding to a higher level of detail, c. acquire new skills through the hands-on application of this knowledge and theory to the development algorithms and the interpretation of their results, d. gain a different perspective on the applicability, usefulness, and scope of selected astrodynamics concepts. You are encouraged to solve the exercise using Python, but no specific programming language is enforced. You are free to choose the programming language you prefer. This course is focused on solving practical assignments individually from a blank slate. The list of Python libraries and toolboxes allowed to solve the practical exercises is published in Brightspace. If you use a computer language other than Python, you must include a section in your reports demonstrating you did not use any external library or toolbox. Support will be given during lectures for solving the practical assignments. The course total study load is 3 ECTS x 28 hours/ECTS =84 hours. With two, two-hour lectures per week for 6 weeks (18 hours total) and 1 hour for preparing for each lecture (12 hours total), you can expect to need 9 hours to solve each of the weekly assignments, including any support you need from the TAs.	
Computer Use	You need to provide (or have access to) a personal computer (desktop or laptop).	
Literature and Study Materials	Lecture notes Flight & Orbital Mechanics (AE2230-I) and Fundamentals of Astrodynamics (AE4874-I)	
Books	Wakker, K.F. 2015. Fundamentals of Astrodynamics. TU Delft Library. http://resolver.tudelft.nl/uuid:3fc91471-8e47-4215-af43-718740e6694e Iaco Veris, Alessandro de. 2018. Practical Astrodynamics. Springer Aerospace Technology. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-62220-0 Vallado, David A, and Wayne D McClain. 2020. Fundamentals of Astrodynamics and Applications. (\$90: https://astrobooks.com/fundamentals-of-astrodynamics-and-applications-4th-edition-vallado-2013-softcover.aspx)	
Reader	Lecture notes AE4984, see course content on Brightspace.	
Assessment	There is no written exam. The final grade is the weighted average of the weekly short individual reports (2-5 pages). The rubrics are available in Brightspace.	
Enrolment / Application	Exclusively through Brightspace. Only students from the profile of Spaceflight Dynamics may enrol.	

AE4S12	Space Systems Engineering	3
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. Specifically, AE1110 + AE1111 + AE2111 + AE3211, or similar	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 2 a. Identifying stakeholder needs 2 b. Generating, evaluating and selecting concepts 2 c. Tutorial 2 d. From stakeholder expectations to technical system requirements 2 e. Logical decomposition and design solution 2 f. Estimating lifecycle costs 2 g. Managing technical risk 3. Integrating Systems 3 a. Integrating the system 3 b. Verifying and validating 3 c. System roll-out & lessons learnt 4. Systems Engineering Approach and Process 4a. Model-Based Systems Engineering 4b. Managing the Systems Engineering Process 4c. Managing the technical effort 4d. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers advanced Space System Engineering demonstrated through many examples from past, current and future space programs. It introduces a process which allows the creation of successful products, such as space applications, space missions and systems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems. The interrelationships between systems engineering and project management, programmatics, cost and risk in the development of space projects are demonstrated at various occasions during the course. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply methods and tools which help to create successful products.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are four high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to use Systems Engineering methods and tools to solve practical real-world problems when engineering a space system. 3. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 4. Participants shall be able to analyze Systems Engineering topics.	
Education Method	Lectures* Tutorial* Individual Homeworks to practise material Group Assignment, to be done by students in small groups The Group Assignment together with graded Individual Homeworks comprise the formal course close-out.	
Literature and Study Materials	Optional: Larson W., Kirkpatrick D., Sellers J.J., Thomas L.D., Verma D. (eds.); Applied Space Systems Engineering, Space Technology Series, McGrawHill (2009) Required: Lecture Slides AE4-S12 Space Systems Engineering.	
Assessment	6 graded Individual Homeworks + 1 Group Assignment. Acceptance of 8 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the secretary of the Chair of Space Systems Engineering (www.tudelft.nl/ae/sse).	
Set-up	The course comprises lectures, tutorials, group work and individual homework. Elements indicated by * are related to the potential impact of the pandemic on the course in 2022/2023 and may be subject to change in the course of the academic year.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5822	Thesis Space	45
Responsible Instructor	Ir. R. Noomen	
Instructor	Dr.ir. P.N.A.M. Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	Every MSc program in aerospace engineering is completed with an MSc thesis which comprises 45 ECTS. It is intended to provide candidates with an opportunity to carry out an in-depth investigation in a particular area of interest and to make a substantive contribution to knowledge in the area. It is expected that the thesis will include a comprehensive and critical synthesis of the relevant literature and will also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution.	
Study Goals	Learning objectives MSc Thesis During the MSc Thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. Major requirements are duration (nominally 7 months) and innovation. The thesis project can be done at the Faculty of Aerospace Engineering, but also at external organizations (e.g. companies, other universities). In any case, the topic has to be approved by the responsible supervisor before the student is allowed to start with his/her project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Thesis supervisor(s) will provide assistance and direction to the student until the thesis has been defended and accepted. Regular progress/review meetings/consultations with the thesis supervisors are required.	
Assessment	At the end of the MSc thesis project, students are required to defend their thesis before an examination committee. To this end, the student is required to present the thesis in the form of a written report and an oral presentation. The written report should describe the work performed in the thesis project. The oral presentation is to be held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion, and is followed by a private discussion with the examination committee. For assessment of the thesis work, the following aspects are considered: content and scientific & engineering approach, report, presentation & defence, creativity & initiative, and project management. The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty will introduce an updated schematic using rubrics for this, more information will be provided at a later moment.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Spaceflight Dynamics

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-construct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments	

AE4499	Space Project	0
Responsible Instructor	Ir. R. Noomen	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Responsible Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	(New) MSc students with a deficit in knowledge on space topics, or who want to do a project with limited duration on a space-related topic.	
Expected prior knowledge	None specific.	
Course Contents	This project covers miscellaneous student assignments within the Space track. Assignments can be a specific research or design assignment or may relate to building a simulation model that can be used for demonstration purposes in lectures. Depending on the size of the exercise, it can be honored with up to 4 ECTS.	
Study Goals	Depends on the specific project.	
Education Method	Homework assignment.	
Books	Depending on the assignment, it could be a general textbook on spaceflight. Examples: SMAD (Wertz et al), The New SMAD (Wertz et al), Spacecraft Systems Engineering (Fortescue et al) and/or the ADSEE course readers and slides. Other input material is also possible, and not necessarily restricted to literature only.	
Reader	Lecture material of space courses BSc AE (depending on the actual assignment)	
Prerequisites	Completed BSc from a degree in engineering.	
Assessment	Grading of deliverables as agreed upon in project statement.	
Enrolment / Application	Email to Profile Coordinator of relevant profile: Barry Zandbergen for Space Engineering, Ron Noomen for Spaceflight Dynamics, or Sebastiaan de Vet for Planetary Exploration.	

AE4889	Special Topics in Astrodynamics	2
Responsible Instructor	Dr.ir. M.J. Heiligers	
Contact Hours / Week x/x/x/x	NOT OFFERED IN 2022-2023 - 0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	You will extend and deepen your insight and analytical abilities based on Fundamentals of Astrodynamics and related coursework, devoting particular attention to their application to a selection of astrodynamics topics.	
Course Contents	Selected topics in astrodynamics	
Study Goals	Within the context of the selected astrodynamics topics: Extend your insight and analytical abilities beyond the Fundamentals of Astrodynamics and related coursework; Actively apply your knowledge and skills accumulated through Fundamentals of Astrodynamics and related coursework; Tackle "open-ended assignments with a hands-on, scientific approach as a first foray into astrodynamics research. Additional detail will be available on the course website (Brightspace).	
Education Method	Workshops, lectures, self-study.	
Computer Use	Yes.	
Prerequisites	- Fundamentals of Astrodynamics	
Assessment	Assignments.	
	Additional detail will be available on the course website (Brightspace).	

AE4895	Measurement Strategies for Planetary Science Missions	3
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice planetary analogue material you will try to find out how the data products and the scale at which it is applied, influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Lab assignments during structured laboratory activities (15%) Maintaining a lab journal (25%) Plenary integration session of your lab results with other students (10%) Final report, in form of short communication paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). Final project (group work) starting mid Q2 with submission deadline end Q2 / beginning Q3 (2/3 of final grade).	
Co-Instructor	K. Cools	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile III: Planetary Exploration

Program Coordinator Dr. S.J. de Vet

Introduction 1 Profile III: "Planetary Exploration"
The Planetary Exploration profile is focused on objects in the Solar System and beyond: planets, moons and asteroids. The emphasis is on science, and its role in driving new space missions and in generating new knowledge from measurements of past and current space missions. Topics that are investigated in this profile are the structure, environment, surface characteristics and dynamics of the objects mentioned here. Your studies can include computer simulations based on analytical or numerical models, but also on a variety of observations: in-situ or remote, from laboratory experiments or field-analogues, and deep space tracking. While his profile can offer a strong theoretical component, it is also possible to include a more practical component.

AE4866-1	Propagation and Optimisation in Astrodynamics	5
Responsible Instructor	D. Dirksen	
Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4868-1 (Numerical Astrodynamics) is required; AE4870B (Re-Entry Systems) is recommended; knowledge of basics of atmospheric flight mechanics is expected.	
Course Contents	1. Introduction. Explanation of course assignment. Introduction to software. 2. Numerical propagation. Formulations of equations of motion. Selection of propagation method. 3. Overview of formulation of acceleration models. Relation between acceleration models and environment models. Influence of acceleration model selection on simulation fidelity/runtime. 4. Introduction to local optimization methods. Constraints in optimization problems. Robustness of optimal solutions. Optimal control. Examples of local optimization in astrodynamics. 5. Global optimization methods (genetic algorithms, particle swarm, etc.). Multi-objective optimization. Examples of global optimization in astrodynamics 6. Pre/post-processing of results. Analysis of numerical propagation/optimization results. Use of numerical tools (e.g. interpolation).	
Study Goals	1. Select and define the settings of a numerical astrodynamics simulation problem, based on top-level description and requirements. 2. Describe the relation between models of the physical environment and the propagated dynamical behaviour. 3. Define and set up local and global optimization problems for typical astrodynamics problems 4. Assess the quality of numerical orbit propagation for typical astrodynamics problems 5. Assess the quality of optimization results for typical astrodynamics problems	
Education Method	Lectures (partly pre-recorded), workshops and open-office hours	
Computer Use	Laptop computer with minimum 4 GB RAM	
Course Relations	AE4870A (Rocket Motion); AE4870B (Re-Entry Systems); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I)	
Literature and Study Materials	Selected articles; lecture notes; lecture slides (details to be announced and distributed during the course).	
Assessment	The assessment is a combination of two individual and one group take-home assignments. The individual assignments are thematically related to the group assignments, and are used as a starting point for the group work . For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 50% Project (individual grade): 25% Project (group grade): 25% Each of these three grades must be ≥ 5.0 to pass the course. In the assignments, students will apply the material covered in the course to the analysis of a real astrodynamics problem. Weekly practical sessions will be held to discuss questions and issues with homework, the practical assignment and the numerical implementation. The resit of the course will be done during the summer period. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	NOTE: To take part in the course, students must submit their preferences for an assignment topic. These must be submitted through Brightspace on the first Wednesday of the course's first lecture week, at the latest. Only enrolling for the course on Brightspace is NOT sufficient to indicate you wish to take part in the course. Instructions for submitting a topic preference will be posted on Brightspace one week before the start of the course, at the latest. Not submitting your topic preference in a timely manner means that you will NOT be able to take part in the course. It is required that students have followed the course AE4868 (Numerical Astrodynamics), and have submitted all assignments, and can still reasonably pass that course, before taking part in this course.	

AE4868-1	Numerical Astrodynamics	4
Responsible Instructor	D. Dirkx	
Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4870B (Re-Entry Systems) is recommended; knowledge of basics of atmospheric flight mechanics is expected.	
Course Contents	Basic orbit propagation/integration methods and options; simulation options in Tudat. Defining input for orbit propagation Architectural design of Tudat software. Creation of user-defined modules for e.g. guidance * Linking analytical and numerical design methods in astrodynamics Details of numerical integration methods: multi-stage, multi-step and extrapolation methods (fixed and variable step size) Selection of numerical integration methods; quantification of integration errors	
Study Goals	Set up and run a numerical orbit propagation using the TUDAT software suite from a given set of simulation settings Critically evaluate and analyze numerical orbit propagation results Interpret numerical propagation results, and relate numerical results to theoretical predictions Select numerical integration scheme for a given astrodynamics problem	
Education Method	Lectures on Tudat, Dynamical Modelling, Analytical vs. Numerical Mission Design and Numerical integration Practical sessions/open-office hours,	
Computer Use	Laptop computer with at least 4 GB of RAM	
Course Relations	AE4866 (Propagation and Optimization in Astrodynamics); AE4870A (Rocket Motion); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4874 II (Astrodynamics II)	
Assessment	Individual take-home assignments. Assignments will consist of setting up, running, analyzing, and documenting several astrodynamics simulations in Tudat. The resit of the course will be done during the summer period. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	

AE4870B	Re-Entry Systems	3
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	IMPORTANT NOTICE: Please read the restrictions under "Assessment" +++++ The course aims at giving a complete overview of re-entry systems, mainly from a mission point of view, but also from the system point of view. To begin with the mission, the fundamental principles of mechanics (Newton's Laws) are applied to derive the equations of motion for translational and rotational motion. Subsequently, simplifications are applied to give a description of planar ballistic, gliding and skipping flight, where the focus will be on analytical solutions for quantifying maximum mechanical and thermal loads. Next, typical mission applications, such as planetary entry and descent with parachutes and/or propulsion systems are discussed, as well as terminal-area flight of winged entry vehicles. To extend the mission applications, attention is also paid to the atmospheric flight aspects during an aero-gravity assist, aerocapture and aerobraking. As many re-entry missions rely on the application of guidance, navigation and control systems, the fundamentals of each of these three sub-systems will be discussed and applied in simple examples, such that after the course the student has a starting point for further study. The relation between vehicle configurations (capsule, winged, lifting body, etc.) and mission aspects, such as flight time and (maximum) system loads is a typical system aspect that will pass review. Finally, all presented theory will be combined in a so-called development plan for building a simulator that can be used for performance analysis of both controlled and uncontrolled re-entry vehicles. How to use such a simulator efficiently is depending on what kind of simulation and analysis technique will be used. Therefore, the course is concluded with an overview of such techniques like Monte-Carlo Analysis, and how to extract the required information from the simulated data.	
Course Contents Continuation	Lecture Topics (the actual topics covered may vary per year): #01 Introduction #02 Entry Environment and Aeroheating #03 Fundamentals of Motion #04 Ballistic entry #05 Gliding Entry #06 Skip Entry #07 Guidance, Navigation and Control #08 Planetary Entry and Descent #09 Advanced Descent and Landing Systems #10 Terminal Area Energy Management	
Study Goals	At the end of this course, the student will be able to (depending on topics covered): 1. Identify the influence of the planetary environment on the motion of and loads on an entry vehicle 2. Derive the equations of planar motion 3. Identify the three unpowered entry mechanisms (ballistic, gliding and skip entry), and derive the corresponding equations of motion 4. Apply a sizing methodology to design an entry descent system (parachute and/or propulsion) 5. Understand the fundamental functions of a Guidance, Navigation and Control System and design a simple system 6. Name the characteristic phases of the Terminal Area flight, and derive the equations for maximum range and maximum dive (steady-state approximation)	
Education Method	Classical lectures (2 x 2 hrs per week). The course is structured in a number of blocks, which each deal with an aspect of re-entry missions and systems. Many real-life examples will be used to highlight certain theoretical aspects, and to show the applicability of the presented theory. Typical exam questions will be used to prepare the student for the exam by testing his/her insight in the presented material. There may be homework assignments to deepen understanding of the material. This will be announced at the beginning of the course. See further details under assessment.	
Course Relations	AE3212-I - Aerospace Flight Dynamics and Simulation AE3212-II - Simulation, Verification and Simulation AE4870A - Rocket Motion	
Literature and Study Materials	1. Lecture Notes and Lecture Slides (Mandatory) Selected references for additional study: 1. Anderson, J. (1989). Hypersonic and High Temperature Gas Dynamics. McGraw-Hill. 2. Bertin, J. (1994). Hypersonic Aerothermodynamics. AIAA. 3. Cornelisse, J., Schoeyer, H., and Wakker, K.F. (1979). Rocket propulsion and spaceflight dynamics. Pitman, London. 4. Hankey, W. (1988). Re-Entry Aerodynamics. AIAA. 5. Mulder, J., van Staveren, W., van der Vaart, J., de Weerd, E., In 't Veld, A. and Mooij, E. (2013). Aerospace Flight Dynamics and Simulation. Delft University of Technology, Lecture Notes AE3-212-I. 6. Regan, F. and Anandakrishnan, S. (1993). Dynamics of Atmospheric Re-Entry. AIAA. 7. Vinh, N. (1981). Optimal trajectories in atmospheric flight. Elsevier. 8. Vinh, N., Busemann, A., and Culp, R. (1980). Hypersonic and planetary entry flight mechanics. The University of Michigan.	
Assessment	Closed-book, written examination. It will be decided at the beginning of the course whether homework assignments are part of the assessment. If that is the case, homework will count for 40% of the grade and the exam for 60%. Both partial grades have to be 6 or higher to pass the course. IMPORTANT NOTICE: The homework and/or exam is only open to students of the following profiles: 1) Spaceflight Dynamics 2) Planetary Exploration 3) Structures in Space Capacity permitting, students from other profiles can send a personal participation request to the course manager	

AE4872	Satellite Orbit Determination	6
Responsible Instructor	Dr.ir. E.J.O. Schrama	
Instructor	Dr.ir. P.N.A.M. Visser	
Instructor	Ir. J.A.A. van den IJssel	
Instructor	C. Siemes	
Responsible for assignments	C. Siemes	
Contact Hours / Week x/x/x/x	0/4/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Course contents 1 Dynamics Introduction to dynamics - o Planetary Gravity field - o Tides and the three-body problem - o Hill radius and Roche limit - o Relation to planetary sciences and astrodynamics Solving Equations of motion - o reformulate orbit problem as a system of ordinary differential equations - o efficiency and accuracy of numerical integration methods - o implementation of numerical integration methods 2 Observations techniques and reference systems Observation techniques - o Laser, Doppler and Camera observations - o Refraction, Electromagnetism, radio- and optical technology - o Tropospheric and ionospheric refraction - o Relativity and the definition of time, - o Classification of time systems (UTC, TAI, etc) - o Light-time effect - o Quality of clocks (Allan Variance behaviour of clocks) Reference systems - o Local and global coordinate systems - o Definition of geoid and reference ellipsoid, height systems - o Precession and nutation, polar motion, polar wander. - o Newton or Einstein, consequences for reference systems 3 Statistics Random variables, probability density functions, moments, hypothesis testing Least squares minimisation - o unconstrained linear parameter estimation, - o data weighting - o nonlinear parameter estimation. Rank deficient equation systems - o compatibility conditions - o general and homogeneous solutions - o constrained linear parameter estimation Mechanisation of parameter estimation algorithms - o Choice of algorithms 4 Orbit determination Perturbation analysis and variational problems - o state transition matrix for initial state vector problems - o partial derivatives for dynamical parameters Parameter estimation - o Identification of parameters - o batch least squares - o Kalman filter, theory and implementation 5 Applications Global Navigation Satellite Systems: - o Technology and terminology, - o various data processing strategies and available software - o Modelling deformation of the solid Earth, - o the Earths gravity field and thermospheric density Satellite laser ranging and Doppler tracking via DORIS: - o technology and terminology, results and applications Observing changes in the cryosphere with satellites Hydrology and Oceanography observed with satellites 6 Homework assignments An exercise related to dynamics, observation systems or reference systems An exercise related to GNSS applied to orbit determination An exercise related to Kalman filtering 7 Numerical Practical Exercises with the Ghost software, typical examples are to solve: - o initial value problems (state vector estimation) - o problems with parameters in a dynamic model (drag parameter estimation) - o problems with time bias parameters	
Study Goals	The candidate should be able 1) To explain the physical and mathematical context of orbit determination (OD) 2) To construct transformations between various coordinate and time systems that play a role in OD	

	3) To examine error sources in satellite tracking data and implement error mitigation strategies 4) To make use of parameter estimation methods in the context of tracking data for OD 5) To apply relevant statistical techniques within the framework of OD 6) To discuss scientific applications of satellite missions that depend on precise OD 6.1) GNSS techniques to model the deformation of the Earth 6.2) Satellite gravimetry to model the gravity field of the Earth 6.3) Satellite altimetry to model the ocean topography 6.4) Satellite Altimeter missions to measure variations of land and sea ice 7) To compute and evaluate precise satellite orbits and dynamic model parameters with state-of-the-art software 8) To work with Linux commands such as ftp, to setup and adapt Python or MATLAB scripts
Education Method	We provide lectures and classroom instructions during quarter 2. In quarter 3 there is a numerical practical to introduce the topics mentioned under the course contents on an MSc level for aerospace engineering students. In quarter 2 all lecture material will be made available, instruction hours will be organised to complement the material presented during the lectures, also, the instruction hours will be used to introduce homework assignments which are part of this course. In quarter 3 we run a numerical practical. During this practical software tools will be used to compute precise satellite orbits and which comes with the estimation of parameters with a state-of-the-art software package (GHOST). In addition a quality assessment of the results will be part of the practical.
Computer Use	Use will be made of Python or MATLAB (preferred) during the assignments, for the numerical practical use will be made of Linux and Fortran.
Literature and Study Materials	Lecture notes will be provided on BrightSpace and they containing most of the material used in this course. Optional reference material: Statistical Orbit Determination, Schutz, Tapley and Born, Academic Press, ISBN: 9780126836301 Montenbruck and Gill, Satellite Orbits, Springer Verlag, doi: doi: 10.1007/978-3-642-58351-3
Assessment	Additional information could be provided via BrightSpace The final grade will come together from: 1) Written examination results in a grade 2) Several individual assignments result in a weighted average grade 3) Numerical practical results in "pass" or "fail". The rules are as follows: 1) The numerical practical result should be: pass 2) The written examination result should be at least a 5,0 3) In order to participate in the numerical practical you should have passed the homework assignments with at least a 5,0 4) No claims can be made with regard to homework assignment results or numerical practical results obtained before the previous course year. For the course in 2022-2023 it means that the results from 2021-2022 are still accepted, however, but we do not accept results from 2020-2021 or earlier years. Note 1: Due to the covid19 pandemic the examination method could be online via remote proctoring in such a way that you obtain the equivalent form of a three hour individual written exam. Note 2: The numerical practical and the assignments were provided in an hybrid format in the academic year 2021-2022, it depends on the university policy and the situation of the country during the pandemic whether we need to stay online in 2022-2023.
Exam Hours	Conform the regulations there are two written exam opportunities within an academic year. These are three hour written exams at the end of quarter 2 and 3.
Permitted Materials during Tests	We provide a closed book examination, required information for the exam (like physical constants or a selected number of equations) will be made available on the examination form.
Tags	Analysis Elektricity & Magnetism Lineair Algebra Matlab Numeric Methods Physics Signals and Systems Technology

AE4874 I	Fundamentals of Astrodynamics	4
Responsible Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ae4866 ae4868 ae4870 ae4872 ae4876 ae4889 ae4894 ae4895	
Summary	Thorough treatment of the fundamentals of astrodynamics; topics are treated in a mathematically consistent way. You will acquire fundamental insights in the theory and methods of astrodynamics as well as working knowledge to compute and analyze spacecraft trajectories.	
Course Contents	Introduction to astrodynamics; two-body and many-body problems; relative motion; coordinates, reference frames, orbital elements, and time; geocentric, cislunar, and interplanetary flight; introduction to orbital perturbations.	
Study Goals	You will be able to: 1. Provide full and accurate descriptions of the fundamental concepts and phenomena of astrodynamics; 2. Explain the relationships between astrodynamics concepts & phenomena and the mathematics & physics theory that describes them; 3. Apply the relevant relationships, models, and methods for astrodynamics analysis; 4. Derive and manipulate the equations of motion for a spacecraft subject to gravitational and other forces; 5. Apply the equations of motion, their solutions, and related equations to analyze the absolute and relative positions and velocities of spacecraft as a function of time. Additional detail will be available in the course syllabus.	
Education Method	Lectures, self study	
Literature and Study Materials	Fundamentals of Astrodynamics, Karel Wakker, 2015. ISBN: 978-946-186-4192 available for download from the TU Delft Library / Institutional Repository also available to order as university reader in hardcopy Astrodynamics course website Orbital Mechanics for Engineering Students, Howard Curtis, 2013, 3rd edition. ISBN: 978-008-097-7478 (print) 978-008-097-7485 (ebook) available for download from the TU Delft Library also available to order from booksellers as hardcopy	
Assessment	Written exam	

AE4876-11	Planetary Sciences II	4
Responsible Instructor Instructor Instructor Instructor Instructor Instructor Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Expected prior knowledge Parts	Dr. D.M. Stam Dr. S.M. Cazaux Dr. A. Menicucci Ir. M.C. Naeije Prof.dr. L.L.A. Vermeersen Dr.ir. W. van der Wal Dr. B. Brandl C. Siemes 0/6/0/0 2 2 none English Core course Planetary Sciences I forms a proper base for the Planetary Sciences II course. Planetary Sciences II is a follow-up, but elective course for core course Planetary Sciences I.	
Course Contents	Intertwined lectures by topical experts give deeper insight and understanding into planetary sciences topics that were only touched upon in Planetary Sciences I, with an emphasis on applied technologies, previous, current and (near) future space missions, and data analysis. Lecture topics focus on: Planetary missions, atmospheric dynamics, magnetic fields, ocean dynamics & oceanography, comets, radiation & space weather, climates & tides, gravitational fields, astronomical missions, and astrobiology. In small groups (4-5 students), students work on the design of a planetary space mission, with a science goal of their choice, supervised by a mentor. Under normal circumstances, the course includes a day at ESTEC, the European Space Agencies facilities at Noordwijk, with a tour of the premises, presentations by ESA mission scientists, and presentations of the planetary space missions designed by the students.	
Study Goals	After completion, the student should be able to reproduce the theoretical backgrounds of the topics taught, and to put them to practice in real life planetary observation problems.	
Education Method Literature and Study Materials	The student should also be able to explain the road map from measured quantities to physical quantities, to think about new developments and to come up with new solutions, i.e., understand and being able to apply the course material. Oral lectures (if necessary, online), homework exercises, planetary mission project, and tutorials. Recommended literature: Fundamental Planetary Science by J. J. Lissauer & I. de Pater, Cambridge University Press, ISBN: 978-0521-61855-7 (paperback) This book is the same book as the one that is used in Planetary Sciences 1. Other books with background information (not required for taking Planetary Sciences 2): C.M.R. Fowler, The solid earth: introduction to Global Geophysics, 2nd ed., Cambridge University Press, 2005, ISBN 0 521 89307 0 W.M. Kaula, Theory of satellite geodesy, Blaisdell, 1966, Open University, Ocean Circulation & Waves, Tides and Shallow-Water Processes editions, Butterworth-Heinemann, 1999 A.E. Gill, Atmosphere-Ocean Dynamics, Int'l geophysics series, vol. 30, Academic Press, INC. (LONDON) LTD., ISBN 0 12 283522 0	
Assessment	Weekly, individual, homework exercises are graded and count towards the final grade. The homework grade must be higher than 5.5. The remainder of the grade is formed by the Planetary Mission Project: In groups of 4-5 students, a mission is designed to investigate a planetary science topic chosen by the students themselves. Each group is guided by a mentor. The Planetary Mission Project is graded through a poster to be presented by each group at the ESTEC day (or, if the ESTEC facilities are closed to outside visitors, though an oral presentation at an online meeting), and a report written by the group. The final grade is computed as follows: 30% homework exercises + 70% planetary mission project	
Remarks	There is no written exam for AE4876-II! Under normal circumstances, towards the end of the course, there will be a visit of a day to ESTEC, the facilities of ESA in Noordwijk. There, the students will attend presentations by ESA personnel about a number of missions. The students will present and defend their poster on the Planetary Mission project, and they get excursions within ESTEC. Attendance of this day is mandatory. The date will be chosen such that there will be minimal to no interference with other lectures.	

AE4880	Optical Space Sensors	4
Responsible Instructor	Ir. J.M. Kuiper	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course is organised from two perspectives: one is the space instrumentation engineering: Sensor physics, anatomy and system design requirements Classes of instruments Geometrical optics Optical design and performance Optical components Opto-mechanical design Detection subsystems Radiometric performance analysis Spectral performance analysis On-ground characterization and calibration Manufacturing, assembly, alignment, test Instrument sizing The other line is the design and performance of different classes of instruments. Spectrometers Optical imagers Radiometers (Space) Interferometers Laser Satellite Communication Deployable optics Miniaturized instruments	
Course Contents	Space instrumentation is a discipline aimed at creating state-of-the-art sensors capable of measuring the environments of the Earth, the planets and their satellites, the Sun, and interplanetary space. This course surveys the physical principles and engineering of instrumentation used throughout the many related fields of space science. Upon completion of the course the students will have a firm grasp of the principles and techniques used to sense and measure photons. The course focuses on the design and engineering process of electro-optical space instruments, starting from the design requirement up till the on-ground functional and performance testing and calibration.	
Study Goals	After the course the student should be able: - To list the candidate measurement techniques for given observation requirements - To list and describe for each type of instrument: - o The specific applications and their importance - o The various elements that make up an instrument and their main function(s) - o The different options that exist to perform this function - o The important performance parameters - To explain how the performance requirements translate into design parameters and operational scenarios - To perform instrument engineering to a level of detail, sufficient to accomplish a feasible concept for the instrument (i.e. conceptual definition and justification of its optical-thermal-mechanical-electrical and detector design including budgeting, sizing and sampling) - To indicate and justify the design drivers - To give the rationale for the critical engineering aspects (e.g. alignment accuracy, straylight, stability, sensitivity to contamination or radiation, testability, manufacturability) - To set up an elementary instrument simulator to predict instrument performance and sensitivities to variations and error sources - To draft an on-ground test flow - To draft a concept for in-orbit calibration - To determine interface requirements on the hosting platform, the orbit and its operation - To give an outlook for future developments and applications - Be able to make a breakdown of a SatCom (laser sat communication) chain and its link budget - Understand the basics of a DST (deployable space telescope) system design and its thermo-mechanical boundary conditions - Understand the basics of a Space Interferometer for exo-planetary research	
Education Method	Lectures are partly given by professionals from space industry. To bridge the knowledge gap in optics from high school level to MSc level (in the absence of relevant preparatory BSc lectures), a series of in-class optical exercises will be given. To support this part the lectures will be partly given on basis of "flipping the classroom". This means that some basic optical lectures on BSc level will be provided with in advance as slides. These will be released on Brightspace at the end of January and need to be studied by the students themselves before the lectures start. In the follow-up classroom exercise sessions questions can be raised on these parts. Elements are: Group assignments of in-class optical exercises. Both the assignments and related homework are part of the formal close-out of the course, in combination with a written exam.	
Literature and Study Materials	Mandatory study material: - BSc level lecture slides to be released in January 2020 as an introduction to the course. - Lecture slides and exercises during the course. - Laser SatCom capita selecta are lectured with slide material only and relevant examples (worked out problems) like link budgets. - The same holds for the topic Deployable Optics - The same holds for the topic Space Interferometers - Reader AE4-880 Space Instrumentation Engineering. Capita selecta of the reader serve as supporting material. Only limited parts of the reader apply to the actual content of the course. These parts will be addressed during the lectures and given in a study reference matrix: to be made actual and released on Brightspace during the course.	
Practical Guide	The Brightspace site AE4880 contains an abundance of information about the course, the content, as well as enrichment data, with links to supporting websites	
Assessment	2 Elements: A) Written exam + B) Group assignments: optical exercises + presentation. If the mark for the exam is 5.0 or higher, the final grade is the weighted average of 80% * mark A + 20 % mark B. These numbers are valid in normal conditions which excludes COVID-19 or similar conditions.	

Permitted Materials during Tests	The exam is an open-book exam. The student is allowed to bring his printed Lecture Slides and Reader to the exam. including the personal notes he made at the lectures (on the slide sheets only). He is NOT allowed to use a laptop, to bring in the exercises, assignments or homework problems, nor their answers, nor previous exams.
Enrolment / Application	Students have to enroll in the first course week or via the Brightspace site of the course.
Remarks	Self activation by the in-class and homework assignments is the key to a successful pass of the exam.
Set-up	The course is organised along two learning ways: one is the space instrumentation engineering that is lectured by the lecturer or guest lecturers. The other line is the design of different classes of instruments. This part is provided by the students in the form of assignments and practicing homework problems.

AE4890-11	Planetary Sciences I	4
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr. D.M. Stam	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Instructor	Dr.ir. W. van der Wal	
Responsible for assignments	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4876 Planetary Sciences II	
Parts	Planetary Sciences: - Origin and Evolution of our Solar System - Planetary Surfaces and Interiors - Planetary Atmospheres - Planetary Magnetism - Planetary Ring Systems - Asteroids and Comets - Extrasolar Planets - Recent Topics in Planetary Science Current topics in planetary science with a focus on a specific topic each time the course is offered, such as Mars, Venus, Galilean satellites, comets, recent discoveries, spacecraft observations, or other developments.	
Course Contents	Planetary science is a major interdisciplinary field, combining aspects of astrophysics with geology, geophysics, meteorology, atmospheric and space science. This close relationship to geophysics, atmospheric and space sciences implies that the study of the planetary bodies in our Solar System and beyond offers the unique opportunity for comparison available to Earth scientists. This course teaches the concepts in the planetary physical sciences and solar system properties. The learning process is greatly enhanced by involving students in solving related problems.	
Study Goals	After the course the student should be able to: 1. Describe the primary physical processes in solar and planetary systems and the properties involved 2. Apply modelling techniques that are commonly used to describe the primary physical processes in solar and planetary systems 3. Demonstrate the use of physical principles to derive knowledge on the state and evolution of (extra-)solar system bodies from observations 4. Assess the quality of evidence of current knowledge of (extra-)solar system bodies 5. Assess the habitability of a (extra-)solar system body based on inferences of its properties and orbit	
Education Method	Classical oral lectures and homework assignments.	
Literature and Study Materials	Jack J. Lissauer and Imke de Pater, Fundamental Planetary Science, Cambridge University Press, ISBN: 9780521618557, 2013. PowerPoint/PDF files of the lectures and lecture notes.	
Books	Jack J. Lissauer & Imke de Pater, Fundamental Planetary Science, Cambridge Univ. Press, 2013.	
Assessment	The final grade for the planetary sciences course is determined from: - mark of a written exam - mark for mandatory homework assignments The mark for the written exam needs to be 5.0 or higher in all cases, no exceptions. If the mark for the exam is 5.0 or higher, grading for this course follows from averaging two parts with equal (so 0.5) weight: the mark for the classroom/take home exercises and the mark for the exam. No opportunity will be given to redo the classroom/take home exercises within the same academic year. The grade for the homework assignments will remain valid for the next consecutive academic year, and so will annul the year thereafter	
Exam Hours	3 hours written exam	
Permitted Materials during Tests	Closed-Book exam, Non-programmable calculator is allowed.	
Set-up	Oral lectures and classroom/take home assignments.	

AE4S12	Space Systems Engineering	3
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. Specifically, AE1110 + AE1111 + AE2111 + AE3211, or similar	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 2 a. Identifying stakeholder needs 2 b. Generating, evaluating and selecting concepts 2 c. Tutorial 2 d. From stakeholder expectations to technical system requirements 2 e. Logical decomposition and design solution 2 f. Estimating lifecycle costs 2 g. Managing technical risk 3. Integrating Systems 3 a. Integrating the system 3 b. Verifying and validating 3 c. System roll-out & lessons learnt 4. Systems Engineering Approach and Process 4a. Model-Based Systems Engineering 4b. Managing the Systems Engineering Process 4c. Managing the technical effort 4d. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers advanced Space System Engineering demonstrated through many examples from past, current and future space programs. It introduces a process which allows the creation of successful products, such as space applications, space missions and systems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems. The interrelationships between systems engineering and project management, programmatics, cost and risk in the development of space projects are demonstrated at various occasions during the course. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply methods and tools which help to create successful products.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are four high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to use Systems Engineering methods and tools to solve practical real-world problems when engineering a space system. 3. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 4. Participants shall be able to analyze Systems Engineering topics.	
Education Method	Lectures* Tutorial* Individual Homeworks to practise material Group Assignment, to be done by students in small groups The Group Assignment together with graded Individual Homeworks comprise the formal course close-out.	
Literature and Study Materials	Optional: Larson W., Kirkpatrick D., Sellers J.J., Thomas L.D., Verma D. (eds.); Applied Space Systems Engineering, Space Technology Series, McGrawHill (2009) Required: Lecture Slides AE4-S12 Space Systems Engineering.	
Assessment	6 graded Individual Homeworks + 1 Group Assignment. Acceptance of 8 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the secretary of the Chair of Space Systems Engineering (www.tudelft.nl/ae/sse).	
Set-up	The course comprises lectures, tutorials, group work and individual homework. Elements indicated by * are related to the potential impact of the pandemic on the course in 2022/2023 and may be subject to change in the course of the academic year.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Officially registered and enrolled as a full time Master student Aerospace Engineering.

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

Enrolment / Application

Special Information

Set-up

AE5822	Thesis Space	45
Responsible Instructor	Ir. R. Noomen	
Instructor	Dr.ir. P.N.A.M. Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	Every MSc program in aerospace engineering is completed with an MSc thesis which comprises 45 ECTS. It is intended to provide candidates with an opportunity to carry out an in-depth investigation in a particular area of interest and to make a substantive contribution to knowledge in the area. It is expected that the thesis will include a comprehensive and critical synthesis of the relevant literature and will also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution.	
Study Goals	Learning objectives MSc Thesis During the MSc Thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. Major requirements are duration (nominally 7 months) and innovation. The thesis project can be done at the Faculty of Aerospace Engineering, but also at external organizations (e.g. companies, other universities). In any case, the topic has to be approved by the responsible supervisor before the student is allowed to start with his/her project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Thesis supervisor(s) will provide assistance and direction to the student until the thesis has been defended and accepted. Regular progress/review meetings/consultations with the thesis supervisors are required.	
Assessment	At the end of the MSc thesis project, students are required to defend their thesis before an examination committee. To this end, the student is required to present the thesis in the form of a written report and an oral presentation. The written report should describe the work performed in the thesis project. The oral presentation is to be held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion, and is followed by a private discussion with the examination committee. For assessment of the thesis work, the following aspects are considered: content and scientific & engineering approach, report, presentation & defence, creativity & initiative, and project management. The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty will introduce an updated schematic using rubrics for this, more information will be provided at a later moment.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Planetary Exploration

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-construct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4499	Space Project	0
Responsible Instructor	Ir. R. Noomen	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Responsible Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	(New) MSc students with a deficit in knowledge on space topics, or who want to do a project with limited duration on a space-related topic.	
Expected prior knowledge	None specific.	
Course Contents	This project covers miscellaneous student assignments within the Space track. Assignments can be a specific research or design assignment or may relate to building a simulation model that can be used for demonstration purposes in lectures. Depending on the size of the exercise, it can be honored with up to 4 ECTS.	
Study Goals	Depends on the specific project.	
Education Method	Homework assignment.	
Books	Depending on the assignment, it could be a general textbook on spaceflight. Examples: SMAD (Wertz et al), The New SMAD (Wertz et al), Spacecraft Systems Engineering (Fortescue et al) and/or the ADSEE course readers and slides. Other input material is also possible, and not necessarily restricted to literature only.	
Reader	Lecture material of space courses BSc AE (depending on the actual assignment)	
Prerequisites	Completed BSc from a degree in engineering.	
Assessment	Grading of deliverables as agreed upon in project statement.	
Enrolment / Application	Email to Profile Coordinator of relevant profile: Barry Zandbergen for Space Engineering, Ron Noomen for Spaceflight Dynamics, or Sebastiaan de Vet for Planetary Exploration.	

AE4893	Physics of Planetary Interiors	4
Responsible Instructor	Dr.ir. B.C. Root	
Instructor	Dr.ir. W. van der Wal	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course focuses on the different aspects of numerical modelling of planetary interiors. The interior of a planet or moon can be studied via observations of its gravity field, shape, surface features, rotation, and tidal deformations. To interpret these observations, the response of the bodies to different forces and heating scenarios need to be modelled. As a student you will get hands-on experience in modelling planetary and exoplanetary bodies with various numerical code packages. Different methodologies will be discussed, ranging from solving the Stokes equation for internal mantle convection to gravity forward modelling and how to solve certain loading scenarios with a finite-element code to the thermal evolution of a planet. You will be able to study a range of internal solid and fluid processes and interpret their surface manifestation. Lecture topics include: 1. Observations related to planetary interiors: Gravity field, rotation, tides, shape (topography, faults) - Example bodies and learn about different internal processes. You will learn how to calculate the internal gravity, density, and pressure of these bodies. 2. How to model fluid-solid mechanics of a planet? - Stokes equations in planetary science in spherical coordinates, rheology - Heat-transport: state equation exercise with different heat regimes - Mantle convection applications: dynamics and ocean flows 3. How to perform gravity field modelling - non-uniqueness, advanced isostasy/flexure models, density anomalies - Forward modelling density anomalies: spectral vs. volumetric methods - Inversion of lithosphere structure - Effect of mantle convection on the gravity field - Lessons learned from seismology on Earth 4. Tidal and loading deformation (Numerical code) - Effects of tidal potential (normal modes, FEM), dissipation - Loading cases: volcanism, meteor impact, ice loading - Surface faulting and the relation to stress, planetary seismicity, gravitational potential theory 5. Special topics on rotation of planetary bodies 6. Interior and planet evolution (combine all material) - Orbital resonance (external effects) - Change in thermal state and effect on tectonic regime (internal effects)	
Study Goals	After the course you will be able to: 1. Recognize the physical processes shaping planetary interiors and understand how they can be approached in a numerical study. 2. Apply fundamental physical laws (Stokes, Poisson equation, etc) in a schematic numerical modelling setup in spherical coordinates to study relevant problems of planetary evolution. 3. Able to operate and assess applicability of state-of-the-art numerical simulations of planetary interiors and their evolution 4. Validate and improve numerical models of planetary interiors with observations. 5. Critically review literature in planetary interior modelling and formulate new research questions in this context.	
Education Method	One lecture and work session per week 3 assignments focussing on different numerical methodologies Numerical modelling exercises will help you to get familiar with different numerical codes in the planetary science field. You will need to perform 3 assignments, which will be a selection of different numerical exercises connected to the lecture material. The (randomized) selection will be made by the course responsible. First numerical exercise: How to model a planetary body assignment (learning goal 1 and 2) Propagate state equation of an icy moon for different heating conditions Apply the heat transport equation to Mercury: conduction vs. convection Second numerical exercise: Gravity field modelling assignment (learning goal 4) Study best way to compute Bouguer anomaly of Mars: spectral vs. volumetric Perform multi-layered inversion of density anomalies in lithosphere. Third numerical exercise: Assignment using existing software to run simulations for one of the following (learning goal 3 and 5) Mantle convection: effect of viscosity on gravity field Meteor impact relaxation Tidal motion of surface Europa (thickness ice/water layer) Relation of Mars on Phobos rotation Diffusion modelling in evolution	
Course Relations	Complementary to other courses: PS1, PS2 and Measurement Strategies Overlap: After rearrangement of PS1 and PS2, there is no overlap anymore CIE4601 Physics of the Earth: rotation and other aspects of solid Earth. This course will be discontinued. CIE4610 Gravity, Geodynamics and Climate Change; Spherical harmonics and basics of potential theory are treated there, but not gravity forward modeling. Several courses at TU Delft discuss Navier-Stokes equation, FEM and inversion, however by focussing on spherical coordinates and spherical shell models we believe this course is much more suited for potential planetary scientists/engineers.	
Literature and Study Materials	Various peer-review journal articles Lecture notes Software and tutorial manual	
Assessment	Numerical assignments: 25% + 25% + 50%	

AE4895	Measurement Strategies for Planetary Science Missions	3
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice planetary analogue material you will try to find out how the data products and the scale at which it is applied, influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Lab assignments during structured laboratory activities (15%) Maintaining a lab journal (25%) Plenary integration session of your lab results with other students (10%) Final report, in form of short communication paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Responsible Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a multidisciplinary Master course at TU Delft/Faculty of Aerospace Engineering (AE). In CSDP students of primarily the Spaceflight track work in groups (6-12 students) on a space mission/spacecraft design identified by interested parties (this may be a space company, space institute, or other) for a total of 112 working hours (4 EC) over a period extending 2-3 months under supervision of faculty staff. Additionally, students spend maximum 28 hr (1 EC) on workshops that provide information and training on the use of a typical process management tool like SCRUM (and/or Kanban) and a Concurrent Design Platform / tool (like e.g. Excel and/or CDP-4). The course aims to develop student skills in multidisciplinary team projects from a challenge-driven perspective. For the students, CSDP brings forward unique characteristics such as: a starting point in a value-driven approach to problem formulation and decision-making, concurrent and collaborative systems engineering to solve engineering problems thereby taking into account realistic constraints and allowing for training of interpersonal skills and/or workplace competencies (teamwork, planning and organizing, creative thinking, etc.). Topics vary year by year and depend on interest of engineers & researchers to define great engineering challenges for a motivated team of students.	
Study Goals	- Students get advanced experience in the use of Concurrent Engineering (CE), as opposed to the more classical sequential engineering (SE), in a design project. Students will learn not only about how to apply CE, but also about the benefits of CE (as opposed to SE, i.e. the waterfall method) and its limitations; - Students will advance their ability in systems engineering/collaborative while also improving their ability in cooperation, collaboration and communication.	
Education Method	Students will perform a design project as part of a multidisciplinary team. Each group is guided by a mentor. Parallel to the design project, students are instructed on the use of CDP4 and SCRUM (and/or Kanban). For each team regular design sessions will be held in a collaborative learning environment. In these design sessions, teams are given the opportunity for brainstorming (requirements generation, creating first solutions), demonstrating progress, setting tasks, and reviewing the design accomplishments. It is the purpose that each team organizes themselves a team of experts (from industry, academia or other) that are willing to advise the team in their project. At various instants, the work accomplished will also be externally reviewed by other CSDP teams as well as staff from the faculty and external experts with as aim to identify strong/weak points in the design.	
Assessment	The design projects are graded based on a design report, a design log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic (e.g. communications, modelling/experimenting and critical and analytical thinking) and personal effectiveness competencies.	
Enrolment / Application	There is no free enrollment for this course. In Q1, an information session will be held informing students of Space Engineering on the projects available and on how they can apply for a project. Students can then apply and only if they are accepted, they can enroll.	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	At the end of this course you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	Brightspace/MATLAB exercises and tests (optional) during education period. The course is concluded through an optimization project and report. Deadlines for handing in the final report will be announced at the start of the course.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 4: Aerospace Structures and Materials

MSc Coordinator S.R. Turteltaub

Contact for students Dr. Sergio Turteltaub (ASM MSc Track Coordinator)

Please use the following email address: MScCoordinator-ASM@tudelft.nl

Official announcements are posted on the Aerospace Structures & Materials Msc Track Brightspace page

Introduction 1

No structure is indestructible. Time, environment, repeated use, and misuse all take their toll on everyday structures, taking advantage of their inherent weaknesses and bringing them closer to failure. This is not the most comforting thought when walking onto an airplane, or driving over a bridge, but it is a reality that structural designers must face; how do you design something that you don't want to fail while accepting that it eventually will? The answer is: very carefully!

The challenge structural engineers are faced with is as follows. First, they must understand the raw materials and the level of engineering already applied in their creation. Second, they need to synthesize material behavior and required structural function into a working design. Third, compromises in the working design need to be made to address manufacturability. Finally, all of this must be completed while continually assessing the impact on the durability and longevity of the final structure.

The Aerospace Structures and Materials (ASM) MSc track aims to equip students with the necessary knowledge and practical skills necessary to tackle this challenge in an industrial or research environment. From an educational standpoint, students will be exposed to a broad range of courses examining this entire process in the context of the design, manufacturing, and analysis of composite aerospace structures. This will provide the foundation for subsequent specialization in one of the thematic profiles mirroring the steps in the design challenge described above: Material, Manufacturing, Structural Analysis and Design. From a research standpoint, students will have the opportunity to carry out high-level research and development in fields directly relevant to aerospace industry and that improve our capability to design the next generation of aerospace structures and materials. Thesis projects can be carried out with any of the research groups within the ASM department, regardless of the thematic profile selected by the student. Additionally, numerous opportunities exist to carry out research within industry under the close supervision of an ASM staff member.

The ASM Track is committed to making your MSc experience a memorable one. Next to offering a world-class education we also offer many opportunities to expand your horizon through industry involvement, visits and guest lectures. Our students have organized themselves in their own ASM student society, Enlighthness which organizes lunch lectures, company visits, drinks and the annual ASM career event.

The track program is built up of a common core of 15 EC followed by a selection from one of our three thematic profiles, after which you can further specialize or widen your horizon by choosing from a large selection of electives. The first year is dedicated mostly to courses in preparation of your second year, in which you first go on your internship and then work on your thesis project.

Our three thematic profiles are:

Profile 1: Materials and manufacturing
Profile 2: Design and safety of structures
Profile 3: Structures for space

You can choose the materials and manufacturing profile if your intention is to develop new aerospace materials and/or work in a production surroundings and translate the needs of the designer to production and vice versa. The design and safety of structures is for you if your interest is on structural design & stress engineering. The structures for space profile is when you want a more in-depth knowledge of structures specifically for space applications, such as space vehicles and launcher structures.

Program Goals

After completing the MSc Track Aerospace Structures & Materials the student will be able to:

- Develop design requirements for materials, manufacturing and structures
- Design a lightweight structure and explain the reasoning and the physics behind the design
- Design a material suitable for aerospace application and explain the reasoning and the physics behind the design
- Carry out a structural analysis using Finite Element Methods and/or other analytical and numerical techniques
- Explain the manufacturing processes and their applications
- Select suitable manufacturing processes
- Manufacture a prototype
- Explain and predict how a design will perform over its life-time and explain how the performance can be monitored

This is executed in a hands-on way with both practical and theoretical learning activities and exciting multi-disciplinary projects within the research framework of the department but also allowing for students' own input and ideas.

Program Structure 1

The 1st period (Q1), core module consists of five themed courses of 3 EC focused on aerospace structures, such as an aircraft wing

AE4ASM001 - Design of lightweight structures I: Composites & Metals
AE4ASM002 - Designing Materials with Aerospace Specific Properties
AE4ASM003 - Linear Modeling (incl. F.E.M)
AE4ASM004 - Manufacturing of Aerospace Structures & Materials
AE4ASM005 - Fatigue of Structures & Materials

Towards the end of the first period (Q1), students choose a thematic profile. The profile courses in the second period (Q2) allow the student to specialize within the chosen profile.

Towards the end of the second period, students choose the direction of graduation and the relevant elective courses for this direction with an aim to further deepen the master thesis work.

Students select electives with together with their preferred thesis supervisor and then finalize their course list with the MSc Track Coordinator.

Program Structure 2

Practical issues: selection and updates in study program and administrative systems

During the MSc the preferred elective courses and thesis direction may change as the interest of the student becomes more clear. This change is facilitated by altering your course selection on the course selection system:

My Study Planning, <https://mystudyplanning.tudelft.nl>

Please make sure to submit your change request for approval by the MSc Track Coordinator who will also consult the thesis supervisor involved. The submit button can be found all the way at the bottom of the page. You will receive an e-mail notification when your request has been processed by the master Track Coordinator. This information is also forwarded by the Track Coordinator to the system managing the administrative information related to your studies (OSIRIS). It is advised to check in OSIRIS whether the change request has been processed by the central administrative office of TUD according to the data from

My Study Planning. Please note that this may take several working days since the course selection system (My Study Planning) is not meant to be an automated process for OSIRIS. The information in the OSIRIS system is always leading.

Keeping your list of electives up-to-date is imperative as it may delay your thesis defense date.

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Materials and Manufacturing

Program Coordinator	S. Teixeira De Freitas
Contact for students	To contact the ASM MSc track coordinator send an email to MScCoordinator-ASM@tudelft.nl. Please indicate the profile (Materials and Manufacturing) for specific questions about the profile itself.
Introduction 1	Profile I Materials and Manufacturing Why? New materials and the way we manufacture them has been linked with human development since pre-historical times (e.g. Stone age, Bronze age). Fast forward 5000 years: Today Climate change is threatening Human Civilization. Critical sectors of our society as mobility, energy and infrastructures must change. New material and manufacturing are a key part of the solution for a new sustainable era. What? In the profile Materials and Manufacturing you will be equipped with the knowledge and tools to create innovative and creative solutions to kick off a new era of sustainable materials. You will learn about polymer science, polymer composite manufacturing, additive manufacturing and characterization techniques of materials and components. Electives with a strong Materials and Manufacturing flavor within the ASM Department are: - Aircraft Manufacturing Laboratory - Trinity exercise - Materials for Space - Functional Coatings - Experimental Techniques & NDT - Design of Self-healing materials - Industrial Composite Manufacturing - Sheet Metal Forming
Program Structure 1	In the ASM Track you do not have to select a profile until October. Use the core module in period 1 to select your final preference. Electives are decided upon in the second period whilst you are taking the profile courses to allow you to orientate yourself on your thesis topic and pick the relevant deepening electives upon the advice of your preferred thesis supervisor. You then register your choice of electives with the MSc track coordinator in December. TU Delft students who intend to start in February are requested to see the Track Coordinator by the end of November or early December to discuss their planning. Be mindful that a February start assumes you will take your internship from April of that year, so plan accordingly and start preparing for your internship in time.

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures. ALTHOUGH COVID-19 MIGHT INTERFERE, IT IS EXPECTED THAT THE LECTURE WILL BE GIVEN IN A LECTURE ROOM.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM002	Designing Materials with Aerospace Specific Properties	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Several courses follow on the general material concepts introduced in this 002 course: -AE4ASM101 Polymer Science; (Q2) -AE4ASM508 self healing materials; (Q3) -AE4ASM103 Functional Coatings. (Q3)	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Ashby, Materials". The students must have completed the materials science courses in the first 3 years of the Aerospace Engineering Curriculum, or any other Engineering curriculum	
Course Contents	In this course an unorthodox approach to materials will be presented. Rather than memorising known routes to reach certain materials properties, the students will be trained to translate these desired properties into material structures and microstructures and to design suitable material production processes to realise these properties. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The course is divided in 7 lectures. The first four lectures cover basic design rules of different material classes and their behaviour. The last three lectures cover material behavior at high temperatures, impact of material degradation on properties and strategies to extend material durability: -introduction to materials and structures -polymers -metals -ceramics and 'smart' materials -materials at high temperatures -materials and damage -materials and lifetime The students will need to deliver a report (assignment). For this they will get a set of questions they need to solve by reading the existing literature.	
Study Goals	The objective of the course is to train the student in reverse material engineering. This skill enables students to initiate and guide new material developments to meet future targets in the industry. By the end of the course, you should be able to: - explain structure and property (inter)relationships of metals, polymers and ceramics - explain the functionalities of aerospace relevant material properties - breakdown and translate these functionalities into material structures and microstructures using reverse engineering methodologies - explain material behaviour and function loss and strategies to decrease the impact of damage in the material function - give examples of new material concepts and explain the underlying concepts.	
Education Method	Lectures	
Literature and Study Materials	A compilation of lecture notes, review papers and assignment.	
Assessment	Assignment report (60%) + oral exam (40%)	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments.	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Ir. J. Sinke	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises/assignments (exercises are stand-alone; assignments are linked)	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Dr.ir. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
Assessment	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	- To understand the possibilities and limitations of various polymer synthesis techniques. - To know and recognise some of the industrial polymer chemical structures. - To be able to explain the impact synthesis and chemical structure has on polymer material properties & their selection for given applications. - To gain insight in the characterisation of polymers via various methods - To be able to estimate performance and processing based on polymer structures - To understand fundamental working principles of polymers in different applications such as aerospace and energy applications - Understand and elucidate the concept of sustainable polymers and biobased materials - To be able to give a convincing pitch talk about a recent development in the field of polymers.	
Education Method	Lectures	
Literature and Study Materials	lecture notes + book: R.J. Young and P.A. Lovell, Introduction to polymers, Chapman & Hall, London, 3rd edition, 2011, ISBN 978-0-8493-3929-5. Please note that the 2nd edition can no longer be used. The newer, 3rd edition, is a complete makeover! Optional/alternative reading: Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective	
Assessment	Written final exam and pitch talk (3 minutes) presentation Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: - In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. - Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. - If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. B. Çalar	
Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: - Have solid knowledge on the mainstream manufacturing processes for polymer composite aerospace structures. This comprises processing steps, tooling and equipment as well as advantages, limitations and applicability. - Understand the relationships between the physics underlying specific manufacturing processes, the part/material quality and the design of the manufacturing process. - Recognise manufacturing defects, to identify their potential sources and to advice on measures to prevent/limit their occurrence.	
Education Method	Interactive lectures with numerous in-class exercises. An assignment is used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). Transition AE4ASM110 -> AE4ASM110-22 - For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM110, the following applies: - In 2022 2023, two opportunities will be offered to complete AE4ASM110. - Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. - If AE4ASM110 is not completed in 2022 2023, students will need to complete AE4ASM110-22 instead.	

AE4ASM515-22	Characterization of materials and components	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof. C.A. Dransfeld	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.	
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various experts in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation (XRD, SAX, NMR) 4. Surface analysis of materials (XPS, AES, AFM) 5. Electrical and electrochemical characterisation of metals and ceramics (Electrochemical Impedance Spectroscopy, dielectrics) 6. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 7. Mechanical characterization (tensile, fracture)	
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important in case of premature failure or degradation or in sub-or above standard material performance. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.	
Education Method	Lectures and assignments	
Literature and Study Materials	Introductory texts and review articles on the various techniques posted on Brightspace.	
Assessment	Oral Exam Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr. J. Wu	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5722	Thesis Aerospace Structures & Materials	45
Responsible Instructor	Prof.dr.ir. R. Benedictus	
Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The final thesis within the Aerospace Structures and Materials track is a 7 month research project in which you proof your competence as an Aerospace Structures & Materials engineer. Your thesis project can be carried out in-house but also in collaboration with industry or external research organization and sometimes even in industry or external research organizations. Ideally the final thesis is combined with the AE4020 Literature Study and the deliverables for AE4010 Research Methodologies into a 9-month full-time thesis project. These 9 months are a nett period and exclude any holidays or other events you may have scheduled. During this project you are supervised by one of our academic staff members sometimes combined with one of their post-docs or PhDs or a supervisor from a company or other research organization.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; Interpret obtained results in a critical manner; To produce results with scientific/technological relevance that can be published in scientific literature; Critically reflect on the work performed at the level of their peers in their particular field; Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; Present work performed in a structured way through an oral presentation to their peers and wider audience; Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; Plan the project efficiently considering resources and methodology; Execute the project controlling the process, effectively using the plan.	
Education Method	In order to complete an MSc Thesis you must have a thesis topic and a supervisor. All available Aerospace Structures and Materials thesis topics can be found on the ASM Track Brightspace community to which all registered ASM students are enrolled. They are listed by supervisor. If you do not have access but are a register ASM student please contact the track coordinator Dr. Gillian Saunders-Smits, by email: MScCoordinator-ASM@tudelft.nl, to request access. Students are free to select a thesis topic from any member of staff within the department providing that the supervisor of their choice feels that students have the right elective and profile courses for the topic. Supervisors may refuse students who they feel lack the right courses for a certain topic. Many staff also offer the opportunity for thesis work in industry and other research organizations through their network. Those opportunities are also offered on the ASM Track Brightspace community. Please note that in some cases you will not be allowed to do your thesis in industry. The Msc Track coordinator has final say in this matter. Students who are registered with the Materials profile are encouraged to contact their profile advisor: dr. Santiago Garcia to discuss the thesis options open to them as not all staff have unlimited places and a division of the students over the supervisors is made based on interest and capacity. Students are welcome to propose their own thesis or acquire a topic outside the university on their own but must be aware that the final approval of such a topic lies with the supervisor and the track coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved. In order for you to start working on your thesis a few conditions have to be met: - You must have finished (almost) all courses of your Master. The only allowable exceptions are AE4010 Research Methodologies and AE4020 Literature Study as they form a coherent whole with the thesis project - Unless your thesis will be a follow up of your internship you must have completed your internship. - You have informed the track coordinator, Dr. Gillian Saunders of your topic and supervisor and ensured your course list in Osiris is correct. If in doubt whether you meet these conditions please contact the track coordinator by email or by visiting the open office hours, every Friday from 09.30-10.30. When you have started your thesis you should organize a kick off meeting with your supervisor(s) to discuss further details and fill out the MSc Kick-off form (available from the AE-Airport) together with your supervisor(s) and send it to the Track Coordinator, Dr. Gillian Saunders, for processing.	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. This assessment matrix is the same for all students who are doing an MSc in Aerospace Engineering. Your assessment will be based upon the work you have carried out during your thesis as well as the way you carried it out; your written thesis report; your oral presentation of your thesis work; and your defense of your thesis work in front of a committee of experts. This committee includes your supervisor, other members of the academic staff of the research group and an external member from another research group as well as any external supervisors and industry guests as applicable. You will receive your diploma directly after your successful defence of your thesis. For the detailed rules on the composition of your MSc thesis committee, please see the MSc Regulations on the AE-Airport.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Materials and Manufacturing
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AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4ASM103		Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	The students must have general knowledge about materials science.		
Course Contents	(Organic) coatings can be defined as all kind of thin (organic) solid phases mechanically or chemically attached to a much thicker structure in order to add a certain functionality not obtainable from the substrate material itself. Color, mechanical, chemical and corrosion protection are obvious traditional functions of coatings but there are other less classical functions that are quickly growing in importance for several fields including aerospace. This course will introduce the basic principles relevant for traditional organic coatings science (e.g. adhesion, barrier, corrosion, mechanics) and those principles relevant for new developments in functional coatings. Special attention will be put on concepts applicable to aerospace field such as self-healing, active corrosion protection by nano and microparticles, anti-icing, self-cleaning, antifouling and drag reduction. Some of the concepts explained can be extended to other domains closely related to coatings such as adhesives and composites.		
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.		
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.		
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.		
Assessment	Written report (60% of final grade) and written exam (40% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.		

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/2/2/2	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Students of the track Aerospace Materials and Structures start in the second period (Q2, November). Students of the Wind energy master start in the third period (Q3, February). All other students start in the second period (Q2, November). Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four students about 14 day sections with support of staff are needed during this course. The planning of these day sections is depending (1) on the other obligatory courses of the track Aerospace Materials and Structures, (2) on the availability of supporting staff and (3) on the availability of the testing equipment. The ratio planned day sections by team over planned by staff is determined at the beginning of this course and notified and clarified at the kick off meeting held in the first week. For two parts of the exercise two consecutive day sections (so a full day) are needed. Finishing period Students started in the second period (Q2, November) are expected to finish at the end of the third period (Q3, April). Students started in the third period (Q3, February) are expected to finish at the end of the fourth period (Q4, June). IT IS EXPECTED THAT THIS LABORATORY COURSE WILL BE HELD AS BEFORE COVID-19. IF NOT ADAPTIONS WILL BE MADE. WHEN THAT DECISION WILL BE TAKEN IS UNCLEAR AT THE MOMENT OF WRITING (JUNE 2021).	
Course Contents	The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Possibly C-scanning of all laminates for determination of laminate quality 4. Estimation of the maximal bending-torsion coupling for a strip of UD composite 5. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 6. Performing mechanical tests 7. Analysing the test results of the mechanical test, including failure analyses 8. Adhesion of thermoset laminates 9. Resistance welding of thermoplastic laminates 10. Possibly preparation of lap shear specimens 11. Performing lap shear tests 12. Estimation of the mechanical properties of a sandwich panel 13. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 14. Performing a bending test on a sandwich panel. 15. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	For this course only one brightspace course will be available. It is called "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)". The Brightspace course "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q3)" should not be used, and is most likely not available for students. SO students starting in Q3 are added to "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)".	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on forehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	In this course the student is introduced to advanced techniques in the analysis and stability of structures. Emphasis is placed on deep understanding of the models, their assumptions, and ranges of applicability. The topics covered include: Principles of structural analysis and stability: stresses, strains, equations of equilibrium, stability criteria Beam theory Stability of beams Plate theory Buckling of plates	
Study Goals	1- Understand the energy formulation of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy methods for the evaluation of structural response 4- Derive buckling load expressions for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	To be announced.	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (70% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open book exam 5 obligatory laboratory assignments done in groups (weeks 2.2-2.6). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	- Classical Lamination Theory - Short Overview of Materials. Composites Design Philosophy - Theory of Elasticity - Engineering Constants - Stress & Strain Transformations / Implications for Testing - Thin laminates / Carpet Plots - Thick Laminates / ABD Matrix / Coupling Effects - Hydrothermal Analysis - Progressive damage analysis - Failure criteria - First ply failure & Last ply failure - Damage tolerance analysis - In-situ strength - Experimental mechanics & in-situ monitoring techniques - Basic Stress Solutions and Buckling of Composite Plates - Typical Airframe Elements - Airframe Design Process, Materials & Damage - Plate Governing Equations / Solution of the PDE - Energy Minimization Methods - Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: - Theory of Elasticity: 2 - Classical Laminar Theory: 4 - Progressive damage analysis: 4 - Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don'ts of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Instructor	Dr.ir. R.C. Alderliesten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	The study goals for this course are: - provide the student knowledge about airworthiness, roles of aerospace stakeholders, maintenance processes during operation, new developments in AC maintenance - the student should understand basic principles in damage tolerance and damage monitoring/inspection, aircraft maintenance - the student should be able to analyse maintenance problems from different angles (technical, regulations, planning). - the student should be able to provide well explained answers/solutions to questions and problems to structural integrity and maintenance issues	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam. Closed book.	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	Combination of Classroom/Online lectures and Practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4ASM508	Design of Self-healing materials	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	Current engineering materials design is based in the well-known concept of damage prevention, which is the improvement of properties necessary to avoid damage. Nevertheless, another concept, employed by natural materials and systems for millions of years, can be applied to man-made engineering materials design. This concept is based on damage management and known as self-repair or self-healing. Under this concept the damage is unavoidable and the systems have to develop mechanisms to autonomously heal the damaged zone to keep the functionality of the system (e.g. skin wound healing). Up to now, material scientists have been able to implement the self-healing concept to polymers, metals, and ceramics in different fields as polymeric and inorganic coatings, high temperature coatings, civil engineering materials as concrete and asphalt, fiber composites, bulk polymers for multipurpose applications, structural aluminium alloys, and others. This course will deal with the self-healing concept, the self-healing materials and systems developed so far, the methods and concepts behind the development of this new class of materials, the techniques employed to evaluate the degree of healing, and the future goals within the self-healing materials field.	
Study Goals	The students will be able to: identify self-healing materials; describe methods to develop self-healing systems; understand the underlying physical and chemical processes behind healing; differentiate advantages and disadvantages of different self-healing approaches; learn how to measure, evaluate and analyze self-healing materials properties.	
Education Method	Lectures and research report. For the research report the students will work in small groups to answer a set of questions related to self-healing materials and their applications.	
Literature and Study Materials	The lectures will be based on the latest developments in the field supported by selected literature. The study material will be the lecture notes and articles facilitated by the instructors.	
Assessment	Oral Exam (40% of final grade) + report (60% of final grade). The report has an individual grade (70%) and a group grade (30%). A minimum grade of 5 over 10 in the exam is mandatory to pass the course.	
Special Information	Assessment may differ depending on the number of registered students.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	S.R. Turteltaub	
Responsible Instructor	Dr.ing. S. Giovanni Pereira Castro	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of structures commonly encountered in aerospace during operation. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM513	Forensic Engineering	3
Responsible Instructor	Ir. M.J. Schuurman	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is only for MSc. Aerospace Engineering students.	
Course Contents	In Forensic Engineering students will get acquainted with forensic principles used in air safety investigations. The principal phases of the investigation process are explained; starting from the fact finding phase through the analysis phase and the final report phase. Examples will be shown of aircraft (serious) incidents and accidents on a case-by-case bases, highlighting different (engineering) failure modes and characteristics. During the course students will learn how to conduct an (accident) investigation and apply forensic concepts to determine the (probable) cause of the event. Finally the drafting of recommendations and its implications are looked at. Apart from theory a couple of practical group exercises are part of the program. The group exercises are aimed to apply the principles learned during the lectures and attendance is mandatory. A report of the group exercises is part of the final grade.	
Study Goals	The goals of the course are: Describe and explain the accident investigation goal and identify and analyse the different investigation phases. Demonstrate and apply accident investigation techniques. Select and use forensic investigation techniques to determine failure causes. Have knowledge of constructing and testing hypothesis and the ability to go through a verification process. Write an Annex 13 accident report with fact, analyses and conclusion including the formulation of recommendations to prevent reoccurrence or diminish the consequence of future events.	
Education Method	The course principles are case and evidence based learning. The course aims at the capability to cooperate with various scientific and design disciplines in a specific case oriented context.	
Prerequisites	- Course is only for MSc. Aerospace Engineering students - AE4ASM513 is an elective for students MSc. program - AE1110-I completed or AE.1110x (MOOC) with certificate	
Assessment	Mandatory lecture attendance is required to do the final exam. The exam consist out of 3 parts: Group field examination Group accident report writing Individual component	
Enrolment / Application	As there is a limit to course participant it is therefore required to register Forensic Engineering as an elective (as soon as possible) through your track coordinator. Each research group has a upper limit of students who can participate. Please let your track coordinator know as soon as possible that you want to take this course and send email to the course coordinator (m.j.schuurman@tudelft.nl) to be registered. Please note; Brightspace registration does not mean you are eligible to take the course. Please also be aware of the course prerequisites.	
Set-up	Forensic Engineering has mandatory attendance - please be aware as it is not possible to sit-in the lectures.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM516	Material Selection in Mechanical Design	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course aims at training the students in the field of material and process selection on the basis of objective and quantitative criteria. The concept of material selection on the basis of so-called Ashby diagrams will be presented in a number of lectures.	
Study Goals	Providing the student with the tools to make material and process selection on the basis of objective and quantitative criteria.	
Education Method	Lectures	
Literature and Study Materials	The book Materials Selection in Mechanical Design by M.F. Ashby. Butterworth Heinemann. 5th edition 2017	
Assessment	Written exam	
Set-up	The course will consist of a number of lectures in which the quantitative selection process involving an increasing number of constraints and objectives will be introduced. During the lectures students are invited to solve a number of simpler cases.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Ir. J. Sinke	
Instructor	Dr. J.A. Poulis	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December 2022 - Week 4.8: 15 May 2023 - Week 1.6: 30 September 2022 The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and	

Set-up	3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility. - Apply quantitative life cycle assessment methods (LCA) to analyse and evaluate industrial composite manufacturing systems	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit. This only applies to the academic year 2021-2022.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 4.1, 4.3, 4.6, 4.8); each day will consist of four lecture hours; weeks will be announced in advance of the course. At the beginning of the course students will group into teams of four to six persons. Each group will design a composite manufacturing process for a given part in the duration of the course. Self-organized work is expected of the teams especially in the weeks without lectures. The results of the team assignment will be documented by the members of each team in a written report and will additionally be presented by each team in the final lecture of the course.	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr. J. Wu	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of new phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Introduction of industrially oriented aeroelastic modelling, loads and certification. The course aims at placing the aeroelastic knowledge into a fully flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced aeroelastic phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Understand which regular and advanced aeroelastic concepts are important for which type of aircraft. 3. Know how to model advanced aeroelastic phenomena in an academic and industrially oriented context. 4. Understand the role of aeroelasticity in the loads process and the design of aircraft.	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. G. Dimitriadis, Introduction to nonlinear aeroelasticity, 2017, Wiley, ISBN:9781118613474	
Reader	Course notes	
Assessment	Take home assignment with oral close-out session.	

AE4ASM523	Design of Spacecraft and Launcher Structures	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Introduction to spacecraft and launcher structures 2. Launch vehicle, re-entry, satellite and deployable structures 3. Environmental requirements for spacecraft and launcher structures 4. Analysis under mechanical and thermal loads 5. Static and dynamic buckling phenomena for launcher structures 6. Typical development and qualification tests	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the spacecraft and launcher structures, their principles, design and analysis methods and tools. Successful students will have achieved the following objectives: They will have gained an advanced understanding in terms of the technologies, principles, methods and tools required to conceive, design, analyse and test spacecraft and launcher structures. They will have obtained capability to transform fundamental technical, economic and societal requirements into an integrated structural solution. They will have established credibility and depth in a discipline focused on current and future technology.	
Education Method	Lectures	
Assessment	To be announced	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4ASM523 Design of Spacecraft and Launcher Structures - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of seven weekly modules. In these modules the following topics will be covered: - Introduction - Thermo-Elastics - Vibrations - Systems Engineering - Verification and Validation The order of the subjects may differ from the above list. The course structure will be explained in detail in the first weekly module. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. The weekly modules will be complemented by a hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students. Time and location of the hang-out sessions will be communicated in the first lecture week. Students are requested to reserve all scheduled timeslots in their agendas. Note that, the course will require group work. Groups will be formed in the first week of the course. Students who are not part of a group cannot participate in the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. The weekly modules will be complemented by hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Computerized test on course content 2. Written feedback on design specification sheet of another group in the course Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, "Introduction to materials science" or books of similar nature. Furthermore, the students are expected to have followed the course AE4ASM002 Designing materials for aerospace relevant conditions	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, Tianwen-1, Hope, ISS, Hubble telescope, James Webb telescope, lightsail-2, Neowise, solar cruiser) will be presented as a carrier for material selection analysis. Current material choice will be analysed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The students will be trained to translate desired properties into material structures and microstructures and to think about suitable material production processes to realize these properties. Students are encouraged to propose alternative materials and reason for their choices. The structure of the lectures will be tailored to maximize the student involvement. Student involvement during and in preparation for lectures will be taken into account for the final grades. The course contains the following topics: week 1: Mars exploration history from Missions to Materials week 2: Material degradation in space - thermal cycling and vacuum week 3: Material degradation in space - UV, AO and space radiation week 4: Material degradation in space - meteoroids and orbital debris week 5: Space mission energy supply week 6: In-situ resource utilisation week 7: Materials testing for space applications (ESA guest speaker) week 8: Q and A week 9: exam week	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 7 lectures of 2h per week, and 7 self-exploratory studying sessions both before and after the lecture takes place. Each week one topic will be covered (shown in table below).	
Literature and Study Materials	A compilation of lecture notes and review papers	
Assessment	Continuous assessments (60%) + take home open book exam (40%). The students are to write three essays on a set of prescribed question/topics with submission deadlines at the end of week 2, 4, 6. The final grade will be the weighted average of reports and final exam. Example questions/topics will be made available on Brightspace. Continuous Assessments (60%) Assignment 1: Essay about relevance of materials research in space missions 15% of the marks Assignment 2: Essay about materials degradation and failure in space missions 20% of the marks Assignment 3: Essay about material design in hypothetical scenarios "think like a materials scientist" 25% of the marks Take-home open-book Exam (40%) a mock exam will be provided on brightspace honor code to be observed DESCRIPTION OF THE ASSIGNMENTS Assignment 1 Theme: Relevance of materials research in space missions (Please include word count after each sub-question) 1. What are the space missions your group would like to analyse? And why you chose these specific missions. (word limit: 50~100 words per mission) 2. What are the space conditions for this particular mission? Please specify at least temperature range, atmospheric condition, radiation and add other aspects that apply. (word limit: 150~200 words per mission) 3. What materials did you find on this space mission both for structural and also functional purposes? (word limit: 300~400 words per mission) Assignment 2 Theme: Materials degradation and failure in space missions (Please include word count after each sub-question) 1. How do you see the materials will degrade in both missions? Please list at least 3 degradation mechanisms and give your reasons (word limit: 500~600 words) 2. What information can you find about actual material failure? How does that differ from your prediction and why is so? (word limit: 500~600 words) Assignment 3 Theme: Think like a materials scientist (Please include word count after each sub-question) 1. What type of materials or specific material properties are needed if the condition of the mission is adjusted for higher radiation requirement, extreme thermal conditions (hot, cold) etc? Please list at least 3 hyperthethic scenarios (word limit: 500~600 words) 2. What structure-property relationship of materials can help you achieve the situation in question 1? (word limit: 500~600 words)	

words)

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Design and Safety of Structures

Program Coordinator	Dr.ir. J.A. Pascoe
Contact for Students	To contact the ASM MSc track coordinator send an email to MScCoordinator-ASM@tudelft.nl. Please indicate the profile (DSS) for specific questions about the profile itself.
Introduction 1	Profile II Design and Safety of Structures The design and safety of structures (DSS) profile is for you if your intention is to become a structural designer or stress engineer. It focusses on accurately calculating and predicting occurring stresses in structures and coming up with clever structural solutions through advanced design techniques and optimization. In the DSS profile you will learn more about bucking & structural analysis; the trinity in structures, materials and manufacturing; the design & optimization of composite reinforced structures, joining methods and experimental techniques. You will also learn about techniques for detecting structural damage, and how to analyse the damage tolerance of aerospace structures. This profile allows you to specialize in structures with an Aerospace focus but also allows you to go into neighbouring fields. Within the different research groups in the Aerospace Structures and Materials Department (ASM) there are plenty of thesis opportunities available. Electives with a strong Design and Safety of Structures flavour within the ASM Department are: - Applied Aircraft Aeroelasticity - Non-linear (structural) modelling (using F.E.M.) - Structural Analysis & Stability of Structures II - Design & Analysis of Composite Structures II - Continuum Mechanics - Forensic Engineering - Aircraft Manufacturing Laboratory - Design of Spacecraft and Launcher Structures - Structural Integrity & Maintenance - Trinity Exercise - Polymer Composite Manufacturing
Program Structure 1	In the ASM Track you do not have to select a profile until October. Use the core module in period 1 to select your final preference. Electives are decided upon in the second period whilst you are taking the profile courses to allow you to orientate yourself on your thesis topic and pick the relevant deepening electives upon the advice of your preferred thesis supervisor. You then register your choice of electives with the MSc track coordinator in December. TU Delft students who intend to start in February are requested to see the Track Coordinator by the end of November or early December to discuss their planning. Be mindful that a February start assumes you will take your internship from April of that year, so plan accordingly and start preparing for your internship in time.

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures. ALTHOUGH COVID-19 MIGHT INTERFERE, IT IS EXPECTED THAT THE LECTURE WILL BE GIVEN IN A LECTURE ROOM.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM002	Designing Materials with Aerospace Specific Properties	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Several courses follow on the general material concepts introduced in this 002 course: -AE4ASM101 Polymer Science; (Q2) -AE4ASM508 self healing materials; (Q3) -AE4ASM103 Functional Coatings. (Q3)	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Ashby, Materials". The students must have completed the materials science courses in the first 3 years of the Aerospace Engineering Curriculum, or any other Engineering curriculum	
Course Contents	In this course an unorthodox approach to materials will be presented. Rather than memorising known routes to reach certain materials properties, the students will be trained to translate these desired properties into material structures and microstructures and to design suitable material production processes to realise these properties. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The course is divided in 7 lectures. The first four lectures cover basic design rules of different material classes and their behaviour. The last three lectures cover material behavior at high temperatures, impact of material degradation on properties and strategies to extend material durability: -introduction to materials and structures -polymers -metals -ceramics and 'smart' materials -materials at high temperatures -materials and damage -materials and lifetime The students will need to deliver a report (assignment). For this they will get a set of questions they need to solve by reading the existing literature.	
Study Goals	The objective of the course is to train the student in reverse material engineering. This skill enables students to initiate and guide new material developments to meet future targets in the industry. By the end of the course, you should be able to: - explain structure and property (inter)relationships of metals, polymers and ceramics - explain the functionalities of aerospace relevant material properties - breakdown and translate these functionalities into material structures and microstructures using reverse engineering methodologies - explain material behaviour and function loss and strategies to decrease the impact of damage in the material function - give examples of new material concepts and explain the underlying concepts.	
Education Method	Lectures	
Literature and Study Materials	A compilation of lecture notes, review papers and assignment.	
Assessment	Assignment report (60%) + oral exam (40%)	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments.	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Ir. J. Sinke	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises/assignments (exercises are stand-alone; assignments are linked)	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Dr.ir. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
Assessment	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	In this course the student is introduced to advanced techniques in the analysis and stability of structures. Emphasis is placed on deep understanding of the models, their assumptions, and ranges of applicability. The topics covered include: - Principles of structural analysis and stability: stresses, strains, equations of equilibrium, stability criteria - Beam theory - Stability of beams - Plate theory - Buckling of plates	
Study Goals	1- Understand the energy formulation of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy methods for the evaluation of structural response 4- Derive buckling load expressions for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	To be announced.	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (70% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open book exam 5 obligatory laboratory assignments done in groups (weeks 2.2-2.6). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates / Carpet Plots f. Thick Laminates / ABD Matrix / Coupling Effects g. Hydrothermal Analysis 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis d. In-situ strength e. Experimental mechanics & in-situ monitoring techniques 3. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Theory of Elasticity: 2 2. Classical Laminate Theory: 4 3. Progressive damage analysis: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Officially registered and enrolled as a full time Master student Aerospace Engineering.

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

Enrolment / Application
Special Information

Set-up

AE5722	Thesis Aerospace Structures & Materials	45
Responsible Instructor	Prof.dr.ir. R. Benedictus	
Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The final thesis within the Aerospace Structures and Materials track is a 7 month research project in which you proof your competence as an Aerospace Structures & Materials engineer. Your thesis project can be carried out in-house but also in collaboration with industry or external research organization and sometimes even in industry or external research organizations. Ideally the final thesis is combined with the AE4020 Literature Study and the deliverables for AE4010 Research Methodologies into a 9-month full-time thesis project. These 9 months are a nett period and exclude any holidays or other events you may have scheduled. During this project you are supervised by one of our academic staff members sometimes combined with one of their post-docs or PhDs or a supervisor from a company or other research organization.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; Interpret obtained results in a critical manner; To produce results with scientific/technological relevance that can be published in scientific literature; Critically reflect on the work performed at the level of their peers in their particular field; Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; Present work performed in a structured way through an oral presentation to their peers and wider audience; Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; Plan the project efficiently considering resources and methodology; Execute the project controlling the process, effectively using the plan.	
Education Method	In order to complete an MSc Thesis you must have a thesis topic and a supervisor. All available Aerospace Structures and Materials thesis topics can be found on the ASM Track Brightspace community to which all registered ASM students are enrolled. They are listed by supervisor. If you do not have access but are a register ASM student please contact the track coordinator Dr. Gillian Saunders-Smits, by email: MSCCoordinator-ASM@tudelft.nl, to request access. Students are free to select a thesis topic from any member of staff within the department providing that the supervisor of their choice feels that students have the right elective and profile courses for the topic. Supervisors may refuse students who they feel lack the right courses for a certain topic. Many staff also offer the opportunity for thesis work in industry and other research organizations through their network. Those opportunities are also offered on the ASM Track Brightspace community. Please note that in some cases you will not be allowed to do your thesis in industry. The Msc Track coordinator has final say in this matter. Students who are registered with the Materials profile are encouraged to contact their profile advisor: dr. Santiago Garcia to discuss the thesis options open to them as not all staff have unlimited places and a division of the students over the supervisors is made based on interest and capacity. Students are welcome to propose their own thesis or acquire a topic outside the university on their own but must be aware that the final approval of such a topic lies with the supervisor and the track coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved. In order for you to start working on your thesis a few conditions have to be met: - You must have finished (almost) all courses of your Master. The only allowable exceptions are AE4010 Research Methodologies and AE4020 Literature Study as they form a coherent whole with the thesis project - Unless your thesis will be a follow up of your internship you must have completed your internship. - You have informed the track coordinator, Dr. Gillian Saunders of your topic and supervisor and ensured your course list in Osiris is correct. If in doubt whether you meet these conditions please contact the track coordinator by email or by visiting the open office hours, every Friday from 09.30-10.30. When you have started your thesis you should organize a kick off meeting with your supervisor(s) to discuss further details and fill out the MSc Kick-off form (available from the AE-Airport) together with your supervisor(s) and send it to the Track Coordinator, Dr. Gillian Saunders, for processing.	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. This assessment matrix is the same for all students who are doing an MSc in Aerospace Engineering. Your assessment will be based upon the work you have carried out during your thesis as well as the way you carried it out; your written thesis report; your oral presentation of your thesis work; and your defense of your thesis work in front of a committee of experts. This committee includes your supervisor, other members of the academic staff of the research group and an external member from another research group as well as any external supervisors and industry guests as applicable. You will receive your diploma directly after your successful defence of your thesis. For the detailed rules on the composition of your MSc thesis committee, please see the MSc Regulations on the AE-Airport.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Design and Safety of Structures
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AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-construct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	- To understand the possibilities and limitations of various polymer synthesis techniques. - To know and recognise some of the industrial polymer chemical structures. - To be able to explain the impact synthesis and chemical structure has on polymer material properties & their selection for given applications. - To gain insight in the characterisation of polymers via various methods - To be able to estimate performance and processing based on polymer structures - To understand fundamental working principles of polymers in different applications such as aerospace and energy applications - Understand and elucidate the concept of sustainable polymers and biobased materials - To be able to give a convincing pitch talk about a recent development in the field of polymers.	
Education Method	Lectures	
Literature and Study Materials	lecture notes + book: R.J. Young and P.A. Lovell, Introduction to polymers, Chapman & Hall, London, 3rd edition, 2011, ISBN 978-0-8493-3929-5. Please note that the 2nd edition can no longer be used. The newer, 3rd edition, is a complete makeover! Optional/alternative reading: Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective	
Assessment	Written final exam and pitch talk (3 minutes) presentation Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM103	Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	(Organic) coatings can be defined as all kind of thin (organic) solid phases mechanically or chemically attached to a much thicker structure in order to add a certain functionality not obtainable from the substrate material itself. Color, mechanical, chemical and corrosion protection are obvious traditional functions of coatings but there are other less classical functions that are quickly growing in importance for several fields including aerospace. This course will introduce the basic principles relevant for traditional organic coatings science (e.g. adhesion, barrier, corrosion, mechanics) and those principles relevant for new developments in functional coatings. Special attention will be put on concepts applicable to aerospace field such as self-healing, active corrosion protection by nano and microparticles, anti-icing, self-cleaning, antifouling and drag reduction. Some of the concepts explained can be extended to other domains closely related to coatings such as adhesives and composites.	
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.	
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.	
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.	
Assessment	Written report (60% of final grade) and written exam (40% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.	

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/2/2/2	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Students of the track Aerospace Materials and Structures start in the second period (Q2, November). Students of the Wind energy master start in the third period (Q3, February). All other students start in the second period (Q2, November). Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four students about 14 day sections with support of staff are needed during this course. The planning of these day sections is depending (1) on the other obligatory courses of the track Aerospace Materials and Structures, (2) on the availability of supporting staff and (3) on the availability of the testing equipment. The ratio planned day sections by team over planned by staff is determined at the beginning of this course and notified and clarified at the kick off meeting held in the first week. For two parts of the exercise two consecutive day sections (so a full day) are needed. Finishing period Students started in the second period (Q2, November) are expected to finish at the end of the third period (Q3, April). Students started in the third period (Q3, February) are expected to finish at the end of the fourth period (Q4, June). IT IS EXPECTED THAT THIS LABORATORY COURSE WILL BE HELD AS BEFORE COVID-19. IF NOT ADAPTIONS WILL BE MADE. WHEN THAT DECISION WILL BE TAKEN IS UNCLEAR AT THE MOMENT OF WRITING (JUNE 2021).	
Course Contents	The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Possibly C-scanning of all laminates for determination of laminate quality 4. Estimation of the maximal bending-torsion coupling for a strip of UD composite 5. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 6. Performing mechanical tests 7. Analysing the test results of the mechanical test, including failure analyses 8. Adhesion of thermoset laminates 9. Resistance welding of thermoplastic laminates 10. Possibly preparation of lap shear specimens 11. Performing lap shear tests 12. Estimation of the mechanical properties of a sandwich panel 13. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 14. Performing a bending test on a sandwich panel. 15. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	For this course only one brightspace course will be available. It is called "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)". The Brightspace course "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q3)" should not be used, and is most likely not available for students. SO students starting in Q3 are added to "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)".	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on forehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. B. Çalar	
Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: - Have solid knowledge on the mainstream manufacturing processes for polymer composite aerospace structures. This comprises processing steps, tooling and equipment as well as advantages, limitations and applicability. - Understand the relationships between the physics underlying specific manufacturing processes, the part/material quality and the design of the manufacturing process. - Recognise manufacturing defects, to identify their potential sources and to advice on measures to prevent/limit their occurrence.	
Education Method	Interactive lectures with numerous in-class exercises. An assignment is used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). Transition AE4ASM110 -> AE4ASM110-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM110, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM110. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM110 is not completed in 2022 2023, students will need to complete AE4ASM110-22 instead.	

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don'ts of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Instructor	Dr.ir. R.C. Alderliesten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	The study goals for this course are: - provide the student knowledge about airworthiness, roles of aerospace stakeholders, maintenance processes during operation, new developments in AC maintenance - the student should understand basic principles in damage tolerance and damage monitoring/inspection, aircraft maintenance - the student should be able to analyse maintenance problems from different angles (technical, regulations, planning). - the student should be able to provide well explained answers/solutions to questions and problems to structural integrity and maintenance issues	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam. Closed book.	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	Combination of Classroom/Online lectures and Practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM508	Design of Self-healing materials	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr.ir. H.E.J.G. Schlagen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	Current engineering materials design is based in the well-known concept of damage prevention, which is the improvement of properties necessary to avoid damage. Nevertheless, another concept, employed by natural materials and systems for millions of years, can be applied to man-made engineering materials design. This concept is based on damage management and known as self-repair or self-healing. Under this concept the damage is unavoidable and the systems have to develop mechanisms to autonomously heal the damaged zone to keep the functionality of the system (e.g. skin wound healing). Up to now, material scientists have been able to implement the self-healing concept to polymers, metals, and ceramics in different fields as polymeric and inorganic coatings, high temperature coatings, civil engineering materials as concrete and asphalt, fiber composites, bulk polymers for multipurpose applications, structural aluminium alloys, and others. This course will deal with the self-healing concept, the self-healing materials and systems developed so far, the methods and concepts behind the development of this new class of materials, the techniques employed to evaluate the degree of healing, and the future goals within the self-healing materials field.	
Study Goals	The students will be able to: identify self-healing materials; describe methods to develop self-healing systems; understand the underlying physical and chemical processes behind healing; differentiate advantages and disadvantages of different self-healing approaches; learn how to measure, evaluate and analyze self-healing materials properties.	
Education Method	Lectures and research report. For the research report the students will work in small groups to answer a set of questions related to self-healing materials and their applications.	
Literature and Study Materials	The lectures will be based on the latest developments in the field supported by selected literature. The study material will be the lecture notes and articles facilitated by the instructors.	
Assessment	Oral Exam (40% of final grade) + report (60% of final grade). The report has an individual grade (70%) and a group grade (30%). A minimum grade of 5 over 10 in the exam is mandatory to pass the course.	
Special Information	Assessment may differ depending on the number of registered students.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	S.R. Turteltaub	
Responsible Instructor	Dr.ing. S. Giovanni Pereira Castro	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of structures commonly encountered in aerospace during operation. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM513	Forensic Engineering	3
Responsible Instructor	Ir. M.J. Schuurman	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is only for MSc. Aerospace Engineering students.	
Course Contents	In Forensic Engineering students will get acquainted with forensic principles used in air safety investigations. The principal phases of the investigation process are explained; starting from the fact finding phase through the analysis phase and the final report phase. Examples will be shown of aircraft (serious) incidents and accidents on a case-by-case bases, highlighting different (engineering) failure modes and characteristics. During the course students will learn how to conduct an (accident) investigation and apply forensic concepts to determine the (probable) cause of the event. Finally the drafting of recommendations and its implications are looked at. Apart from theory a couple of practical group exercises are part of the program. The group exercises are aimed to apply the principles learned during the lectures and attendance is mandatory. A report of the group exercises is part of the final grade.	
Study Goals	The goals of the course are: Describe and explain the accident investigation goal and identify and analyse the different investigation phases. Demonstrate and apply accident investigation techniques. Select and use forensic investigation techniques to determine failure causes. Have knowledge of constructing and testing hypothesis and the ability to go through a verification process. Write an Annex 13 accident report with fact, analyses and conclusion including the formulation of recommendations to prevent reoccurrence or diminish the consequence of future events.	
Education Method	The course principles are case and evidence based learning. The course aims at the capability to cooperate with various scientific and design disciplines in a specific case oriented context.	
Prerequisites	- Course is only for MSc. Aerospace Engineering students - AE4ASM513 is an elective for students MSc. program - AE1110-I completed or AE.1110x (MOOC) with certificate	
Assessment	Mandatory lecture attendance is required to do the final exam. The exam consist out of 3 parts: Group field examination Group accident report writing Individual component	
Enrolment / Application	As there is a limit to course participant it is therefore required to register Forensic Engineering as an elective (as soon as possible) through your track coordinator. Each research group has a upper limit of students who can participate. Please let your track coordinator know as soon as possible that you want to take this course and send email to the course coordinator (m.j.schuurman@tudelft.nl) to be registered. Please note; Brightspace registration does not mean you are eligible to take the course. Please also be aware of the course prerequisites.	
Set-up	Forensic Engineering has mandatory attendance - please be aware as it is not possible to sit-in the lectures.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM515-22	Characterization of materials and components	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof. C.A. Dransfeld	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.	
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various experts in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation (XRD, SAX, NMR) 4. Surface analysis of materials (XPS, AES, AFM) 5. Electrical and electrochemical characterisation of metals and ceramics (Electrochemical Impedance Spectroscopy, dielectrics) 6. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 7. Mechanical characterization (tensile, fracture)	
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important in case of premature failure or degradation or in sub-or above standard material performance. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.	
Education Method	Lectures and assignments	
Literature and Study Materials	Introductory texts and review articles on the various techniques posted on Brightspace.	
Assessment	Oral Exam Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.	

AE4ASM516	Material Selection in Mechanical Design	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course aims at training the students in the field of material and process selection on the basis of objective and quantitative criteria. The concept of material selection on the basis of so-called Ashby diagrams will be presented in a number of lectures.	
Study Goals	Providing the student with the tools to make material and process selection on the basis of objective and quantitative criteria.	
Education Method	Lectures	
Literature and Study Materials	The book Materials Selection in Mechanical Design by M.F. Ashby. Butterworth Heinemann. 5th edition 2017	
Assessment	Written exam	
Set-up	The course will consist of a number of lectures in which the quantitative selection process involving an increasing number of constraints and objectives will be introduced. During the lectures students are invited to solve a number of simpler cases.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Ir. J. Sinke	
Instructor	Dr. J.A. Poulis	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December 2022 - Week 4.8: 15 May 2023 - Week 1.6: 30 September 2022 The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and	

Set-up	3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility. - Apply quantitative life cycle assessment methods (LCA) to analyse and evaluate industrial composite manufacturing systems	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit. This only applies to the academic year 2021-2022.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 4.1, 4.3, 4.6, 4.8); each day will consist of four lecture hours; weeks will be announced in advance of the course. At the beginning of the course students will group into teams of four to six persons. Each group will design a composite manufacturing process for a given part in the duration of the course. Self-organized work is expected of the teams especially in the weeks without lectures. The results of the team assignment will be documented by the members of each team in a written report and will additionally be presented by each team in the final lecture of the course.	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr. J. Wu	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of new phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Introduction of industrially oriented aeroelastic modelling, loads and certification. The course aims at placing the aeroelastic knowledge into a fully flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced aeroelastic phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Understand which regular and advanced aeroelastic concepts are important for which type of aircraft. 3. Know how to model advanced aeroelastic phenomena in an academic and industrially oriented context. 4. Understand the role of aeroelasticity in the loads process and the design of aircraft.	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. G. Dimitriadis, Introduction to nonlinear aeroelasticity, 2017, Wiley, ISBN:9781118613474	
Reader	Course notes	
Assessment	Take home assignment with oral close-out session.	

AE4ASM523	Design of Spacecraft and Launcher Structures	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Introduction to spacecraft and launcher structures 2. Launch vehicle, re-entry, satellite and deployable structures 3. Environmental requirements for spacecraft and launcher structures 4. Analysis under mechanical and thermal loads 5. Static and dynamic buckling phenomena for launcher structures 6. Typical development and qualification tests	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the spacecraft and launcher structures, their principles, design and analysis methods and tools. Successful students will have achieved the following objectives: They will have gained an advanced understanding in terms of the technologies, principles, methods and tools required to conceive, design, analyse and test spacecraft and launcher structures. They will have obtained capability to transform fundamental technical, economic and societal requirements into an integrated structural solution. They will have established credibility and depth in a discipline focused on current and future technology.	
Education Method	Lectures	
Assessment	To be announced	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4ASM523 Design of Spacecraft and Launcher Structures - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of seven weekly modules. In these modules the following topics will be covered: - Introduction - Thermo-Elastics - Vibrations - Systems Engineering - Verification and Validation The order of the subjects may differ from the above list. The course structure will be explained in detail in the first weekly module. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. The weekly modules will be complemented by a hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students. Time and location of the hang-out sessions will be communicated in the first lecture week. Students are requested to reserve all scheduled timeslots in their agendas. Note that, the course will require group work. Groups will be formed in the first week of the course. Students who are not part of a group cannot participate in the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. The weekly modules will be complemented by hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Computerized test on course content 2. Written feedback on design specification sheet of another group in the course Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, "Introduction to materials science" or books of similar nature. Furthermore, the students are expected to have followed the course AE4ASM002 "Designing materials for aerospace relevant conditions"	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, Tianwen-1, Hope, ISS, Hubble telescope, James Webb telescope, lightsail-2, Neowise, solar cruiser) will be presented as a carrier for material selection analysis. Current material choice will be analysed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The students will be trained to translate desired properties into material structures and microstructures and to think about suitable material production processes to realize these properties. Students are encouraged to propose alternative materials and reason for their choices. The structure of the lectures will be tailored to maximize the student involvement. Student involvement during and in preparation for lectures will be taken into account for the final grades. The course contains the following topics: week 1: Mars exploration history from Missions to Materials week 2: Material degradation in space - thermal cycling and vacuum week 3: Material degradation in space - UV, AO and space radiation week 4: Material degradation in space - meteoroids and orbital debris week 5: Space mission energy supply week 6: In-situ resource utilisation week 7: Materials testing for space applications (ESA guest speaker) week 8: Q and A week 9: exam week	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 7 lectures of 2h per week, and 7 self-exploratory studying sessions both before and after the lecture takes place. Each week one topic will be covered (shown in table below).	
Literature and Study Materials	A compilation of lecture notes and review papers	
Assessment	Continuous assessments (60%) + take home open book exam (40%). The students are to write three essays on a set of prescribed question/topics with submission deadlines at the end of week 2, 4, 6. The final grade will be the weighted average of reports and final exam. Example questions/topics will be made available on Brightspace. Continuous Assessments (60%) Assignment 1: Essay about relevance of materials research in space missions 15% of the marks Assignment 2: Essay about materials degradation and failure in space missions 20% of the marks Assignment 3: Essay about material design in hypothetical scenarios "think like a materials scientist" 25% of the marks Take-home open-book Exam (40%) a mock exam will be provided on brightspace honor code to be observed DESCRIPTION OF THE ASSIGNMENTS Assignment 1 Theme: Relevance of materials research in space missions (Please include word count after each sub-question) 1. What are the space missions your group would like to analyse? And why you chose these specific missions. (word limit: 50~100 words per mission) 2. What are the space conditions for this particular mission? Please specify at least temperature range, atmospheric condition, radiation and add other aspects that apply. (word limit: 150~200 words per mission) 3. What materials did you find on this space mission both for structural and also functional purposes? (word limit: 300~400 words per mission) Assignment 2 Theme: Materials degradation and failure in space missions (Please include word count after each sub-question) 1. How do you see the materials will degrade in both missions? Please list at least 3 degradation mechanisms and give your reasons (word limit: 500~600 words) 2. What information can you find about actual material failure? How does that differ from your prediction and why is so? (word limit: 500~600 words) Assignment 3 Theme: Think like a materials scientist (Please include word count after each sub-question) 1. What type of materials or specific material properties are needed if the condition of the mission is adjusted for higher radiation requirement, extreme thermal conditions (hot, cold) etc? Please list at least 3 hyperthethic scenarios (word limit: 500~600 words) 2. What structure-property relationship of materials can help you achieve the situation in question 1? (word limit: 500~600 words)	

words)

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile III: Structures for Space

Program Coordinator	I. Uriol Balbin
Contact for students	To contact the ASM MSc track coordinator send an email to MScCoordinator-ASM@tudelft.nl. Please indicate the profile (S4S) for specific questions about the profile itself.
Introduction 1	Profile III Structures for Space The Structures for Space (S4S) profile is for you if your intention is to work in the rapidly growing space sector as a structural engineer in the design, analysis and verification of spacecraft and launcher structures. In this profile you will learn the technologies and principles required to design, analyze and test the different types of space structures. You will also become familiar with spacecraft and launcher structures and will develop strategies necessary to transform the technical, economic and societal requirements into an integrated structural solution. Finally, you will be able to use these methods and tools to solve practical structural problems for present and future space exploration. Electives with a strong Structures for Space flavour within the ASM Department are: - Additive Manufacturing - Fundamentals of Aeroelasticity - Trinity Exercise - Stability & Analysis of Structures II - Materials for Space - Spacecraft Thermal Design Electives with a strong Structures for Space flavour within the Aerospace Faculty are: - MicroSat Engineering - Collaborative Space System Design Project - Spacecraft Attitude Dynamics and Control
Program Structure 1	In the ASM Track you do not have to select a profile until October. Use the core module in period 1 to select your final preference. Electives are decided upon in the second period whilst you are taking the profile courses to allow you to orientate yourself on your thesis topic and pick the relevant deepening electives upon the advice of your preferred thesis supervisor. You then register your choice of electives with the MSc track coordinator in December. TU Delft students who intend to start in February are requested to see the Track Coordinator by the end of November or early December to discuss their planning. Be mindful that a February start assumes you will take your internship from April of that year, so plan accordingly and start preparing for your internship in time.

AE4870A	Rocket Motion	3
Responsible Instructor	Ir. M.C. Naeije	
Instructor	Ir. W.J.F. Simons	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	Course Overview	
Study Goals	1. Introduction and recap previous courses on orbital mechanics 2. Fundamentals of rocket motion 3. Launch trajectories in a homogeneous gravity field, in vacuum and in an atmosphere (2D); vertical flight, constant pitch angle, gravity, and sounding rockets. 4. Theory of the multi-stage rocket; optimal mass distribution. 5. Ballistic flight over the Earth; 2D, and 3D over spherical (non-rotating and rotating Earth). 6. Launch systems design, unconventional launchers like air launch 7. Summary and question hour (workout exam question example) This course introduces the fundamentals of rocket motion by studying the motion of rockets under different circumstances. Emphasising on analytical solutions of the equations of motion, it will provide insight into the qualitative and quantitative aspects of launch trajectories of rockets in a homogeneous gravitational field, the performance of single and multi-stage rockets, and the exoatmospheric ballistic flight over the Earth. It discusses launcher design considerations and unconventional launch systems like air launch. At the end of the course you will be able to assess the performance of existing space launchers, and set up the (preliminary) design of such systems for specific applications. You also should be able to design a rocket motion simulator (e.g. in Matlab).	
Education Method	Lectures	
Literature and Study Materials	To be studied: Lecture slides AE4870A Rocket Motion Lecture notes AE4-870A Rocket Motion (selected chapters) Further reading: Turner, Martin J.L., Rocket and Spacecraft propulsion: principles, practice and new developments. 3rd Edition. ISBN 978-3-540-69202-7 (selected parts)	
Assessment	Closed-book written exam	

AE4870B	Re-Entry Systems	3
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	IMPORTANT NOTICE: Please read the restrictions under "Assessment" +++++ The course aims at giving a complete overview of re-entry systems, mainly from a mission point of view, but also from the system point of view. To begin with the mission, the fundamental principles of mechanics (Newton's Laws) are applied to derive the equations of motion for translational and rotational motion. Subsequently, simplifications are applied to give a description of planar ballistic, gliding and skipping flight, where the focus will be on analytical solutions for quantifying maximum mechanical and thermal loads. Next, typical mission applications, such as planetary entry and descent with parachutes and/or propulsion systems are discussed, as well as terminal-area flight of winged entry vehicles. To extend the mission applications, attention is also paid to the atmospheric flight aspects during an aero-gravity assist, aerocapture and aerobraking. As many re-entry missions rely on the application of guidance, navigation and control systems, the fundamentals of each of these three sub-systems will be discussed and applied in simple examples, such that after the course the student has a starting point for further study. The relation between vehicle configurations (capsule, winged, lifting body, etc.) and mission aspects, such as flight time and (maximum) system loads is a typical system aspect that will pass review. Finally, all presented theory will be combined in a so-called development plan for building a simulator that can be used for performance analysis of both controlled and uncontrolled re-entry vehicles. How to use such a simulator efficiently is depending on what kind of simulation and analysis technique will be used. Therefore, the course is concluded with an overview of such techniques like Monte-Carlo Analysis, and how to extract the required information from the simulated data.	
Course Contents Continuation	Lecture Topics (the actual topics covered may vary per year): #01 Introduction #02 Entry Environment and Aeroheating #03 Fundamentals of Motion #04 Ballistic entry #05 Gliding Entry #06 Skip Entry #07 Guidance, Navigation and Control #08 Planetary Entry and Descent #09 Advanced Descent and Landing Systems #10 Terminal Area Energy Management	
Study Goals	At the end of this course, the student will be able to (depending on topics covered): 1. Identify the influence of the planetary environment on the motion of and loads on an entry vehicle 2. Derive the equations of planar motion 3. Identify the three unpowered entry mechanisms (ballistic, gliding and skip entry), and derive the corresponding equations of motion 4. Apply a sizing methodology to design an entry descent system (parachute and/or propulsion) 5. Understand the fundamental functions of a Guidance, Navigation and Control System and design a simple system 6. Name the characteristic phases of the Terminal Area flight, and derive the equations for maximum range and maximum dive (steady-state approximation)	
Education Method	Classical lectures (2 x 2 hrs per week). The course is structured in a number of blocks, which each deal with an aspect of re-entry missions and systems. Many real-life examples will be used to highlight certain theoretical aspects, and to show the applicability of the presented theory. Typical exam questions will be used to prepare the student for the exam by testing his/her insight in the presented material. There may be homework assignments to deepen understanding of the material. This will be announced at the beginning of the course. See further details under assessment.	
Course Relations	AE3212-I - Aerospace Flight Dynamics and Simulation AE3212-II - Simulation, Verification and Simulation AE4870A - Rocket Motion	
Literature and Study Materials	1. Lecture Notes and Lecture Slides (Mandatory) Selected references for additional study: 1. Anderson, J. (1989). Hypersonic and High Temperature Gas Dynamics. McGraw-Hill. 2. Bertin, J. (1994). Hypersonic Aerothermodynamics. AIAA. 3. Cornelisse, J., Schoeyer, H., and Wakker, K.F. (1979). Rocket propulsion and spaceflight dynamics. Pitman, London. 4. Hankey, W. (1988). Re-Entry Aerodynamics. AIAA. 5. Mulder, J., van Staveren, W., van der Vaart, J., de Weerd, E., In 't Veld, A. and Mooij, E. (2013). Aerospace Flight Dynamics and Simulation. Delft University of Technology, Lecture Notes AE3-212-I. 6. Regan, F. and Anandakrishnan, S. (1993). Dynamics of Atmospheric Re-Entry. AIAA. 7. Vinh, N. (1981). Optimal trajectories in atmospheric flight. Elsevier. 8. Vinh, N., Busemann, A., and Culp, R. (1980). Hypersonic and planetary entry flight mechanics. The University of Michigan.	
Assessment	Closed-book, written examination. It will be decided at the beginning of the course whether homework assignments are part of the assessment. If that is the case, homework will count for 40% of the grade and the exam for 60%. Both partial grades have to be 6 or higher to pass the course. IMPORTANT NOTICE: The homework and/or exam is only open to students of the following profiles: 1) Spaceflight Dynamics 2) Planetary Exploration 3) Structures in Space Capacity permitting, students from other profiles can send a personal participation request to the course manager	

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures. ALTHOUGH COVID-19 MIGHT INTERFERE, IT IS EXPECTED THAT THE LECTURE WILL BE GIVEN IN A LECTURE ROOM.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM002	Designing Materials with Aerospace Specific Properties	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Several courses follow on the general material concepts introduced in this 002 course: -AE4ASM101 Polymer Science; (Q2) -AE4ASM508 self healing materials; (Q3) -AE4ASM103 Functional Coatings. (Q3)	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Ashby, Materials". The students must have completed the materials science courses in the first 3 years of the Aerospace Engineering Curriculum, or any other Engineering curriculum	
Course Contents	In this course an unorthodox approach to materials will be presented. Rather than memorising known routes to reach certain materials properties, the students will be trained to translate these desired properties into material structures and microstructures and to design suitable material production processes to realise these properties. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The course is divided in 7 lectures. The first four lectures cover basic design rules of different material classes and their behaviour. The last three lectures cover material behavior at high temperatures, impact of material degradation on properties and strategies to extend material durability: -introduction to materials and structures -polymers -metals -ceramics and 'smart' materials -materials at high temperatures -materials and damage -materials and lifetime The students will need to deliver a report (assignment). For this they will get a set of questions they need to solve by reading the existing literature.	
Study Goals	The objective of the course is to train the student in reverse material engineering. This skill enables students to initiate and guide new material developments to meet future targets in the industry. By the end of the course, you should be able to: - explain structure and property (inter)relationships of metals, polymers and ceramics - explain the functionalities of aerospace relevant material properties - breakdown and translate these functionalities into material structures and microstructures using reverse engineering methodologies - explain material behaviour and function loss and strategies to decrease the impact of damage in the material function - give examples of new material concepts and explain the underlying concepts.	
Education Method	Lectures	
Literature and Study Materials	A compilation of lecture notes, review papers and assignment.	
Assessment	Assignment report (60%) + oral exam (40%)	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments.	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Ir. J. Sinke	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises/assignments (exercises are stand-alone; assignments are linked)	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Dr.ir. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
Assessment	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	In this course the student is introduced to advanced techniques in the analysis and stability of structures. Emphasis is placed on deep understanding of the models, their assumptions, and ranges of applicability. The topics covered include: - Principles of structural analysis and stability: stresses, strains, equations of equilibrium, stability criteria - Beam theory - Stability of beams - Plate theory - Buckling of plates	
Study Goals	1- Understand the energy formulation of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy methods for the evaluation of structural response 4- Derive buckling load expressions for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	To be announced.	

AE4ASM523	Design of Spacecraft and Launcher Structures	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Introduction to spacecraft and launcher structures 2. Launch vehicle, re-entry, satellite and deployable structures 3. Environmental requirements for spacecraft and launcher structures 4. Analysis under mechanical and thermal loads 5. Static and dynamic buckling phenomena for launcher structures 6. Typical development and qualification tests	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the spacecraft and launcher structures, their principles, design and analysis methods and tools. Successful students will have achieved the following objectives: They will have gained an advanced understanding in terms of the technologies, principles, methods and tools required to conceive, design, analyse and test spacecraft and launcher structures. They will have obtained capability to transform fundamental technical, economic and societal requirements into an integrated structural solution. They will have established credibility and depth in a discipline focused on current and future technology.	
Education Method	Lectures	
Assessment	To be announced	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4ASM523 Design of Spacecraft and Launcher Structures - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of seven weekly modules. In these modules the following topics will be covered: - Introduction - Thermo-Elastics - Vibrations - Systems Engineering - Verification and Validation The order of the subjects may differ from the above list. The course structure will be explained in detail in the first weekly module. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. The weekly modules will be complemented by a hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students. Time and location of the hang-out sessions will be communicated in the first lecture week. Students are requested to reserve all scheduled timeslots in their agendas. Note that, the course will require group work. Groups will be formed in the first week of the course. Students who are not part of a group cannot participate in the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. The weekly modules will be complemented by hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Computerized test on course content 2. Written feedback on design specification sheet of another group in the course Grading criteria will be defined in the first lecture.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5722	Thesis Aerospace Structures & Materials	45
Responsible Instructor	Prof.dr.ir. R. Benedictus	
Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The final thesis within the Aerospace Structures and Materials track is a 7 month research project in which you proof your competence as an Aerospace Structures & Materials engineer. Your thesis project can be carried out in-house but also in collaboration with industry or external research organization and sometimes even in industry or external research organizations. Ideally the final thesis is combined with the AE4020 Literature Study and the deliverables for AE4010 Research Methodologies into a 9-month full-time thesis project. These 9 months are a nett period and exclude any holidays or other events you may have scheduled. During this project you are supervised by one of our academic staff members sometimes combined with one of their post-docs or PhDs or a supervisor from a company or other research organization.	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; Interpret obtained results in a critical manner; To produce results with scientific/technological relevance that can be published in scientific literature; Critically reflect on the work performed at the level of their peers in their particular field; Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; Present work performed in a structured way through an oral presentation to their peers and wider audience; Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; Plan the project efficiently considering resources and methodology; Execute the project controlling the process, effectively using the plan.	
Education Method	In order to complete an MSc Thesis you must have a thesis topic and a supervisor. All available Aerospace Structures and Materials thesis topics can be found on the ASM Track Brightspace community to which all registered ASM students are enrolled. They are listed by supervisor. If you do not have access but are a register ASM student please contact the track coordinator Dr. Gillian Saunders-Smits, by email: MScCoordinator-ASM@tudelft.nl, to request access. Students are free to select a thesis topic from any member of staff within the department providing that the supervisor of their choice feels that students have the right elective and profile courses for the topic. Supervisors may refuse students who they feel lack the right courses for a certain topic. Many staff also offer the opportunity for thesis work in industry and other research organizations through their network. Those opportunities are also offered on the ASM Track Brightspace community. Please note that in some cases you will not be allowed to do your thesis in industry. The Msc Track coordinator has final say in this matter. Students who are registered with the Materials profile are encouraged to contact their profile advisor: dr. Santiago Garcia to discuss the thesis options open to them as not all staff have unlimited places and a division of the students over the supervisors is made based on interest and capacity. Students are welcome to propose their own thesis or acquire a topic outside the university on their own but must be aware that the final approval of such a topic lies with the supervisor and the track coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved. In order for you to start working on your thesis a few conditions have to be met: - You must have finished (almost) all courses of your Master. The only allowable exceptions are AE4010 Research Methodologies and AE4020 Literature Study as they form a coherent whole with the thesis project - Unless your thesis will be a follow up of your internship you must have completed your internship. - You have informed the track coordinator, Dr. Gillian Saunders of your topic and supervisor and ensured your course list in Osiris is correct. If in doubt whether you meet these conditions please contact the track coordinator by email or by visiting the open office hours, every Friday from 09.30-10.30. When you have started your thesis you should organize a kick off meeting with your supervisor(s) to discuss further details and fill out the MSc Kick-off form (available from the AE-Airport) together with your supervisor(s) and send it to the Track Coordinator, Dr. Gillian Saunders, for processing.	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. This assessment matrix is the same for all students who are doing an MSc in Aerospace Engineering. Your assessment will be based upon the work you have carried out during your thesis as well as the way you carried it out; your written thesis report; your oral presentation of your thesis work; and your defense of your thesis work in front of a committee of experts. This committee includes your supervisor, other members of the academic staff of the research group and an external member from another research group as well as any external supervisors and industry guests as applicable. You will receive your diploma directly after your successful defence of your thesis. For the detailed rules on the composition of your MSc thesis committee, please see the MSc Regulations on the AE-Airport.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Structures for Space

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism. There are five deadlines for handing in each year.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	- To understand the possibilities and limitations of various polymer synthesis techniques. - To know and recognise some of the industrial polymer chemical structures. - To be able to explain the impact synthesis and chemical structure has on polymer material properties & their selection for given applications. - To gain insight in the characterisation of polymers via various methods - To be able to estimate performance and processing based on polymer structures - To understand fundamental working principles of polymers in different applications such as aerospace and energy applications - Understand and elucidate the concept of sustainable polymers and biobased materials - To be able to give a convincing pitch talk about a recent development in the field of polymers.	
Education Method	Lectures	
Literature and Study Materials	lecture notes + book: R.J. Young and P.A. Lovell, Introduction to polymers, Chapman & Hall, London, 3rd edition, 2011, ISBN 978-0-8493-3929-5. Please note that the 2nd edition can no longer be used. The newer, 3rd edition, is a complete makeover! Optional/alternative reading: Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective	
Assessment	Written final exam and pitch talk (3 minutes) presentation Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM103	Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	(Organic) coatings can be defined as all kind of thin (organic) solid phases mechanically or chemically attached to a much thicker structure in order to add a certain functionality not obtainable from the substrate material itself. Color, mechanical, chemical and corrosion protection are obvious traditional functions of coatings but there are other less classical functions that are quickly growing in importance for several fields including aerospace. This course will introduce the basic principles relevant for traditional organic coatings science (e.g. adhesion, barrier, corrosion, mechanics) and those principles relevant for new developments in functional coatings. Special attention will be put on concepts applicable to aerospace field such as self-healing, active corrosion protection by nano and microparticles, anti-icing, self-cleaning, antifouling and drag reduction. Some of the concepts explained can be extended to other domains closely related to coatings such as adhesives and composites.	
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.	
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.	
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.	
Assessment	Written report (60% of final grade) and written exam (40% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.	

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/2/2/2	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Students of the track Aerospace Materials and Structures start in the second period (Q2, November). Students of the Wind energy master start in the third period (Q3, February). All other students start in the second period (Q2, November). Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four students about 14 day sections with support of staff are needed during this course. The planning of these day sections is depending (1) on the other obligatory courses of the track Aerospace Materials and Structures, (2) on the availability of supporting staff and (3) on the availability of the testing equipment. The ratio planned day sections by team over planned by staff is determined at the beginning of this course and notified and clarified at the kick off meeting held in the first week. For two parts of the exercise two consecutive day sections (so a full day) are needed. Finishing period Students started in the second period (Q2, November) are expected to finish at the end of the third period (Q3, April). Students started in the third period (Q3, February) are expected to finish at the end of the fourth period (Q4, June). IT IS EXPECTED THAT THIS LABORATORY COURSE WILL BE HELD AS BEFORE COVID-19. IF NOT ADAPTIONS WILL BE MADE. WHEN THAT DECISION WILL BE TAKEN IS UNCLEAR AT THE MOMENT OF WRITING (JUNE 2021).	
Course Contents	The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Possibly C-scanning of all laminates for determination of laminate quality 4. Estimation of the maximal bending-torsion coupling for a strip of UD composite 5. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 6. Performing mechanical tests 7. Analysing the test results of the mechanical test, including failure analyses 8. Adhesion of thermoset laminates 9. Resistance welding of thermoplastic laminates 10. Possibly preparation of lap shear specimens 11. Performing lap shear tests 12. Estimation of the mechanical properties of a sandwich panel 13. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 14. Performing a bending test on a sandwich panel. 15. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	For this course only one brightspace course will be available. It is called "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)". The Brightspace course "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q3)" should not be used, and is most likely not available for students. SO students starting in Q3 are added to "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)".	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on forehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (70% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open book exam 5 obligatory laboratory assignments done in groups (weeks 2.2-2.6). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	- Classical Lamination Theory - Short Overview of Materials. Composites Design Philosophy - Theory of Elasticity - Engineering Constants - Stress & Strain Transformations / Implications for Testing - Thin laminates / Carpet Plots - Thick Laminates / ABD Matrix / Coupling Effects - Hydrothermal Analysis - Progressive damage analysis - Failure criteria - First ply failure & Last ply failure - Damage tolerance analysis - In-situ strength - Experimental mechanics & in-situ monitoring techniques - Basic Stress Solutions and Buckling of Composite Plates - Typical Airframe Elements - Airframe Design Process, Materials & Damage - Plate Governing Equations / Solution of the PDE - Energy Minimization Methods - Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: - Theory of Elasticity: 2 - Classical Laminar Theory: 4 - Progressive damage analysis: 4 - Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM110-22		Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen		
Instructor	Dr.ir. B.C.P. Jongbloed		
Instructor	Prof. C.A. Dransfeld		
Instructor	Dr. B. Çalar		
Instructor	D.M.J. Peeters		
Contact Hours / Week x/x/x/x	0/4/0/0		
Education Period	2		
Start Education	2		
Exam Period	none		
Course Language	English		
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.		
Study Goals	At the end of this course, the student should be able to: Have solid knowledge on the mainstream manufacturing processes for polymer composite aerospace structures. This comprises processing steps, tooling and equipment as well as advantages, limitations and applicability. Understand the relationships between the physics underlying specific manufacturing processes, the part/material quality and the design of the manufacturing process. Recognise manufacturing defects, to identify their potential sources and to advice on measures to prevent/limit their occurrence.		
Education Method	Interactive lectures with numerous in-class exercises. An assignment is used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.		
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). Transition AE4ASM110 -> AE4ASM110-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM110, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM110. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM110 is not completed in 2022 2023, students will need to complete AE4ASM110-22 instead.		

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don't's of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Instructor	Dr.ir. R.C. Alderliesten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	The study goals for this course are: - provide the student knowledge about airworthiness, roles of aerospace stakeholders, maintenance processes during operation, new developments in AC maintenance - the student should understand basic principles in damage tolerance and damage monitoring/inspection, aircraft maintenance - the student should be able to analyse maintenance problems from different angles (technical, regulations, planning). - the student should be able to provide well explained answers/solutions to questions and problems to structural integrity and maintenance issues	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam. Closed book.	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	Combination of Classroom/Online lectures and Practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4ASM508	Design of Self-healing materials	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr.ir. H.E.J.G. Schlagen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	Current engineering materials design is based in the well-known concept of damage prevention, which is the improvement of properties necessary to avoid damage. Nevertheless, another concept, employed by natural materials and systems for millions of years, can be applied to man-made engineering materials design. This concept is based on damage management and known as self-repair or self-healing. Under this concept the damage is unavoidable and the systems have to develop mechanisms to autonomously heal the damaged zone to keep the functionality of the system (e.g. skin wound healing). Up to now, material scientists have been able to implement the self-healing concept to polymers, metals, and ceramics in different fields as polymeric and inorganic coatings, high temperature coatings, civil engineering materials as concrete and asphalt, fiber composites, bulk polymers for multipurpose applications, structural aluminium alloys, and others. This course will deal with the self-healing concept, the self-healing materials and systems developed so far, the methods and concepts behind the development of this new class of materials, the techniques employed to evaluate the degree of healing, and the future goals within the self-healing materials field.	
Study Goals	The students will be able to: identify self-healing materials; describe methods to develop self-healing systems; understand the underlying physical and chemical processes behind healing; differentiate advantages and disadvantages of different self-healing approaches; learn how to measure, evaluate and analyze self-healing materials properties.	
Education Method	Lectures and research report. For the research report the students will work in small groups to answer a set of questions related to self-healing materials and their applications.	
Literature and Study Materials	The lectures will be based on the latest developments in the field supported by selected literature. The study material will be the lecture notes and articles facilitated by the instructors.	
Assessment	Oral Exam (40% of final grade) + report (60% of final grade). The report has an individual grade (70%) and a group grade (30%). A minimum grade of 5 over 10 in the exam is mandatory to pass the course.	
Special Information	Assessment may differ depending on the number of registered students.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	S.R. Turteltaub	
Responsible Instructor	Dr.ing. S. Giovanni Pereira Castro	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of structures commonly encountered in aerospace during operation. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM513	Forensic Engineering	3
Responsible Instructor	Ir. M.J. Schuurman	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is only for MSc. Aerospace Engineering students.	
Course Contents	In Forensic Engineering students will get acquainted with forensic principles used in air safety investigations. The principal phases of the investigation process are explained; starting from the fact finding phase through the analysis phase and the final report phase. Examples will be shown of aircraft (serious) incidents and accidents on a case-by-case bases, highlighting different (engineering) failure modes and characteristics. During the course students will learn how to conduct an (accident) investigation and apply forensic concepts to determine the (probable) cause of the event. Finally the drafting of recommendations and its implications are looked at. Apart from theory a couple of practical group exercises are part of the program. The group exercises are aimed to apply the principles learned during the lectures and attendance is mandatory. A report of the group exercises is part of the final grade.	
Study Goals	The goals of the course are: Describe and explain the accident investigation goal and identify and analyse the different investigation phases. Demonstrate and apply accident investigation techniques. Select and use forensic investigation techniques to determine failure causes. Have knowledge of constructing and testing hypothesis and the ability to go through a verification process. Write an Annex 13 accident report with fact, analyses and conclusion including the formulation of recommendations to prevent reoccurrence or diminish the consequence of future events.	
Education Method	The course principles are case and evidence based learning. The course aims at the capability to cooperate with various scientific and design disciplines in a specific case oriented context.	
Prerequisites	- Course is only for MSc. Aerospace Engineering students - AE4ASM513 is an elective for students MSc. program - AE1110-I completed or AE.1110x (MOOC) with certificate	
Assessment	Mandatory lecture attendance is required to do the final exam. The exam consist out of 3 parts: Group field examination Group accident report writing Individual component	
Enrolment / Application	As there is a limit to course participant it is therefore required to register Forensic Engineering as an elective (as soon as possible) through your track coordinator. Each research group has a upper limit of students who can participate. Please let your track coordinator know as soon as possible that you want to take this course and send email to the course coordinator (m.j.schuurman@tudelft.nl) to be registered. Please note; Brightspace registration does not mean you are eligible to take the course. Please also be aware of the course prerequisites.	
Set-up	Forensic Engineering has mandatory attendance - please be aware as it is not possible to sit-in the lectures.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM515-22	Characterization of materials and components	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof. C.A. Dransfeld	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.	
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various experts in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation (XRD, SAX, NMR) 4. Surface analysis of materials (XPS, AES, AFM) 5. Electrical and electrochemical characterisation of metals and ceramics (Electrochemical Impedance Spectroscopy, dielectrics) 6. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 7. Mechanical characterization (tensile, fracture)	
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important in case of premature failure or degradation or in sub-or above standard material performance. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.	
Education Method	Lectures and assignments	
Literature and Study Materials	Introductory texts and review articles on the various techniques posted on Brightspace.	
Assessment	Oral Exam Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.	

AE4ASM516	Material Selection in Mechanical Design	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course aims at training the students in the field of material and process selection on the basis of objective and quantitative criteria. The concept of material selection on the basis of so-called Ashby diagrams will be presented in a number of lectures.	
Study Goals	Providing the student with the tools to make material and process selection on the basis of objective and quantitative criteria.	
Education Method	Lectures	
Literature and Study Materials	The book Materials Selection in Mechanical Design by M.F. Ashby. Butterworth Heinemann. 5th edition 2017	
Assessment	Written exam	
Set-up	The course will consist of a number of lectures in which the quantitative selection process involving an increasing number of constraints and objectives will be introduced. During the lectures students are invited to solve a number of simpler cases.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Ir. J. Sinke	
Instructor	Dr. J.A. Poulis	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December 2022 - Week 4.8: 15 May 2023 - Week 1.6: 30 September 2022 The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and	

Set-up	3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility. - Apply quantitative life cycle assessment methods (LCA) to analyse and evaluate industrial composite manufacturing systems	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit. This only applies to the academic year 2021-2022.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 4.1, 4.3, 4.6, 4.8); each day will consist of four lecture hours; weeks will be announced in advance of the course. At the beginning of the course students will group into teams of four to six persons. Each group will design a composite manufacturing process for a given part in the duration of the course. Self-organized work is expected of the teams especially in the weeks without lectures. The results of the team assignment will be documented by the members of each team in a written report and will additionally be presented by each team in the final lecture of the course.	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr. J. Wu	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of new phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Introduction of industrially oriented aeroelastic modelling, loads and certification. The course aims at placing the aeroelastic knowledge into a fully flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced aeroelastic phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Understand which regular and advanced aeroelastic concepts are important for which type of aircraft. 3. Know how to model advanced aeroelastic phenomena in an academic and industrially oriented context. 4. Understand the role of aeroelasticity in the loads process and the design of aircraft.	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. G. Dimitriadis, Introduction to nonlinear aeroelasticity, 2017, Wiley, ISBN:9781118613474	
Reader	Course notes	
Assessment	Take home assignment with oral close-out session.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, "Introduction to materials science" or books of similar nature. Furthermore, the students are expected to have followed the course AE4ASM002 "Designing materials for aerospace relevant conditions"	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, Tianwen-1, Hope, ISS, Hubble telescope, James Webb telescope, lightsail-2, Neowise, solar cruiser) will be presented as a carrier for material selection analysis. Current material choice will be analysed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The students will be trained to translate desired properties into material structures and microstructures and to think about suitable material production processes to realize these properties. Students are encouraged to propose alternative materials and reason for their choices. The structure of the lectures will be tailored to maximize the student involvement. Student involvement during and in preparation for lectures will be taken into account for the final grades. The course contains the following topics: week 1: Mars exploration history from Missions to Materials week 2: Material degradation in space - thermal cycling and vacuum week 3: Material degradation in space - UV, AO and space radiation week 4: Material degradation in space - meteoroids and orbital debris week 5: Space mission energy supply week 6: In-situ resource utilisation week 7: Materials testing for space applications (ESA guest speaker) week 8: Q and A week 9: exam week	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 7 lectures of 2h per week, and 7 self-exploratory studying sessions both before and after the lecture takes place. Each week one topic will be covered (shown in table below).	
Literature and Study Materials	A compilation of lecture notes and review papers	
Assessment	Continuous assessments (60%) + take home open book exam (40%). The students are to write three essays on a set of prescribed question/topics with submission deadlines at the end of week 2, 4, 6. The final grade will be the weighted average of reports and final exam. Example questions/topics will be made available on Brightspace. Continuous Assessments (60%) Assignment 1: Essay about relevance of materials research in space missions 15% of the marks Assignment 2: Essay about materials degradation and failure in space missions 20% of the marks Assignment 3: Essay about material design in hypothetical scenarios "think like a materials scientist" 25% of the marks Take-home open-book Exam (40%) a mock exam will be provided on brightspace honor code to be observed DESCRIPTION OF THE ASSIGNMENTS Assignment 1 Theme: Relevance of materials research in space missions (Please include word count after each sub-question) 1. What are the space missions your group would like to analyse? And why you chose these specific missions. (word limit: 50~100 words per mission) 2. What are the space conditions for this particular mission? Please specify at least temperature range, atmospheric condition, radiation and add other aspects that apply. (word limit: 150~200 words per mission) 3. What materials did you find on this space mission both for structural and also functional purposes? (word limit: 300~400 words per mission) Assignment 2 Theme: Materials degradation and failure in space missions (Please include word count after each sub-question) 1. How do you see the materials will degrade in both missions? Please list at least 3 degradation mechanisms and give your reasons (word limit: 500~600 words) 2. What information can you find about actual material failure? How does that differ from your prediction and why is so? (word limit: 500~600 words) Assignment 3 Theme: Think like a materials scientist (Please include word count after each sub-question) 1. What type of materials or specific material properties are needed if the condition of the mission is adjusted for higher radiation requirement, extreme thermal conditions (hot, cold) etc? Please list at least 3 hyperthethic scenarios (word limit: 500~600 words) 2. What structure-property relationship of materials can help you achieve the situation in question 1? (word limit: 500~600 words)	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Both passive as well as active thermal control systems are discussed. Passive devices include: radiators, baffles, heat pipes, multi layer insulation, coatings and finishes. Active devices include: louvers, controlled heaters, cryogenic coolers, Peltier elements and fluid loops.	
Study Goals	The student is able to generate a thermal design of a satellite thermal control system using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - distinguish design hardware options - generate a (concept of a) thermal design - generate analysis models for hand calculation methods and/or computer tools - analyze a thermal design using hand calculation methods and/or computer tools - distinguish thermal environmental test methods	
Education Method	Lectures + homework Lectures are given by an experienced thermal design engineer from Airbus DS NL. Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Asnwers to the questions may be given by fellow students and/or the lecturer(s).	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%)	

AE4S10	Microsat Engineering	4
Responsible Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Lectures from experts on various aspects of small satellites and group/individual projects for practicing.	
Course Contents	The course covers the engineering of small satellites. The concept and definition of micro-, nano-, pico-, and femto-satellites is introduced along with the past, present, and future of small satellites in general. The course comprises seven to eight lectures, partly from professionals in the space industry. The lectures are intended to familiarise the students with current industry trends and state-of the art research in small satellite engineering. The lectures vary each year and potential lecture topics include communications, AOCS, onboard computer, distributed space systems, miniature payloads, small satellite missions, etc.	
Study Goals	The course will enable students to design and engineer small satellite missions. The high-level learning objectives of the course are: The students shall be able to - explain key aspects of small satellite missions - explain and describe elements in the architecture of a spacecraft mission - discuss in detail at least one element in the architecture of a spacecraft mission - present and defend their design of an element in the architecture of a spacecraft mission - show familiarity with current trends in space industry and small satellite research	
Education Method	Lectures (Provided partly by professionals from the space industry). Group work (Students work on assignment in groups; Coaching will be provided from experts to guide the students). Individual work (Students work on small individual assignments).	
Literature and Study Materials	Optional: Wertz J.R., Larson W.J. (eds.); Space Mission Analysis and Design; 3rd edition Kluwer Academic Publishers (2003). Mandatory: Lecture Notes AE4-S10 MicroSatellite Engineering.	
Prerequisites	General mathematical, physical and engineering knowledge from completed BSc study. In particular, AE1101 + AE1102 + AE3201, or similar	
Assessment	There will be two assessment components A Group Assignment and an individual assignment. Group assignment The final demonstration or report will be graded on the following aspects: - Planning - Presentation, argumentation and defence of the design - Functionality and Usability of the deliverable - Context of Small satellite (How well has the student team incorporated current small satellite research and industry trends/needs in the design) - Independence - Creativity Individual assignment This assignment will be graded on the following aspects: - Understanding and critical review of articles in a specified technical field of small satellites	
Permitted Materials during Tests	N/A	
Enrolment / Application	Students have to enroll via the BrightSpace site of the course. Professionals are welcome to the course. Conditions for participation and registration are available with the secretary of the Chair of Space Systems Engineering (www.sse.lr.tudelft.nl).	
Set-up	The course comprises lectures, guest lectures, team sessions, group work and individual work.	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 5: Flight Performance and Propulsion

MSc Coordinator

Dr.ir. G. la Rocca

Introduction 1

The main goal of the Flight Performance and Propulsion Master track is to contribute to the improvement of flight performance and propulsion. To achieve that we focus on innovative aircraft configurations and novel propulsion concepts. Therefore, aircraft -engine integration is also an important topic of research. These research areas are highly relevant if we want to contribute to the ambitious targets to reduce the environmental impact of aviation, as proposed by the Advisory Council for Aviation Research and Innovation Research in Europe (ACARE). Our mission is to advance the design of innovative aircraft configurations and propulsion concepts, by exploration of:

1. New technologies and unconventional airplane configurations,
2. Advanced flight physics modeling to improve the prediction and simulation of air-vehicle performance,
3. New methods and tools to improve the quality and effectiveness of the design process.

The staff members of the Flight Performance and Propulsion Master track have a range of different expertise, varying from air-breathing propulsion systems, flight mechanics, aircraft design and integration, combustion, thermodynamics, systems engineering and multidisciplinary design optimization. The knowledge and competences within the Flight Performance and Propulsion section are therefore grouped in three key areas:

1. Propulsion & Power
2. Flight Mechanics
3. Aircraft Design & Design Methodologies

The FPP Master Track offers an educational program consisting of two profiles and a shared set of core courses, relevant to both profiles. FPP students can choose one of the two profiles, according to their interests and development goals.

Profile I (FP) is focused on flight performance and includes mandatory courses in the areas of aircraft aerodynamic design, experimental simulations, aeroelastics and digital design methodologies.

Profile II (PP) offers a different set of compulsory courses dedicated to propulsion & power, such as turbomachinery, internal flows, combustion, heat transfer and modeling and simulation methods for propulsion and power.

The common set of courses provides knowledge that is relevant to both profiles and includes subjects such as aircraft aerodynamics, CFD, aero engine technology and advanced design methods, such as multidisciplinary design optimization (MDO)

Next to the track and profile courses, students can choose among a large number of elective courses, in consultation with the master track coordinator, such to develop in other relevant scientific and technological domains, as well as to acquire or reinforce skills for personal and professional development.

The competence acquired during the first year of the Master program is put to fruition in the final year during the research period dedicated to the thesis. During the Master thesis, the student is engaged in a project most often within one of the active research programs. The project is carried out under the supervision of one of the academic staff members, but often the student also benefits from the closer supervision of one of the PhD candidates involved in the same project. Thesis projects in cooperation with industry and research institutes are also very common.

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Flight Performance

Program Coordinator	Dr.ir. G. la Rocca
Introduction 1	Education: The profile program consists of courses that deal with aerodynamic design of high subsonic transport aircraft, aeroelasticity, experimental simulations and digital design automation (KBE). Some of the courses have a more theoretical approach and aim to provide a fundamental understanding of the governing physics whereas other courses are more applied (e.g. include experimental sessions in the wind tunnel), others require good programming skills. Elective courses are grouped in three areas, Flight mechanics, Propulsion integration and Aircraft Design & Design Methodologies. A selection of elective courses is made in consultation with the profile coordinator. The competence acquired during the first year of the Master program is put to fruition in the final year during the research period dedicated to the thesis. During the Master thesis, the student is engaged in a project most often within one of the active research programs. The project is carried out under the supervision of one of the academic staff members, but often the student also benefits from the closer supervision of one of the PhD candidates involved in the same project. Below, an overview of the research activities is provided. Research: We focus our research on advanced and innovative aircraft configurations, novel propulsion concepts and aircraft engine integration. Our mission is to advance the design of innovative aircraft configurations and propulsion concepts, by exploration of new technologies to obtain novel or improved solutions, advances in flight physics to improve the prediction and simulation of air-vehicle performance, and new methods and tools to improve the quality and effectiveness of the design process. The knowledge and competences are therefore clustered in three key areas: aircraft design & design methodologies, flight mechanics, and propulsion integration. Aircraft design & design methodologies Major technological advances in flight performance can be realized by new unconventional airplane designs. To support the development of such novel configurations, advanced design methodologies (e.g. Multidisciplinary Design Optimization and Knowledge Based Engineering) are required that go beyond statistical and analytical methods. To that extent, design support tools are developed that are able to sustain the evolutionary improvement of current aircraft design, as well as to support the investigation of novel aircraft configurations and assess the impact of new technologies such as hybrid-electric propulsion, morphing structures or flow control. Flight mechanics It is our objective to focus on two specific aspects of flight mechanics and thereby to advance the current state of the art: - Investigation of the inherent flying qualities of novel fixed wing aircraft configurations and scaled models to enable scaled flight testing - Automatic generation of flight mechanics simulation models for application in multi-disciplinary design optimization frameworks Propulsion integration Apart from the optimisation of isolated aircraft engines, the efficient integration of engines to the airframe is also of crucial importance to maximise aircraft performance and minimise noise emissions. One major ongoing project within this domain focusses on the aerodynamic and acoustic performance of propellers integrated on the airframe. Apart from the analysis and optimization of the isolated propeller system, also the interactions between the airframe and the propeller need to be considered. The goal of the investigations is to reduce the performance and noise penalties associated with these airframe interaction effects. Students involved in the project perform experiments in the faculty's various wind tunnel facilities using powered propeller test rigs and advanced measurement techniques, while also CFD tools are applied for numerical evaluations.

AE4115	Experimental Simulations	3
Responsible Instructor	T. Sinnige	
Instructor	Prof.dr.ing. G. Eitelberg	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling. Fractional analysis, symmetry and similarity, application to specific problems from aerodynamics, BL-approximations, S/BL interactions; 3 Dimensional analysis: Buckingham's Pi-theorem, rigorous methods of obtaining dimensionless parameters, simplifications of Navier-Stokes equations; 4 Subsonic and supersonic wind tunnels: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 5 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 6 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 7 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 8 Design of experiments: factorial designs, response-surface models; 9 Reynolds-number effects and simulation; 10 Pre-test support session; 11 Engine integration: engine simulators, performance of those. Thrust reverser testing for re-ingestion and stability and control; 12 Acoustic testing. Noise generation mechanisms for submerged bodies, edges and jets. Appropriate scaling. Definition of sound pressure levels. Measurement requirements and devices. Far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session. (Excursion) Ground test facilities of DNW (organized by AWEP society, limited places available).	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports and oral exam.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter. The lectures are supported by an excursion to the testing facilities of DNW, organized by the AWEP society.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and (to limited extend) viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Typically Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during at the beginning of Q1).	
Assessment	The assessment is done through 5 personal assignments.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written (proctored online) exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4rt quarter of the academic year) and not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Remarks	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr. F. Oliviero	
Instructor	Dr.ir. M.F.M. Hoogreef	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 - The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4238	Aero Engine Technology	4
Responsible Instructor	A. Gangoli Rao	
Instructor	Ir. M. Pini	
Instructor	F. Yin	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery"; "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Turbomachinery - o New Engine /propulsion Concepts - o Turbine Cooling - o Inlets and Exhaust - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an hybrid course in which the lectures are pre-recorded and some for tutorials will be solved in the classroom. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13 Analyse the cooling requirements and compute the effect of cooling on engine performance 14. Evaluate the calorific values for various fuels 15. Understand the basics of combustion and emission formation in Aero engines 16. Understand the working of gas turbine combustor 17. Understand the basic principles of gas turbine controls and evaluate different strategies for control 18. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a hybrid teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Reader	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on several homework assignments (5-6) given during the course and a final exam. All assignments together will form 50% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015 Optional textbooks: - Airplane Design, Prt I-VII, Roskam, J, DARCorp, 1990-2003 - Synthesis of Subsonic Airplane Design, Torenbeek, E, Kluwer Academic Publishers, 1982 - Fluid Dynamic Drag, Hoerner, S. F., Hoerner Fluid Dynamics, 1965 - Fluid Dynamic Lift, Hoerner, S. F. Hoerner Fluid Dynamics, 1965	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4ASM506		Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker		
Contact Hours / Week x/x/x/x	0/4/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.		
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.		
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.		
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.		
Assessment	1. Written exam. 2. Take home assignment.		

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5222	Thesis Flight Performance & Propulsion	45
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Instructor	Dr.ir. G. la Rocca	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. http://studenten.tudelft.nl/fileadmin/Files/studentenportal/os/LRspecifiek/docs/Rubrics_MSc-LR_version_14-Oct-13_met_naam_en_studienummer.pdf.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Flight Performance

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4120	Viscous Flows	3
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Responsible Instructor	M. Kotsonis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE3-W02 (4P720 TUE) plus basic aerodynamics.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade element/momentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsen's Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4136-22	CFD 2: Discretization techniques	3
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results.	
Study Goals continuation	After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Special Information	M.I.Gerritma@tudelft.nl	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Dr. S.J. Hulshoff	
Responsible Instructor	Prof.dr.ir. S. Hickel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	Prof.dr. F. Scarano	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4206	Turbomachinery	3
Responsible Instructor	Ir. M. Pini	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course covers fundamental concepts of internal flows in turbomachines and related design aspects. The usual course schedule consists of the following lectures: Lecture 0: Introduction course overview: topics, structure and assessment Lecture 1: Recall of Turbomachinery Fundamentals & Principles integral governing equations with applications, relative motion, thermodynamic transformations in turbomachinery stages, velocity triangles with practical examples; Lecture 2: Unsteady Flows in Turbomachinery concept of reduced frequency with applications; work exchange in turbomachinery rows; Lecture 3: Similarity in Turbomachinery similarity analysis for incompressible flows, Balje and Smith diagrams, accounting for compressibility, stage dimensionless performance parameters; Lecture 4: Fluid-dynamic Design of Turbomachinery design procedure, mean-line design model, radial equilibrium model, CFD-based models Lecture 5-6: Axial and Radial Turbines examples of axial and radial turbines, typical turbine stages, blade profiles for turbines, flow characteristics in subsonic, transonic, and supersonic cascades, basic concepts of diffusers, aerodynamics of radial impellers Lecture 7-8: Axial and Radial Compressors examples of axial and radial compressors, typical compressor stages, blade profiles for compressors, flow characteristics and unique incidence, stall and surge instability, slip factor, diffusers for radial compressors, design maps Seminar(s): topic(s) may vary but usually include turbomachinery aero-mechanics, shape optimisation with CFD methods, rocket engine turbo-pumps, and advanced turbomachinery components for aero-engines Tutorial: training session on the software packages used for the group assignment	
Course Contents	This course educates aerospace students to apply theoretical knowledge in the field of fluid- mechanics and thermodynamics for analyzing and designing a thermal turbomachinery. Typical examples are: axial and radial compressors of gas turbine engines, industrial gas and steam turbines, radial turbines of automotive turbochargers, turbo-expanders and compressors used in refrigeration and liquefaction industry, micro-turbines for power generation. Since the turbomachinery field is intrinsically interdisciplinary, it is assumed that students already possess basic knowledge of thermodynamics, fluid-mechanics, applied mathematics, propulsion systems, internal flows, acquired in previous courses. Prerequisite of the course are therefore bachelor and master courses addressing fundamental (e.g. Thermodynamics, Internal Flows) and applied aspects (e.g. Aero-Engines) of air-breathing engines. Starting from basic concepts of gas-dynamics and fluid-machinery, the course brings the student to the level that he is able to select a turbomachinery for a specific application, perform the conceptual design of the stages, and eventually predict the flow characteristics within the blade passages by means of conceptual physical models and more sophisticated simulation tools.	
Study Goals	After the course the student will be able to: - Classify and illustrate the main features of a turbomachine - Select the most appropriate turbomachine for a given aerospace application - Apply first principles to simplified turbomachinery configurations - Identify the most prominent loss sources of a given turbomachinery stage - Define the aerodynamic blade design of turbomachinery cascades - Predict the performance of a turbomachine with analytical and numerical methods - Devise solutions for performance improvement of turbomachinery cascades based on physical understanding - Perform the conceptual and detailed fluid-dynamic design of a turbomachine for propulsion systems using analytical and numerical tools	
Education Method	Lectures, concept quizzes, concept exercises, and tutorials in preparation to the team assignment.	
Books	- S.L. Dixon, C. A. Hall Fluid-Mechanics and Thermodynamics of Turbomachinery, 7th edition, 2013 - M. Schobeiri, Turbomachinery Flow Physics and Dynamic Performance, 2nd edition, Springer, 2012 - E.A. Baskharone, "Principles of Turbomachinery in Air-Breathing Engine", Cambridge University Press	
Assessment	The final course grade results from the weighted average of: - the assessment of the final project assignment, counting for 40% of the final grade (individual score 1-10, minimum threshold: 6); - the assessment of the concept exercises, counting for 10% of the final grade. No minimum threshold; - the assessment of the written exam, counting for 50% of the final grade (individual score 1-10, minimum threshold: 6); The written exam takes place at the end of course. The exam consists of one or two concept exercises covering the material discussed during classes. The final assignment is meant to challenge students with the real engineering problem of designing a turbomachine for propulsion or power generation application. The concept exercises are numerical exercises that are formulated to stimulate critical thinking on turbomachinery problems of practical application. They are also meant as diagnostic tool to help students self-assessing their level of understanding.	
Permitted Materials during Tests	Lectures notes, slides, formula sheets and textbooks are allowed during the written exam. Calculator can be used.	
Set-up	The course is organized in 8 teaching sessions, 2/3 seminar(s), and a possible visit to a turbomachinery company. Lectures usually consist of a theoretical part integrated with in-class exercises and quizzes to stimulate critical thinking and learning-by-doing. During classes, the students, autonomously or in groups, are engaged in a series of learning activities, supervised by the team of instructors. Their solution is briefly discussed at the beginning of the next class. The seminar(s) are devoted to show students current research topics in turbomachinery and are given by recognised experts in the field.	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Course Contents	Dr. D. Ragni Prof.dr.ir. M. Snellen 0/0/2/0 3 3 Exam by appointment English	
Study Goals	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources. The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resourcen/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4261	Internal Flows	3
Responsible Instructor	Ir. M. Pini	
Instructor	C.M. De Servi	
Responsible for assignments	T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course we address fluid dynamic phenomena of interest in internal flow situations for aerospace propulsion applications. The goal of the course is to develop physical insight into the phenomena that characterize internal flow in fluid machinery. As such we will discuss not just what happens, but why it happens and what are the consequences for propulsion systems. The course covers topic, which are not dealt with in courses or texts about external fluid dynamics. The flows described in Internal Flows are generally rotational, often three-dimensional, unsteady, and sometimes occurring in non-inertial (e.g., rotating) coordinate systems. The course can be viewed as the development of flow models and ideas to enable allow physical insight into the behavior of three-dimensional and unsteady flows. One of the objectives of the course is to provide an increased ability to interpret computational and experimental results and hence to effectively extract conclusions about the key features of complex internal flows. This course is inspired by 16.540 Internal Flow from the Department of Aeronautics and Astronautics at MIT. Much of the material is based on the lectures of Profesor E. M. Greitzer, with permission.	
Study Goals	1) Development of physical insight into the phenomena, which characterize internal flow in fluid machinery (not just what happened, but why it happened) 2) Development of the critical thinking needed to define, in a rigorous manner, the levels of modeling needed for useful descriptions of internal flow situations 3) Development of the ability to interpret numerical simulations and experimental results in terms of concepts and first principles	
Education Method	This course will use innovative didactic methods that emphasize active learning. Students will be engaged through active feedback systems and online material. Given that a textbook exists for the course material, it is the intent of the lectures to discuss the material rather than to present it for the first time in class. As such the students will be expected to read the material before class and to engage in in-class discussions of the reading material by answering questions and elaborating concepts. Here we list a high-level description of the course contents. More detailed information are available in the complete syllabus on Brightspace Lecture 1 Structure and Content of the Course, Introduction to Flow Regimes Lectures 1-2 Useful Basic Ideas Pressure fields and streamline curvature Upstream influence in turbomachines Control volume description of propulsion systems Asymmetry of real fluid motions Behavior of an inlet and turbomachinery system Lectures 3-6 Vorticity and Circulation Kinematics of vorticity Dynamics of vorticity Behavior of vortex lines Rotational flow description in terms of vorticity and virculation Mixing enhancement due to streamwise vorticity Lectures 7-9 Flow in Rotating Passages Non-dimensional parameters in a rotating flow Conserved quantities in a steady rotating flow Phenomena in flows where rotation dominates Rotating Channel Flow in Constant Area Straight Passages Boundary layers in rotating straight channels Rotating Flow in Turbomachinery Passages Secondary flow in rotating blade rows Lectures 10-12 Loss Sources and Loss Accounting Appropriate metrics for loss Boundary layer losses Mixing Losses Averaging of a mon-uniform flow Effect of pressure level on average properties and mixing losses Loss characterization in turbomachinery cascades Lectures 13-14 Compressible Flows in fluid machinery Corrected mass per unit area Heat and friction in compressible flows Influence coefficients Compressible fluids in fluid devices	
Literature and Study Materials	Internal Flow: Concept and Applications, E. M. Greitzer, C. S. Tan, M. B. Graf, Cambridge University Press http://www.cambridge.org/gb/academic/subjects/engineering/thermal-fluids-engineering/internal-flow-concepts-and-applications?format=PB#MciShGFvqgyx6G4p.97	
Assessment	The student will be asked to develop concept questions and corresponding answers which help to illustrate the physical principles and approximations that are made. This will be done in small (six or seven person) teams so you can discuss the questions among yourselves. One of these questions is to be developed each week by each team. The concept questions will count 20% of the final grade. Approximately five short (~20 minute) "concept quizzes" will be given during class, roughly equally spaced throughout the term. These may make use of the questions generated by students. The total of these quizzes will count for 30% of the final course	

grade.

A written and oral final exam will be held. The final exam will count for 50% of the final course grade.

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	A. Gangoli Rao	
Responsible Instructor	Dr. I. Langella	
Instructor	Prof.dr. D.J.E.M. Roekaerts	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Radiation and transport processes 4. Premixed and diffusion flames 5. Turbulent reactive flows 6. Combustion experiments and diagnostics 7. Laboratory Tutorial & Exercise 8. Practical combustion systems for aerospace applications	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation- and adiabatic flame temperature. Apply principles of thermodynamics to analyze reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames. Derive equations related to turbulent reacting flows and chemistry interaction. Explain laser diagnostics and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions. In addition, the student is required to complete project assignments and perform a laboratory exercise. The project assignments and associated numerical problems are relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with laminar flame speed measurements of the simplest hydrocarbon fuel, methane. The student will gain hands-on experience in assembling the experimental setup, acquiring data, performing data analysis and writing a technical report.	
Course Relations	Aero Engine Technology	
Literature and Study Materials	Slides and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the student must pass the oral examination, the evaluation of the laboratory exercise, and successfully complete the assignments given during the course. The work on the laboratory exercise and assignments should be summarized in reports, to be written by a single student or in a group of two. The grade is dictated by the result of the oral examination (50%) and the grades obtained in the assignments and lab exercise (50%).	

AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
Responsible Instructor	Prof.dr.ir. P. Colonna di Paliano	
Instructor	C.M. De Servi	
Instructor	Dr.ir. W.P.J. Visser	
Instructor	Prof. F. Casella	
Contact Hours / Week x/x/x/x	0/x/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE1240-I Thermodynamics AE2230-II Propulsion and Power AE1205 Programming and Scientific Computing in Python for Aerospace AE4238 Aero Engine Technology ME45000 Advanced Heat Transfer	
Course Contents	2nd EDUCATION PERIOD Part 1 (2 ECTS) Module 1 Introduction, Context, Foundations Module 2 Conservation equations Module 3 Modeling paradigms Module 4 Numerical methods and software Module 5 Modelica Module 6 Constitutive equations Module 7 Components and system modeling Module 8 Verification and validation 3rd EDUCATION PERIOD Part 1 (1 ECTS) Module 9 Model-based control Take Home exam Part 2 (2 ECTS) Module 10 - Team Project to be chosen among a Aero engine with GSP or GTPsim b Power or propulsion system with Modelica	
Study Goals	After learning the content of the course, the student will be able to: Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results. In particular, this overarching objective can be obtained by developing the following theoretical capabilities: 1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose. 2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.). 3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model. 4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables). 5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation) 6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.	
Education Method	Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents) Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software. Students are strongly advised to make use of the BrightSpace Forum (Q&A) to post questions and interact with instructors and other students. In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.	
Computer Use	Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim. More information on these programs is provided on BrightSpace: Course Information -> Resources and Tools -> Open Modelica and GSP	
Literature and Study Materials	Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules	
Books	Reference text for thermodynamics: W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018. Reference text for numerical methods:	

Reader
Assessment

P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.

The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's).

Part 1: Exercises

The first module is tested with a Quiz.

The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded.

Part 1: Written exam

Take home exam is graded and covers the material of the 9 Modules of Part 1 of the course. Duration: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace.

Part 2: Simulation project

Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola

Graded deliverables: technical report, model files.

A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam.

Grading

the final grade is calculated as

Final Grade = $0.2 \times \text{Exercises Grade} + 0.4 \times \text{Take Home Exam Part 1 Grade} + 0.4 \times (\text{Simulation Project} + \text{Oral Exam Grade})$.

A final grade can be obtained only if the grade of the Take Home Exam Part 1 and of Part 2 are sufficient (≥ 6).

The Part 2 oral exam can only take place after a sufficient grade (≥ 6) on Part 1 has been obtained.

Resit exams

One opportunity to resit the Take Home Exam Part 1 is provided during the second quarter the course is running in.

The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory a. Laplace transformation b. Elementary closed loop systems c. Transfer functions 2. Poles and zeros a. First order systems b. Second order systems c. Pole placement for simple systems 3. Root locus method a. Characteristic equation b. Angle and magnitude conditions 4. State space formulation a. Controllability, observability b. Ackerman's formula c. LQR 5. Basic controllers: P,PI,PD,PID 6. Frequency response (Bode plots) 7. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 8. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 9. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 10. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 11. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 12. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 13. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance - substantiate the function of a Flight Control System(FCS) in civil aviation.	

	- apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 AE3212-I AE2235-I	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace	
Assessment	Exercise	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hover performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aerospace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course. One final take-home assignment at the end of the course.	
Special Information	A helicopter flight in SIMONA simulator will be given to the students following this course. Note: This may differ due to COVID-19 measures. Changes to this will be communicated via Brightspace.	
Set-up	Online education and Classical lecturing	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Self-supervised learning - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4431	Aircraft noise and emissions	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Responsible Instructor	Prof.dr. D.G. Simons	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The aircraft noise part of this course is required for AE4463P 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics - Dutch aircraft noise model NRM - acoustic imaging Education method: The practicing is in the form of concrete assignments, which are individually graded.	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and terts band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. (Acoustic imaging: Principle of beamforming. Imaging aircraft noise data.) Aircraft noise sources: Propeller noise (mechanism and directional properties) and blade passage frequency. Turbo engine noise and directionality of the fan and exhaust jet noise source. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise (and the modelling of it). (Aero-acoustics (Lighthill theory).) The topics in parentheses (acoustic imaging, airframe noise modelling and aero-acoustics) are part of the reader, but do not belong to this course and will be treated in the follow-up course AE4463P. Aircraft emissions and climate effects Aircraft emissions: Gas emissions from aircraft engines. Complete and actual combustion. Combustion products from burning kerosene. Emission Index. Carbon dioxide emissions of transport (aviation and, for comparison, road traffic). Atmospheric radiation physics: Concept of radiation flux. Solar spectrum and solar constant. Radiative properties of natural surfaces. Thermal emission. Black body radiation and the laws of Planck, Wien, Stefan-Boltzmann and Kirchoff. Atmospheric transmission and emission. Effects of ozone. Climate effects: A climate model for the Earth. Greenhouse effect. Radiative forcing. Example calculations of global warming due to various carbon emission scenarios. Contribution of aviation. Feedback in the climate system.	
Study Goals	Understand the relationship between aviation and the environment.	
Education Method	Lectures and exercises.	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions Readers:	

	Course AE4431 Aircraft Noise and Emissions Part A: Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022 Course AE4431 Aircraft Noise and Emissions Part B: Introduction to aircraft emissions and climate effects Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022
Assessment	Written exam (students may bring their own formula sheet to the exam)
Set-up	Lectures and exercises

AE4462-17	Aircraft Emissions and Climate Effects	4
Responsible Instructor	Dr. V. Grewe	
Instructor	Dr. I.C. Dedoussi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics. Aircraft noise and emissions (AE4431) highly recommended.	
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture slides and hand-outs	
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	AE4260 B (Experimental Applications of Aeroacoustics)	
Expected prior knowledge	AE4431 (Aircraft noise and emissions)	
Course Contents	AE4463P aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) will be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B and F are required. A choice can be made between C, D and E.	
Study Goals	The course aims at providing the students with an thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to:	
	1. Explain the different steps in a finite element analysis and apply them to practical engineering problems	
	2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function)	
	3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up	
	4. Use and interpret results from 1D and 2D elements in commercial finite element software	
	5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace.	
	Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments.	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	- Classical Lamination Theory - Short Overview of Materials. Composites Design Philosophy - Theory of Elasticity - Engineering Constants - Stress & Strain Transformations / Implications for Testing - Thin laminates / Carpet Plots - Thick Laminates / ABD Matrix / Coupling Effects - Hydrothermal Analysis - Progressive damage analysis - Failure criteria - First ply failure & Last ply failure - Damage tolerance analysis - In-situ strength - Experimental mechanics & in-situ monitoring techniques - Basic Stress Solutions and Buckling of Composite Plates - Typical Airframe Elements - Airframe Design Process, Materials & Damage - Plate Governing Equations / Solution of the PDE - Energy Minimization Methods - Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: - Theory of Elasticity: 2 - Classical Laminar Theory: 4 - Progressive damage analysis: 4 - Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Ir. J. Sinke	
Instructor	Dr. J.A. Poulis	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December 2022 - Week 4.8: 15 May 2023 - Week 1.6: 30 September 2022 The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and	

Set-up	3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3
	There is an intentional overlap so that continuity and handover is guaranteed

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Bachelor program, preferably electrical engineering	
Course Contents	The course gives an overview of different types of electrical machines and drives. Different types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	Written exam (100% Weightage), Closed book, Scientific (non-graphic) calculator can be used, No formula sheets Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	In order to participate in the written exam, students need to complete the three half day lab practicals. During corona period, three remote practicals are offered that are done individually. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments, computer demonstrations as well as instruction. In 2021, all lectures will be online. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction/office hours, during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. Weekly instruction is likely to be on campus.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45025	Introduction to Multiphase Flow	5
Responsible Instructor	Dr.ir. W.P. Breugem	
Instructor	Prof.dr.ir. R.A.W.M. Henkes	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are gas and oil transport, dredging, steel making, refineries and chemical plants, sediment transport in rivers, cloud formation, etc. This course gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of gas/liquid two-phase flows, spatial/statistical averaging of the Navier-Stokes equations and continuum modelling, flow pattern maps and mechanistic modelling of multiphase pipe flows, physics of particles (droplets, bubbles, solids), dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.	
Study Goals	At the end of this course the student is able to: 1.a classify multiphase flows into separated and dispersed multiphase flows. 1.b formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface. 1.c formulate relevant dimensionless numbers for multiphase flows. 1.d explain the concept of dynamic similarity of multiphase flows. 1.e reflect on scaling choices when designing a laboratory experiment or setting up a numerical simulation. 2.a apply a linear stability analysis to two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability). 2.b give physical mechanisms for the transition of stratified to non-stratified (bubble or slug) multiphase flows in both (tilted) horizontal and vertical pipes. 3.a apply averaging methods (such as statistical, spatial and temporal averaging methods) to the Navier-Stokes equations for describing the macroscopic behavior of multiphase flows. 3.b formulate appropriate closure models needed to solve the averaged Navier-Stokes equations. 4.a classify gas/liquid pipe flows into stratified, slug and bubble flows, dependent on the superficial gas and liquid velocities. 4.b apply simplified 1D (cross-section averaged) models to each of the flow regimes. 4.c formulate appropriate closure models in order to compute the streamwise pressure drop and liquid holdup in two-phase (gas/liquid) pipe flow. 5.a classify dispersed multiphase flows into flows with solid particles, gas bubbles and/or liquid droplets. 5.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets. 5.c apply the Newton-Euler equations of motion to an analysis of the dynamics of solid particles. 5.d describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio. 5.e describe possible interactions between solid particles and turbulence of the carrier phase. 6.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Haberman-Morton number. 6.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag. 6.c give physical mechanisms for the breakup of bubbles/droplets in turbulence. 7. describe the dynamics of some particle-laden flows such as sediment transport through a pipe and/or sediment transport in rivers. 8. describe the merits and shortcomings of different experimental methods for multiphase flows. 9. describe the merits and shortcomings of different numerical methods for simulating multiphase flows. 10.a give examples of multiphase flow in industrial processes. 10.b describe the types of models used to design multiphase industrial processes.	
Education Method	Oral lecture 0/0/2/2; Plenary discussion of homework assignments 0/0/2/2	
Assessment	Written exam. The student can earn bonus points by making homework assignments. The final grade is the sum of the grade for the written exam and the bonus points from the homework.	
Department	3mE Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	At the end of this course you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	Brightspace/MATLAB exercises and tests (optional) during education period. The course is concluded through an optimization project and report. Deadlines for handing in the final report will be announced at the start of the course.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Propulsion and Power

Program Coordinator	A. Gangoli Rao
Program Title	Master of Science in Propulsion and Power
Director of Education	Ir. Joris Melkert
Contact for Students	If you have specific question about the profile, you can send your queries to Prof. Arvind Gangoli Rao (A.Gangoirao@tudelft.nl)
Introduction 1	Education. The MSc profile of propulsion and power is a part of the MSc track Flight Performance and Propulsion. In this profile the students will be able to specialize in courses related to aircraft propulsion. Several mandatory profile courses form the core of the knowledge that specializes our graduate students, covering relevant aspects in turbomachinery, combustion, propulsion system, component design and their integration. Elective courses are chosen in consultation with the Profile Coordinator, and these are grouped in Methods, Advanced Topics, and Aircraft Design, so that the student can choose one of these aspects. The competence acquired during the first year of the Master program is put to fruition in the final year during the research period dedicated to the thesis during which the student is engaged in a project. The project is carried out under the supervision of one of the academic staff members, but often the student also benefits from the closer supervision of one of the PhD candidates involved in the same project. Below, an overview of the research activities is provided. Research: In our stride to make aviation sustainable, Aircraft propulsion is the best way to reduce fuel consumption and emissions. The use of alternative fuels and their usage in an aircraft is determined by the propulsion system. The propulsion system account for more than 90% of the life-cycle emissions and around 30% of the life-cycle cost. Education in topics related to propulsion is at the core of all aerospace Master programs in the world, and TU Delft is no exception. The courses of our profile create the knowledge needed to effectively perform thesis work within one of the running research programs. They provide an exciting opportunity to learn about cutting-edge technologies that will dramatically improve the efficiency and sustainability of aviation. NextGen Aircraft Propulsion Systems: We are currently pursuing the study of three disruptive concepts with the potential to leapfrog propulsion systems to a much higher efficiency level and much lower emissions. For example, new engine architectures like hybrid aircraft engines, multi-fuel engines, electrically assisted engines, etc. Novel Fuels: One of the enablers for sustainable aviation is alternative fuels like biofuels, hydrogen, synthetic fuels, etc. The Hybrid Engine proposed in the AHEAD project is a novel propulsion system with a different architecture as compared to the conventional turbofan engine. The hybrid engine concept is based on several unique technologies like shrouded contra-rotating fans, bleed cooling, and a dual hybrid combustion system. Such system uses cryogenic fuel (like LH2 or LNG), and biofuel under flameless conditions to reduce CO2 and NOx emission respectively. These technologies allow for a leap forward in terms of environmental friendliness, thanks to the adoption of sustainable fuels. Investigations are theoretical, numerical and also experimental, thanks to the newly established Clean Combustion Lab. In the Clean Combustion lab, we are investigating low NOx combustion technologies for novel energy carriers like Hydrogen, biofuels, methanol, or a mixture of different fuels. Waste Heat Recovery: we were possibly the first to conceive the Combined-Cycle Engine (CC Engine) configuration from the idea that even in modern and future gas turbine engines, approximately half of the energy of the fuel is dissipated into the atmosphere as thermal energy. In the CC-Engine, the exhaust from a gas turbine provides thermal energy to closed-cycle supercritical CO2 turbine, thereby increasing the efficiency. Combustion Research: We carry out fundamental investigations on low emission combustion techniques (such as flameless combustion) using conventional and alternative fuels. The clean combustion lab is equipped with latest diagnostics system that provides insights in combustion physics.

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and (to limited extend) viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Typically Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during at the beginning of Q1).	
Assessment	The assessment is done through 5 personal assignments.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written (proctored online) exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr. F. Oliviero	
Instructor	Dr.ir. M.F.M. Hoogreef	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszczanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4206	Turbomachinery	3
Responsible Instructor	Ir. M. Pini	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course covers fundamental concepts of internal flows in turbomachines and related design aspects. The usual course schedule consists of the following lectures: Lecture 0: Introduction course overview: topics, structure and assessment Lecture 1: Recall of Turbomachinery Fundamentals & Principles integral governing equations with applications, relative motion, thermodynamic transformations in turbomachinery stages, velocity triangles with practical examples; Lecture 2: Unsteady Flows in Turbomachinery concept of reduced frequency with applications; work exchange in turbomachinery rows; Lecture 3: Similarity in Turbomachinery similarity analysis for incompressible flows, Balje and Smith diagrams, accounting for compressibility, stage dimensionless performance parameters; Lecture 4: Fluid-dynamic Design of Turbomachinery design procedure, mean-line design model, radial equilibrium model, CFD-based models Lecture 5-6: Axial and Radial Turbines examples of axial and radial turbines, typical turbine stages, blade profiles for turbines, flow characteristics in subsonic, transonic, and supersonic cascades, basic concepts of diffusers, aerodynamics of radial impellers Lecture 7-8: Axial and Radial Compressors examples of axial and radial compressors, typical compressor stages, blade profiles for compressors, flow characteristics and unique incidence, stall and surge instability, slip factor, diffusers for radial compressors, design maps Seminar(s): topic(s) may vary but usually include turbomachinery aero-mechanics, shape optimisation with CFD methods, rocket engine turbo-pumps, and advanced turbomachinery components for aero-engines Tutorial: training session on the software packages used for the group assignment	
Course Contents	This course educates aerospace students to apply theoretical knowledge in the field of fluid- mechanics and thermodynamics for analyzing and designing a thermal turbomachinery. Typical examples are: axial and radial compressors of gas turbine engines, industrial gas and steam turbines, radial turbines of automotive turbochargers, turbo-expanders and compressors used in refrigeration and liquefaction industry, micro-turbines for power generation. Since the turbomachinery field is intrinsically interdisciplinary, it is assumed that students already possess basic knowledge of thermodynamics, fluid-mechanics, applied mathematics, propulsion systems, internal flows, acquired in previous courses. Prerequisite of the course are therefore bachelor and master courses addressing fundamental (e.g. Thermodynamics, Internal Flows) and applied aspects (e.g. Aero-Engines) of air-breathing engines. Starting from basic concepts of gas-dynamics and fluid-machinery, the course brings the student to the level that he is able to select a turbomachinery for a specific application, perform the conceptual design of the stages, and eventually predict the flow characteristics within the blade passages by means of conceptual physical models and more sophisticated simulation tools.	
Study Goals	After the course the student will be able to: - Classify and illustrate the main features of a turbomachine - Select the most appropriate turbomachine for a given aerospace application - Apply first principles to simplified turbomachinery configurations - Identify the most prominent loss sources of a given turbomachinery stage - Define the aerodynamic blade design of turbomachinery cascades - Predict the performance of a turbomachine with analytical and numerical methods - Devise solutions for performance improvement of turbomachinery cascades based on physical understanding - Perform the conceptual and detailed fluid-dynamic design of a turbomachine for propulsion systems using analytical and numerical tools	
Education Method	Lectures, concept quizzes, concept exercises, and tutorials in preparation to the team assignment.	
Books	- S.L. Dixon, C. A. Hall Fluid-Mechanics and Thermodynamics of Turbomachinery, 7th edition, 2013 - M. Schobeiri, Turbomachinery Flow Physics and Dynamic Performance, 2nd edition, Springer, 2012 - E.A. Baskharone, "Principles of Turbomachinery in Air-Breathing Engine", Cambridge University Press	
Assessment	The final course grade results from the weighted average of: - the assessment of the final project assignment, counting for 40% of the final grade (individual score 1-10, minimum threshold: 6); - the assessment of the concept exercises, counting for 10% of the final grade. No minimum threshold; - the assessment of the written exam, counting for 50% of the final grade (individual score 1-10, minimum threshold: 6); The written exam takes place at the end of course. The exam consists of one or two concept exercises covering the material discussed during classes. The final assignment is meant to challenge students with the real engineering problem of designing a turbomachine for propulsion or power generation application. The concept exercises are numerical exercises that are formulated to stimulate critical thinking on turbomachinery problems of practical application. They are also meant as diagnostic tool to help students self-assessing their level of understanding.	
Permitted Materials during Tests	Lectures notes, slides, formula sheets and textbooks are allowed during the written exam. Calculator can be used.	
Set-up	The course is organized in 8 teaching sessions, 2/3 seminar(s), and a possible visit to a turbomachinery company. Lectures usually consist of a theoretical part integrated with in-class exercises and quizzes to stimulate critical thinking and learning-by-doing. During classes, the students, autonomously or in groups, are engaged in a series of learning activities, supervised by the team of instructors. Their solution is briefly discussed at the beginning of the next class. The seminar(s) are devoted to show students current research topics in turbomachinery and are given by recognised experts in the field.	

AE4238	Aero Engine Technology	4
Responsible Instructor	A. Gangoli Rao	
Instructor	Ir. M. Pini	
Instructor	F. Yin	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery"; "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Turbomachinery - o New Engine /propulsion Concepts - o Turbine Cooling - o Inlets and Exhaust - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an hybrid course in which the lectures are pre-recorded and some for tutorials will be solved in the classroom. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13 Analyse the cooling requirements and compute the effect of cooling on engine performance 14. Evaluate the calorific values for various fuels 15. Understand the basics of combustion and emission formation in Aero engines 16. Understand the working of gas turbine combustor 17. Understand the basic principles of gas turbine controls and evaluate different strategies for control 18. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a hybrid teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen
	2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Reader	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on several homework assignments (5-6) given during the course and a final exam. All assignments together will form 50% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4261	Internal Flows	3
Responsible Instructor	Ir. M. Pini	
Instructor	C.M. De Servi	
Responsible for assignments	T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course we address fluid dynamic phenomena of interest in internal flow situations for aerospace propulsion applications. The goal of the course is to develop physical insight into the phenomena that characterize internal flow in fluid machinery. As such we will discuss not just what happens, but why it happens and what are the consequences for propulsion systems. The course covers topic, which are not dealt with in courses or texts about external fluid dynamics. The flows described in Internal Flows are generally rotational, often three-dimensional, unsteady, and sometimes occurring in non-inertial (e.g., rotating) coordinate systems. The course can be viewed as the development of flow models and ideas to enable allow physical insight into the behavior of three-dimensional and unsteady flows. One of the objectives of the course is to provide an increased ability to interpret computational and experimental results and hence to effectively extract conclusions about the key features of complex internal flows.	
Study Goals	This course is inspired by 16.540 Internal Flow from the Department of Aeronautics and Astronautics at MIT. Much of the material is based on the lectures of Profesor E. M. Greitzer, with permission. 1) Development of physical insight into the phenomena, which characterize internal flow in fluid machinery (not just what happened, but why it happened) 2) Development of the critical thinking needed to define, in a rigorous manner, the levels of modeling needed for useful descriptions of internal flow situations 3) Development of the ability to interpret numerical simulations and experimental results in terms of concepts and first principles	
Education Method	This course will use innovative didactic methods that emphasize active learning. Students will be engaged through active feedback systems and online material. Given that a textbook exists for the course material, it is the intent of the lectures to discuss the material rather than to present it for the first time in class. As such the students will be expected to read the material before class and to engage in in-class discussions of the reading material by answering questions and elaborating concepts.	
	Here we list a high-level description of the course contents. More detailed information are available in the complete syllabus on Brightspace	
	Lecture 1 Structure and Content of the Course, Introduction to Flow Regimes	
	Lectures 1-2 Useful Basic Ideas	
	Pressure fields and streamline curvature Upstream influence in turbomachines Control volume description of propulsion systems Asymmetry of real fluid motions Behavior of an inlet and turbomachinery system	
	Lectures 3-6 Vorticity and Circulation	
	Kinematics of vorticity Dynamics of vorticity Behavior of vortex lines Rotational flow description in terms of vorticity and virculation Mixing enhancement due to streamwise vorticity	
	Lectures 7-9 Flow in Rotating Passages	
	Non-dimensional parameters in a rotating flow Conserved quantities in a steady rotating flow Phenomena in flows where rotation dominates Rotating Channel Flow in Constant Area Straight Passages Boundary layers in rotating straight channels Rotating Flow in Turbomachinery Passages Secondary flow in rotating blade rows	
	Lectures 10-12 Loss Sources and Loss Accounting	
	Appropriate metrics for loss Boundary layer losses Mixing Losses Averaging of a mon-uniform flow Effect of pressure level on average properties and mixing losses Loss characterization in turbomachinery cascades	
Literature and Study Materials	Internal Flow: Concept and Applications, E. M. Greitzer, C. S. Tan, M. B. Graf, Cambridge University Press http://www.cambridge.org/gb/academic/subjects/engineering/thermal-fluids-engineering/internal-flow-concepts-and-applications?format=PB#MciShGFvqgyx6G4p.97	
Assessment	The student will be asked to develop concept questions and corresponding answers which help to illustrate the physical principles and approximations that are made. This will be done in small (six or seven person) teams so you can discuss the questions among yourselves. One of these questions is to be developed each week by each team. The concept questions will count 20% of the final grade. Approximately five short (~20 minute) "concept quizzes" will be given during class, roughly equally spaced throughout the term. These may make use of the questions generated by students. The total of these quizzes will count for 30% of the final course	

grade.

A written and oral final exam will be held. The final exam will count for 50% of the final course grade.

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	A. Gangoli Rao	
Responsible Instructor	Dr. I. Langella	
Instructor	Prof.dr. D.J.E.M. Roekaerts	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Radiation and transport processes 4. Premixed and diffusion flames 5. Turbulent reactive flows 6. Combustion experiments and diagnostics 7. Laboratory Tutorial & Exercise 8. Practical combustion systems for aerospace applications	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation- and adiabatic flame temperature. Apply principles of thermodynamics to analyze reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames. Derive equations related to turbulent reacting flows and chemistry interaction. Explain laser diagnostics and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions. In addition, the student is required to complete project assignments and perform a laboratory exercise. The project assignments and associated numerical problems are relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with laminar flame speed measurements of the simplest hydrocarbon fuel, methane. The student will gain hands-on experience in assembling the experimental setup, acquiring data, performing data analysis and writing a technical report.	
Course Relations	Aero Engine Technology	
Literature and Study Materials	Slides and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the student must pass the oral examination, the evaluation of the laboratory exercise, and successfully complete the assignments given during the course. The work on the laboratory exercise and assignments should be summarized in reports, to be written by a single student or in a group of two. The grade is dictated by the result of the oral examination (50%) and the grades obtained in the assignments and lab exercise (50%).	

AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
Responsible Instructor	Prof.dr.ir. P. Colonna di Paliano	
Instructor	C.M. De Servi	
Instructor	Dr.ir. W.P.J. Visser	
Instructor	Prof. F. Casella	
Contact Hours / Week x/x/x/x	0/x/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE1240-I Thermodynamics AE2230-II Propulsion and Power AE1205 Programming and Scientific Computing in Python for Aerospace AE4238 Aero Engine Technology ME45000 Advanced Heat Transfer	
Course Contents	2nd EDUCATION PERIOD Part 1 (2 ECTS) Module 1 Introduction, Context, Foundations Module 2 Conservation equations Module 3 Modeling paradigms Module 4 Numerical methods and software Module 5 Modelica Module 6 Constitutive equations Module 7 Components and system modeling Module 8 Verification and validation 3rd EDUCATION PERIOD Part 1 (1 ECTS) Module 9 Model-based control Take Home exam Part 2 (2 ECTS) Module 10 - Team Project to be chosen among a Aero engine with GSP or GTPsim b Power or propulsion system with Modelica	
Study Goals	After learning the content of the course, the student will be able to: Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results. In particular, this overarching objective can be obtained by developing the following theoretical capabilities: 1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose. 2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.). 3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model. 4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables). 5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation) 6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.	
Education Method	Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents) Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software. Students are strongly advised to make use of the BrightSpace Forum (Q&A) to post questions and interact with instructors and other students. In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.	
Computer Use	Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim. More information on these programs is provided on BrightSpace: Course Information -> Resources and Tools -> Open Modelica and GSP	
Literature and Study Materials	Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules	
Books	Reference text for thermodynamics: W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018. Reference text for numerical methods:	

Reader
Assessment

P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.

The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's).

Part 1: Exercises

The first module is tested with a Quiz.

The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded.

Part 1: Written exam

Take home exam is graded and covers the material of the 9 Modules of Part 1 of the course. Duration: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace.

Part 2: Simulation project

Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola

Graded deliverables: technical report, model files.

A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam.

Grading

the final grade is calculated as

Final Grade = $0.2 \times \text{Exercises Grade} + 0.4 \times \text{Take Home Exam Part 1 Grade} + 0.4 \times (\text{Simulation Project} + \text{Oral Exam Grade})$.

A final grade can be obtained only if the grade of the Take Home Exam Part 1 and of Part 2 are sufficient (≥ 6).

The Part 2 oral exam can only take place after a sufficient grade (≥ 6) on Part 1 has been obtained.

Resit exams

One opportunity to resit the Take Home Exam Part 1 is provided during the second quarter the course is running in.

The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. I.A. Houthuijsen	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Learning objectives are: 1. Demonstrate engineering knowledge and skills: technical skills, interpreting results, creativity, usability for company/research institute; 2. Familiarize and apply with analytical academic approach: experimental work/computer skills/design skills, extending knowledge/methods, argumentation, ambition to learn; 3. Show ability to write a technical report: which is structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (more detailed information can be found in the guidelines on writing the internship report on Brightspace AE5051); 4. Demonstrate behavioural competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to adapt to different cultures, a thorough self-reflection is given in the reflection chapter of the report; 5. Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx. 40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Coordinator. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in	

online.
Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included.

The internship report is assessed at three criteria:

1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);
2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can be forwarded to an assessor. It is then assessed and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is forwarded to the assessor by the Internship Office. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. Perfect for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com). Please visit the preparation tab for a manual on OnStage.

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same subject, and require different deliverables. Students need to finish their internship before starting their thesis. If this is applicable students must describe the two tasks (thesis and internship) in detail in the approval form.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5222	Thesis Flight Performance & Propulsion	45
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Instructor	Dr.ir. G. la Rocca	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. http://studenten.tudelft.nl/fileadmin/Files/studentenportal/os/LRspecifiek/docs/Rubrics_MSc-LR_version_14-Oct-13_met_naam_en_studienummer.pdf.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments, computer demonstrations as well as instruction. In 2021, all lectures will be online. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction/office hours, during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. Weekly instruction is likely to be on campus.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Propulsion and Power

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-construct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4115	Experimental Simulations	3
Responsible Instructor	T. Sinnige	
Instructor	Prof.dr.ing. G. Eitelberg	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling. Fractional analysis, symmetry and similarity, application to specific problems from aerodynamics, BL-approximations, S/BL interactions; 3 Dimensional analysis: Buckingham's Pi-theorem, rigorous methods of obtaining dimensionless parameters, simplifications of Navier-Stokes equations; 4 Subsonic and supersonic wind tunnels: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 5 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 6 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 7 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 8 Design of experiments: factorial designs, response-surface models; 9 Reynolds-number effects and simulation; 10 Pre-test support session; 11 Engine integration: engine simulators, performance of those. Thrust reverser testing for re-ingestion and stability and control; 12 Acoustic testing. Noise generation mechanisms for submerged bodies, edges and jets. Appropriate scaling. Definition of sound pressure levels. Measurement requirements and devices. Far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session. (Excursion) Ground test facilities of DNW (organized by AWEP society, limited places available).	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports and oral exam.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter. The lectures are supported by an excursion to the testing facilities of DNW, organized by the AWEP society.	

AE4120	Viscous Flows	3
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Responsible Instructor	M. Kotsonis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4136-22	CFD 2: Discretization techniques	3
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results.	
Study Goals continuation	After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignment and oral examination.	
Enrolment / Application	Brightspace	
Special Information	M.I.Gerritma@tudelft.nl	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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6	Variational inverses	[Inversion]	Finalize []	--																																						
7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Dr. S.J. Hulshoff	
Responsible Instructor	Prof.dr.ir. S. Hickel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	Prof.dr. F. Scarano	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszczanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4rt quarter of the academic year) and not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Remarks	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015 Optional textbooks: - Airplane Design, Prt I-VII, Roskam, J, DARCorp, 1990-2003 - Synthesis of Subsonic Airplane Design, Torenbeek, E, Kluwer Academic Publishers, 1982 - Fluid Dynamic Drag, Hoerner, S. F., Hoerner Fluid Dynamics, 1965 - Fluid Dynamic Lift, Hoerner, S. F. Hoerner Fluid Dynamics, 1965	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas:	
Study Goals	1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources. The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resourcen/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Self-supervised learning - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4431	Aircraft noise and emissions	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Responsible Instructor	Prof.dr. D.G. Simons	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The aircraft noise part of this course is required for AE4463P 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics - Dutch aircraft noise model NRM - acoustic imaging Education method: The practicing is in the form of concrete assignments, which are individually graded.	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and tert's band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phono and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. (Acoustic imaging: Principle of beamforming. Imaging aircraft noise data.) Aircraft noise sources: Propeller noise (mechanism and directional properties) and blade passage frequency. Turbo engine noise and directionality of the fan and exhaust jet noise source. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise (and the modelling of it). (Aero-acoustics (Lighthill theory).) The topics in parentheses (acoustic imaging, airframe noise modelling and aero-acoustics) are part of the reader, but do not belong to this course and will be treated in the follow-up course AE4463P. Aircraft emissions and climate effects Aircraft emissions: Gas emissions from aircraft engines. Complete and actual combustion. Combustion products from burning kerosene. Emission Index. Carbon dioxide emissions of transport (aviation and, for comparison, road traffic). Atmospheric radiation physics: Concept of radiation flux. Solar spectrum and solar constant. Radiative properties of natural surfaces. Thermal emission. Black body radiation and the laws of Planck, Wien, Stefan-Boltzmann and Kirchoff. Atmospheric transmission and emission. Effects of ozone. Climate effects: A climate model for the Earth. Greenhouse effect. Radiative forcing. Example calculations of global warming due to various carbon emission scenarios. Contribution of aviation. Feedback in the climate system.	
Study Goals	Understand the relationship between aviation and the environment.	
Education Method	Lectures and exercises.	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions Readers:	

	Course AE4431 Aircraft Noise and Emissions Part A: Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022 Course AE4431 Aircraft Noise and Emissions Part B: Introduction to aircraft emissions and climate effects Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022
Assessment	Written exam (students may bring their own formula sheet to the exam)
Set-up	Lectures and exercises

AE4462-17	Aircraft Emissions and Climate Effects	4
Responsible Instructor	Dr. V. Grewe	
Instructor	Dr. I.C. Dedoussi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics. Aircraft noise and emissions (AE4431) highly recommended.	
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture slides and hand-outs	
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	AE4260 B (Experimental Applications of Aeroacoustics)	
Expected prior knowledge	AE4431 (Aircraft noise and emissions)	
Course Contents	AE4463P aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) will be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B and F are required. A choice can be made between C, D and E.	
Study Goals	The course aims at providing the students with an thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4ASM003		Linear Modeling (incl. F.E.M)	3
Responsible Instructor	D.M.J. Peeters		
Contact Hours / Week x/x/x/x	4/0/0/0		
Education Period	1		
Start Education	1		
Exam Period	none		
Course Language	English		
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills		
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.		
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles		
Education Method	Weekly lecture and practical		
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments.		

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	- Classical Lamination Theory - Short Overview of Materials. Composites Design Philosophy - Theory of Elasticity - Engineering Constants - Stress & Strain Transformations / Implications for Testing - Thin laminates / Carpet Plots - Thick Laminates / ABD Matrix / Coupling Effects - Hydrothermal Analysis - Progressive damage analysis - Failure criteria - First ply failure & Last ply failure - Damage tolerance analysis - In-situ strength - Experimental mechanics & in-situ monitoring techniques - Basic Stress Solutions and Buckling of Composite Plates - Typical Airframe Elements - Airframe Design Process, Materials & Damage - Plate Governing Equations / Solution of the PDE - Energy Minimization Methods - Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: - Theory of Elasticity: 2 - Classical Laminar Theory: 4 - Progressive damage analysis: 4 - Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Ir. J. Sinke	
Instructor	Dr. J.A. Poulis	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December 2022 - Week 4.8: 15 May 2023 - Week 1.6: 30 September 2022 The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and	

Set-up	3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

AE4S01	Thermal Rocket Propulsion	4
Responsible Instructor	Ir. B.T.C. Zandbergen	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Propulsion material contained in the BSc courses AE1222-II and AE2230-I. Course material will be made available via Brightspace.	
Course Contents	The course focuses on thermo-(chemical) rocket propulsion system analysis and design. Topics dealt with include: 1. Fundamentals of (thermo-chemical) rocket propulsion; 2. Ideal rocket motor/nozzle: Ideal performances, optimum thrust, characteristic velocity and thrust coefficient, and quality factors; 3. Nozzles: Types of nozzles (conical, bell, etc.), nozzle dimensions, flow divergence, boundary layers, under- and over-expansion, and Summerfield criterion; 4. Chemical propellants: Molar mass, specific heat ratio and adiabatic flame temperature calculation for gas mixtures (based on known reaction equation), mass density, dynamic viscosity, thermal conductivity; 5. Chemical equilibrium calculations, hemical equilibrium flow, frozen flow and chemical kinetics; Introduction to program for calculation of chemical equilibrium gas composition and gas properties (in tutorial); 6. Heat transfer & cooling: Convection, radiation and conduction; 7. Cooling: Thermal insulation, ablation, radiation, film, dump and regenerative cooling; 8. Liquid rocket engine combustor design: Steady state internal ballistics, liquid injection, operating pressure, chamber pressure drop, and characteristic length; 9. Solid rocket motor combustor design*: solid regression, grain shape and internal ballistics including operating pressure, necessary condition(s) for stable operation, pressure sensitivity for initial temperature and change in Klemmung, local conditions (flow velocity, pressure, etc.)), and two phase flow; 10. Hybrid rocket motor combustor design*: Solid regression, grain shape, (quasi-)steady operation (operating pressure, and local conditions (flow velocity, pressure, etc.)); 11-13 Liquid propellant storage and feed systems including gas-pressure and pump fed systems, motor cycles and propellant distribution; 14. Capita Selecta * Only one of the two will be dealt with. This varies from year to year.	
Study Goals	At the end of this course, the student shall be able to perform important steps in the analysis and design of thermo-(chemical) rocket propulsion systems using basic methods that allow for taking into account fluid flow, heat addition, propellant thermo-chemistry, heat transfer and cooling, liquid, solid or hybrid ballistics, (liquid) propellant feeding, and propellant storage.	
Education Method	Lecture + exercises (for feedback); During the lectures extensive use is made of existing data to allow for verification of the methods presented.	
Literature and Study Materials	Lecture notes "Thermal Rocket Propulsion" + hand-outs. Supporting information and tools are available through Brightspace. Recommended literature: G.P. Sutton, Rocket propulsion elements, Wiley Inter Science, 6th edition (or higher).	
Assessment	Assessment is through a number of small homework assignments (20% of final grade) and an hybrid (written and online) exam (80% of final grade). In case of a fail grade there is one resit exam opportunity that makes up 100% of the final grade (no homework assignments). Resit exam, like the normal exam, is a hybrid exam and will be scheduled in Q3 (exam period).	
Remarks	Enrolment for the course is via Brightspace. The course will be made available on Brightspace from early September.	
Set-up	Lectures (14 x 2 hours) possibly with 1 tutorial on the use of RPA-Lite and/or NASA CEA for rocket motor performance estimation.	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Bachelor program, preferably electrical engineering	
Course Contents	The course gives an overview of different types of electrical machines and drives. Different types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	Written exam (100% Weightage), Closed book, Scientific (non-graphic) calculator can be used, No formula sheets Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	In order to participate in the written exam, students need to complete the three half day lab practicals. During corona period, three remote practicals are offered that are done individually. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ME45025	Introduction to Multiphase Flow	5
Responsible Instructor	Dr.ir. W.P. Breugem	
Instructor	Prof.dr.ir. R.A.W.M. Henkes	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are gas and oil transport, dredging, steel making, refineries and chemical plants, sediment transport in rivers, cloud formation, etc. This course gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of gas/liquid two-phase flows, spatial/statistical averaging of the Navier-Stokes equations and continuum modelling, flow pattern maps and mechanistic modelling of multiphase pipe flows, physics of particles (droplets, bubbles, solids), dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.	
Study Goals	At the end of this course the student is able to: 1.a classify multiphase flows into separated and dispersed multiphase flows. 1.b formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface. 1.c formulate relevant dimensionless numbers for multiphase flows. 1.d explain the concept of dynamic similarity of multiphase flows. 1.e reflect on scaling choices when designing a laboratory experiment or setting up a numerical simulation. 2.a apply a linear stability analysis to two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability). 2.b give physical mechanisms for the transition of stratified to non-stratified (bubble or slug) multiphase flows in both (tilted) horizontal and vertical pipes. 3.a apply averaging methods (such as statistical, spatial and temporal averaging methods) to the Navier-Stokes equations for describing the macroscopic behavior of multiphase flows. 3.b formulate appropriate closure models needed to solve the averaged Navier-Stokes equations. 4.a classify gas/liquid pipe flows into stratified, slug and bubble flows, dependent on the superficial gas and liquid velocities. 4.b apply simplified 1D (cross-section averaged) models to each of the flow regimes. 4.c formulate appropriate closure models in order to compute the streamwise pressure drop and liquid holdup in two-phase (gas/liquid) pipe flow. 5.a classify dispersed multiphase flows into flows with solid particles, gas bubbles and/or liquid droplets. 5.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets. 5.c apply the Newton-Euler equations of motion to an analysis of the dynamics of solid particles. 5.d describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio. 5.e describe possible interactions between solid particles and turbulence of the carrier phase. 6.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Haberman-Morton number. 6.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag. 6.c give physical mechanisms for the breakup of bubbles/droplets in turbulence. 7. describe the dynamics of some particle-laden flows such as sediment transport through a pipe and/or sediment transport in rivers. 8. describe the merits and shortcomings of different experimental methods for multiphase flows. 9. describe the merits and shortcomings of different numerical methods for simulating multiphase flows. 10.a give examples of multiphase flow in industrial processes. 10.b describe the types of models used to design multiphase industrial processes.	
Education Method	Oral lecture 0/0/2/2; Plenary discussion of homework assignments 0/0/2/2	
Assessment	Written exam. The student can earn bonus points by making homework assignments. The final grade is the sum of the grade for the written exam and the bonus points from the homework.	
Department	3mE Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	At the end of this course you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	Brightspace/MATLAB exercises and tests (optional) during education period. The course is concluded through an optimization project and report. Deadlines for handing in the final report will be announced at the start of the course.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

MS43310	Materials at High Temperature	4
Responsible Instructor	V. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Department	3mE Department Materials Science & Engineering

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 6: European Wind Energy Master Track Rotor Design	
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MSc Coordinator	Dr. J.J.E. Teuwen
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Year

Organization

Education

2022/2023

Aerospace Engineering

Master Aerospace Engineering

Profile I: Aerodynamics	
Program Coordinator	W. Yu

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE3-W02 (4P720 TUe) plus basic aerodynamics.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade element/momentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsen's Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	Prof.dr. F. Scarano	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control Blade materials and assessment of fatigue damage Wind data and description The use of standards for load calculations	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): AE4W13 Site conditions for wind turbine design AE4135 Rotor / wake aerodynamics AE4W21-14 Wind turbine aeroelasticity ET4117 Electrical machines and drives OE44135 Offshore wind support structures	

AE4W13	Site Conditions for Wind Turbine Design	3
Responsible Instructor	Dr.ir. W.A.A.M. Bierbooms	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Course Contents	Introduction Atmospheric Boundary Layer Meteorology Fluid Dynamics Turbulence Wind power meteorology Measurement instruments Statistics Methods in turbulence Offshore wind climate External conditions in standards Wave and current climate and loading Example analysis site data Turbulence spectra Satellite remote sensing techniques Simulation of turbulent wind fields and gusts	
Study Goals	After successfully completing the course, students will be able to: Discuss the relevance of particular site data for wind turbine design, operation and siting. Analyse simple problems related to wind power meteorology Interpret scientific dissemination by experts on main points Apply analysis methods to convert basic site data to wind turbine design parameters	
Education Method	Lecture + assignments + compulsory practical	
Course Relations	Follow up courses AE4-W09, AE4135	
Literature and Study Materials	www.eos.ubc.ca/books/Practical_Meteorology Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier. Lecture notes and reader	
Prerequisites	AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Assessment	The written exam consists out of two parts, one for each period (scheduled in the exam period). The final grade is a weighted average of these 2 exams and the grade for one of the assignments (which one will be specified during the course). In order to take part in Exam2 one should join the compulsory practical The resit is an oral exam (to be scheduled with the teacher) on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).	
Set-up	The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	W. Yu	
Gast Instructor	Dr.ir. A.C. Viré	
Gast Instructor	Prof. R. Beuker	
Gast Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MANDATORY: AE4ASM506 (Aeroelasticity). The student should have completed or is attending this course. An equivalent course can be accepted upon discussion with the instructor.	
Course Contents	The course will cover main aerodynamic and structural phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. Possible resonances on wind turbines. What are the excitation frequencies, what are the natural frequencies? 2. What are wind turbine aeroelasticity tools based on? Linearised tools in frequency domain and nonlinear time domain simulation tools. How accurate are these tools and where are inaccuracies coming from? 3. Wind turbine mode shapes, whirling modes. 4. Wind turbine instabilities. 5. Dynamic inflow and dynamic stall.	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand which aeroelastic principles are important for wind turbines. 2. explain the contribution of the different wind turbine components on the resulting time responses. 3. apply the knowledge to program their own models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE5912	Thesis Wind Energy Rotor Design	45
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Course Coordinator	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The thesis project engages the student in innovative research on a topic related to wind energy. The student addresses a specific research question, and is responsible for the planning and execution of the project. The project may be carried out in industry (preferably one of the EWEM partners), but most are carried out within the Wind Energy research group or at DTU, where the student is normally engaged as part of a research team working within a particular area. Please notice that DTU has a strict deadline for submitting your final thesis report (deadline for delivery).	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Frequent meetings with the graduation supervisors, where feedback is given on the students progress and performance, and also suggestions for improvement. More information can be found on Brightspace; "organizations" - Wind Energy: https://brightspace.tudelft.nl/d2l/home/43780 (Aerodynamics Profile) - Aerospace Structures & Materials MSc Track: https://brightspace.tudelft.nl/d2l/home/43786 (Structures and Composites Profile) See also: http://www.windenergy.lr.tudelft.nl/ and https://www.tudelft.nl/lr/organisatie/afdelingen/aerospace-structures-and-materials	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Courses at DTU

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Aerodynamics

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism. There are five deadlines for handing in each year.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

AE5055	Professional Training	6
Responsible Instructor	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	N.A.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The aim of the professional training is to expose students to a real work environment for a period of 4 weeks (consecutive) on a full-time basis.	
Study Goals	- To increase your understanding of employment options after graduation - To get a better perception of your tasks and responsibilities within a professional environment - To create an understanding of the context in which wind energy is practiced by industry, institutes and organizations.	
Education Method	This is an elective. You have to arrange the internship yourself. Before you start you should ask me for approval about the specific topic (specify it in 1 to 2 pages). Deadline for topic approval: 15 April (afterwards you are not allowed to do an internship anymore) Here you can find contact persons: https://www.tudelft.nl/en/ae/organisation/departments/aerodynamics-wind-energy-flight-performance-and-propulsion/wind-energy/education/master-projects/msc-possibilities-industry You can of course also contact one of the EWEM partners. Also keep in mind that DTU nowadays does not provide much room for delay in your MSc: you have to start your thesis project in November and finish it in August. This implies that you have to do your internship during the summer.	
Assessment	After your internship the grade will be settled by your supervisor at the company and me based on your report of about 25 pages.	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Structures and Composites	
Responsible Program	Dr. J.J.E. Teuwen
Employee	

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/2/2/2	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Students of the track Aerospace Materials and Structures start in the second period (Q2, November). Students of the Wind energy master start in the third period (Q3, February). All other students start in the second period (Q2, November). Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four students about 14 day sections with support of staff are needed during this course. The planning of these day sections is depending (1) on the other obligatory courses of the track Aerospace Materials and Structures, (2) on the availability of supporting staff and (3) on the availability of the testing equipment. The ratio planned day sections by team over planned by staff is determined at the beginning of this course and notified and clarified at the kick off meeting held in the first week. For two parts of the exercise two consecutive day sections (so a full day) are needed. Finishing period Students started in the second period (Q2, November) are expected to finish at the end of the third period (Q3, April). Students started in the third period (Q3, February) are expected to finish at the end of the fourth period (Q4, June). IT IS EXPECTED THAT THIS LABORATORY COURSE WILL BE HELD AS BEFORE COVID-19. IF NOT ADAPTIONS WILL BE MADE. WHEN THAT DECISION WILL BE TAKEN IS UNCLEAR AT THE MOMENT OF WRITING (JUNE 2021).	
Course Contents	The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Possibly C-scanning of all laminates for determination of laminate quality 4. Estimation of the maximal bending-torsion coupling for a strip of UD composite 5. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 6. Performing mechanical tests 7. Analysing the test results of the mechanical test, including failure analyses 8. Adhesion of thermoset laminates 9. Resistance welding of thermoplastic laminates 10. Possibly preparation of lap shear specimens 11. Performing lap shear tests 12. Estimation of the mechanical properties of a sandwich panel 13. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 14. Performing a bending test on a sandwich panel. 15. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	For this course only one brightspace course will be available. It is called "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)". The Brightspace course "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q3)" should not be used, and is most likely not available for students. SO students starting in Q3 are added to "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)".	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on forehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	- Classical Lamination Theory - Short Overview of Materials. Composites Design Philosophy - Theory of Elasticity - Engineering Constants - Stress & Strain Transformations / Implications for Testing - Thin laminates / Carpet Plots - Thick Laminates / ABD Matrix / Coupling Effects - Hydrothermal Analysis - Progressive damage analysis - Failure criteria - First ply failure & Last ply failure - Damage tolerance analysis - In-situ strength - Experimental mechanics & in-situ monitoring techniques - Basic Stress Solutions and Buckling of Composite Plates - Typical Airframe Elements - Airframe Design Process, Materials & Damage - Plate Governing Equations / Solution of the PDE - Energy Minimization Methods - Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: - Theory of Elasticity: 2 - Classical Laminar Theory: 4 - Progressive damage analysis: 4 - Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	Combination of Classroom/Online lectures and Practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control Blade materials and assessment of fatigue damage Wind data and description The use of standards for load calculations	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): AE4W13 Site conditions for wind turbine design AE4135 Rotor / wake aerodynamics AE4W21-14 Wind turbine aeroelasticity ET4117 Electrical machines and drives OE44135 Offshore wind support structures	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	W. Yu	
Gast Instructor	Dr.ir. A.C. Viré	
Gast Instructor	Prof. R. Beuker	
Gast Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MANDATORY: AE4ASM506 (Aeroelasticity). The student should have completed or is attending this course. An equivalent course can be accepted upon discussion with the instructor.	
Course Contents	The course will cover main aerodynamic and structural phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. Possible resonances on wind turbines. What are the excitation frequencies, what are the natural frequencies? 2. What are wind turbine aeroelasticity tools based on? Linearised tools in frequency domain and nonlinear time domain simulation tools. How accurate are these tools and where are inaccuracies coming from? 3. Wind turbine mode shapes, whirling modes. 4. Wind turbine instabilities. 5. Dynamic inflow and dynamic stall.	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand which aeroelastic principles are important for wind turbines. 2. explain the contribution of the different wind turbine components on the resulting time responses. 3. apply the knowledge to program their own models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE5912	Thesis Wind Energy Rotor Design	45
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Course Coordinator	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The thesis project engages the student in innovative research on a topic related to wind energy. The student addresses a specific research question, and is responsible for the planning and execution of the project. The project may be carried out in industry (preferably one of the EWEM partners), but most are carried out within the Wind Energy research group or at DTU, where the student is normally engaged as part of a research team working within a particular area. Please notice that DTU has a strict deadline for submitting your final thesis report (deadline for delivery).	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Frequent meetings with the graduation supervisors, where feedback is given on the students progress and performance, and also suggestions for improvement. More information can be found on Brightspace; "organizations" - Wind Energy: https://brightspace.tudelft.nl/d2l/home/43780 (Aerodynamics Profile) - Aerospace Structures & Materials MSc Track: https://brightspace.tudelft.nl/d2l/home/43786 (Structures and Composites Profile) See also: http://www.windenergy.lr.tudelft.nl/ and https://www.tudelft.nl/lr/organisatie/afdelingen/aerospace-structures-and-materials	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	Dr. P.E. Vermaas	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also write essays.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Working book for tutorial sessions: 'Ethics and Technology for Aerospace Engineering', available on brightspace.	
Assessment	The grade will be based on the essay.	
Enrolment / Application	The weekly assignments are compulsory, students will not receive their grade if they have not passed these assignments. TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Courses at DTU

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Structures and Composites
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AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc.) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	S.R. Turteltaub	
Responsible Instructor	Dr.ing. S. Giovanni Pereira Castro	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of structures commonly encountered in aerospace during operation. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE5055	Professional Training	6
Responsible Instructor	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	N.A.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The aim of the professional training is to expose students to a real work environment for a period of 4 weeks (consecutive) on a full-time basis.	
Study Goals	- To increase your understanding of employment options after graduation - To get a better perception of your tasks and responsibilities within a professional environment - To create an understanding of the context in which wind energy is practiced by industry, institutes and organizations.	
Education Method	This is an elective. You have to arrange the internship yourself. Before you start you should ask me for approval about the specific topic (specify it in 1 to 2 pages). Deadline for topic approval: 15 April (afterwards you are not allowed to do an internship anymore) Here you can find contact persons: https://www.tudelft.nl/en/ae/organisation/departments/aerodynamics-wind-energy-flight-performance-and-propulsion/wind-energy/education/master-projects/msc-possibilities-industry You can of course also contact one of the EWEM partners. Also keep in mind that DTU nowadays does not provide much room for delay in your MSc: you have to start your thesis project in November and finish it in August. This implies that you have to do your internship during the summer.	
Assessment	After your internship the grade will be settled by your supervisor at the company and me based on your report of about 25 pages.	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

Year	2022/2023
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Master LR Keuzevakken

AE4006	Sustainability in Engineering	4
Responsible Instructor	C.D. Rans	
Instructor	Dr.ir. M.M. Rutten	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond ones own horizon and an (inter-)active scrutiny of ones social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); Week 2: Responsibility & Ethical Codes (Engineers Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. Cooperate with others to analyse and evaluate those relations. 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of two 3-hour seminars, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Permitted Materials during Tests	Not applicable	
Enrolment / Application	The course AE4006 is the 4 ECTS version of the Sustainability in Engineering course. There is also a 5 ECTS version of the course, with course code AE4006E. Further details about the two versions will be given in the first class. Students can indicate whether they wish to follow the 4 ECTS or the 5 ECTS version of the course in the first week. Enrollment takes place via Brightspace. There is one joint Brightspace page for both versions of the course.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	MSc students in Aerospace Engineering. Students from other masters are also welcome to take part in the course.	
Expected prior knowledge	Students who take part in this course must have completed almost all courses of their MSc. Ideally they should be at the beginning of their MSc thesis work.	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents with handing deadline at the end of each period (1 - 4) and over the summer (period 5). The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gannt Charts 6. Academic Writing - Learn the peculiarities of academic writing.	
Study Goals	The course has been designed to be suitable for distant learning At the end of the course you will be able to: - formulate a research question and aim - correctly cite the literature relevant in your research field - select the appropriate methodologies and tools for your research - set up a clear research plan	
Education Method	(Online) Lectures, Assignments and voluntary Peer review of each others research plans.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Prerequisites	Students must be starting their Literature Study or their Thesis project.	
Assessment	For this course you will have to write a (draft) Research Plan according to a prescribed format based on the topics of your Literature Study or ideally your MSc thesis. All work MUST be submitted digitally in PDF-format via the plagiarism tool on the BrightSpace page of the course. All assignments will be checked for plagiarism.	
Enrolment / Application	There are five deadlines for handing in each year. Please enroll via BrightSpace. The course contents are always open. The submission deadlines will open progressively through the year.	

AE4115	Experimental Simulations	3
Responsible Instructor	T. Sinnige	
Instructor	Prof.dr.ing. G. Eitelberg	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling. Fractional analysis, symmetry and similarity, application to specific problems from aerodynamics, BL-approximations, S/BL interactions; 3 Dimensional analysis: Buckingham's Pi-theorem, rigorous methods of obtaining dimensionless parameters, simplifications of Navier-Stokes equations; 4 Subsonic and supersonic wind tunnels: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 5 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 6 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 7 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 8 Design of experiments: factorial designs, response-surface models; 9 Reynolds-number effects and simulation; 10 Pre-test support session; 11 Engine integration: engine simulators, performance of those. Thrust reverser testing for re-ingestion and stability and control; 12 Acoustic testing. Noise generation mechanisms for submerged bodies, edges and jets. Appropriate scaling. Definition of sound pressure levels. Measurement requirements and devices. Far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session. (Excursion) Ground test facilities of DNW (organized by AWEP society, limited places available).	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports and oral exam.	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter. The lectures are supported by an excursion to the testing facilities of DNW, organized by the AWEP society.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	

AE4120	Viscous Flows	3
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Responsible Instructor	M. Kotsonis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Dr.ir. L.L.M. Veldhuis	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and (to limited extend) viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Typically Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during at the beginning of Q1).	
Assessment	The assessment is done through 5 personal assignments.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE3-W02 (4P720 TUE) plus basic aerodynamics.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade element/momentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsen's Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4136-22	CFD 2: Discretization techniques	3
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results.	
Study Goals continuation	After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Special Information	M.I.Gerritma@tudelft.nl	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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6	Variational inverses	[Inversion]	Finalize []	--																																						
7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Dr. S.J. Hulshoff	
Responsible Instructor	Prof.dr.ir. S. Hickel	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	Prof.dr. F. Scarano	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4185	Fluid Flow Data Processing and Visualization	3
Responsible Instructor	W.J. Baars	
Responsible Instructor	Dr. D. Modesti	
Responsible Instructor	Dr. N.A.K. Doan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is a general introduction to data processing techniques used in fluid mechanics to analyse unsteady flows. Specifically, techniques related to the flow visualization and the postprocessing of numerical data will be presented. The overall content of the course is: 1. Review of fundamentals of turbulent flows (energy cascade, Reynolds-averaged equations, Reynolds stress, turbulent kinetic energy, vorticity equation, turbulence spectrum, energy budget, ...) 2. Characteristics of flow datasets (Nyquist criterion, sampling and resolution, first principle of visualization, vortex/Q criteria, ...) 3. Statistical analysis in flows - Deterministic/random data - Averaging (ensemble, condition, phase) - Probability: PDFs, statistical moments, conditional PDFs - Correlation function - Ergodic hypothesis - Statistical convergence and uncertainty 4. Temporal and spectral data processing - Fourier transform - Physical/spectral space and multi-dimensional FFTs - Auto/cross spectra - Integral length/time scale - Convolution (filtering) and transfer function - RANS/LES averaging/filtering 5. Modal analysis and coherent structure education - Base of Proper Orthogonal Decomposition (and some extensions)	
Study Goals	At the end of this course, the student should be able to: - Visualize fluid mechanics datasets using relevant quantities - Analyse fluid mechanics datasets using statistical methods - Apply temporal and spectral analysis methods to analyse flow datasets - Postprocess fluid mechanics datasets using modal-based methods	
Education Method	The course will be a combination of lectures, scheduled tutorials and homework assignments.	
Assessment	The course will be assessed with a final assignment (80%) and an oral exam (20%).	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4rt quarter of the academic year) and not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Remarks	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4206	Turbomachinery	3
Responsible Instructor	Ir. M. Pini	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course covers fundamental concepts of internal flows in turbomachines and related design aspects. The usual course schedule consists of the following lectures: Lecture 0: Introduction course overview: topics, structure and assessment Lecture 1: Recall of Turbomachinery Fundamentals & Principles integral governing equations with applications, relative motion, thermodynamic transformations in turbomachinery stages, velocity triangles with practical examples; Lecture 2: Unsteady Flows in Turbomachinery concept of reduced frequency with applications; work exchange in turbomachinery rows; Lecture 3: Similarity in Turbomachinery similarity analysis for incompressible flows, Balje and Smith diagrams, accounting for compressibility, stage dimensionless performance parameters; Lecture 4: Fluid-dynamic Design of Turbomachinery design procedure, mean-line design model, radial equilibrium model, CFD-based models Lecture 5-6: Axial and Radial Turbines examples of axial and radial turbines, typical turbine stages, blade profiles for turbines, flow characteristics in subsonic, transonic, and supersonic cascades, basic concepts of diffusers, aerodynamics of radial impellers Lecture 7-8: Axial and Radial Compressors examples of axial and radial compressors, typical compressor stages, blade profiles for compressors, flow characteristics and unique incidence, stall and surge instability, slip factor, diffusers for radial compressors, design maps Seminar(s): topic(s) may vary but usually include turbomachinery aero-mechanics, shape optimisation with CFD methods, rocket engine turbo-pumps, and advanced turbomachinery components for aero-engines Tutorial: training session on the software packages used for the group assignment	
Course Contents	This course educates aerospace students to apply theoretical knowledge in the field of fluid- mechanics and thermodynamics for analyzing and designing a thermal turbomachinery. Typical examples are: axial and radial compressors of gas turbine engines, industrial gas and steam turbines, radial turbines of automotive turbochargers, turbo-expanders and compressors used in refrigeration and liquefaction industry, micro-turbines for power generation. Since the turbomachinery field is intrinsically interdisciplinary, it is assumed that students already possess basic knowledge of thermodynamics, fluid-mechanics, applied mathematics, propulsion systems, internal flows, acquired in previous courses. Prerequisite of the course are therefore bachelor and master courses addressing fundamental (e.g. Thermodynamics, Internal Flows) and applied aspects (e.g. Aero-Engines) of air-breathing engines. Starting from basic concepts of gas-dynamics and fluid-machinery, the course brings the student to the level that he is able to select a turbomachinery for a specific application, perform the conceptual design of the stages, and eventually predict the flow characteristics within the blade passages by means of conceptual physical models and more sophisticated simulation tools.	
Study Goals	After the course the student will be able to: - Classify and illustrate the main features of a turbomachine - Select the most appropriate turbomachine for a given aerospace application - Apply first principles to simplified turbomachinery configurations - Identify the most prominent loss sources of a given turbomachinery stage - Define the aerodynamic blade design of turbomachinery cascades - Predict the performance of a turbomachine with analytical and numerical methods - Devise solutions for performance improvement of turbomachinery cascades based on physical understanding - Perform the conceptual and detailed fluid-dynamic design of a turbomachine for propulsion systems using analytical and numerical tools	
Education Method	Lectures, concept quizzes, concept exercises, and tutorials in preparation to the team assignment.	
Books	- S.L. Dixon, C. A. Hall Fluid-Mechanics and Thermodynamics of Turbomachinery, 7th edition, 2013 - M. Schobeiri, Turbomachinery Flow Physics and Dynamic Performance, 2nd edition, Springer, 2012 - E.A. Baskharone, "Principles of Turbomachinery in Air-Breathing Engine", Cambridge University Press	
Assessment	The final course grade results from the weighted average of: - the assessment of the final project assignment, counting for 40% of the final grade (individual score 1-10, minimum threshold: 6); - the assessment of the concept exercises, counting for 10% of the final grade. No minimum threshold; - the assessment of the written exam, counting for 50% of the final grade (individual score 1-10, minimum threshold: 6); The written exam takes place at the end of course. The exam consists of one or two concept exercises covering the material discussed during classes. The final assignment is meant to challenge students with the real engineering problem of designing a turbomachine for propulsion or power generation application. The concept exercises are numerical exercises that are formulated to stimulate critical thinking on turbomachinery problems of practical application. They are also meant as diagnostic tool to help students self-assessing their level of understanding.	
Permitted Materials during Tests	Lectures notes, slides, formula sheets and textbooks are allowed during the written exam. Calculator can be used.	
Set-up	The course is organized in 8 teaching sessions, 2/3 seminar(s), and a possible visit to a turbomachinery company. Lectures usually consist of a theoretical part integrated with in-class exercises and quizzes to stimulate critical thinking and learning-by-doing. During classes, the students, autonomously or in groups, are engaged in a series of learning activities, supervised by the team of instructors. Their solution is briefly discussed at the beginning of the next class. The seminar(s) are devoted to show students current research topics in turbomachinery and are given by recognised experts in the field.	

AE4238	Aero Engine Technology	4
Responsible Instructor	A. Gangoli Rao	
Instructor	Ir. M. Pini	
Instructor	F. Yin	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery"; "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Turbomachinery - o New Engine /propulsion Concepts - o Turbine Cooling - o Inlets and Exhaust - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an hybrid course in which the lectures are pre-recorded and some for tutorials will be solved in the classroom. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13 Analyse the cooling requirements and compute the effect of cooling on engine performance 14. Evaluate the calorific values for various fuels 15. Understand the basics of combustion and emission formation in Aero engines 16. Understand the working of gas turbine combustor 17. Understand the basic principles of gas turbine controls and evaluate different strategies for control 18. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a hybrid teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen
	2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Reader	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on several homework assignments (5-6) given during the course and a final exam. All assignments together will form 50% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015 Optional textbooks: - Airplane Design, Prt I-VII, Roskam, J, DARCorp, 1990-2003 - Synthesis of Subsonic Airplane Design, Torenbeek, E, Kluwer Academic Publishers, 1982 - Fluid Dynamic Drag, Hoerner, S. F., Hoerner Fluid Dynamics, 1965 - Fluid Dynamic Lift, Hoerner, S. F. Hoerner Fluid Dynamics, 1965	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr. F. Avallone	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resourcen/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4261	Internal Flows	3
Responsible Instructor	Ir. M. Pini	
Instructor	C.M. De Servi	
Responsible for assignments	T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course we address fluid dynamic phenomena of interest in internal flow situations for aerospace propulsion applications. The goal of the course is to develop physical insight into the phenomena that characterize internal flow in fluid machinery. As such we will discuss not just what happens, but why it happens and what are the consequences for propulsion systems. The course covers topic, which are not dealt with in courses or texts about external fluid dynamics. The flows described in Internal Flows are generally rotational, often three-dimensional, unsteady, and sometimes occurring in non-inertial (e.g., rotating) coordinate systems. The course can be viewed as the development of flow models and ideas to enable allow physical insight into the behavior of three-dimensional and unsteady flows. One of the objectives of the course is to provide an increased ability to interpret computational and experimental results and hence to effectively extract conclusions about the key features of complex internal flows. This course is inspired by 16.540 Internal Flow from the Department of Aeronautics and Astronautics at MIT. Much of the material is based on the lectures of Profesor E. M. Greitzer, with permission.	
Study Goals	1) Development of physical insight into the phenomena, which characterize internal flow in fluid machinery (not just what happened, but why it happened) 2) Development of the critical thinking needed to define, in a rigorous manner, the levels of modeling needed for useful descriptions of internal flow situations 3) Development of the ability to interpret numerical simulations and experimental results in terms of concepts and first principles	
Education Method	This course will use innovative didactic methods that emphasize active learning. Students will be engaged through active feedback systems and online material. Given that a textbook exists for the course material, it is the intent of the lectures to discuss the material rather than to present it for the first time in class. As such the students will be expected to read the material before class and to engage in in-class discussions of the reading material by answering questions and elaborating concepts. Here we list a high-level description of the course contents. More detailed information are available in the complete syllabus on Brightspace Lecture 1 Structure and Content of the Course, Introduction to Flow Regimes Lectures 1-2 Useful Basic Ideas Pressure fields and streamline curvature Upstream influence in turbomachines Control volume description of propulsion systems Asymmetry of real fluid motions Behavior of an inlet and turbomachinery system Lectures 3-6 Vorticity and Circulation Kinematics of vorticity Dynamics of vorticity Behavior of vortex lines Rotational flow description in terms of vorticity and virculation Mixing enhancement due to streamwise vorticity Lectures 7-9 Flow in Rotating Passages Non-dimensional parameters in a rotating flow Conserved quantities in a steady rotating flow Phenomena in flows where rotation dominates Rotating Channel Flow in Constant Area Straight Passages Boundary layers in rotating straight channels Rotating Flow in Turbomachinery Passages Secondary flow in rotating blade rows Lectures 10-12 Loss Sources and Loss Accounting Appropriate metrics for loss Boundary layer losses Mixing Losses Averaging of a mon-uniform flow Effect of pressure level on average properties and mixing losses Loss characterization in turbomachinery cascades Lectures 13-14 Compressible Flows in fluid machinery Corrected mass per unit area Heat and friction in compressible flows Influence coefficients Compressible fluids in fluid devices	
Literature and Study Materials	Internal Flow: Concept and Applications, E. M. Greitzer, C. S. Tan, M. B. Graf, Cambridge University Press http://www.cambridge.org/gb/academic/subjects/engineering/thermal-fluids-engineering/internal-flow-concepts-and-applications?format=PB#MciShGFvqgyx6G4p.97	
Assessment	The student will be asked to develop concept questions and corresponding answers which help to illustrate the physical principles and approximations that are made. This will be done in small (six or seven person) teams so you can discuss the questions among yourselves. One of these questions is to be developed each week by each team. The concept questions will count 20% of the final grade. Approximately five short (~20 minute) "concept quizzes" will be given during class, roughly equally spaced throughout the term. These may make use of the questions generated by students. The total of these quizzes will count for 30% of the final course	

grade.

A written and oral final exam will be held. The final exam will count for 50% of the final course grade.

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	A. Gangoli Rao	
Responsible Instructor	Dr. I. Langella	
Instructor	Prof.dr. D.J.E.M. Roekaerts	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Radiation and transport processes 4. Premixed and diffusion flames 5. Turbulent reactive flows 6. Combustion experiments and diagnostics 7. Laboratory Tutorial & Exercise 8. Practical combustion systems for aerospace applications	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation- and adiabatic flame temperature. Apply principles of thermodynamics to analyze reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames. Derive equations related to turbulent reacting flows and chemistry interaction. Explain laser diagnostics and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions. In addition, the student is required to complete project assignments and perform a laboratory exercise. The project assignments and associated numerical problems are relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with laminar flame speed measurements of the simplest hydrocarbon fuel, methane. The student will gain hands-on experience in assembling the experimental setup, acquiring data, performing data analysis and writing a technical report.	
Course Relations	Aero Engine Technology	
Literature and Study Materials	Slides and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the student must pass the oral examination, the evaluation of the laboratory exercise, and successfully complete the assignments given during the course. The work on the laboratory exercise and assignments should be summarized in reports, to be written by a single student or in a group of two. The grade is dictated by the result of the oral examination (50%) and the grades obtained in the assignments and lab exercise (50%).	

AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
Responsible Instructor	Prof.dr.ir. P. Colonna di Paliano	
Instructor	C.M. De Servi	
Instructor	Dr.ir. W.P.J. Visser	
Instructor	Prof. F. Casella	
Contact Hours / Week x/x/x/x	0/x/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE1240-I Thermodynamics AE2230-II Propulsion and Power AE1205 Programming and Scientific Computing in Python for Aerospace AE4238 Aero Engine Technology ME45000 Advanced Heat Transfer	
Course Contents	2nd EDUCATION PERIOD Part 1 (2 ECTS) Module 1 Introduction, Context, Foundations Module 2 Conservation equations Module 3 Modeling paradigms Module 4 Numerical methods and software Module 5 Modelica Module 6 Constitutive equations Module 7 Components and system modeling Module 8 Verification and validation 3rd EDUCATION PERIOD Part 1 (1 ECTS) Module 9 Model-based control Take Home exam Part 2 (2 ECTS) Module 10 - Team Project to be chosen among a Aero engine with GSP or GTPsim b Power or propulsion system with Modelica	
Study Goals	After learning the content of the course, the student will be able to: Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results. In particular, this overarching objective can be obtained by developing the following theoretical capabilities: 1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose. 2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.). 3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model. 4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables). 5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation) 6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.	
Education Method	Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents) Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software. Students are strongly advised to make use of the BrightSpace Forum (Q&A) to post questions and interact with instructors and other students. In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.	
Computer Use	Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim. More information on these programs is provided on BrightSpace: Course Information -> Resources and Tools -> Open Modelica and GSP	
Literature and Study Materials	Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules	
Books	Reference text for thermodynamics: W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018. Reference text for numerical methods:	

Reader	P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.
Assessment	The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's). Part 1: Exercises The first module is tested with a Quiz. The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded. Part 1: Written exam Take home exam is graded and covers the material of the 9 Modules of Part 1 of the course. Duration: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace. Part 2: Simulation project Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola Graded deliverables: technical report, model files. A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam. Grading the final grade is calculated as Final Grade = 0.2 x Exercises Grade + 0.4 x Take Home Exam Part 1 Grade + 0.4 x (Simulation Project + Oral Exam Grade). A final grade can be obtained only if the grade of the Take Home Exam Part 1 and of Part 2 are sufficient (≥ 6). The Part 2 oral exam can only take place after a sufficient grade (≥ 6) on Part 1 has been obtained. Resit exams One opportunity to resit the Take Home Exam Part 1 is provided during the second quarter the course is running in. The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

AE4270	Control and Operations Project	4
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Dr. O.A. Sharpans'kykh	
Contact Hours / Week x/x/x/x	///	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course code is a placeholder for miscellaneous student assignments within the Control and Operations track. For example, a lecturer would like to have some data processed or have a nice simulation model that can be used for demonstration purposes in the lectures. Through this course code, we now have a way to compensate students for their efforts by considering it as an elective of 4 ECTS. If there are such assignments available, they will be announced.	
Study Goals	n/a, depends on the nature of the assignment	
Education Method	n/a	
Assessment	A (short) report needs to be written about the assignment. The content needs to be discussed with the supervisor of the assignment.	
Elective	Yes	

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory a. Laplace transformation b. Elementary closed loop systems c. Transfer functions 2. Poles and zeros a. First order systems b. Second order systems c. Pole placement for simple systems 3. Root locus method a. Characteristic equation b. Angle and magnitude conditions 4. State space formulation a. Controllability, observability b. Ackerman's formula c. LQR 5. Basic controllers: P,PI,PD,PID 6. Frequency response (Bode plots) 7. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 8. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 9. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 10. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 11. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 12. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 13. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance - substantiate the function of a Flight Control System(FCS) in civil aviation.	

	- apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 AE3212-I AE2235-I	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace	
Assessment	Exercise	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4311	Nonlinear and Adaptive Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I AE4301 AE4301P AE4312/AE4320	
Course Contents	Safety and survivability of modern civil and military aircraft when airframe and structural failures have occurred in flight have become a key issue in the aerospace industry nowadays. This course introduces nonlinear and adaptive control techniques to enhance reconfigurabilities of aircraft flight control systems. Advanced nonlinear and adaptive control techniques such as Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion (INDI), Adaptive Backstepping (ABS) and Incremental Backstepping are described together with applications to fault tolerant and reconfigurable aircraft flight control system designs. Unconventional yet available control effectors are used through control allocation algorithms to achieve required performance.	
Study Goals	Students will be able to acquire the skills in designing fault tolerant and reconfigurable flight control systems using Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion, Adaptive Backstepping (ABS) and Incremental Backstepping control techniques.	
Education Method	Lectures altered with assisted computer exercises	
Literature and Study Materials	Handouts of lecture slides (available on Brightspace)	
	Recommended literature: J.-J.E. Slotine & Weiping Li, Applied non-linear control, Englewood Cliffs : Prentice-Hall, 1991, ISBN 0-13-040890-5 M. Krstic, et al, Nonlinear and adaptive control design, John Wiley & Sons, Inc., ISBN 0-471-12732-9	
Assessment	Individual Take-home assignment	
Remarks	Hand-in exercise and a report of the design of an advanced controller. The report must be handed in at the request date given by the course instructor.	
Set-up	Course contents Introduction to advanced flight control Nonlinear aircraft flight control based on Nonlinear Dynamic Inversion Adaptive Nonlinear Dynamic Inversion Incremental Nonlinear Dynamic Inversion Aircraft fault tolerant flight control based on Adaptive and Incremental Nonlinear Dynamic Inversion Nonlinear control based on backstepping Nonlinear aircraft flight control based on backstepping Adaptive Backstepping Incremental Backstepping Aircraft fault tolerant flight control based on Adaptive and Incremental Backstepping Control allocation	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hover performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aerospace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course. One final take-home assignment at the end of the course.	
Special Information	A helicopter flight in SIMONA simulator will be given to the students following this course. Note: This may differ due to COVID-19 measures. Changes to this will be communicated via Brightspace.	
Set-up	Online education and Classical lecturing	

AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3
Responsible Instructor	D.M. Pool	
Instructor	Dr.ir. C. Borst	
Instructor	Dr.ir. M.M. van Paassen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is an elective extension to the AE4316 (core) course for the C&S master profile. Prior knowledge from AE4316 is expected and mandatory.	
Course Contents	This course further explores the various aspects of supervisory control and manual control (cybernetics) and their applications in aerospace as introduced in the AE4316 lecture series. Through a number of lectures in the first weeks, this course provides detailed practical information on the current advanced state-of-the art methods used for measuring, analyzing, and modeling human performance and behavior in supervisory and manual control tasks and how such methods are applied in applications relevant to aerospace. Furthermore, through extensive practical work performed in small groups, students will gain practical experience with the advanced analysis of human-machine system performance through an experiment they need to plan and perform themselves. In this experiment, which will be performed in one of the simulator setups available at Control & Simulation (i.e., the Air Traffic Management Laboratory, the Human-Machine Laboratory or the SIMONA Research Simulator), students will collect data for an experiment of their choice, analyze the collected data, and reflect on the conclusions drawn from their experimental measurements.	
Study Goals	1. Formulate a good research question and hypotheses for a supervisory or manual control experiment. 2. Demonstrate understanding of the direct link between an experiments independent, dependent and control variables and the data analysis that can be performed on its data. 3. Analyze data from a supervisory or manual control experiment using state-of-the-art data analysis techniques and statistical tests. 4. Compare experimental findings with theoretical constructs and elementary models of human performance or control behavior. 5. Reflect on the scientific conclusions drawn from a set of human subject experiment data and their reliability. 6. Reflect on the consequences of experiment design choices on the scope and validity of human subject experiment results.	
Education Method	Classroom lectures, twice a week in the first three lecture weeks (3.1-3.3). After that, students will work in groups on the design and execution of an experiment they will perform in one of the simulation facilities at Control & Simulation.	
Literature and Study Materials	The presentation slides of all lectures will be made available through Brightspace Recommended reading for this course: 1. Jagacinski, Richard J., and John Flach. Control theory for humans: Quantitative approaches to modeling performance. Taylor & Francis, 2011. 2. Wickens, Christopher D., Lee, John D., Liu, Yili, and Gordon Becker, Sallie E. An Introduction to Human Factors Engineering, Second Edition, Pearson Education Inc., 2004. 3. Field, Andy. Discovering statistics using SPSS. Sage publications, 2009.	
Assessment	No written exam. Instead, grading for AE4316P will be based on the quality of two group deliverables: an Experiment plan (deadline end week 3.4, 20% of final grade) and a final Scientific Article (deadline end of Q3, 80% of final grade).	
Elective	Yes	
Set-up	Classroom lectures will be provided in the first three weeks. The following topics will be addressed in the lectures: 1. Introduction and experiment descriptions. 2. Human subject experiment design. 3. Experimental data analysis and statistics. 4. Cognitive Work Analysis. 5. Multi-channel manual control behavior identification. In addition to the lectures, students work in small groups on a practical assignment. This effort includes: developing an experimental plan, collecting and analyzing data, and writing a scientific article.	

AE4317	Autonomous Flight of Micro Air Vehicles	4
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. C. de Wagter	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Programming experience (any language) If the student has no previous background in C / C++, we expect the student to do an online tutorial before / at the beginning of the course. For example: Desired prior knowledge: Programming experience with C / C++ AE4393 (Avionics and Operations)	
Course Contents	This course covers the challenges and existing state-of-the-art methods for enabling autonomous flight of Micro Air Vehicles (MAVs), ranging from 20-gram flapping wings to 1 kg quad rotors. The emphasis is on computationally efficient, bio-inspired approaches to MAV autonomous flight.	
Study Goals	The theoretical knowledge will be applied by the students in the practical assignment, in which student groups program quad rotors in order to avoid obstacles in TU Delft's Cyberzoo. (1) Students can explain the differences between MAVs and larger aircraft in terms of control, design, aerodynamics, and electronics. (2) Students can explain how the above-mentioned differences affect the task of achieving autonomous flight with MAVs. (3) Students can list and explain Artificial Intelligence (AI) and control techniques useful for autonomous flight of Micro Air Vehicles (MAVs). a) Bio-inspired robotics b) Vision-based navigation (4) Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task. (5) Students can reproduce the current state of regulations for MAV operations. In addition, they can assess and discuss on the safety and ethical aspects of MAV operations.	
Education Method	The course will consist of theory (7 lectures) and a practical assignment (6 practical sessions and a competition session). Brief description practical assignment: Student teams will develop an efficient, purely vision-based approach to autonomously navigate through an obstacle field. The algorithm will have to run onboard the Parrot AR drone 2, a commercially available quad rotor equipped with a range of sensors. Importantly for the project, it has a single frontal camera. Hence, the developed algorithm concerns a monocular vision solution to obstacle avoidance.	
Assessment	The assessment of the course will consist for 50% of the theoretical exam, and 50% of the practical assignment. Assessment of the practical assignment is based on the performance of the drone on the task, the followed approach as explained in a presentation, and the code produced by the team.	

AE4320	System Identification of Aerospace Vehicles	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	D.M. Pool	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301.	
Parts	Week arrangement 1. Introduction. A global overview of the complete system identification cycle applied to aerospace vehicles, from sensor data acquisition and state estimation to aircraft aerodynamic model identification and validation. Introduction to sensor data acquisition and management, state and parameter estimation, and the two-step method for aerodynamic model identification. Finally an overview is give of advanced modelling techniques. 2. State estimation. Introduction to the essential role of the Kalman Filter (KF) in aircraft state estimation. The extension of the standard KF to nonlinear systems will be introduced in the form of the Extended Kalman Filter/Smotherer (EKF/EKS), and iterated extended Kalman filter (IEKF). Applications of the Kalman filters in aerospace navigation, attitude determination, on-board sensor calibration, sensor integration, and data fusion will be discussed. 3. Parameter estimation. Introduction to the process of parameter estimation and its role in creating accurate aerodynamic models of aircraft. Introduction to four of the most widely used parameter estimation techniques; Least squares (LS), Weighted Least squares (WLS), Total Least Squares (TLS), and Maximum Likelihood (ML) parameter estimation. Additionally, recursive parameter estimation methods like Recursive Least Squares (RLS) will be introduced. 4. Advanced optimization techniques. Some problems cannot be solved adequately with standard linear optimization techniques. In that case, a nonlinear optimization method must be used. In this part of the course, a powerful global nonlinear optimization technique based on interval analysis will be introduced. 5. Advanced modelling methods: Neural Networks. Neural Networks are a class of black-box function approximators that can be used to model nonlinear systems like Micro Air Vehicles and high performance aircraft. The application of neural networks to aircraft system identification is discussed, and it will be shown how neural networks can be used inside a framework for parameter estimation. 6. Advanced modelling methods: Multivariate Splines. Multivariate Simplex B-Splines are a powerful type of white-box function approximators that can be used to create accurate models of aircraft with highly nonlinear aerodynamics such as MAVs and high performance aircraft. A number of unique methods for assessing the quality of the created models are discussed, and pointers towards future research are given. 7. Demonstration: Aircraft system identification with multivariate splines. An in-depth demonstration will be given of the process of Aircraft system identification with multivariate splines using real flight test data from the Cessna Citation II laboratory aircraft. The goal is not only to create an accurate flight dynamics model of the Citation II, but also to apply various model validation techniques to assess the quality of the created model.	
Course Contents	Accurate aerodynamic models play a crucial role in the design and operation of flight simulators and flight control systems. The creation of accurate aerodynamic models from CFD, wind tunnel, and flight test data has historically been a highly challenging task. This is a direct result of the nonlinear nature of aircraft (aero)dynamics and the fact that not all aircraft states can be measured directly. As a consequence, the aerospace vehicle parameter identification problem constitutes a joint parameter-state estimation problem. It is the aim of this course to provide the student with a complete overview of the system identification cycle as it is currently applied to aerospace systems, and to introduce the student to the current state of the art in the field of aerodynamic model identification. The course consists of 7 parts covering the entire system identification cycle. In the first part of the course, the process of data acquisition using on-board sensors including accelerometers, gyroscopes, GPS and various air data sensors is discussed. It will be demonstrated that measurements made by real-world sensors are contaminated with noise, and are sometimes biased. Additionally, some aircraft states, like the true angle of attack, cannot be measured directly and must be reconstructed by combining sensor measurements. The second part of the course introduces the concept of state estimation in which prior physical knowledge of the system is used in a Kalman filter (KF) to estimate the true aircraft states from the measured states. Next to the ordinary Kalman filter, the Extended Kalman filter (EKF) and the Iterated extended Kalman filter (IEKF) will be introduced. In the third part of the course various methods for the estimation of model parameters will be discussed. It will be shown how the results from the state estimation are used in combination with a parameter estimator. This part not only focuses on offline batch parameter estimation methods like ordinary least squares (OLS), weighted least squares (WLS), total least squares (TLS), and maximum likelihood (ML) estimators, but also on recursive parameter estimation methods like recursive least squares (RLS) that can be used online. Special attention will be paid on the process of choosing a parameter estimator that is right for the job. The fourth part of the course introduces an advanced global nonlinear optimization method based on interval analysis. In aerospace system identification, so-called non-convex and nonlinear optimization problems are often encountered. Such problems can be solved with global nonlinear optimization methods like interval analysis. The fifth and sixth part of the course focuses on two advanced model structures that can be used in combination with the earlier introduced parameter estimation methods. In the fifth part of the course the neural network black-box function approximator is introduced. It is shown how neural networks are used to approximate scattered multidimensional data. In the sixth part of the course a new method for aerodynamic model identification based on multivariate simplex B-splines is introduced. This method was recently developed at the Faculty of Aerospace Engineering of the TU-Delft, and has a number of advantages over existing methods. For example, the simplex B-splines have a transparent model structure, are general in any number of dimensions, and can be computed efficiently in real-time.	

Study Goals	In the final part of the course, all theory introduced in the first six parts is used in a demonstration that uses real flight data from a real-life system like the Cessna Citation II laboratory aircraft, or a Micro Air Vehicle to identify a multivariate spline based aerodynamic model. The created model is validated using various techniques. Finally, pointers are given towards further research in the field of aerodynamic model validation. After completing the course, students will have a thorough understanding of the theoretical elements of aircraft system identification. Students will be able to apply these elements to go from real CFD, wind tunnel, and flight test data to an accurate aerodynamic model that can be used in simulator or flight control system applications.
Education Method	Lectures.
Literature and Study Materials	Q.P. Chu, Modern flight test technologies and system identification, lecture notes, Faculty of Aerospace Engineering, Delft University of technology. C.C. de Visser, Global Nonlinear Model Identification with Multivariate Splines, PhD thesis, Faculty of Aerospace Engineering, Delft University of technology. Recommended literature V. Klein and E. A. Morelli, Aircraft System Identification, AIAA, 2006. J.A. Mulder, J.K.Sridhar, J.Breeman, Nonlinear Analysis and Manoeuvre Design, Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 2, , May 1994 R.E. Maine, K.W.Illiff, AGARD Flight Test Techniques Series on Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 1, , December 1986 . Eric Walter and Luc Pronzato, Identification of parametric models, Springer, ISBN 3-540-76119-5.
Assessment	Individual Take-home assignments

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written (or on-line remote exam, depending on COVID-19 situation) (50%) and assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4323	Real-time Distributed Flight and Space Simulation	3
Responsible Instructor	Ir. O. Stroosma	
Instructor	Ing. E.H.H. Thung	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE1205 (Programming & Scientific Computing in Python, or equivalent introduction to programming) AE4322 (Piloted Flight Simulation)	
Course Contents	Lecture topics, not necessarily in chronological order 1) C++ introduction 2) Real-time software engineering 3) Delft University Environment for Communication and Activation (DUECA) 4) Modelling tools 5) Verification and Validation 6) Software project management Programming assignments, individually and in a team. The course is completed with a practical project in which students program, demonstrate and document a distributed simulation with an aerospace theme. Attendance in the project meetings is mandatory.	
Study Goals	1) The student can identify the software issues related to real-time, distributed simulation. 2) The student can actively apply the theory of rigid body dynamics in real-time simulation. 3) The student can combine the modelling of dynamical systems with numerical integration routines. 4) The student can use quaternions to represent rigid body attitudes. 5) The student can implement simulation models using different software languages and tools (C++, Simulink). 6) The student can create 3D models for computer visualisation and optimise these for real-time simulation. 7) The student can (in a team) implement a real-time simulation on a distributed platform, using advanced software tools.	
Education Method	Lectures, programming instruction, and programming assignments.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace.	
Assessment	Individual assignment during 2nd/3rd week and team project report. Attendance at the team project meetings in week 3 and later is mandatory.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Self-supervised learning - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4351	Robust Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	S. Theodoulis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I WI2056LR AE4301 AE4301P	
Course Contents	The vast majority of aerospace vehicles, spanning from UAVs and rotorcrafts to military or commercial aircraft and all the way up to space launchers, need to fly in a secure, predictable and repeatable way. To achieve this, the on-board flight control system needs to be designed to withstand (i.e. to be robust against) a whole array of uncertainties, disturbances, noise, rapidly-varying flight conditions and imperfect components. Furthermore, the designer needs to provide with solid guarantees and verifiable certificates that the flight control system must meet. This course is aimed at offering to the students hands-on theoretical and practical knowledge on the modelling, design and analysis of robust flight control laws, paying special attention to the industrial applicability of the tools albeit with a research twist.	
Study Goals	This course builds upon traditional control design tools already acquired by the students (such as pole placement, root locus or Bode diagram techniques, etc.), extending them to modern multivariable optimal and robust control techniques extensively used in industry and research. The path towards a robust flight controller goes through three steps: modelling, design and finally analysis. The modelling part of the course deals with uncertainty modelling that may stem from various sources such as poorly known aerodynamics, varying flight conditions or imperfect actuators and sensors. The linear fractional representation (LFR) concept for linear systems is introduced eventually leading to a detailed description of what we know about what we don't know. The design part of the course gradually moves from optimal control methods (such as LQR, LQG, LQG/LTR, H2) towards robust control methods (such as Hinf, Hinf by loop shaping) in a state or output feedback configuration. Controllers of predefined structure (such as PID, lead/lag, etc.) but designed using robustness considerations are also presented. The final analysis part of the course tests the controllers designed for the uncertain model of the system and evaluates in a systematic way their robust stability and performance. This is done via traditional tools such as multivariable stability margins but most importantly using the structural singular value (SSV) and the underlying mu analysis theory and tools. To illustrate the above concepts the longitudinal and lateral dynamics of a highly maneuverable fighter aircraft are considered and various control and stability augmentation systems are designed. The results and various simulations are done in MATLAB/Simulink.	
Education Method	Lectures together with assisted computer exercises in MATLAB/Simulink	
Books	Some books that may be used from the students for reference are: - Multivariable Feedback Control: Analysis and Design, Ian Postlethwaite and Sigurd Skogestad - Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems, Brian L. Stevens, Eric N. Johnson, and Frank L. Lewis - Robust Multivariable Control of Aerospace Systems, Declan Bates and Ian Postlethwaite	
Assessment	Take-home assignment	

AE4352	Mathematical and human-inspired decision making	3
Responsible Instructor	Dr. A. Jamshidnejad	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course you will learn about model predictive control (MPC) as well as human-inspired decision making (particularly, how fuzzy logic is used for decision making)	
Study Goals	After finishing this course successfully, the student should be able to: - Formulate model predictive control (MPC) problem for mathematical decision making - Develop fuzzy logic controllers for human-inspired decision making - Implement MPC controllers and fuzzy logic controllers for case study problems	
Education Method	Lectures	
Assessment	Group project (report + presentation of results)	

AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
Responsible Instructor	Dr. O.A. Sharfans'kykh	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Introduction. Agents and Multiagent systems. Agent-based modelling architectures. Examples from air transportation. Emergence in Multiagent systems. Agent-based simulation. Agent-based modeling and simulation tools. Agent-based coordination, planning, and scheduling in air transportation. Nature-inspired approaches to solve optimization problems. Swarm intelligence. Adaptive behavior and learning in agent-based systems. Collaborative decision making in air transportation. Negotiation, auctions, game-theoretic approaches. Agent-based model analysis: sensitivity, uncertainty, robustness. Validation of agent-based models.	
Study Goals	The student has to be able: - to formulate a practical air transportation problem as an agent or a multiagent system model; - to identify appropriate agent-based methods and to apply them; - to implement agent-based models; - to perform agent-based simulation, interpret and analyze simulation results; - to be able to apply agent-based optimization techniques.	
Education Method	Lectures and practical sessions	
Assessment	70% of the final grade for the course is the grade for a group assignment; 30% is the grade for the individual defense.	
Tags	Adventurous Algorithmics Analysis Artificial intelligence Group work Intensive Modelling Programming Small groups Transport & Logistics Transport phenomena	

AE4431	Aircraft noise and emissions	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Responsible Instructor	Prof.dr. D.G. Simons	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The aircraft noise part of this course is required for AE4463P 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics - Dutch aircraft noise model NRM - acoustic imaging Education method: The practicing is in the form of concrete assignments, which are individually graded.	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and tert's band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phono and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. (Acoustic imaging: Principle of beamforming. Imaging aircraft noise data.) Aircraft noise sources: Propeller noise (mechanism and directional properties) and blade passage frequency. Turbo engine noise and directionality of the fan and exhaust jet noise source. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise (and the modelling of it). (Aero-acoustics (Lighthill theory).) The topics in parentheses (acoustic imaging, airframe noise modelling and aero-acoustics) are part of the reader, but do not belong to this course and will be treated in the follow-up course AE4463P. Aircraft emissions and climate effects Aircraft emissions: Gas emissions from aircraft engines. Complete and actual combustion. Combustion products from burning kerosene. Emission Index. Carbon dioxide emissions of transport (aviation and, for comparison, road traffic). Atmospheric radiation physics: Concept of radiation flux. Solar spectrum and solar constant. Radiative properties of natural surfaces. Thermal emission. Black body radiation and the laws of Planck, Wien, Stefan-Boltzmann and Kirchhoff. Atmospheric transmission and emission. Effects of ozone. Climate effects: A climate model for the Earth. Greenhouse effect. Radiative forcing. Example calculations of global warming due to various carbon emission scenarios. Contribution of aviation. Feedback in the climate system.	
Study Goals	Understand the relationship between aviation and the environment.	
Education Method	Lectures and exercises.	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions Readers:	

	Course AE4431 Aircraft Noise and Emissions Part A: Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022 Course AE4431 Aircraft Noise and Emissions Part B: Introduction to aircraft emissions and climate effects Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2022
Assessment	Written exam (students may bring their own formula sheet to the exam)
Set-up	Lectures and exercises

AE4439	Air Transport Operations Modelling Project	2
Responsible Instructor	Dr. O.A. Sharpans'kykh	
Instructor	Dr.ir. B.F. Lopes dos Santos	
Instructor	A. Bombelli	
Instructor	Dr. M.A. Mitici	
Instructor	Dr. M. Lourenço Baptista	
Instructor	Dr. L. Li	
Instructor	Ir. P.C. Roling	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	All mandatory ATO courses.	
Course Contents	The course has a project format. It is meant to deepen knowledge and skills in mathematical and computational modelling in application to specific air transport problems. The course can be done under supervision of any ATO lecturer, who is willing to supervise such a project. Specific project themes can be very diverse, but should be in line with the research areas of ATO. A specific theme can be defined in consultation with a supervisor. It is advisable to work on a project in Q3 or Q4, when the basic modelling techniques have been learned in Q1 and Q2.	
Study Goals	n/a, depend on the project	
Education Method	The course has a project format.	
Assessment	A report needs to be written about the project. The content needs to be discussed with the supervisor.	

AE4446	Airport and Cargo Operations	4
Responsible Instructor	A. Bombelli	
Instructor	Dr. L. Li	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course consists of the following lectures: Lecture 1 Introduction Lecture 2 System planning Lecture 3 Airside capacity and runway design Lecture 4 Landside operations Lecture 5 Gate planning Lecture 6 Air cargo supply chain Lecture 7 Bin packing problems in air cargo operations Lecture 8 Cross-docking and ULD build-up scheduling Lecture 9-14 Guest lectures	
Study Goals	A short quiz is given after lectures 1 till 8. A bonus point for the test can be earned by taking part in these quizzes. At the end of the course, students will be able to: - Understand and critically assess key airport characteristics. - Analyze and describe an airport in terms of operations. - Understand the air cargo supply chain in terms of stakeholders and operations - Understand and reproduce mathematical models describing different operations occurring inside or around airports	
Education Method	The course consist of about 8 base lectures and 6 guest lectures (based on availability of guests). Students work in a group of 4 students to assess and reproduce airport/cargo related research and write a report. They also have to review about 3-4 other groups work. A closed book computer test is used to test and grade the individual knowledge of the students.	
Literature and Study Materials	Slides and other information and supplementary material are posted on Brightspace	
Books	Airport Systems: Planning, Design, and Management, Richard de Neufville and Amadeo Odoni, McGraw-Hill	
Assessment	40 % Individual computer test (must be 5.5 or higher) 40 % Group report (must be 5.5 or higher) 10 % Group review of other groups reports 10 % Homework quizzes	
Permitted Materials during Tests	Non graphical calculator	

AE4462-17		Aircraft Emissions and Climate Effects	4
Responsible Instructor	Dr. V. Grewe		
Instructor	Dr. I.C. Dedoussi		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics. Aircraft noise and emissions (AE4431) highly recommended.		
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories		
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.		
Education Method	Lectures and exercises		
Literature and Study Materials	Lecture slides and hand-outs		
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.		
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.		

AE4463P	Advanced Aircraft Noise Modeling and Measurement	3
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	AE4260 B (Experimental Applications of Aeroacoustics)	
Expected prior knowledge	AE4431 (Aircraft noise and emissions)	
Course Contents	AE4463P aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) will be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B and F are required. A choice can be made between C, D and E.	
Study Goals	The course aims at providing the students with an thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4465	Maintenance Modeling & Analysis	4
Responsible Instructor	Dr.ir. B.F. Lopes dos Santos	
Responsible Instructor	Dr. M. Lourenço Baptista	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Aircraft maintenance is one of the most important operational activities of the aviation industry. It covers all the planning and execution steps that ensure that the airline fleet is safe and in proper working conditions to fly, being a key part to achieving certification and compliance of air travel. This course aims to help students develop skills necessary to: 1 - Acquire the theoretical concepts necessary to model the maintenance needs of present and future aircraft 2 - Analyze data with both classical reliability methods and advanced predictive techniques in order to predict and forecast the present and future maintenance needs of aeronautical equipment On the successful completion of the course, the student should be able to: I Preventive maintenance aka Reliability engineering (classical reliability methods): 1. List, relate, and structure concepts from preventive maintenance 2. Apply life data analysis tools (e.g. Weibull distribution) to determine aircraft system and component reliability characteristics. 3. Apply stochastic demand arrival (e.g. point processes) models to determine and predict aircraft system and component reliability characteristics. II Predictive maintenance: 1. List, relate, and structure concepts from predictive maintenance 2. Select and apply forecasting techniques to model the reliability of complex systems 3. Select and apply predictive models (e.g. physics-of-failure, data-driven) models to address aeronautical problems that involve the estimation of the remaining useful life Apply simple models from physics to predict failure behavior Apply data-driven techniques from artificial intelligence such as neural networks (machine learning) to produce predictive results 4. Apply appropriate methodologies to evaluate a predictive maintenance solution (measures such as mean absolute error, root mean square error, scoring functions).	
Study Goals	This course has two main learning objectives, described below. 1. Apply concepts, models and strategies from preventive maintenance, including lifetime distribution models to model the reliability of non-repairable systems stochastic point processes to model the reliability of repairable systems 2. Understand and apply concepts, models and strategies from predictive maintenance, including forecasting techniques to model the reliability of complex systems machine learning models to model the reliability of complex systems evaluation techniques to assess the reliability of complex systems.	
Education Method	The content of the course will be introduced via on-campus lectures. The student will also have access to exercises and quizzes on Brightspace. There will be assignments to be solved in group or individually. The assignments may involve small coding problems, open questions, or mathematical exercises. The main bibliography for this course will be: The presentation slides of all lectures will be made available through Brightspace. Reference literature (covering the third part of the course): Goebel, K., M. Daigle, A. Saxena, S. Sankararaman, I. Roychoudhury, and J. R. Celaya. "Prognostics: the science of predictions." ISBN-13 (2017): 978-1539074830.	
Assessment	The final grade of the course will be composed of the individual exam grade (60%) and individual or group assignments (40% - 4 to 6 assignments). The exercises and quizzes on Brightspace are formative only. It is required that students need to have a grade equal to or higher than 5.5 in the individual exam and assignment grade to pass the course. Transition AE4426-19 -> AE4465 1. The course Stochastic Processes and Simulation, profile course for SAT, is cancelled. 2. For the profile courses, the replacement is Maintenance Modelling & Analysis (AE4465, 4 EC). 3. For students from the 2021 2022 cohort and earlier who have not yet completed AE4426-19, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4426-19. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4426-19 is not completed in 2022 2023, students will need to complete AE4465.	

AE4889	Special Topics in Astrodynamics	2
Responsible Instructor	Dr.ir. M.J. Heiligers	
Contact Hours / Week x/x/x/x	NOT OFFERED IN 2022-2023 - 0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	You will extend and deepen your insight and analytical abilities based on Fundamentals of Astrodynamics and related coursework, devoting particular attention to their application to a selection of astrodynamics topics.	
Course Contents	Selected topics in astrodynamics	
Study Goals	Within the context of the selected astrodynamics topics: Extend your insight and analytical abilities beyond the Fundamentals of Astrodynamics and related coursework; Actively apply your knowledge and skills accumulated through Fundamentals of Astrodynamics and related coursework; Tackle "open-ended assignments with a hands-on, scientific approach as a first foray into astrodynamics research.	
Education Method	Additional detail will be available on the course website (Brightspace).	
Computer Use	Workshops, lectures, self-study.	
Prerequisites	Yes.	
Assessment	- Fundamentals of Astrodynamics	
	Assignments.	
	Additional detail will be available on the course website (Brightspace).	

AE4893	Physics of Planetary Interiors	4
Responsible Instructor	Dr.ir. B.C. Root	
Instructor	Dr.ir. W. van der Wal	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course focuses on the different aspects of numerical modelling of planetary interiors. The interior of a planet or moon can be studied via observations of its gravity field, shape, surface features, rotation, and tidal deformations. To interpret these observations, the response of the bodies to different forces and heating scenarios need to be modelled. As a student you will get hands-on experience in modelling planetary and exoplanetary bodies with various numerical code packages. Different methodologies will be discussed, ranging from solving the Stokes equation for internal mantle convection to gravity forward modelling and how to solve certain loading scenarios with a finite-element code to the thermal evolution of a planet. You will be able to study a range of internal solid and fluid processes and interpret their surface manifestation. Lecture topics include: 1. Observations related to planetary interiors: Gravity field, rotation, tides, shape (topography, faults) - Example bodies and learn about different internal processes. You will learn how to calculate the internal gravity, density, and pressure of these bodies. 2. How to model fluid-solid mechanics of a planet? - Stokes equations in planetary science in spherical coordinates, rheology - Heat-transport: state equation exercise with different heat regimes - Mantle convection applications: dynamics and ocean flows 3. How to perform gravity field modelling - non-uniqueness, advanced isostasy/flexure models, density anomalies - Forward modelling density anomalies: spectral vs. volumetric methods - Inversion of lithosphere structure - Effect of mantle convection on the gravity field - Lessons learned from seismology on Earth 4. Tidal and loading deformation (Numerical code) - Effects of tidal potential (normal modes, FEM), dissipation - Loading cases: volcanism, meteor impact, ice loading - Surface faulting and the relation to stress, planetary seismicity, gravitational potential theory 5. Special topics on rotation of planetary bodies 6. Interior and planet evolution (combine all material) - Orbital resonance (external effects) - Change in thermal state and effect on tectonic regime (internal effects)	
Study Goals	After the course you will be able to: 1. Recognize the physical processes shaping planetary interiors and understand how they can be approached in a numerical study. 2. Apply fundamental physical laws (Stokes, Poisson equation, etc) in a schematic numerical modelling setup in spherical coordinates to study relevant problems of planetary evolution. 3. Able to operate and assess applicability of state-of-the-art numerical simulations of planetary interiors and their evolution 4. Validate and improve numerical models of planetary interiors with observations. 5. Critically review literature in planetary interior modelling and formulate new research questions in this context.	
Education Method	One lecture and work session per week 3 assignments focussing on different numerical methodologies Numerical modelling exercises will help you to get familiar with different numerical codes in the planetary science field. You will need to perform 3 assignments, which will be a selection of different numerical exercises connected to the lecture material. The (randomized) selection will be made by the course responsible. First numerical exercise: How to model a planetary body assignment (learning goal 1 and 2) Propagate state equation of an icy moon for different heating conditions Apply the heat transport equation to Mercury: conduction vs. convection Second numerical exercise: Gravity field modelling assignment (learning goal 4) Study best way to compute Bouguer anomaly of Mars: spectral vs. volumetric Perform multi-layered inversion of density anomalies in lithosphere. Third numerical exercise: Assignment using existing software to run simulations for one of the following (learning goal 3 and 5) Mantle convection: effect of viscosity on gravity field Meteor impact relaxation Tidal motion of surface Europa (thickness ice/water layer) Relation of Mars on Phobos rotation Diffusion modelling in evolution	
Course Relations	Complementary to other courses: PS1, PS2 and Measurement Strategies Overlap: After rearrangement of PS1 and PS2, there is no overlap anymore CIE4601 Physics of the Earth: rotation and other aspects of solid Earth. This course will be discontinued. CIE4610 Gravity, Geodynamics and Climate Change; Spherical harmonics and basics of potential theory are treated there, but not gravity forward modeling. Several courses at TU Delft discuss Navier-Stokes equation, FEM and inversion, however by focussing on spherical coordinates and spherical shell models we believe this course is much more suited for potential planetary scientists/engineers.	
Literature and Study Materials	Various peer-review journal articles Lecture notes Software and tutorial manual	
Assessment	Numerical assignments: 25% + 25% + 50%	

AE4895	Measurement Strategies for Planetary Science Missions	3
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice planetary analogue material you will try to find out how the data products and the scale at which it is applied, influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Lab assignments during structured laboratory activities (15%) Maintaining a lab journal (25%) Plenary integration session of your lab results with other students (10%) Final report, in form of short communication paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	- To understand the possibilities and limitations of various polymer synthesis techniques. - To know and recognise some of the industrial polymer chemical structures. - To be able to explain the impact synthesis and chemical structure has on polymer material properties & their selection for given applications. - To gain insight in the characterisation of polymers via various methods - To be able to estimate performance and processing based on polymer structures - To understand fundamental working principles of polymers in different applications such as aerospace and energy applications - Understand and elucidate the concept of sustainable polymers and biobased materials - To be able to give a convincing pitch talk about a recent development in the field of polymers.	
Education Method	Lectures	
Literature and Study Materials	lecture notes + book: R.J. Young and P.A. Lovell, Introduction to polymers, Chapman & Hall, London, 3rd edition, 2011, ISBN 978-0-8493-3929-5. Please note that the 2nd edition can no longer be used. The newer, 3rd edition, is a complete makeover! Optional/alternative reading: Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective	
Assessment	Written final exam and pitch talk (3 minutes) presentation Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM103		Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	The students must have general knowledge about materials science.		
Course Contents	(Organic) coatings can be defined as all kind of thin (organic) solid phases mechanically or chemically attached to a much thicker structure in order to add a certain functionality not obtainable from the substrate material itself. Color, mechanical, chemical and corrosion protection are obvious traditional functions of coatings but there are other less classical functions that are quickly growing in importance for several fields including aerospace. This course will introduce the basic principles relevant for traditional organic coatings science (e.g. adhesion, barrier, corrosion, mechanics) and those principles relevant for new developments in functional coatings. Special attention will be put on concepts applicable to aerospace field such as self-healing, active corrosion protection by nano and microparticles, anti-icing, self-cleaning, antifouling and drag reduction. Some of the concepts explained can be extended to other domains closely related to coatings such as adhesives and composites.		
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.		
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.		
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.		
Assessment	Written report (60% of final grade) and written exam (40% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.		

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/2/2/2	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Students of the track Aerospace Materials and Structures start in the second period (Q2, November). Students of the Wind energy master start in the third period (Q3, February). All other students start in the second period (Q2, November). Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four students about 14 day sections with support of staff are needed during this course. The planning of these day sections is depending (1) on the other obligatory courses of the track Aerospace Materials and Structures, (2) on the availability of supporting staff and (3) on the availability of the testing equipment. The ratio planned day sections by team over planned by staff is determined at the beginning of this course and notified and clarified at the kick off meeting held in the first week. For two parts of the exercise two consecutive day sections (so a full day) are needed. Finishing period Students started in the second period (Q2, November) are expected to finish at the end of the third period (Q3, April). Students started in the third period (Q3, February) are expected to finish at the end of the fourth period (Q4, June). IT IS EXPECTED THAT THIS LABORATORY COURSE WILL BE HELD AS BEFORE COVID-19. IF NOT ADAPTIONS WILL BE MADE. WHEN THAT DECISION WILL BE TAKEN IS UNCLEAR AT THE MOMENT OF WRITING (JUNE 2021).	
Course Contents	The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Possibly C-scanning of all laminates for determination of laminate quality 4. Estimation of the maximal bending-torsion coupling for a strip of UD composite 5. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 6. Performing mechanical tests 7. Analysing the test results of the mechanical test, including failure analyses 8. Adhesion of thermoset laminates 9. Resistance welding of thermoplastic laminates 10. Possibly preparation of lap shear specimens 11. Performing lap shear tests 12. Estimation of the mechanical properties of a sandwich panel 13. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 14. Performing a bending test on a sandwich panel. 15. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	For this course only one brightspace course will be available. It is called "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)". The Brightspace course "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q3)" should not be used, and is most likely not available for students. SO students starting in Q3 are added to "AE4ASM105 Trinity Exercise (20xx-20xx+1 Q2)".	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on forehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	In this course the student is introduced to advanced techniques in the analysis and stability of structures. Emphasis is placed on deep understanding of the models, their assumptions, and ranges of applicability. The topics covered include: Principles of structural analysis and stability: stresses, strains, equations of equilibrium, stability criteria Beam theory Stability of beams Plate theory Buckling of plates	
Study Goals	1- Understand the energy formulation of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy methods for the evaluation of structural response 4- Derive buckling load expressions for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	To be announced.	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (70% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open book exam 5 obligatory laboratory assignments done in groups (weeks 2.2-2.6). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Design & Analysis of Composite Structures I	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	- Classical Lamination Theory - Short Overview of Materials. Composites Design Philosophy - Theory of Elasticity - Engineering Constants - Stress & Strain Transformations / Implications for Testing - Thin laminates / Carpet Plots - Thick Laminates / ABD Matrix / Coupling Effects - Hydrothermal Analysis - Progressive damage analysis - Failure criteria - First ply failure & Last ply failure - Damage tolerance analysis - In-situ strength - Experimental mechanics & in-situ monitoring techniques - Basic Stress Solutions and Buckling of Composite Plates - Typical Airframe Elements - Airframe Design Process, Materials & Damage - Plate Governing Equations / Solution of the PDE - Energy Minimization Methods - Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: - Theory of Elasticity: 2 - Classical Laminar Theory: 4 - Progressive damage analysis: 4 - Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory and progressive damage analysis. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 20 points A2. Progressive damage and damage tolerance analysis 40 points Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method). B1. Design problem 40 points The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least	

be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. B. Çalar	
Instructor	D.M.J. Peeters	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: Have solid knowledge on the mainstream manufacturing processes for polymer composite aerospace structures. This comprises processing steps, tooling and equipment as well as advantages, limitations and applicability. Understand the relationships between the physics underlying specific manufacturing processes, the part/material quality and the design of the manufacturing process. Recognise manufacturing defects, to identify their potential sources and to advice on measures to prevent/limit their occurrence.	
Education Method	Interactive lectures with numerous in-class exercises. An assignment is used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade).	
	Transition AE4ASM110 -> AE4ASM110-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM110, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM110. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM110 is not completed in 2022 2023, students will need to complete AE4ASM110-22 instead.	

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don't's of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Instructor	Dr.ir. R.C. Alderliesten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	The study goals for this course are: - provide the student knowledge about airworthiness, roles of aerospace stakeholders, maintenance processes during operation, new developments in AC maintenance - the student should understand basic principles in damage tolerance and damage monitoring/inspection, aircraft maintenance - the student should be able to analyse maintenance problems from different angles (technical, regulations, planning). - the student should be able to provide well explained answers/solutions to questions and problems to structural integrity and maintenance issues	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam. Closed book.	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	Combination of Classroom/Online lectures and Practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. Parallel to that, there also exists an equivalent online version of the course on Brightspace.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	1. Written exam. 2. Take home assignment.	

AE4ASM508	Design of Self-healing materials	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr.ir. H.E.J.G. Schlagen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	Current engineering materials design is based in the well-known concept of damage prevention, which is the improvement of properties necessary to avoid damage. Nevertheless, another concept, employed by natural materials and systems for millions of years, can be applied to man-made engineering materials design. This concept is based on damage management and known as self-repair or self-healing. Under this concept the damage is unavoidable and the systems have to develop mechanisms to autonomously heal the damaged zone to keep the functionality of the system (e.g. skin wound healing). Up to now, material scientists have been able to implement the self-healing concept to polymers, metals, and ceramics in different fields as polymeric and inorganic coatings, high temperature coatings, civil engineering materials as concrete and asphalt, fiber composites, bulk polymers for multipurpose applications, structural aluminium alloys, and others. This course will deal with the self-healing concept, the self-healing materials and systems developed so far, the methods and concepts behind the development of this new class of materials, the techniques employed to evaluate the degree of healing, and the future goals within the self-healing materials field.	
Study Goals	The students will be able to: identify self-healing materials; describe methods to develop self-healing systems; understand the underlying physical and chemical processes behind healing; differentiate advantages and disadvantages of different self-healing approaches; learn how to measure, evaluate and analyze self-healing materials properties.	
Education Method	Lectures and research report. For the research report the students will work in small groups to answer a set of questions related to self-healing materials and their applications.	
Literature and Study Materials	The lectures will be based on the latest developments in the field supported by selected literature. The study material will be the lecture notes and articles facilitated by the instructors.	
Assessment	Oral Exam (40% of final grade) + report (60% of final grade). The report has an individual grade (70%) and a group grade (30%). A minimum grade of 5 over 10 in the exam is mandatory to pass the course.	
Special Information	Assessment may differ depending on the number of registered students.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	S.R. Turteltaub	
Responsible Instructor	Dr.ing. S. Giovanni Pereira Castro	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of structures commonly encountered in aerospace during operation. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM513	Forensic Engineering	3
Responsible Instructor	Ir. M.J. Schuurman	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is only for MSc. Aerospace Engineering students.	
Course Contents	In Forensic Engineering students will get acquainted with forensic principles used in air safety investigations. The principal phases of the investigation process are explained; starting from the fact finding phase through the analysis phase and the final report phase. Examples will be shown of aircraft (serious) incidents and accidents on a case-by-case bases, highlighting different (engineering) failure modes and characteristics. During the course students will learn how to conduct an (accident) investigation and apply forensic concepts to determine the (probable) cause of the event. Finally the drafting of recommendations and its implications are looked at. Apart from theory a couple of practical group exercises are part of the program. The group exercises are aimed to apply the principles learned during the lectures and attendance is mandatory. A report of the group exercises is part of the final grade.	
Study Goals	The goals of the course are: Describe and explain the accident investigation goal and identify and analyse the different investigation phases. Demonstrate and apply accident investigation techniques. Select and use forensic investigation techniques to determine failure causes. Have knowledge of constructing and testing hypothesis and the ability to go through a verification process. Write an Annex 13 accident report with fact, analyses and conclusion including the formulation of recommendations to prevent reoccurrence or diminish the consequence of future events.	
Education Method	The course principles are case and evidence based learning. The course aims at the capability to cooperate with various scientific and design disciplines in a specific case oriented context.	
Prerequisites	- Course is only for MSc. Aerospace Engineering students - AE4ASM513 is an elective for students MSc. program - AE1110-I completed or AE.1110x (MOOC) with certificate	
Assessment	Mandatory lecture attendance is required to do the final exam. The exam consist out of 3 parts: Group field examination Group accident report writing Individual component	
Enrolment / Application	As there is a limit to course participant it is therefore required to register Forensic Engineering as an elective (as soon as possible) through your track coordinator. Each research group has a upper limit of students who can participate. Please let your track coordinator know as soon as possible that you want to take this course and send email to the course coordinator (m.j.schuurman@tudelft.nl) to be registered. Please note; Brightspace registration does not mean you are eligible to take the course. Please also be aware of the course prerequisites.	
Set-up	Forensic Engineering has mandatory attendance - please be aware as it is not possible to sit-in the lectures.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM515-22	Characterization of materials and components	4
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof. C.A. Dransfeld	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.	
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various experts in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation (XRD, SAX, NMR) 4. Surface analysis of materials (XPS, AES, AFM) 5. Electrical and electrochemical characterisation of metals and ceramics (Electrochemical Impedance Spectroscopy, dielectrics) 6. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 7. Mechanical characterization (tensile, fracture)	
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important in case of premature failure or degradation or in sub-or above standard material performance. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.	
Education Method	Lectures and assignments	
Literature and Study Materials	Introductory texts and review articles on the various techniques posted on Brightspace.	
Assessment	Oral Exam Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.	

AE4ASM516	Material Selection in Mechanical Design	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course aims at training the students in the field of material and process selection on the basis of objective and quantitative criteria. The concept of material selection on the basis of so-called Ashby diagrams will be presented in a number of lectures.	
Study Goals	Providing the student with the tools to make material and process selection on the basis of objective and quantitative criteria.	
Education Method	Lectures	
Literature and Study Materials	The book Materials Selection in Mechanical Design by M.F. Ashby. Butterworth Heinemann. 5th edition 2017	
Assessment	Written exam	
Set-up	The course will consist of a number of lectures in which the quantitative selection process involving an increasing number of constraints and objectives will be introduced. During the lectures students are invited to solve a number of simpler cases.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Ir. J. Sinke	
Instructor	Dr. J.A. Poulis	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December 2022 - Week 4.8: 15 May 2023 - Week 1.6: 30 September 2022 The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and	

Set-up	3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility. - Apply quantitative life cycle assessment methods (LCA) to analyse and evaluate industrial composite manufacturing systems	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Special Information	Assessment may differ due to COVID-19 measures. In the addendum of the regulations for 2020-2021 (see: https://www.tudelft.nl/studenten/faculteiten/lr-studentenportal/onderwijs/master/regulations-msc/) are the regulations described for alternative assessment. Changes to the assessment method will be communicated via Brightspace.	
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit. This only applies to the academic year 2021-2022.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 4.1, 4.3, 4.6, 4.8); each day will consist of four lecture hours; weeks will be announced in advance of the course. At the beginning of the course students will group into teams of four to six persons. Each group will design a composite manufacturing process for a given part in the duration of the course. Self-organized work is expected of the teams especially in the weeks without lectures. The results of the team assignment will be documented by the members of each team in a written report and will additionally be presented by each team in the final lecture of the course.	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr. J. Wu	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Dr.ir. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of new phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Introduction of industrially oriented aeroelastic modelling, loads and certification. The course aims at placing the aeroelastic knowledge into a fully flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced aeroelastic phenomena such as nonlinear aeroelasticity and aeroelastic control. 2. Understand which regular and advanced aeroelastic concepts are important for which type of aircraft. 3. Know how to model advanced aeroelastic phenomena in an academic and industrially oriented context. 4. Understand the role of aeroelasticity in the loads process and the design of aircraft.	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. G. Dimitriadis, Introduction to nonlinear aeroelasticity, 2017, Wiley, ISBN:9781118613474	
Reader	Course notes	
Assessment	Take home assignment with oral close-out session.	

AE4ASM523	Design of Spacecraft and Launcher Structures	3
Responsible Instructor	Prof. C. Bisagni	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Introduction to spacecraft and launcher structures 2. Launch vehicle, re-entry, satellite and deployable structures 3. Environmental requirements for spacecraft and launcher structures 4. Analysis under mechanical and thermal loads 5. Static and dynamic buckling phenomena for launcher structures 6. Typical development and qualification tests	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the spacecraft and launcher structures, their principles, design and analysis methods and tools. Successful students will have achieved the following objectives: They will have gained an advanced understanding in terms of the technologies, principles, methods and tools required to conceive, design, analyse and test spacecraft and launcher structures. They will have obtained capability to transform fundamental technical, economic and societal requirements into an integrated structural solution. They will have established credibility and depth in a discipline focused on current and future technology.	
Education Method	Lectures	
Assessment	To be announced	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4ASM523 Design of Spacecraft and Launcher Structures - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of seven weekly modules. In these modules the following topics will be covered: - Introduction - Thermo-Elastics - Vibrations - Systems Engineering - Verification and Validation The order of the subjects may differ from the above list. The course structure will be explained in detail in the first weekly module. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. The weekly modules will be complemented by a hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students. Time and location of the hang-out sessions will be communicated in the first lecture week. Students are requested to reserve all scheduled timeslots in their agendas. Note that, the course will require group work. Groups will be formed in the first week of the course. Students who are not part of a group cannot participate in the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. The weekly modules will be complemented by hangout-sessions, during which it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Computerized test on course content 2. Written feedback on design specification sheet of another group in the course Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, "Introduction to materials science" or books of similar nature. Furthermore, the students are expected to have followed the course AE4ASM002 "Designing materials for aerospace relevant conditions"	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, Tianwen-1, Hope, ISS, Hubble telescope, James Webb telescope, lightsail-2, Neowise, solar cruiser) will be presented as a carrier for material selection analysis. Current material choice will be analysed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The students will be trained to translate desired properties into material structures and microstructures and to think about suitable material production processes to realize these properties. Students are encouraged to propose alternative materials and reason for their choices. The structure of the lectures will be tailored to maximize the student involvement. Student involvement during and in preparation for lectures will be taken into account for the final grades. The course contains the following topics: week 1: Mars exploration history from Missions to Materials week 2: Material degradation in space - thermal cycling and vacuum week 3: Material degradation in space - UV, AO and space radiation week 4: Material degradation in space - meteoroids and orbital debris week 5: Space mission energy supply week 6: In-situ resource utilisation week 7: Materials testing for space applications (ESA guest speaker) week 8: Q and A week 9: exam week	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 7 lectures of 2h per week, and 7 self-exploratory studying sessions both before and after the lecture takes place. Each week one topic will be covered (shown in table below).	
Literature and Study Materials	A compilation of lecture notes and review papers	
Assessment	Continuous assessments (60%) + take home open book exam (40%). The students are to write three essays on a set of prescribed question/topics with submission deadlines at the end of week 2, 4, 6. The final grade will be the weighted average of reports and final exam. Example questions/topics will be made available on Brightspace. Continuous Assessments (60%) Assignment 1: Essay about relevance of materials research in space missions 15% of the marks Assignment 2: Essay about materials degradation and failure in space missions 20% of the marks Assignment 3: Essay about material design in hypothetical scenarios "think like a materials scientist" 25% of the marks Take-home open-book Exam (40%) a mock exam will be provided on brightspace honor code to be observed DESCRIPTION OF THE ASSIGNMENTS Assignment 1 Theme: Relevance of materials research in space missions (Please include word count after each sub-question) 1. What are the space missions your group would like to analyse? And why you chose these specific missions. (word limit: 50~100 words per mission) 2. What are the space conditions for this particular mission? Please specify at least temperature range, atmospheric condition, radiation and add other aspects that apply. (word limit: 150~200 words per mission) 3. What materials did you find on this space mission both for structural and also functional purposes? (word limit: 300~400 words per mission) Assignment 2 Theme: Materials degradation and failure in space missions (Please include word count after each sub-question) 1. How do you see the materials will degrade in both missions? Please list at least 3 degradation mechanisms and give your reasons (word limit: 500~600 words) 2. What information can you find about actual material failure? How does that differ from your prediction and why is so? (word limit: 500~600 words) Assignment 3 Theme: Think like a materials scientist (Please include word count after each sub-question) 1. What type of materials or specific material properties are needed if the condition of the mission is adjusted for higher radiation requirement, extreme thermal conditions (hot, cold) etc? Please list at least 3 hyperthethic scenarios (word limit: 500~600 words) 2. What structure-property relationship of materials can help you achieve the situation in question 1? (word limit: 500~600 words)	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Both passive as well as active thermal control systems are discussed. Passive devices include: radiators, baffles, heat pipes, multi layer insulation, coatings and finishes. Active devices include: louvers, controlled heaters, cryogenic coolers, Peltier elements and fluid loops.	
Study Goals	The student is able to generate a thermal design of a satellite thermal control system using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - distinguish design hardware options - generate a (concept of a) thermal design - generate analysis models for hand calculation methods and/or computer tools - analyze a thermal design using hand calculation methods and/or computer tools - distinguish thermal environmental test methods	
Education Method	Lectures + homework Lectures are given by an experienced thermal design engineer from Airbus DS NL. Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Asnwers to the questions may be given by fellow students and/or the lecturer(s).	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%)	

AE4S01	Thermal Rocket Propulsion	4
Responsible Instructor	Ir. B.T.C. Zandbergen	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Propulsion material contained in the BSc courses AE1222-II and AE2230-I. Course material will be made available via Brightspace.	
Course Contents	The course focuses on thermo-(chemical) rocket propulsion system analysis and design. Topics dealt with include: 1. Fundamentals of (thermo-chemical) rocket propulsion; 2. Ideal rocket motor/nozzle: Ideal performances, optimum thrust, characteristic velocity and thrust coefficient, and quality factors; 3. Nozzles: Types of nozzles (conical, bell, etc.), nozzle dimensions, flow divergence, boundary layers, under- and over-expansion, and Summerfield criterion; 4. Chemical propellants: Molar mass, specific heat ratio and adiabatic flame temperature calculation for gas mixtures (based on known reaction equation), mass density, dynamic viscosity, thermal conductivity; 5. Chemical equilibrium calculations, hemical equilibrium flow, frozen flow and chemical kinetics; Introduction to program for calculation of chemical equilibrium gas composition and gas properties (in tutorial); 6. Heat transfer & cooling: Convection, radiation and conduction; 7. Cooling: Thermal insulation, ablation, radiation, film, dump and regenerative cooling; 8. Liquid rocket engine combustor design: Steady state internal ballistics, liquid injection, operating pressure, chamber pressure drop, and characteristic length; 9. Solid rocket motor combustor design*: solid regression, grain shape and internal ballistics including operating pressure, necessary condition(s) for stable operation, pressure sensitivity for initial temperature and change in Klemmung, local conditions (flow velocity, pressure, etc.)), and two phase flow; 10. Hybrid rocket motor combustor design*: Solid regression, grain shape, (quasi-)steady operation (operating pressure, and local conditions (flow velocity, pressure, etc.)); 11-13 Liquid propellant storage and feed systems including gas-pressure and pump fed systems, motor cycles and propellant distribution; 14. Capita Selecta * Only one of the two will be dealt with. This varies from year to year.	
Study Goals	At the end of this course, the student shall be able to perform important steps in the analysis and design of thermo-(chemical) rocket propulsion systems using basic methods that allow for taking into account fluid flow, heat addition, propellant thermo-chemistry, heat transfer and cooling, liquid, solid or hybrid ballistics, (liquid) propellant feeding, and propellant storage.	
Education Method	Lecture + exercises (for feedback); During the lectures extensive use is made of existing data to allow for verification of the methods presented.	
Literature and Study Materials	Lecture notes "Thermal Rocket Propulsion" + hand-outs. Supporting information and tools are available through Brightspace. Recommended literature: G.P. Sutton, Rocket propulsion elements, Wiley Inter Science, 6th edition (or higher).	
Assessment	Assessment is through a number of small homework assignments (20% of final grade) and an hybrid (written and online) exam (80% of final grade). In case of a fail grade there is one resit exam opportunity that makes up 100% of the final grade (no homework assignments). Resit exam, like the normal exam, is a hybrid exam and will be scheduled in Q3 (exam period).	
Remarks	Enrolment for the course is via Brightspace. The course will be made available on Brightspace from early September.	
Set-up	Lectures (14 x 2 hours) possibly with 1 tutorial on the use of RPA-Lite and/or NASA CEA for rocket motor performance estimation.	

AE4S10	Microsat Engineering	4
Responsible Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Lectures from experts on various aspects of small satellites and group/individual projects for practicing.	
Course Contents	The course covers the engineering of small satellites. The concept and definition of micro-, nano-, pico-, and femto-satellites is introduced along with the past, present, and future of small satellites in general. The course comprises seven to eight lectures, partly from professionals in the space industry. The lectures are intended to familiarise the students with current industry trends and state-of the art research in small satellite engineering. The lectures vary each year and potential lecture topics include communications, AOCS, onboard computer, distributed space systems, miniature payloads, small satellite missions, etc.	
Study Goals	The course will enable students to design and engineer small satellite missions. The high-level learning objectives of the course are: The students shall be able to - explain key aspects of small satellite missions - explain and describe elements in the architecture of a spacecraft mission - discuss in detail at least one element in the architecture of a spacecraft mission - present and defend their design of an element in the architecture of a spacecraft mission - show familiarity with current trends in space industry and small satellite research	
Education Method	Lectures (Provided partly by professionals from the space industry). Group work (Students work on assignment in groups; Coaching will be provided from experts to guide the students). Individual work (Students work on small individual assignments).	
Literature and Study Materials	Optional: Wertz J.R., Larson W.J. (eds.); Space Mission Analysis and Design; 3rd edition Kluwer Academic Publishers (2003). Mandatory: Lecture Notes AE4-S10 MicroSatellite Engineering.	
Prerequisites	General mathematical, physical and engineering knowledge from completed BSc study. In particular, AE1101 + AE1102 + AE3201, or similar	
Assessment	There will be two assessment components A Group Assignment and an individual assignment. Group assignment The final demonstration or report will be graded on the following aspects: - Planning - Presentation, argumentation and defence of the design - Functionality and Usability of the deliverable - Context of Small satellite (How well has the student team incorporated current small satellite research and industry trends/needs in the design) - Independence - Creativity Individual assignment This assignment will be graded on the following aspects: - Understanding and critical review of articles in a specified technical field of small satellites	
Permitted Materials during Tests	N/A	
Enrolment / Application	Students have to enroll via the BrightSpace site of the course. Professionals are welcome to the course. Conditions for participation and registration are available with the secretary of the Chair of Space Systems Engineering (www.sse.lr.tudelft.nl).	
Set-up	The course comprises lectures, guest lectures, team sessions, group work and individual work.	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Responsible Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a multidisciplinary Master course at TU Delft/Faculty of Aerospace Engineering (AE). In CSDP students of primarily the Spaceflight track work in groups (6-12 students) on a space mission/spacecraft design identified by interested parties (this may be a space company, space institute, or other) for a total of 112 working hours (4 EC) over a period extending 2-3 months under supervision of faculty staff. Additionally, students spend maximum 28 hr (1 EC) on workshops that provide information and training on the use of a typical process management tool like SCRUM (and/or Kanban) and a Concurrent Design Platform / tool (like e.g Excel and/or CDP-4). The course aims to develop student skills in multidisciplinary team projects from a challenge-driven perspective. For the students, CSDP brings forward unique characteristics such as: a starting point in a value-driven approach to problem formulation and decision-making, concurrent and collaborative systems engineering to solve engineering problems thereby taking into account realistic constraints and allowing for training of interpersonal skills and/or workplace competencies (teamwork, planning and organizing, creative thinking, etc.). Topics vary year by year and depend on interest of engineers & researchers to define great engineering challenges for a motivated team of students.	
Study Goals	- Students get advanced experience in the use of Concurrent Engineering (CE), as opposed to the more classical sequential engineering (SE), in a design project. Students will learn not only about how to apply CE, but also about the benefits of CE (as opposed to SE, i.e. the waterfall method) and its limitations; - Students will advance their ability in systems engineering/collaborative while also improving their ability in cooperation, collaboration and communication.	
Education Method	Students will perform a design project as part of a multidisciplinary team. Each group is guided by a mentor. Parallel to the design project, students are instructed on the use of CDP4 and SCRUM (and/or Kanban). For each team regular design sessions will be held in a collaborative learning environment. In these design sessions, teams are given the opportunity for brainstorming (requirements generation, creating first solutions), demonstrating progress, setting tasks, and reviewing the design accomplishments. It is the purpose that each team organizes themselves a team of experts (from industry, academia or other) that are willing to advise the team in their project. At various instants, the work accomplished will also be externally reviewed by other CSDP teams as well as staff from the faculty and external experts with as aim to identify strong/weak points in the design.	
Assessment	The design projects are graded based on a design report, a design log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic (e.g. communications, modelling/experimenting and critical and analytical thinking) and personal effectiveness competencies.	
Enrolment / Application	There is no free enrollment for this course. In Q1, an information session will be held informing students of Space Engineering on the projects available and on how they can apply for a project. Students can then apply and only if they are accepted, they can enroll.	

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: describe the components and configurations of wind turbines describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. identify the drivers of dynamic behaviour and calculate their typical values describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Special Information	Assessment may differ due to COVID-19 measures. Changes to the assessment method will be communicated via Brightspace.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	W. Yu	
Gast Instructor	Dr.ir. A.C. Viré	
Gast Instructor	Prof. R. Beuker	
Gast Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MANDATORY: AE4ASM506 (Aeroelasticity). The student should have completed or is attending this course. An equivalent course can be accepted upon discussion with the instructor.	
Course Contents	The course will cover main aerodynamic and structural phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. Possible resonances on wind turbines. What are the excitation frequencies, what are the natural frequencies? 2. What are wind turbine aeroelasticity tools based on? Linearised tools in frequency domain and nonlinear time domain simulation tools. How accurate are these tools and where are inaccuracies coming from? 3. Wind turbine mode shapes, whirling modes. 4. Wind turbine instabilities. 5. Dynamic inflow and dynamic stall.	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand which aeroelastic principles are important for wind turbines. 2. explain the contribution of the different wind turbine components on the resulting time responses. 3. apply the knowledge to program their own models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy).	
Course Contents	Note: Without some affinity/experience with python (and numpy) programming it might still be possible to pass the course, but will become quite difficult. In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics, following the book; building the backbone. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ Research papers will be made available through Brightspace.	
Assessment	Assignments are based on PyTorch: https://pytorch.org/ 1. 0.1 weight: Presentation: during the seminar a small group of students presents a paper. You will have to present once. 2. 0.9 weight: Exam about the papers, assignments, and the theory. 2. Pass/Fail: Project: in a small group of students you work on a deep learning paper reproducibility project. Final grade: (pass project) * (0.1 * Presentation + 0.9 * Exam) *** Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) ***	

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam will be multiple choice. The logistics will depend on the Corona regulations.	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Bachelor program, preferably electrical engineering	
Course Contents	The course gives an overview of different types of electrical machines and drives. Different types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	Written exam (100% Weightage), Closed book, Scientific (non-graphic) calculator can be used, No formula sheets Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	In order to participate in the written exam, students need to complete the three half day lab practicals. During corona period, three remote practicals are offered that are done individually. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments, computer demonstrations as well as instruction. In 2021, all lectures will be online. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction/office hours, during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. Weekly instruction is likely to be on campus.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45025	Introduction to Multiphase Flow	5
Responsible Instructor	Dr.ir. W.P. Breugem	
Instructor	Prof.dr.ir. R.A.W.M. Henkes	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are gas and oil transport, dredging, steel making, refineries and chemical plants, sediment transport in rivers, cloud formation, etc. This course gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of gas/liquid two-phase flows, spatial/statistical averaging of the Navier-Stokes equations and continuum modelling, flow pattern maps and mechanistic modelling of multiphase pipe flows, physics of particles (droplets, bubbles, solids), dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.	
Study Goals	At the end of this course the student is able to: 1.a classify multiphase flows into separated and dispersed multiphase flows. 1.b formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface. 1.c formulate relevant dimensionless numbers for multiphase flows. 1.d explain the concept of dynamic similarity of multiphase flows. 1.e reflect on scaling choices when designing a laboratory experiment or setting up a numerical simulation. 2.a apply a linear stability analysis to two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability). 2.b give physical mechanisms for the transition of stratified to non-stratified (bubble or slug) multiphase flows in both (tilted) horizontal and vertical pipes. 3.a apply averaging methods (such as statistical, spatial and temporal averaging methods) to the Navier-Stokes equations for describing the macroscopic behavior of multiphase flows. 3.b formulate appropriate closure models needed to solve the averaged Navier-Stokes equations. 4.a classify gas/liquid pipe flows into stratified, slug and bubble flows, dependent on the superficial gas and liquid velocities. 4.b apply simplified 1D (cross-section averaged) models to each of the flow regimes. 4.c formulate appropriate closure models in order to compute the streamwise pressure drop and liquid holdup in two-phase (gas/liquid) pipe flow. 5.a classify dispersed multiphase flows into flows with solid particles, gas bubbles and/or liquid droplets. 5.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets. 5.c apply the Newton-Euler equations of motion to an analysis of the dynamics of solid particles. 5.d describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio. 5.e describe possible interactions between solid particles and turbulence of the carrier phase. 6.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Haberman-Morton number. 6.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag. 6.c give physical mechanisms for the breakup of bubbles/droplets in turbulence. 7. describe the dynamics of some particle-laden flows such as sediment transport through a pipe and/or sediment transport in rivers. 8. describe the merits and shortcomings of different experimental methods for multiphase flows. 9. describe the merits and shortcomings of different numerical methods for simulating multiphase flows. 10.a give examples of multiphase flow in industrial processes. 10.b describe the types of models used to design multiphase industrial processes.	
Education Method	Oral lecture 0/0/2/2; Plenary discussion of homework assignments 0/0/2/2	
Assessment	Written exam. The student can earn bonus points by making homework assignments. The final grade is the sum of the grade for the written exam and the bonus points from the homework.	
Department	3mE Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	3mE Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	At the end of this course you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	Brightspace/MATLAB exercises and tests (optional) during education period. The course is concluded through an optimization project and report. Deadlines for handing in the final report will be announced at the start of the course.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

MS43310	Materials at High Temperature	4
Responsible Instructor	V. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Department	3mE Department Materials Science & Engineering

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of control theory	
Course Contents	Theory and application of intelligent control system i.e. the combination of machine learning and control theory. Topics: * Introduction to nonlinear control * Introduction to intelligent control: model learning, control learning * Fuzzy sets and systems * Artificial neural networks, deep learning (Reservoir Computing, CNN, Geometric Deep Learning) * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of real-world applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely fuzzy logic, iterative control, Gaussian processes, artificial neural networks, etc. LO2 Implement 'intelligent control' techniques in MatLab and Simulink or Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Group work Lineair Algebra Mathematics Matlab Modelling	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42035	Integration Project Systems and Control	5
Responsible Instructor	D. Boskos	
Instructor	S.C. Bregman	
Instructor	Dr. L. Laurenti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	SC42015 Control Theory SC42155 Modeling of Dynamical Systems SC42095 Control Engineering SC42025 Filtering and Identification	
Course Contents	The course is based on practical laboratory sessions, in which students will gain hands-on experience with the application of control theory to real-world systems. Matlab and Simulink are used as the primary software environment for the design, analysis and real-time implementation of the algorithms. Students work in groups of two in the lab, with a setup of their choice. The setups include for instance the inverted pendulum, helicopter model, inverted wedge, rotational pendulum, flexible link, magnetic levitation.	
Study Goals	The goal of this course is to integrate and apply the theoretical knowledge gained in the compulsory courses of the Systems and Control curriculum. These course cover the subjects of Control Theory, Physical Modeling, Filtering and Identification. The concepts and tools to be used include mechanistic modeling (based on principles like mass balances, Lagrange equations, etc.), filtering and estimation (e.g., Kalman filtering), linear control design and performance analysis, system identification in open and closed loop. It is assumed that students already know these concepts or are able to look them up in the literature. No theoretical lectures are given in this course.	
Education Method	Project, demonstration	
Assessment	There is no written exam. The final grade is determined on the basis of a written report (50% of the grade), the final discussion of the results with the lecturers (50%) and the performance during the entire course period (rounding off the grade).	
Remarks	Old course code: SC52035	
Department	3mE Department Delft Center for Systems and Control	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (WI3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (WI3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R² and in R³. Vibrations of membranes. Bessel functions. Shock waves. Lectures and Maple practical work (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Remarks	The contents of WI3150TU + WI3151TU is the same as TW2070.	
Co-Instructor	Dr.ir. W.T. van Horssen	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R ² and in R ³ . Vibrations of membranes. Bessel functions. Shock waves. Lectures and Maple practical work (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and practical work.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Practical work and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Remarks	The contents of WI3150TU + WI3151TU is the same as TW2070.	
Judgement	The practical work is mandatory, and as long as the practical work is not completed no final grading will be given.	
Co-Instructor	Dr.ir. W.T. van Horssen	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports.	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	- The student is able to recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - The student is able to solve the shortest path problem and the transportation problem as well as some small flow shop problems. - The student is able to solve the travelling salesman problem by the Branch and Bound algorithm. - The student knows several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - The student knows some basic theorems concerning the mentioned problems and is able to prove some of these theorems. - The student has knowledge about methods to solve large scale problems, especially shortest path and vehicle routing problems. - The student has basic knowledge about scheduling problems and complexity theory - The student is able to implement several of the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam.	
Tags	Optimisation Transport & Logistics	
Co-Instructor	Dr. K.S. Postek	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The assessment consists of three parts: homework exercises deadline start of Q2 leads to grade G1, take home exam deadline half of January grade G2 and a written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5.	
Co-Instructor	disclaimer: information may change depending on the developments around the coronavirus. Prof.dr.ir. M.B. van Gijzen	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High-resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	In the first period a set of exercises is given. These should be worked out, strict deadline 26-3-2019 (Grade: G_1). The exercise should be handed in individually. In the middle of April a take home exam including practical exercises is given. This exam can be done by groups of two students. The report of this exam should be returned to us before 11-6-2019 (Grade: G_2). Thereafter an appointment can be made for an oral examination over Chapter 18 (including exercise 18.1, 18.2, and 18.3), 19, and 20 (Grade: G_3). Note that material discussed in Chapter 1 to 9, which is relevant for the material in Chapter 18 to 20 can also be part of the oral examination. The oral exam must be passed with a 6 or higher. Then, provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: $(G_1 + G_2 + 2 G_3)/4$. disclaimer: information may change depending on the developments around the coronavirus.	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). Final project (group work) starting mid Q2 with submission deadline end Q2 / beginning Q3 (2/3 of final grade).	
Co-Instructor	K. Cools	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

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